

Intellectual Property and E-commerce Security in Cyber Laws

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David Padua

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TABLE OF CONTENTS

Preface

xi

Chapter 1	Developing Cyber Law	1
	An Analogy for Cyber Law	1
	Spontaneously Evolving Institutions in Cyberspace	3
	Substantive Criminal Law	4
	Illegal Access (Hacking)	4
	Data Espionage	7
	Illegal Interception	9
	Data Interference	12
	Restructuring the Virtual Space	15
	The Cyber Law Platform	17
	Cyber Laws and Legal Issues	18
	Jurisdiction and Sovereignty	18
	Net Neutrality	19
	Free Speech in Cyberspace	19
	Governance	20
	Conclusions: Potential and Applicability	22
Chapter 2	Intellectual Property	24
	Introduction	24
	History	24
	Types	25
	Objectives	27
	Financial Incentive	27
	Infringement, Misappropriation, and Enforcement	29

Criticisms	30
Objections to Overbroad Intellectual Property Laws	32
Access Knowledge of Intellectual Property	33
The Historical Evolution of Enclosure and A2K	34
Frame Mobilization in IP Industries	43
Frame Mobilization in A2K	48
Expansion in Nature and Scope of Intellectual Property Laws	52
Chapter 3 Patents, Trademarks and Copyrights	54
Patents	54
What is a Patent?	54
The Indian Patents and Design Act of 1911 and Its Amendments	58
Coverage	58
Trademarks	60
Justification	63
History	63
Scope of Copyright	65
Obtaining and Enforcing Copyright	66
Exclusive Rights	68
Limitations and Exceptions to Copyright	68
Transfer and Licensing, and Assignment	70
Duration	72
Copyright Infringement	73
Chapter 4 The Evolution of Property Rights in Cyberspace	74
The Security of Property Rights in Cyberspace	75
Factors Influencing the Evolution of Property Rights in Cyberspace	76
Externalities and the Evolution of Property Rights	81
The Security of Property Rights	84
Assurance Problems and Sources of Credibility in Contracting	86
Evolving Property Rights in Cyberspace	98
Chapter 5 Intellectual Property Rights and Agreement	104
The Origins of Intellectual Property Rights	104
IPR Protection Score Analysis	105
Static Inefficiency	109
Duration of Intellectual Property Rights	112
Scope of Intellectual Property Rights	113
Licensing	113
Transition Period for Less Developed Countries	113
Dispute Settlement Mechanisms	114
Reinvigourating Multilateralism	118

The TRIPs Agreement on International Trade	119
Countering Views on the Effects of the TRIPs Agreement	121
On the Economies of Developing Countries	123
Empirical Studies	123
International Agreement on Protection of Intellectual Property Rights	125
The TRIPs Agreement	128
Views on the Effect of the Trips Agreement	130
The Economic Status of Developing Countries	130
Theories on the Key to Economic Growth	131
The Ideology of Private Rights I Intellectual Property	133
IP and Personality	137
The Utilitarian Defence	138
The Structure of the TRIPs Agreement	141
Basic Principles and Obligations of the TRIPs Agreement	142
Standards	142
Copyrights	142
Trademarks	144
Other Standards	146
Patents	147
The Weakening of the Patent Regime	151
Chapter 6 Cyber Security as Science	152
Security Degrades Over Time	153
UNIX Passwords	153
Cyber Security: A Problem of National Importance	155
Trusting Systems in a Dangerous World	155
The Information Technology Infrastructure is ‘Critical’	155
Ubiquitous Interconnectivity = Widespread Vulnerability	156
Software is a Major Vulnerability	158
Attacks and Vulnerabilities are Growing Rapidly	158
Endless Patching is not the Answer	159
Fundamentally New Security Models, Methods Needed	160
Central Role for Federal R&D	161
Cyber Security Threats in Counter- Attacks	162
History of Cyber Security	163
Cyber-security Risks	166
A Network Model for Cyber Security	166
Promoting Automation of Security	168
Computer Security	169
Use Security Software that Updates Automatically	169
Treat Your Personal Information Like Cash	169
Check Out Companies	170

Personal Information	170
Protect Your Passwords	170
Hardware Mechanisms that Protect Computers and Data	170
Computer Security Risks to Home Users	170
Computer Security Basics	171
Network Security	177
Brief History of Internet	177
Security Timeline	177
Security in Networks	178
Network Concepts	179
The Cyber Security Strategy	193
Protection of Critical Information Infrastructures	193
Strengthening IT Security in the Public Administration	194
National Cyber Response Centre	194
Cyber Protection/Cyber Security Liability	195
Theft of Digital Assets	195
The Future of Cyber Security	196
The Global Online Population Expands	197
Chapter 7 Jurisdiction and Sovereignty	200
Net Neutrality	201
Free Speech in Cyberspace	202
Legal Aspects	202
New Dimensions and Legal Reforms	203
Competition Law	203
Principles of Jurisdiction	206
Effectiveness of Judicial System	216
Chapter 8 E-commerce: Contracts, Legal Issues and Agreements	221
Introduction	221
Legal Issues in E-Commerce	222
E-Commerce Global Strategies	222
The Risks of Global Web Marketing	224
Changing Patterns in Global Distribution	224
Measuring E-Commerce Success	226
Measurement Practices in e-Commerce	229
Customer Relationships	231
Information Practices in E-Commerce	233
Architecture of E-Commerce Systems	235
Internet, E-Commerce Systems and Information Systems	236
Databases	236
E-Commerce Systems and the Customer	236

Web Site Evaluation Software	237
Internet Service Providers	237
Example of E-Commerce Architecture	237
E-Commerce Security Technology	238
E-commerce Contracts	239
Unfair Terms	241
The Distance Selling Regulations	241
The Cooling-off Period	242
Web Site Linking Agreements	243
Structure of E-Commerce	245
Strategy of E-Commerce	248
Development of E-Commerce	251
Three Approaches to E-Service	252
British Library	255
A Second Taxonomy of E-Service	255
The Self-Service Myth	255
E-Service Performance	256
A Critical Incident Approach to E-Service Performance	257
Positive Critical Incidents	257
Negative Critical Incidents	258
Logistic Regression of High-Low Service Sites	259
E-Service Metrics: A Management Tool	260
E-Retail Service Quality: An Alternative Performance Metric	261
Additional Guidance on Practical E-Service Provision	262
Chapter 9 The Evolution of Cyber Crime	263
Evolution of Motivation	264
Categories of Cyber crime	265
Cyber Crime Tools	266
Typology of Cyber crime	268
Economic Impact of Cyber crime	269
Results of Selected Surveys	269
Difficulties Related to Cyber Crime Statistics	270
Evolution and Profile of the Attacker	271
Malware and Threat Evolution	271
Evolution of Exploit Frameworks	274
Defence Evolution	274
Cyber Victims	275
Current Situation	276
Consequences for Developing Countries	276
International Cyber Crime	276
Future Trends in Legislation	278

The Emerging Threat of Internet Bots	279
Profiling and Target Selection	280
Attack Techniques	281
Criminal Use of Bots and Botnets	282
Defending against Bots and Botnets	284
Content-related Offences	286
Erotic or Pornographic Material (Excluding Child Pornography)	286
Child Pornography	287
Racism, Hate Speech, Glorification of Violence	288
Religious Offences	289
Illegal Gambling and Online Games	290
Libel and False Information	291
Spam and Related Threats	292
Other Forms of Illegal Content	293
The Future of Cyber Crime Activity	294
Future Trends in Cyber Crime Activity	294
Mitigating Cyber Crime Activity	297

Chapter 10 Intellectual Property and Computer Crime 300

Types of Economic Crimes and Development of Statutory Law	301
Mail and Wire Fraud: Generic Frauds and Swindles	302
Victims of Economic Crime	307
Consumers	307
Industry	308
Waging the War on Economic Crime	309
Law Enforcement	309
National Fraud Center	310
Economic Crime Investigation Institute	310
National White Collar Crime Center	310
The National Coalition for the Prevention of Economic Crime	311
Internet Fraud Council	311
Internet Fraud Complaint Center	311
Independent Corporations and Private Sector Industry Coalitions	312
Future Needs and Challenges	312
Law Enforcement Training	312
Laws, Regulations and Reporting Systems	312
Public-Private Partnerships	313
Balancing Privacy Interests	313

***Bibliography* 316**

***Index* 320**

Preface

Cyber laws and intellectual property offers a fascinating exploration into the legal framework governing the digital landscape and the protection of creative and innovative works. This book aims to provide you with comprehensive insights into the dynamic field of cyber laws and intellectual property rights, their significance in the digital age, and their impact on individuals, businesses, and society.

In our rapidly evolving digital era, the internet has become an indispensable part of our daily lives. With the proliferation of online activities, the need for robust legal protections in cyberspace has become more crucial than ever. Cyber laws encompass a wide range of legal principles and regulations that govern online interactions, data protection, privacy, cyber security, and digital commerce. Understanding cyber laws is essential for both individuals and organizations to navigate the complexities of the digital world responsibly and ethically.

We will explore the realm of intellectual property (IP) and its significance in fostering creativity and innovation. Intellectual property encompasses the legal rights granted to individuals or entities for their original creations, including patents for inventions, copyrights for literary and artistic works, trademarks for brands, and trade secrets for valuable confidential information. Understanding intellectual property rights is essential for creators, innovators, and businesses to protect their ideas and investments in a digital and globalized world.

As you embark on this exploration of cyber laws and intellectual property, may this book inspire you to embrace the significance of legal protections in the digital world. I hope that this book serves as a valuable resource and guide in your quest to understand and appreciate the complexities of cyber laws and intellectual property.

Author

Developing Cyber Law

The main feature of an application that would ideally realise the above principles and successfully implement them online would be the direct incorporation of law in the network's software protocol mechanisms. An example could be the TCP/IP set of protocols for Internet communications, which are structured on mathematical logic and patterns chosen for their technical adequacy. However, the latter are not immutable, and new, different models of communication protocols have been proposed and are currently under development. Two different elements will contribute to achieving the aspired goals. The first is the careful selection of legal rules that will regulate appropriately online environments. The second is the effective, accurate transfer of legal methodology into computer language.

AN ANALOGY FOR CYBER LAW

The vast majority of contract disputes in international trade are resolved through negotiation, perhaps with the help of mediators, but in the event that a voluntary solution cannot be achieved, virtually all international trade contracts have clauses that refer any dispute to arbitration. International arbitration is attractive relative to national courts for a number of procedural reasons, but arbitration is also attractive because it provides a means of supporting the contracting party's choice of legal jurisdiction. Böckstiegal (1984: 23) points out that:

When, in relation to arbitration, judges, arbitrators or authors speak of the 'applicable law' they do not always mean the same thing. The term is sometimes used rather indifferently with regard to three separate questions: What is the applicable procedural

law? What are the applicable conflict of law rules? What is the applicable substantive law? All three of these questions have their definitive relevance in any kind of international commercial arbitration.”

A large number of international trade associations have their own conflict resolution procedures, using arbitrators with special expertise in trade matters of concern to association members, including the community’s “practices and usage” (*i.e.*, customary rules) [over three decades ago, for example, Lazarus, *et al.* (1965) discussed more than 120 such tribunals], but other sources of arbitration are also available. The International Chamber of Commerce’s (ICC) arbitration institution provides a substantial list of arbitrators with expertise in international commerce, for instance, as do nation-specific organizations around the world which are eager to provide arbitrators for international trade disputes.

Ad hoc arbitration is also widely used, and in this regard, there is a rapidly growing market in private dispute resolution services provided by for-profit firms, at least in the United States. Procedural rules vary across these institutions, some of which offer different procedures depending upon the wishes of the disputing parties, but more importantly, a contract can also specify the substantive law and conflict-of-law rules under which any dispute should be resolved. A contract might designate the contract law of the seat of arbitration or of some other national legal system be applied, for instance.

The “usual way” of determining the relevant substantive law for international commercial arbitration, however, is to decide cases “exclusively on the interpretation of contracts and the relevance of trade usages so that very little depends on the question of the applicable. Lew’s (1978: 581) detailed analysis of available records reveals that in principle, “The answer to every dispute is to be found *prima facie* in the contract itself. What did the parties intend, what did they agree and what did they expect?” When an arbitrator cannot discover the parties’ intent in the contract, however, the focus turns to consideration of what the parties expected or should have expected, and in this regard, international arbitrators generally intentionally “denationalize” their awards, making them acceptable by showing their consistency with accepted traditional “practices and usage” (customary law) of the relevant business community.

Contracts might explicitly state that the practices and usages of a particular commercial community (*e.g.*, a trade association, an informal group of traders who deal in the same products) should be applied, or this may simply be understood. Business custom provides the default rule, at any rate, as arbitrators apply the customary rules which are commonly recognized within the “private international law systems from which the parties come”, unless a particular nation’s substantive law has been specified in the contract.

In this regard, there are in fact many different commercial communities, and their customary laws can be quite different. The wide variety of activities and relationships that exist in a modern world mean that many rules that are effective for one type of

transaction or one group may not be effective for another. Consider the diamond traders discussed by Bernstein (1992) and the oil traders discussed by Trakman (1983), for instance. The products being traded within these commercial communities are very different, suggesting that very different contractual issues are likely to be relevant, but the trading communities are also very different. Diamond merchants share common ethnic and religious backgrounds, creating an environment of mutual understanding (*e.g.*, of common trade practices and usage) and trust, for example, thus reducing the need for highly technical and specific contracts.

On the other hand, oil traders display much greater ethnic and religious diversity as well as differences in motivations (a number of oil producing states have nationalized production, for instance, so political considerations can have major impacts of decision-making), possibly reducing the level of common understanding and undermining trust relationships, thus dictating much more specific and complex contracts. These two commercial groups are likely to share many of the same rules but they are also likely to be some important differences in traditions and practices.

Indeed, imposition of the diamond merchants contractual rules and governance institutions on the oil traders would probably lead to much higher transactions costs for these traders, including more contract disputes, while imposition of the oil traders rules on the diamond merchants would add unnecessary complexity and costs to their contracting process. Thus, as Cooter (1994: 216) explains, decentralized lawmaking is actually desirable in the increasingly complex international economy. The benefits of specialization that are anticipated in economic activity also can arise in the legal arena, making polycentric law more effective than monocentric (monopolized) law.

While various commentators have suggested analogies for the analysis of law in cyberspace, including the frontier west, and feudalism, the contention here is that the best analogy is international commerce. Since international commerce operates under a polycentric system of customary law, the hypothesis is that similar arrangements can provide effective governance in cyberspace. To see if this hypothesis might be supportable, let us consider evidence of the evolving property rights, contracting arrangements (sources of trust and recourse), and customary law in cyberspace.

SPONTANEOUSLY EVOLVING INSTITUTIONS IN CYBERSPACE

The purpose of this chapter is to illustrate that the economic theory of spontaneously evolving institutions of property, contract, and customary law that was detailed in the preceding chapter can predict and explain developments in cyberspace. This will be done by applying the analysis in each of the subsections and sub-subsections of Section II in corresponding subsections and sub-subsections to explain recent and/or currently-arising developments in cyberspace. Cyberspace is a very dynamic environment, however, so any attempt to describe its institutions is going to be incomplete and inaccurate. Many of the innovations in the technology of property rights delineation and in institutional development may not even be relevant by the time this chapter

is published. The innovations may prove to be ineffective, for instance, or new even more effective developments may displace them. Thus, considered as simply an incomplete sampling of the kinds of things the entrepreneurial participants in the dynamic and evolving socioeconomic and institutional landscape of cyberspace are trying or considering as they search for ways to internalize externalities and facilitate cooperative interaction.

SUBSTANTIVE CRIMINAL LAW

ILLEGAL ACCESS (HACKING)

Since the development of computer networks, their ability to connect computers and offer users access to other computer systems, computers have been used by hackers for criminal purposes. There is substantial variation in hackers' motivations. Hackers need not be present at the crime scene; they just need to circumvent the protection securing the network. In many cases of illegal access, the security systems protecting the physical location of network hardware are more sophisticated than the security systems protecting sensitive information on networks, even in the same building. The illegal access to computer systems hinders computer operators from managing, operating and controlling their systems in an undisturbed and uninhibited manner. The aim of protection is to maintain the integrity of computer systems.

It is vital to distinguish between illegal access and subsequent offences (such as data espionage), as legal provisions have a different focus of protection. In most cases, illegal access (where law seeks to protect the integrity of the computer system itself) is not the end-goal, but rather a first step towards further crimes, such as modifying or obtaining stored data (where law seeks to protect the integrity and confidentiality of the data). The question is whether the act of illegal access should be criminalised, in addition to subsequent offences? Analysis of the various approaches to the criminalisation of illegal computer access at the national level shows that enacted provisions sometimes confuse illegal access with subsequent offences, or seek to limit the criminalisation of the illegal access to grave violations only.

Some countries criminalise mere access, while others limit criminalisation to offences only in cases where the accessed system is protected by security measures, or where the perpetrator has harmful intentions, or where data was obtained, modified or damaged. Other countries do not criminalise the access itself, but only subsequent offences. Opponents to the criminalisation of illegal access refer to situations where no dangers were created by mere intrusion, or where acts of "hacking" have led to the detection of loopholes and weaknesses in the security of targeted computer systems.

Convention on Cyber crime

The Convention on Cyber crime includes a provision on illegal access protecting the integrity of the computer systems by criminalising the unauthorised access to a system.

Noting inconsistent approaches at the national level, the Convention offers the possibility of limitations that – at least in most cases – enable countries without legislation to retain more liberal laws on illegal access.

The Provision

- Article 2 – Illegal access: Each Party shall adopt such legislative and other measures as may be necessary to establish as criminal offences under its domestic law, when committed intentionally, the access to the whole or any part of a computer system without right. A Party may require that the offence be committed by infringing security measures, with the intent of obtaining computer data or other dishonest intent, or in relation to a computer system that is connected to another computer system.

The Covered Acts

The term “access” does not specify a certain means of communication, but is open-ended and open to further technical developments. It shall include all means of entering another computer system, including Internet attacks, as well as illegal access to wireless networks. Even unauthorised access to computers that are not connected to any network (*e.g.*, by circumventing a password protection) are covered by the provision.

This broad approach means that illegal access not only covers future technical developments, but is also covers secret data accessed by insiders and employees. The second sentence of Article 2 offers the possibility of limiting the criminalisation of illegal access to access over a network. The illegal acts and protected systems are thus defined in a way that remains open to future developments. The Explanatory Report lists hardware, components, stored data, directories, traffic and content-related data as examples of the parts of computer systems that can be accessed.

Mental Element

Like all other offences defined by the Convention on Cyber crime Art. 2 requires that the offender is carrying out the offences intentionally. The Convention does not contain a definition of the term “internationally”. In the Explanatory Report the drafters pointed out that the definition of “intentionally” should happen on a national level.

Without Right

Access to a Computer can only be prosecuted under Article 2 of the Convention, if it should happen “without right”. Access to a system permitting free and open access by the public or access to a system with the authorisation of the owner or other rights-holder is not “without right”. In addition to the subject of free access, the legitimacy of security testing procedures is also addressed. Network administrators and security companies that test the protection of computer systems in order to identify potential gaps in the security measures were wary of the possibility of criminalisation under

illegal access. Despite the fact that these professionals generally work with the permission of the owner and therefore act legally, the drafters of the Convention emphasised that “testing or protection of the security of a computer system authorised by the owner or operator, are with right”. The fact, that the victim of the crime handed out a password or similar access code to the offender does not necessary mean that the offender then acted with right when he accessed the computer system of the victim.

If the offender persuaded the victim to disclose a password or access code due to a successful social engineering approach it is necessary to verify if the authorisation given by the victim does cover the act carried out by the offender. In general this is not the case and the offender therefore acts without right.

Restrictions and Reservations

As an alternative to the broad approach, the Convention offers the possibility of restricting criminalisation with additional elements, listed in the second sentence. The procedure of how to utilise this reservation is laid down in Article 42 of the Convention.

Possible reservations relate to security measures, special intent to obtain computer data, other dishonest intent that justifies criminal culpability, or requirements that the offence be committed against a computer system through a network. A similar approach can be found in the EU Framework Decision on Attacks against Information Systems.

Commonwealth Computer and Computer Related Crimes Model Law

A similar approach can be found in Sec. 5 of the 2002 Commonwealth Model Law:

- Sec. 5.: A person who intentionally, without lawful excuse or justification, accesses the whole or any part of a computer system commits an offence punishable, on conviction, by imprisonment for a period not exceeding, or a fine not exceeding or both.

The main difference to the Convention on Cyber crime is the fact that Sec. 5 of the Commonwealth Model Law does, unlike Art. 2 Convention on Cyber crime, not contain options to make reservations.

Stanford Draft Convention

The informal 1999 Stanford Draft Convention recognises illegal access as one of those offences the signatory states should criminalise.

The Provision

- Art. 3 – Offences:
 - 1. Offences under this Convention are committed if any person unlawfully and intentionally engages in any of the following conduct without legally recognised authority, permission, or consent:
 - (c) enters into a cyber system for which access is restricted in a conspicuous and unambiguous manner;

The Covered Acts

The draft provision shows a number of similarities to Art. 2 of the Convention on Cyber crime. Both require an intentional act that is committed without right/without authority. In this context requirement of the draft provision (“*without legally recognised authority, permission, or consent*”) is more precise than the term “without right” used Convention on Cyber crime and explicitly aims to incorporate the concept of selfdefence. The main difference to the Convention is the fact that the draft provision uses the term “cyber system”. The cyber system is defined in Art. 1, paragraph 3 of the Draft Convention. It covers any computer or network of computers used to relay, transmit, coordinate, or control communications of data or Programmes.

This definition shows many similarities to the definition of the term ‘computer system’ provided by Art. 1 (a) Convention on Cyber crime. Although the Draft Convention refers to acts related to the exchange of data and does therefore primarily focus on network based computer systems both definitions include interconnected computer as well as stand alone machines.

DATA ESPIONAGE

The Convention on Cyber crime as well as the Commonwealth Model Law and the Stanford Draft Convention provide legal solutions for illegal interception only. It is questionable whether Article 3 of the Convention on Cyber crime applies to other cases than those where offences are carried out by intercepting data transfer processes. The question of whether illegal access to information stored on a hard disk is covered by the Convention was discussed with great interest. Since a transfer process is needed, it is likely that Art. 3 of the Convention on Cyber crime does not cover forms of data espionage other than the interception of transfer processes.

One issue frequently discussed in this context is the question if the criminalisation of illegal accesses renders the criminalisation of data espionage unnecessary. In those cases where the offender has legitimate access to a computer system (*e.g.* because he is ordered to repair it) and on this occasion (in violation of the limited legitimating) copies files from the system, the act is in general not covered by the provisions criminalising illegal access. Given that much vital data is today stored in computer systems, it is essential to evaluate whether existing mechanisms to protect data are adequate or whether other criminal law provision are necessary to protect the user from data espionage. Today, computer users can use various hardware devices and software tools in order to protect secret information.

They can install firewalls, access control systems or encrypt stored information and by this decrease the risk of data espionage. Although user-friendly devices are available, requiring only limited knowledge by users, truly effective protection of data on a computer system often requires knowledge that few users have. Especially data stored on private computer systems is often not adequately protected against data espionage. Therefore criminal law provisions can offer an additional protection.

Examples

Some countries have decided to extend the protection that is available through technical measures by criminalising data espionage. There are two main approaches. Some countries follow a narrow approach and criminalise data espionage, only where specific secret information is obtained - an example is 18 U.S.C § 1831, that criminalises economic espionage. The provision does not only cover data espionage, but other ways of obtaining secret information as well.

Economic Espionage

- In General — Whoever, intending or knowing that the Offence will benefit any foreign government, foreign instrumentality, or foreign agent, knowingly—
 - Steals, or without authorization appropriates, takes, carries away, or conceals, or by fraud, artifice, or deception obtains a trade secret;
 - Without authorization copies, duplicates, sketches, draws, photographs, downloads, uploads, alters, destroys, photocopies, replicates, transmits, delivers, sends, mails, communicates, or conveys a trade secret;
 - Receives, buys, or possesses a trade secret, knowing the same to have been stolen or appropriated, obtained, or converted without authorization;
 - Attempts to commit any Offence described in any of paragraphs (1) through (3); or
 - Conspires with one or more other persons to commit any Offence described in any of paragraphs (1) through (3), and one or more of such persons do any act to effect the object of the conspiracy, shall, except as provided in subsection (b), be fined not more than \$500,000 or imprisoned not more than 15 years, or both.
- Organizations — Any organization that commits any Offence described in subsection (a) shall be fined not more than \$10,000,000.

Other countries have adopted a broader approach and criminalised the act of obtaining stored computer data, even if they do not contain economic secrets.

Section 202a. Data Espionage

- Any person who obtains without authorization, for himself or for another, data which are not meant for him and which are specially protected against unauthorised access, shall be liable to imprisonment for a term not exceeding three years or to a fine
- Data within the meaning of subsection 1 are only such as are stored or transmitted electronically or magnetically or in any form not directly visible.

This provision not only covers economic secrets, but stored computer data in general. In terms of its objects of protection, this approach is broader compared to § 1831 USC, but the application of the provision is limited as obtaining data is only criminalised where data are specially protected against unauthorised access. The protection of stored

computer data under German criminal law is thus limited to persons or businesses that have taken measures to avoid falling victim to such offences.

Relevance of Such Provision

The implementation of such provision is especially relevant with regard to cases, where the offender was authorised to access a computer system (*e.g.*, because he was ordered to fix a computer problem) and then abused the authorisation to illegally obtain information stored on the computer system. With regard to the fact that the permission covers the access to the computer system it is in general not possible to cover with provisions criminalising the illegal access.

Without Right

The application of data espionage provisions in general requires that the data was obtained without the consent of the victim. The success of phishing attacks clearly demonstrates the success of scams based on the manipulation of users. Due to the consent of the victim offenders who succeed in manipulating of users to disclose secret information cannot be prosecuted on the basis of the above mentioned provisions.

ILLEGAL INTERCEPTION

The use of ICTs is accompanied by several risks related to the security of information transfer. Unlike classic mail order operations within a country, data transfer processes over the Internet involve numerous providers and different points where the data transfer process could be intercepted. The weakest point for intercept remains the user, especially users of private home computers, who are often inadequately protected against external attacks.

As offenders generally always aim for the weakest point, the risk of attacks against private users is great, all the more so given:

- The development of vulnerable technologies; and
- The rising relevance of personal information for offenders.

New network technologies (such as “wireless LAN”) offer several advantages for Internet access. Setting up a wireless network in a private home, for example, allows families to connect to the Internet from anywhere inside a given radius, without the need for cable connections. But the popularity of this technology and resulting comfort is accompanied by serious risks to network security.

If an unprotected wireless network is available perpetrators can log on to this network and use it for criminal purposes without the need to get access to a building. They simply need to get inside the radius of the wireless network to launch an attack. Field tests suggest that in some areas as many as 50 per cent of private wireless networks are not protected against unauthorised interception or access. In most cases, lack of protection arises from a lack of knowledge as to how to configure protection measures. In the past, perpetrators concentrated mainly on business networks for illegal interceptions.

Interception of corporate communications was more likely to yield useful information, than data transferred within private networks. The rising number of identity thefts of private personal data suggests that the focus of the perpetrators may have changed. Private data such as credit card numbers, social security numbers, passwords and bank account information are now of great interest to offenders.

The Convention on Cyber Crime

The Convention on Cyber crime includes a provision protecting the integrity of non-public transmissions by criminalising their unauthorised interception. This provision aims to equate the protection of electronic transfers with the protection of voice conversations against illegal tapping and/or recording that currently already exists in most legal systems.

The Provision

- Article 3 – Illegal interception: Each Party shall adopt such legislative and other measures as may be necessary to establish as criminal offences under its domestic law, when committed intentionally, the interception without right, made by technical means, of non-public transmissions of computer data to, from or within a computer system, including electromagnetic emissions from a computer system carrying such computer data. A Party may require that the offence be committed with dishonest intent, or in relation to a computer system that is connected to another computer system.

The Covered Acts

The applicability of Article 3 is limited to the interception of transmissions realised by technical measures. Interceptions related to electronic data can be defined as any act of acquiring data during a transfer process. The question if illegal access to information stored on a hard disk is covered by the provision is controversially discussed. In general the provision only applies to the interception of transmissions - access to stored information is not considered as an interception of a transmission.

The fact that the application of the provision is discussed even in cases where the offender physically access a standalone computer system partly arises as a result of the fact, that the Convention on Cyber crime does not contain a provision related to data espionage and the Explanatory Report to the Convention contains two slightly imprecise explanations with regard to the application of Art. 3:

- The Explanatory Report first of all points out that the provision covers communication processes taking place within a computer system. However, this still leaves open the question of whether the provision should only apply in cases where victims send data that are then intercepted by offenders or whether it should apply also when the offender himself operates the computer.
- The guide points out that interception can be committed either indirectly

through the use of tapping devices or “through access and use of the computer system”. If offenders gain access to a computer system and use it to make unauthorised copies of stored data on an external disc drive, where the act leads to a data transfer (sending data from the internal to the external hard disc), this process is not *intercepted*, but rather *initiated*, by offenders. The missing element of technical interception is a strong argument against the application of the provision in cases of illegal access to stored information.

The term “transmission” covers all data transfers, whether by telephone, fax, e-mail or file transfer. The offence established under Article 3 applies only to non-public transmissions. A transmission is “non-public”, if the transmission process is confidential. The vital element to differentiate between public and non-public transmissions is not the nature of the data transmitted, but the nature of the transmission process itself. Even the transfer of publicly available information can be considered criminal, if the parties involved in the transfer intend to keep the content of their communications secret. Use of public networks does not exclude “non-public” communications.

Mental Element

Like all other offences defined by the Convention on Cyber crime, Article 3 requires that the offender is carrying out the offences intentionally. The Convention does not contain a definition of the term “internationally”. In the Explanatory Report the drafters pointed out that the definition of “intentionally” should happen on a national level.

Without Right

The interception of communication can only be prosecuted under Article 3 of the Convention, if it should happen “without right”.

The drafters of the Convention provided a set of examples for interceptions that are not carried out without right:

- Action on the basis instructions or by authorisation of the participants of the transmission;
- Authorised testing or protection activities agreed to by the participants;
- Lawful interception on the basis of criminal law provisions or in the interests of national security.

Another issue raised within the negotiation of the Convention was the question if the use of cookies would lead to criminal sanctions based on Art. 3. The drafters pointed out that common commercial practices (such as cookies) are not considered to be interceptions without right.

Restrictions and Reservations

Article 3 offers the option of restricting criminalisation by requiring additional elements listed in the second sentence, including a “dishonest intent” or relation to a computer system connected to another computer system.

Commonwealth Computer and Computer Related Crimes Model Law

A similar approach can be found in Sec. 8 of the 2002 Commonwealth Model Law.

- *Sec. 8.:* A person who, intentionally without lawful excuse or justification, intercepts by technical means:
 - Any non-public transmission to, from or within a computer system; or
 - Electromagnetic emissions from a computer system that are carrying computer data; commits an offence punishable, on conviction, by imprisonment for a period not exceeding, or a fine not exceeding, or both.

Stanford Draft Convention

The informal 1999 Stanford Draft Convention does not explicitly criminalise the interception of computer data.

DATA INTERFERENCE

The protection of tangible, or physical, objects against intentional damage is a classic element of national penal legislation. With continuing digitalisation, more critical business information is stored as data. Attacks or obtaining of this information can result in financial losses. Besides deletion, the alteration of such information could also have major consequences. Previous legislation has in some not completely brought the protection of data in line with the protection of tangible objects. This enabled offenders to design scams that do not lead to criminal sanctions.

Convention on Cyber Crime

In Article 4, the Convention on Cyber crime includes a provision that protects the integrity of data against unauthorised interference. The aim of the provision is to fill existing gaps in some national penal laws and to provide computer data and computer programmes with protections similar to those enjoyed by tangible objects against the intentional infliction of damage.

The Provision

- *Article 4 – Data interference:*
 - Each Party shall adopt such legislative and other measures as may be necessary to establish as criminal offences under its domestic law, when committed intentionally, the damaging, deletion, deterioration, alteration or suppression of computer data without right.
 - A Party may reserve the right to require that the conduct described in paragraph 1 result in serious harm.
- *The covered acts:*
 - The terms “damaging” and “deterioration” mean any act related to the negative alteration of the integrity of information content of data and programmes;

- “Deleting” covers acts where information is removed from storage media and is considered comparable to the destruction of a tangible object. While providing the definition the drafters of the Convention did not differentiate between the various ways data can be deleted. Dropping a file to the virtual trash bin does not remove the file from the hard disk. Even “emptying” the trash bin does not necessarily remove the file. It is therefore uncertain if the ability to recover a deleted file hinders the application of the provision.
- “Suppression” of computer data denotes an action that affects the availability of data to the person with access to the medium, where the information is stored in a negative way. The application of the provision is especially discussed with regard to Denial-of-Service attacks. During the attack the data provided on the targeted computer system are not available anymore for potential user as well as the owner of the computer system.
- The term “alteration” covers the modification of existing data, without necessarily lowering the serviceability of the data. This act is especially covering the installation of malicious software like spyware, viruses or adware on the victim’s computer.

Mental Element

Like all other offences defined by the Convention on Cyber crime Article 4 requires that the offender is carrying out the offences intentionally. The Convention does not contain a definition of the term “internationally”. In the Explanatory Report the drafters pointed out that the definition of “intentionally” should happen on a national level.

Without Right

The acts must be committed “without right”. The right to alter data was discussed, especially in the context of “re-mailers”. Re-mailers are used to modify certain data for the purpose of facilitating anonymous communications. The Explanatory Reports mention that, in principle, these acts are considered a legitimate protection of privacy and can thus be considered as being undertaken with authorisation.

Restrictions and Reservations

Article 4 offers the option of restricting criminalisation by limiting it to cases where serious harm arises, a similar approach to the EU Framework Decision on Attacks against Information Systems, which enables Member States to limit the applicability of the substantive criminal law provision to “cases which are not minor”.

Commonwealth Computer and Computer Related Crimes Model Law

An approach in line with Art. 4 Convention on Cyber crime can be found in Sec. 8 of the 2002 Commonwealth Model Law.

Sec. 6

- A person who, intentionally or recklessly, without lawful excuse or justification, does any of the following acts:
 - Destroys or alters data; or
 - Renders data meaningless, useless or ineffective; or
 - Obstructs, interrupts or interferes with the lawful use of data; or
 - Obstructs, interrupts or interferes with any person in the lawful use of data; or
 - Denies access to data to any person entitled to it; commits an offence punishable, on conviction, by imprisonment for a period not exceeding or a fine not exceeding [amount], or both.
- Subsection (1) applies whether the person's act is of temporary or permanent effect.

Stanford Draft Convention

The informal 1999 Stanford Draft Convention contains two provisions that criminalise acts related to interference with computer data.

The Provision***Art. 3:***

- Offences under this Convention are committed if any person unlawfully and intentionally engages in any of the following conduct without legally recognised authority, permission, or consent:
 - Creates, stores, alters, deletes, transmits, diverts, misroutes, manipulates, or interferes with data or Programmes in a cyber system with the purpose of causing, or knowing that such activities would cause, said cyber system or another cyber system to cease functioning as intended, or to perform functions or activities not intended by its owner and considered illegal under this Convention;
 - Creates, stores, alters, deletes, transmits, diverts, misroutes, manipulates, or interferes with data in a cyber system for the purpose and with the effect of providing false information in order to cause substantial damage to persons or property;

The Covered Acts

The main difference between the Convention on Cyber crime and the Commonwealth Model Law and the approach of the Draft Convention is the fact, that Draft Convention does only criminalise the interference with data if this interferes with the functioning of a computer system (Art. 3, paragraph 1a) or if the act is committed with the purpose of providing false information in order to causing damage to a person or property (Art. 3,

paragraph 1b). Therefore the draft law does not criminalise the deletion of a regular text document of a data storage device as this does neither influence the functioning of a computer nor does it provide false information. The Convention on Cyber crime and the Commonwealth Model Law both follow a broader approach by protecting the integrity of computer data without the mandatory requirement of further effects.

RESTRUCTURING THE VIRTUAL SPACE

Before going through these successive steps, it is crucial to define, in advance, the regulated area's dimensions, as successful application of law is indissolubly bound up with both the natural and social parameters of a given and potentially regulated space. The community is, therefore, assumed to operate on a shared online platform, open in membership, though closed in access (meaning, active via typing allocated passwords), and in general performing similarly to a typical Web browser. This then allows e-mail communications and stereotypical data exchange, graphical on-screen representation of web sites and an additional set of assisting applications and accessory tools. Additionally, it is important to identify entities that represent subjects and objects within this electronic world.

The online social structure, as a real-world metaphor, includes active participants and static objects in analogously operating digitised projections. Both users and their autonomous software, which may execute a variety of prearranged tasks on the formers' behalf, are defined as "actors". According to typical online functionalism, documents, pictures, sounds, programmes and contiguous exchangeable information are classified as "movable objects", while IP addresses, including single web-sites and hosted secondary platforms, constitute a class of "immovable objects".

Codification

Common use of the word "codification" originally captured the lexical confusion between a certain method for legislating and the graphic expression of written law, leading today to a dual-conception of both the method of codification and the instrument of code.

In the legal world, codification serves a number of substantial needs, offering:

- Accessibility,
- Completeness and cohesion in representation,
- Consistency and
- Certainty.

However, in developing a networking protocol that will articulate legal norms through automated transmissions, "Codification" means actually "encoding" into computer language the existing legal norms, changing them from passive text into an active, automated regulatory framework. In other words, it means cutting the text of law into pieces and reinstating it in the form of processible code.

Objectives and Method of Implementation

Codification requires the development of a translatable legal logic that preserves law's desired regulatory impact on social behaviour and interaction. As these pre-set rules reflect the distinctive character of the electronic reality and, at the same time, direct its operation, codification's objectives may be summarised as follows:

- “Scalable” clarity, thus requiring a complete and comprehensive regulatory framework that maximises internal compatibility between the variety of induced legal themes,
- Open provisional structure for assimilating future file formats and models of data distribution,
- Precise representation of real-world legal knowledge, in processible, mathematically arranged logical sequences, and
- Creation of a uniquely identifiable and well ordered interactive setting, capable of preventing online participants' evasive behaviour while exercising their exchange practices.

The first two objectives are realised through the first step of the codification procedures, and define the selection of law that will eventually run through the online community. As online activities substantially remodel real-world patterns of contact and association, the claim is made here that the required framework, from which the most suitable legal rules will be extracted, already exists in the form of statute law. Continental codifications carry the advantage of exhibiting ordered text-layout in concise logical sequences, which can easily be transcribed in computer language patterns as shown previously with the “obligations” example. Moreover, they bring to the point common universal “abstract rules and principles which apply to all of the specific circumstances.”

A scheme for efficiently running online environments would incorporate the core protocol of these codifications. At the same time, however, it would allow the online community to evolve through more relaxed common law practices that take place within the platform's software sequences.

While the platform's legal evolution will be implemented through online interactivity, with the extended use of autonomous agents, the computer code will progressively integrate precedents with the potential for altering the community's “legal” standards. This issue will be explained and addressed briefly in the last part of the paper.

The third of the above objectives reflects the second step of the codification procedure.

In short, it includes formation of a legal ontology, assessed by evaluating the induced concepts along with the individual objects and variables that they prescribe (persons, items, quantitative elements *etc.*) Following the analysis and adaptation of legal concepts to the community's standards, a finite number of common key notions are revealed to the protocol's designers/creators. These key notions are essentially

determined by the frequency and functionality of the used terminology. Therefore, simple or complex terms consisting of different sub-notions are easily marked down to complete a legal ontology that will eventually be transcribed as computer code.

An idea of what this ontology would look like can be gained through the previous Boolean rephrasing of “obligation”. “Obligation” shares certain aspects with concepts like “contract” or “ownership”. Moreover, variables like “item” (varying in the electronic environment from audio files to executable programmes) or “creditor” (emerging from an e-commerce agreement as the seller or the buyer), frequently appear in everyday online communications. A concept itself may include another concept, or may also use shared legal terminology.

While determining the applied law’s structure involves mainly assistance from legal experts, the second step requires additional technological support, A.I. expertise in the field of ontological modelling and the relevant computer programming skills. Practically, this is the part of the procedure where the law becomes essentially encoded.

THE CYBER LAW PLATFORM

Users participate in the community by accessing the platform through their browsers. The platform itself constitutes the common network application, which operates under the protocol’s standards and rules. Browsers, with the appropriate software development, allow users to observe current online communal legal preferences and to customise their participation by becoming fully or partially subject to the protocol’s normative standards.

For example, B selects to activate the concepts of ownership and possession *erga omnes*, with regard to the content of his personal web-page. This page includes pictures of famous expressionist paintings (*possession*) and one original article (written by B) on “The power of coloured lineament in the art of Van Gogh” (*ownership*). Either acquisition of his uploaded material is blocked or, supposing that this option lies technically unfeasible, pre-set conditions generate the appropriate automatic procedure (i.e. tort) as prescribed by the law/protocol in case of breach.

Conversely, he might also reject “ownership”, meaning that he declines rights of property over his article, and nonetheless opts to remain within the general regulatory setting of possession. However, his personal refusal of “ownership” does not authorise him to infringe the rights of other participants who have subscribed to this specific concept. If private law practices are adopted as community standards they take on public law significance and create legal preferences and obligations *erga omnes*, thus limiting the users’ range of options in customising their online profiles. Hypothetically, if the “liability of possessor as against owner” concept becomes part of the system’s *jus cogens*, B will not be allowed to exclude himself from its community-wide compulsory effect, although he may choose to deactivate his software from pursuing the prescribed legal effect against others automatically.

Users may alter their preferences at any instance provided, of course, that no existing cyberrule or regulation will block their request. The platform offers the possibility of legal cross-fertilisation through the exchange of practices between users. Individual modifications and subsequent *inter partes* agreements between the community's members are spread across the platform. The system's collective intelligence spots popular selections and may convey them to users for approval. Once a practice becomes widely embraced, the platform has flexibility to adopt it and establish it as a system preference, thus partially standardising it as *jus dispositivum*, optional law that does not defy the original protocol's operational principles.

Uniform imposition of these procedures over humans and autonomous software ultimately substantiates code both as natural and positive law. We have also examined the community's evolution through common law practices. Finally, the delicate issue of enforcing the appropriate measures for applying justice will be determined by the community, since this form of "justice" is established to serve the collective perception, not a centralised governing entity, thus becoming one of CyberLaw's "selling points".

CYBER LAWS AND LEGAL ISSUES

Cyberlaw is a term that encapsulates the legal issues related to use of communicative, transactional, and distributive aspects of networked information devices and technologies. It is less a distinct field of law in the way that property or contract are, as it is a domain covering many areas of law and regulation. Some leading topics include intellectual property, privacy, freedom of expression, and jurisdiction.

JURISDICTION AND SOVEREIGNTY

Jurisdiction is an aspect of state sovereignty and it refers to judicial, legislative and administrative competence. Although jurisdiction is an aspect of sovereignty, it is not coextensive with it. The laws of a nation may have extraterritorial impact extending the jurisdiction beyond the sovereign and territorial limits of that nation.

This is particularly problematic as the medium of the Internet does not explicitly recognize sovereignty and territorial limitations. There is no uniform, international jurisdictional law of universal application, and such questions are generally a matter of conflict of laws, particularly private international law. An example would be where the contents of a web site are legal in one country and illegal in another. In the absence of a uniform jurisdictional code, legal practitioners are generally left with a conflict of law issue.

Another major problem of cyberlaw lies in whether to treat the Internet as if it were physical space (and thus subject to a given jurisdiction's laws) or to act as if the Internet is a world unto itself (and therefore free of such restraints). Those who favour the latter view often feel that government should leave the Internet community to self-regulate. John Perry Barlow, for example, has addressed the governments of the world

and stated, “Where there are real conflicts, where there are wrongs, we will identify them and address them by our means. We are forming our own Social Contract. This governance will arise according to the conditions of our world, not yours. Our world is different”. A more balanced alternative is the Declaration of Cyber secession: “Human beings possess a mind, which they are absolutely free to inhabit with no legal constraints. Human civilization is developing its own (collective) mind. All we want is to be free to inhabit it with no legal constraints. Since you make sure we cannot harm you, you have no ethical right to intrude our lives. So stop intruding!” Other scholars argue for more of a compromise between the two notions, such as Lawrence Lessig’s argument that “The problem for law is to work out how the norms of the two communities are to apply given that the subject to whom they apply may be in both places at once” (Lessig, Code 190).

Though rhetorically attractive, cyber secession initiatives have had little real impact on the Internet or the laws governing it. In practical terms, a user of the Internet is subject to the laws of the state or nation within which he or she goes online. Thus, in the U.S., Jake Baker faced criminal charges for his e-conduct, and numerous users of peer-to-peer file-sharing software were subject to civil lawsuits for copyright infringement. This system runs into conflicts, however, when these suits are international in nature. Simply put, legal conduct in one nation may be decidedly illegal in another. In fact, even different standards concerning the burden of proof in a civil case can cause jurisdictional problems. For example, an American celebrity, claiming to be insulted by an online American magazine, faces a difficult task of winning a lawsuit against that magazine for libel. But if the celebrity has ties, economic or otherwise, to England, he or she can sue for libel in the British court system, where the standard of “libelous speech” is far lower.

NET NEUTRALITY

Another major area of interest is net neutrality, which affects the regulation of the infrastructure of the Internet. Though not obvious to most Internet users, every packet of data sent and received by every user on the Internet passes through routers and transmission infrastructure owned by a collection of private and public entities, including telecommunications companies, universities, and governments, suggesting that the Internet is not as independent as Barlow and others would like to believe. This is turning into one of the most critical aspects of cyberlaw and has immediate jurisdictional implications, as laws in force in one jurisdiction have the potential to have dramatic effects in other jurisdictions when host servers or telecommunications companies are affected.

FREE SPEECH IN CYBERSPACE

In comparison to traditional print-based media, the accessibility and relative anonymity of cyber space has torn down traditional barriers between an individual

and his or her ability to publish. Any person with an internet connection has the potential to reach an audience of millions with little-to-no distribution costs. Yet this new form of highly-accessible authorship in cyber space raises questions and perhaps magnifies legal complexities relating to the freedom and regulation of speech in cyberspace. Recently, these complexities have taken many forms, three notable examples being the Jake Baker incident, in which the limits of obscene Internet postings were at issue, the controversial distribution of the DeCSS code, and *Gutnick v Dow Jones*, in which libel laws were considered in the context of online publishing. The last example was particularly significant because it epitomized the complexities inherent to applying one country's laws (nation-specific by definition) to the internet (international by nature). In 2003, Jonathan Zittrain considered this issue in his paper, "Be Careful What You Ask For: Reconciling a Global Internet and Local Law".

In many countries, speech through cyberspace has proven to be another means of communication which has been regulated by the government.

The Open Net Initiative, whose mission statement is "to investigate and challenge state filtration and surveillance practices" in order to "...generate a credible picture of these practices," has released numerous reports documenting the filtration of internet-speech in various countries. While China has thus far proven to be the most rigorous in its attempts to filter unwanted parts of the internet from its citizens, many other countries-including Australia, Singapore, Iran, Saudi Arabia, and Tunisia-have engaged in similar practices of internet censorship. In one of the most vivid examples of information-control, the Chinese government for a short time transparently forwarded requests to the Google search engine to its own, state-controlled search engines.

These examples of filtration bring to light many underlying questions concerning the freedom of speech, namely, does the government have a legitimate role in limiting access to information? And if so, what forms of regulation are acceptable? The recent blocking of "blogspot" and other websites in India failed to reconcile the conflicting interests of speech and expression on the one hand and legitimate government concerns on the other hand. In the UK the case of *Keith-Smith v Williams* confirmed that existing libel laws applied to internet discussions.

GOVERNANCE

The unique structure of the Internet has raised several judicial concerns. While grounded in physical computers and other electronic devices, the Internet is independent of any geographic location. While real individuals connect to the Internet and interact with others, it is possible for them to withhold personal information and make their real identities anonymous. If there are laws that could govern the Internet, then it appears that such laws would be fundamentally different from laws that geographic nations use today. In their essay "Law and Borders — The Rise of Law in Cyberspace", David Johnson and David Post offer a solution to the problem of Internet governance. Given the Internet's unique situation, with respect to geography and identity, Johnson

and Post believe that it becomes necessary for the Internet to govern itself.

Instead of obeying the laws of a particular country, Internet citizens will obey the laws of electronic entities like service providers. Instead of identifying as a physical person, Internet citizens will be known by their usernames or email addresses. Since the Internet defies geographical boundaries, national laws will no longer apply. Instead, an entirely new set of laws will be created to address concerns like intellectual property and individual rights. In effect, the Internet will exist as its own sovereign nation.

Even if the Internet represents a legal paradigm shift, Johnson and Post do not make clear exactly how or by whom the law of the Internet will be enforced. Instead, the authors see market mechanisms, like those that Medieval merchants used, guiding Internet citizens' actions like Adam Smith's invisible hand. Yet, as more physical locations go online, the greater the potential for physical manifestation of electronic misdeeds. What do we do when someone electronically turns off the hospital lights?

However, there is also substantial literature and commentary that the internet is not only "regulable," but is already subject to substantial regulation, both public and private, by many parties and at many different levels. Leaving aside the most obvious examples of internet filtering in nations like China or Saudi Arabia or Iran (that monitor content), there are four primary modes of regulation of the internet described by Lawrence Lessig in his book, *Code and Other Laws of Cyberspace*:

1. Law: Standard East Coast Code, and the most self-evident of the four modes of regulation. As the numerous statutes, evolving case law and precedents make clear, many actions on the internet are already subject to conventional legislation (both with regard to transactions conducted on the internet and images posted). Areas like gambling, child pornography, and fraud are regulated in very similar ways online as off-line. While one of the most controversial and unclear areas of evolving laws is the determination of what forum has subject matter jurisdiction over activity (economic and other) conducted on the internet, particularly as cross border transactions affect local jurisdictions, it is certainly clear that substantial portions of internet activity are subject to traditional regulation, and that conduct that is unlawful off-line is presumptively unlawful online, and subject to similar laws and regulations. Scandals with major corporations led to US legislation rethinking corporate governance regulations such as the Sarbanes-Oxley Act.
2. Architecture: West Coast Code: these mechanisms concern the parameters of how information can and cannot be transmitted across the internet. Everything from internet filtering software (which searches for keywords or specific URLs and blocks them before they can even appear on the computer requesting them), to encryption programmes, to the very basic architecture of TCP/IP protocol, falls within this category of regulation. It is arguable that all other modes of regulation either rely on, or are significantly supported by, regulation via West Coast Code.

3. Norms: As in all other modes of social interaction, conduct is regulated by social norms and conventions in significant ways. While certain activities or kinds of conduct online may not be specifically prohibited by the code architecture of the internet, or expressly prohibited by applicable law, nevertheless these activities or conduct will be invisibly regulated by the inherent standards of the community, in this case the internet “users.” And just as certain patterns of conduct will cause an individual to be ostracised from our real world society, so too certain actions will be censored or self-regulated by the norms of whatever community one chooses to associate with on the internet.
4. Markets: Closely allied with regulation by virtue of social norms, markets also regulate certain patterns of conduct on the internet. While economic markets will have limited influence over non-commercial portions of the internet, the internet also creates a virtual marketplace for information, and such information affects everything from the comparative valuation of services to the traditional valuation of stocks. In addition, the increase in popularity of the internet as a means for transacting all forms of commercial activity, and as a forum for advertisement, has brought the laws of supply and demand in cyberspace.

CONCLUSIONS: POTENTIAL AND APPLICABILITY

These simplified examples underline the key element of this distinctive merging of law into technology: the automation of regulating interactivity.

From the lawyer's perspective, within this new system resides inherent potential:

- For overcoming legislative ambiguities that emerge from transnational networking communications, and
- For shaping global regulatory frameworks.

The CyberLaw protocol's structural logic facilitates the introduction of indirect legal harmonisation for the online community's various participants, regardless of their national origin or status. In this respect, typical conflicts of international private law that emerge from the global online context (*e.g.*, jurisdiction) give way to the CyberLaw's main operation of precluding disputes through computer code by keeping behavioural activity in compliance with the law at the precise moment of its execution.

Additionally, as the adopted legal policies are merely a minimalist restatement of the fundamental principles of private law, they are easily understandable across the network community. Exploring the protocol's functionality on this basis, a wide range of areas and issues of law that have required additional legislation for their Internet manifestations (and the institution of further, more complex and ambiguous terminology) revert to familiar legal concepts that outline and prescribe the fundamental proportions of fixed social systems, *e.g.*, copyright in Cyber Law is ideally dealt with in the traditional ownership context.

However, the creation of electronic-legal hybrids that would substitute and eventually circumvent the effect of real-world law, is not the intent of the CyberLaw protocol.

On the contrary, the protocol provides a practical prevention mechanism against the increasing Internet-related litigiousness that places unsuspecting individual users in costly judicial battles with gigantic international entities.

Moreover, the transparency of the automatically imposed behavioural limits and restrictions over online interactivity is brought under narrow scrutiny. Law enforcement within democratic structures is characterised by clarity and its availability to the public for criticism and debate, a principle social institution, which the “invisible” computer language sequences presumably challenge in practice. However, the protocol does not constitute an administration of justice mechanism but, rather, an interactivity management system which replicates private law methodology.

Since CyberLaw acknowledges the community system in a Luhmannian sense the CyberLaw protocol’s explicit incorporation of law enhances the respect for the human participant as legal actor, unlike the arbitrarily created software platforms, which currently monopolise the global electronic market. Protection and security through the protocol, though, are provided equally for both the individual and the service provider, reflecting the universality of Law in the electronic environment. Therefore, in assessing the added value for international online transactions, the private sector as well as state authorities are invited to contribute to the development of the CyberLaw protocol, in pursuit of satisfactory regulatory schemes for the Internet.

Referring initially to basic legal modules, the law through the protocol becomes alive: a collective autonomous innovation that, by hosting a series of small cross-operational tasks, evolves ultimately only through the initiative of its masters/subjects. In this respect, cyber-democracy, capable of presenting its own tailor-made law, is not just a digital utopia. Beyond the above-explored option of building local, independent communities, the Internet as a whole is the protocol’s ultimate and most ambitious challenge. It is arguable whether the Web has the capacity to integrate globally new socio-legal solutions, especially since a framework of universally shared formal laws was not *ab initio* cultivated across the users’ community. In addition, the commercialisation of the Web has led to the fragmented administration of cyberspace by innumerable, and thus uncontrollable, private operators.

Encouragingly, however, as long as many, and technically more advanced, electronic network constructions are still underway, the CyberLaw protocol proposal offers the promising potential for future interactive systems to construct their operational basis concretely towards human law, predicting and escaping the already encountered ambiguities on the “two-dimensional” Internet.

Intellectual Property

INTRODUCTION

Intellectual property (IP) is a legal concept which refers to creations of the mind for which exclusive rights are recognized. Under intellectual property law, owners are granted certain exclusive rights to a variety of intangible assets, such as musical, literary, and artistic works; discoveries and inventions; and words, phrases, symbols, and designs. Common types of intellectual property rights include copyright, trademarks, patents, industrial design rights, trade dress, and in some jurisdictions trade secrets.

Although many of the legal principles governing intellectual property rights have evolved over centuries, it was not until the 19th century that the term *intellectual property* began to be used, and not until the late 20th century that it became commonplace in the majority of the world. The British Statute of Anne (1710) and the Statute of Monopolies (1624) are now seen as the origins of copyright and patent law respectively.

HISTORY

Modern usage of the term *intellectual property* goes back at least as far as 1867 with the founding of the North German Confederation whose constitution granted legislative power over the protection of intellectual property (*Schutz des geistigen Eigentums*) to the confederation. When the administrative secretariats established by the Paris Convention (1883) and the Berne Convention (1886) merged in 1893, they located in Berne, and also adopted the term intellectual property in their new combined title, the United International Bureaux for the Protection of Intellectual Property.

The organisation subsequently relocated to Geneva in 1960, and was succeeded in 1967 with the establishment of the World Intellectual Property Organization (WIPO) by treaty as an agency of the United Nations. According to Lemley, it was only at this point that the term really began to be used in the United States (which had not been a party to the Berne Convention), and it did not enter popular usage until passage of the Bayh-Dole Act in 1980. “The history of patents does not begin with inventions, but rather with royal grants by Queen Elizabeth I (1558–1603) for monopoly privileges... Approximately 200 years after the end of Elizabeth’s reign, however, a patent represents a legal [right] obtained by an inventor providing for exclusive control over the production and sale of his mechanical or scientific invention... [demonstrating] the evolution of patents from royal prerogative to common-law doctrine.” In an 1818 collection of his writings, the French liberal theorist, Benjamin Constant, argued against the recently introduced idea of “property which has been called intellectual.”

The term *intellectual property* can be found used in an October 1845 Massachusetts Circuit Court ruling in the patent case *Davoll et al. v. Brown.*, in which Justice Charles L. Woodbury wrote that “only in this way can we protect intellectual property, the labours of the mind, productions and interests are as much a man’s own...as the wheat he cultivates, or the flocks he rears.”

The statement that “discoveries are...property” goes back earlier. Section 1 of the French law of 1791 stated, “All new discoveries are the property of the author; to assure the inventor the property and temporary enjoyment of his discovery, there shall be delivered to him a patent for five, ten or fifteen years.” In Europe, French author A. Nion mentioned *propriété intellectuelle* in his *Droits civils des auteurs, artistes et inventeurs*, published in 1846. Until recently, the purpose of intellectual property law was to give as little protection possible in order to encourage innovation. Historically, therefore, they were granted only when they were necessary to encourage invention, limited in time and scope. The concept’s origins can potentially be traced back further. Jewish law includes several considerations whose effects are similar to those of modern intellectual property laws, though the notion of intellectual creations as property does not seem to exist – notably the principle of *Hasagat Ge’vul* (unfair encroachment) was used to justify limited-term publisher (but not author) copyright in the 16th century. In 500 BCE, the government of the Greek state of Sybaris offered one year’s patent “to all who should discover any new refinement in luxury.”

TYPES

Common types of intellectual property rights include patents, copyright, industrial design rights, trademarks, trade dress, and in some jurisdictions trade secrets. There are also more specialized varieties of *sui generis* exclusive rights, such as circuit design rights (called mask work rights in USA law, protected under the Integrated Circuit Topography Act in Canadian law, and in European Union law by Directive 87/54/EEC of 16 December 1986 on the legal protection of topographies of semiconductor products),

plant breeders' rights, plant variety rights, industrial design rights, supplementary protection certificates for pharmaceutical products and database rights (in European law).

Patents

A patent grants an inventor exclusive rights to make, use, sell, and import an invention for a limited period of time, in exchange for the public disclosure of the invention. An invention is a solution to a specific technological problem, which may be a product or a process.

Copyright

A copyright gives the creator of an original work exclusive rights to it, usually for a limited time. Copyright may apply to a wide range of creative, intellectual, or artistic forms, or "works". Copyright does not cover ideas and information themselves, only the form or manner in which they are expressed.

Industrial Design Rights

An industrial design right protects the visual design of objects that are not purely utilitarian. An industrial design consists of the creation of a shape, configuration or composition of pattern or colour, or combination of pattern and colour in three dimensional form containing aesthetic value. An industrial design can be a two- or three-dimensional pattern used to produce a product, industrial commodity or handicraft.

Trademarks

A trademark is a recognizable sign, design or expression which identifies products or services of a particular source from those of others.

Trade Dress

Trade dress is a legal term of art that generally refers to characteristics of the visual appearance of a product or its packaging (or even the design of a building) that signify the source of the product to consumers.

Trade Secrets

A trade secret is a formula, practice, process, design, instrument, pattern, or compilation of information which is not generally known or reasonably ascertainable, by which a business can obtain an economic advantage over competitors or customers. In the US, trade secret law is primarily handled at the state level under the Uniform Trade Secrets Act, which most states have adopted, and a federal law, the Economic Espionage Act of 1996, which makes the theft or misappropriation of a trade secret a federal crime. This law contains two provisions criminalizing two sorts of activity. The first, 18 U.S.C. § 1831(a), criminalizes the theft of trade secrets to benefit foreign powers.

The second, 18 U.S.C. § 1832, criminalizes their theft for commercial or economic purposes. (The statutory penalties are different for the two offences.) Trade secret law varies from country to country.

OBJECTIVES

The stated objective of most intellectual property law (with the exception of trademarks) is to “Promote progress.” By exchanging limited exclusive rights for disclosure of inventions and creative works, society and the patentee/copyright owner mutually benefit, and an incentive is created for inventors and authors to create and disclose their work. Some commentators have noted that the objective of intellectual property legislators and those who support its implementation appears to be “absolute protection.” “If some intellectual property is desirable because it encourages innovation, they reason, more is better. The thinking is that creators will not have sufficient incentive to invent unless they are legally entitled to capture the full social value of their inventions.” This absolute protection or full value view treats intellectual property as another type of ‘real’ property, typically adopting its law and rhetoric. Other recent developments in intellectual property law, such as the America Invents Act, stress international harmonization.

FINANCIAL INCENTIVE

These exclusive rights allow owners of intellectual property to benefit from the property they have created, providing a financial incentive for the creation of an investment in intellectual property, and, in case of patents, pay associated research and development costs. Some commentators, such as David Levine and Michele Boldrin, dispute this justification. In 2013 the United States Patent and Trademark Office claimed that the worth of intellectual property to the U.S. economy is more than US\$5 trillion and creates employment for an estimated 18 million American people. The value of intellectual property is considered similarly high in other developed nations, such as those in the European Union.

Economic Growth

The WIPO treaty and several related international agreements are premised on the notion that the protection of intellectual property rights are essential to maintaining economic growth. The *WIPO Intellectual Property Handbook* gives two reasons for intellectual property laws:

One is to give statutory expression to the moral and economic rights of creators in their creations and the rights of the public in access to those creations. The second is to promote, as a deliberate act of Government policy, creativity and the dissemination and application of its results and to encourage fair trading which would contribute to economic and social development.

The Anti-Counterfeiting Trade Agreement (ACTA) states that “effective enforcement of intellectual property rights is critical to sustaining economic growth across all industries and globally”. Economists estimate that two-thirds of the value of large businesses in the U.S. can be traced to intangible assets. “IP-intensive industries” are estimated to generate 72 per cent more value added (price minus material cost) per employee than “non-IP-intensive industries”.

A joint research project of the WIPO and the United Nations University measuring the impact of IP systems on six Asian countries found “a positive correlation between the strengthening of the IP system and subsequent economic growth.” Economists have also shown that IP can be a disincentive to innovation when that innovation is drastic. IP makes excludable non-rival intellectual products that were previously non-excludable. This creates economic inefficiency as long as the monopoly is held. A disincentive to direct resources towards innovation can occur when monopoly profits are less than the overall welfare improvement to society. This situation can be seen as a market failure, and an issue of appropriability.

Morality

According to Article 27 of the Universal Declaration of Human Rights, “everyone has the right to the protection of the moral and material interests resulting from any scientific, literary or artistic production of which he is the author”. Although the relationship between intellectual property and human rights is a complex one, there are moral arguments for intellectual property. The arguments that justify intellectual property fall into three major categories. Personality theorists believe intellectual property is an extension of an individual. Utilitarians believe that intellectual property stimulates social progress and pushes people to further innovation. Lockeans argue that intellectual property is justified based on deservedness and hard work.

Various moral justifications for private property can be used to argue in favour of the morality of intellectual property, such as:

1. *Natural Rights/Justice Argument:* this argument is based on Locke’s idea that a person has a natural right over the labour and/or products which is produced by his/her body. Appropriating these products is viewed as unjust. Although Locke had never explicitly stated that natural right applied to products of the mind, it is possible to apply his argument to intellectual property rights, in which it would be unjust for people to misuse another’s ideas. Locke’s argument for intellectual property is based upon the idea that laborers have the right to control that which they create. They argue that we own our bodies which are the laborers, this right of ownership extends to what we create. Thus, intellectual property ensures this right when it comes to production.
2. *Utilitarian-Pragmatic Argument:* according to this rationale, a society that protects private property is more effective and prosperous than societies that do not. Innovation and invention in 19th century America has been said to be

attributed to the development of the patent system. By providing innovators with “durable and tangible return on their investment of time, labour, and other resources”, intellectual property rights seek to maximize social utility. The presumption is that they promote public welfare by encouraging the “creation, production, and distribution of intellectual works”. Utilitarians argue that without intellectual property there would be a lack of incentive to produce new ideas. Systems of protection such as Intellectual property optimize social utility.

3. “*Personality*” *Argument*: this argument is based on a quote from Hegel: “Every man has the right to turn his will upon a thing or make the thing an object of his will, that is to say, to set aside the mere thing and recreate it as his own”. European intellectual property law is shaped by this notion that ideas are an “extension of oneself and of one’s personality”. Personality theorists argue that by being a creator of something one is inherently at risk and vulnerable for having their ideas and designs stolen and/or altered. Intellectual property protects these moral claims that have to do with personality.

Lysander Spooner (1855) argues “that a man has a natural and absolute right—and if a natural and absolute, then necessarily a perpetual, right—of property, in the ideas, of which he is the discoverer or creator; that his right of property, in ideas, is intrinsically the same as, and stands on identically the same grounds with, his right of property in material things; that no distinction, of principle, exists between the two cases”.

Writer Ayn Rand argued in her book *Capitalism: The Unknown Ideal* that the protection of intellectual property is essentially a moral issue. The belief is that the human mind itself is the source of wealth and survival and that all property at its base is intellectual property. To violate intellectual property is therefore no different morally than violating other property rights which compromises the very processes of survival and therefore constitutes an immoral act.

INFRINGEMENT, MISAPPROPRIATION, AND ENFORCEMENT

Unauthorized use of intellectual property rights, called “infringement” with respect to patents, copyright, and trademarks, and “misappropriation” with respect to trade secrets, may be a breach of civil law or criminal law, depending on the type of intellectual property, jurisdiction, and the nature of the action.

Patent infringement typically is caused by using or selling a patented invention without permission from the patent holder. The scope of the patented invention or the extent of protection is defined in the claims of the granted patent. There is safe harbour in many jurisdictions to use a patented invention for research. This safe harbour does not exist in the US unless the research is done for purely philosophical purposes, or in order to gather data in order to prepare an application for regulatory approval of a drug. In

general, patent infringement cases are handled under civil law (*e.g.*, in the US) but several jurisdictions incorporate infringement in criminal law also (for example, Argentina, China, France, Japan, Russia, South Korea).

Copyright infringement is reproducing, distributing, displaying or performing a work, or to make derivative works, without permission from the copyright holder, which is typically a publisher or other business representing or assigned by the work's creator. It is often called "piracy". While copyright is created the instance a work is fixed, generally the copyright holder can only get money damages if the owner registers the copyright. Enforcement of copyright is generally the responsibility of the copyright holder. The ACTA trade agreement, signed in May 2011 by the United States, Japan, Switzerland, and the EU, requires that its parties add criminal penalties, including incarceration and fines, for copyright and trademark infringement, and obligated the parties to active police for infringement. There is a safe harbour to use copyrighted works under the fair use doctrine.

Trademark infringement occurs when one party uses a trademark that is identical or confusingly similar to a trademark owned by another party, in relation to products or services which are identical or similar to the products or services of the other party. As with copyright, there are common law rights protecting a trademark, but registering a trademark provides legal advantages for enforcement. Infringement can be addressed by civil litigation and, in several jurisdictions, under criminal law. In the US, the Trademark Counterfeiting Act of 1984 criminalized the intentional trade in counterfeit goods and services and ACTA amplified the penalties.

Trade secret misappropriation is different from violations of other intellectual property laws, since by definition trade secrets are secret, while patents and registered copyrights and trademarks are publicly available. In the US, trade secrets are protected under state law, and states have nearly universally adopted the Uniform Trade Secrets Act. The US also has federal law in the form of the Economic Espionage Act of 1996 (18 U.S.C. §§ 1831–1839), which makes the theft or misappropriation of a trade secret a federal crime. This law contains two provisions criminalizing two sorts of activity. The first, 18 U.S.C. § 1831(a), criminalizes the theft of trade secrets to benefit foreign powers. The second, 18 U.S.C. § 1832, criminalizes their theft for commercial or economic purposes. (The statutory penalties are different for the two offences.) In Commonwealth common law jurisdictions, confidentiality and trade secrets are regarded as an equitable right rather than a property right but penalties for theft are roughly the same as the US. As of 2011 trade in counterfeit copyrighted and trademarked works was a \$600 billion industry worldwide and accounted for 5 per cent–7 per cent of global trade.

CRITICISMS

Some critics of intellectual property, such as those in the free culture movement, characterize it as intellectual protectionism, intellectual monopoly or government-granted monopoly, and argue the public interest is harmed by protectionist legislation

such as copyright extension, software patents and business method patents. A critique against the idea of *intellectual property* has been formulated by Eben Moglen in his *dot Communist Manifesto*:

Society confronts the simple fact that when everyone can possess every intellectual work of beauty and utility—reaping all the human value of every increase of knowledge—at the same cost that any one person can possess them, it is no longer moral to exclude. If Rome possessed the power to feed everyone amply at no greater cost than that of Caesar’s own table, the people would sweep Caesar violently away if anyone were left to starve. But the bourgeois system of ownership demands that knowledge and culture be rationed by the ability to pay.

—Eben Moglen *dot Communist Manifesto*

The Term Itself

Free Software Foundation founder Richard Stallman argues that, although the term *intellectual property* is in wide use, it should be rejected altogether, because it “systematically distorts and confuses these issues, and its use was and is promoted by those who gain from this confusion.” He claims that the term “operates as a catch-all to lump together disparate laws [which] originated separately, evolved differently, cover different activities, have different rules, and raise different public policy issues” and that it creates a “bias” by confusing these monopolies with ownership of limited physical things, likening them to “property rights”. Stallman advocates referring to copyrights, patents and trademarks in the singular and warns against abstracting disparate laws into a collective term. Similarly, Boldrin and Levine prefer to use the term “intellectual monopoly” as a more appropriate and clear definition of the concept.

Lawrence Lessig, along with many other copyleft and free software activists, has criticized the implied analogy with physical property (like land or an automobile). They argue such an analogy fails because physical property is generally rivalrous while intellectual works are non-rivalrous (that is, if one makes a copy of a work, the enjoyment of the copy does not prevent enjoyment of the original). Other arguments along these lines claim that unlike the situation with tangible property, there is no natural scarcity of a particular idea or information: once it exists at all, it can be re-used and duplicated indefinitely without such re-use diminishing the original. Stephan Kinsella has objected to *intellectual property* on the grounds that the word “property” implies scarcity, which may not be applicable to ideas.

Alternative Terms

In civil law jurisdictions, intellectual property has often been referred to as intellectual rights, traditionally a somewhat broader concept that has included moral rights and other personal protections that cannot be bought or sold. Use of the term *intellectual rights* has declined since the early 1980s, as use of the term *intellectual property* has increased.

Alternative terms *monopolies on information* and *intellectual monopoly* have emerged among those who argue against the “property” or “intellect” or “rights” assumptions, notably Richard Stallman. The backronyms *intellectual protectionism* and *intellectual poverty*, whose initials are also *IP*, have found supporters as well, especially among those who have used the backronym *digital restrictions management*. The argument that an intellectual property right should (in the interests of better balancing of relevant private and public interests) be termed an *intellectual monopoly privilege* (IMP) has been advanced by several academics including Birgitte Andersen and Thomas Alured Faunce.

OBJECTIONS TO OVERBROAD INTELLECTUAL PROPERTY LAWS

Some critics of intellectual property, such as those in the free culture movement, point at intellectual monopolies as harming health (in the case of pharmaceutical patents), preventing progress, and benefiting concentrated interests to the detriment of the masses, and argue that the public interest is harmed by ever expansive monopolies in the form of copyright extensions, software patents, and business method patents. More recently scientists and engineers are expressing concern that patent thickets are undermining technological development even in high-tech fields such as nanotechnology. Peter Drahos notes that “Property rights confer authority over resources. When authority is granted to the few over resources on which many depend, the few gain power over the goals of the many. This has consequences for both political and economic freedoms with in a society.”

The World Intellectual Property Organization (WIPO) recognizes that conflicts may exist between the respect for and implementation of current intellectual property systems and other human rights. In 2001 the UN Committee on Economic, Social and Cultural Rights issued a document called “Human rights and intellectual property” that argued that intellectual property tends to be governed by economic goals when it should be viewed primarily as a social product; in order to serve human well-being, intellectual property systems must respect and conform to human rights laws. According to the Committee, when systems fail to do so they risk infringing upon the human right to food and health, and to cultural participation and scientific benefits. In 2004 the General Assembly of WIPO adopted *The Geneva Declaration on the Future of the World Intellectual Property Organization* which argues that WIPO should “focus more on the needs of developing countries, and to view IP as one of many tools for development—not as an end in itself”.

Further along these lines, The ethical problems brought up by IP rights are most pertinent when it is socially valuable goods like life-saving medicines are given IP protection. While the application of IP rights can allow companies to charge higher than the marginal cost of production in order to recoup the costs of research and development, the price may exclude from the market anyone who cannot afford the

cost of the product, in this case a life-saving drug. “An IPR driven regime is therefore not a regime that is conducive to the investment of R&D of products that are socially valuable to predominately poor populations....”

Some libertarian critics of intellectual property have argued that allowing property rights in ideas and information creates artificial scarcity and infringes on the right to own tangible property. Stephan Kinsella uses the following scenario to argue this point: [I]magine the time when men lived in caves. One bright guy—let’s call him Galt-Magnon—decides to build a log cabin on an open field, near his crops. To be sure, this is a good idea, and others notice it. They naturally imitate Galt-Magnon, and they start building their own cabins. But the first man to invent a house, according to IP advocates, would have a right to prevent others from building houses on their own land, with their own logs, or to charge them a fee if they do build houses. It is plain that the innovator in these examples becomes a partial owner of the tangible property (*e.g.*, land and logs) of others, due not to first occupation and use of that property (for it is already owned), but due to his coming up with an idea. Clearly, this rule flies in the face of the first-user homesteading rule, arbitrarily and groundlessly overriding the very homesteading rule that is at the foundation of all property rights.

Thomas Jefferson once said in a letter to Isaac McPherson on August 13, 1813:

“If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea, which an individual may exclusively possess as long as he keeps it to himself; but the moment it is divulged, it forces itself into the possession of every one, and the receiver cannot dispossess himself of it. Its peculiar character, too, is that no one possesses the less, because every other possesses the whole of it. He who receives an idea from me, receives instruction himself without lessening mine; as he who lights his taper at mine, receives light without darkening me.”

In 2005 the RSA launched the Adelphi Charter, aimed at creating an international policy statement to frame how governments should make balanced intellectual property law. Another limitation of current U.S. Intellectual Property legislation is its focus on individual and joint works; thus, copyright protection can only be obtained in ‘original’ works of authorship. This definition excludes any works that are the result of community creativity, for example Native American songs and stories; current legislation does not recognize the uniqueness of indigenous cultural ‘property’ and its ever-changing nature. Simply asking native cultures to ‘write down’ their cultural artifacts on tangible mediums ignores their necessary orality and enforces a Western bias of the written form as more authoritative.

ACCESS KNOWLEDGE OF INTELLECTUAL PROPERTY

Intellectual property law has been the location of tremendous conflict and flux in recent years. IP rights have become significantly stronger over the past thirty years, in

both the domestic and international realms. The most widely accepted explanation for this trend is derived from public choice theory. IP rights, the argument goes, create opportunities for potentially lucrative rents. Businesses that could benefit from such rents recognize this fact and will generally be willing to spend up to the amount of their potential rents in order to secure these rights. Those most hurt by stronger IP are industries based upon copying, which do not enjoy monopoly rents, and average consumers, each of whom may be hurt in small ways and/or far in the future. In the “market” for law, then, IP industries purportedly enjoy a significant advantage.

How, then, are we to understand the recent countermovement that has emerged, and the recent shift in the political valence of IP law? This Part describes the recent strengthening of IP law and the emerging countermobilization and explains why public choice theories do not, in fact, fully and satisfactorily explain either event. Acts of framing have been central to both contexts and have permitted those involved to interpret their interests, forge common cause with others, and justify the legal action they have sought in terms that can persuade others. The frames adopted in the process of this mobilization and countermobilization matter because frames are not merely resources that can be wielded to serve their makers. They also generate opportunities for a group’s opponents and make possible unpredictable chains of argument and counterargument.

THE HISTORICAL EVOLUTION OF ENCLOSURE AND A2K

As many scholars have noted, “By virtually any measure, intellectual property rights have expanded dramatically in the last three decades.” Yochai Benkler and James Boyle have analogized the shift to a new “enclosure movement.” Whatever it is called, the nature of the trend is clear: over the past thirty years, exclusive rights over information have grown broader (to cover more kinds of information), deeper (giving IP owners more robust rights of exclusion), and more severe (imposing greater penalties on infringers).

This worldwide phenomenon has been driven significantly by developments in the United States, so we can begin our discussion here. In recent years, the scope of patentability expanded significantly, standards for non-obviousness diminished, the experimental-use exemption was weakened, and patents became significantly more likely to be upheld in the face of challenges in the courts.

The protectable subject matter of copyright law likewise expanded, most prominently to include software. The term of copyright protection was extended repeatedly, and enforcement actions against private, non-commercial copiers became more common. Penalties for violating copyright became harsher, and the rights accorded to owners, for example to prevent derivative uses, became more robust. Finally, entirely new forms of protection emerged, such as a *sui generis* system to protect plant varieties and exclusive rights in data used to register pharmaceutical products.

Interesting questions can be raised about whether all of these changes in law should be categorized together and about the appropriate baseline against which to measure

the “strengthening” of IP law. There is clearly room for further specification of the claim, but on balance, it is fair to say, as many scholars have, that IP law has become stronger, both at a formal level and practically in many respects.

New technologies also make it possible for rights holders to exert more control over information at the code level. “Digital rights management” (DRM) tools or “technical protection measures” have become an important part of the contemporary appropriation strategies of the information industries. Adobe uses DRM, for example, to prevent readers of its eBooks from copying text or using read-aloud programmes. Sony has developed “sterile” CDs, which permit users to make only one copy, which in turn cannot be copied. Agriculture has seen the development of its own analogue to DRM technologies, so-called terminator genes and other genetic use restriction technologies designed to enhance the excludability of proprietary plant varieties.

Like DRM, these technologies frequently prevent uses that are permitted in patent law. In a sense, information industries are using technical protection measures to alter the genetic nature of information, manufacturing scarcity where before there was reproducibility. Although these changes are technical in nature, they have been accompanied by a supportive legal regime.

Terminator technologies could obviously be the subject of government regulation, and digital encryption schemes and copy protection measures rely upon laws such as the Digital Millennium Copyright Act (DMCA), which make it illegal in most circumstances to attempt to circumvent DRM technologies. The trend of expansion in the international realm is even more striking. The TRIPS Agreement requires all members of the WTO to implement high levels of substantive IP protection. It requires, for example, patents in all fields of technology and of no less than twenty years in duration, some form of plant-variety protection, adherence to most of the Berne Convention on Copyright, and copyright protection for software object and source code. Moreover, TRIPS is just a floor.

The WIPO Copyright Treaty (WCT), which came into force in 2002, is an “optional” WIPO agreement that the United States is pressing a growing number of countries to join through bilateral trade agreements. It adds to the substantive protections in the Berne Convention, for example by requiring signatories to create criminal as well as civil sanctions for the “removal or alteration of any electronic rights management information without authority.”

The United States and European Union have also used bilateral and regional trade agreements to add to the substantive protections required by TRIPS. Since 2000, the United States has completed free trade agreements with more than a dozen countries, and it is pursuing agreements with many more. Such agreements typically require signatories to increase IP protection in a host of areas, for example, by providing sui generis protection for pharmaceutical registration data, limiting the grounds on which compulsory licenses can be granted, providing for the extension of patent terms to compensate for delays arising from regulatory approval processes, providing for patents

on life forms, adhering to the International Union for the Protection of New Varieties of Plants (UPOV) and the WIPO Copyright Treaty, enacting local versions of the DMCA, and extending copyright terms. Over the last ten to fifteen years, however, numerous groups have emerged to contest the recent expansion of intellectual property. They have recently begun to forge alliances with one another and, jointly and severally, have had a substantial effect on both the substance and political valence of IP law.

Some of the earliest moments in this recent mobilization came on the heels of the negotiations over the TRIPS Agreement. In 1993, for example, more than five hundred thousand farmers demonstrated in Bangalore, India, demanding that their government reject the TRIPS Agreement and exclusive rights in seed stocks for multinational firms.

It was only one in a series of similar actions around India in the surrounding months. The Indian protests were the most visible manifestation of a network of farmers in developing countries and NGOs organizing around the subject, united with “striking uniformity” by their sense that “the international IPR regime is heavily weighted against farmers.” Campaigns such as these rapidly built cross-border networks in order to contest newly internationalizing IP law. In 1993, for example, farmers’ rights advocates built an international campaign against one company’s efforts to obtain U.S. and E.U. patents related to extractions of an insecticidal compound from the Indian neem tree.

The fact that the tree could be used to produce a pesticide had been known for many years in India. Activists filed a patent reexamination request in the United States accompanied by signatures “of more than 100,000 Indians, as well as by more than 225 agricultural, scientific, and trade groups in 45 countries.” Explaining the action, a leader of the group said: “The real battle is whether the genetic resources of the planet will be maintained as a shared commons or whether this common inheritance will be commercially enclosed and become the intellectual property of a few big corporations.” The U.S. reexamination appeal failed, but the opposition to the European patent succeeded, providing what campaigners pronounced the “First Legal Defeat of a Biopiracy Patent.”

In 1995, the Clinton Administration launched an initiative to strengthen copyright law and to introduce sui generis protections for databases both in the United States and (with the help of the European Union) at the international level through WIPO. Almost immediately, the initiatives met strong opposition from scientific, academic, and consumer rights circles. The opposition quickly internationalized, as domestic opponents to the Clinton Administration’s plan went to WIPO to press their case. The amount and intensity of attention to copyright and database protection issues among academic and public interest groups was unprecedented, as was the international public interest coalition that emerged at WIPO.

Their efforts met with significant success: “In the end, none of the original U.S.-sponsored digital agenda proposals emerged unscathed from the negotiation process, and at least one—the proposed database treaty—did not emerge at all.” A few years later, the access-to-medicines campaign was born, when AIDS activists and humanitarian

organizations joined forces to demand antiretroviral medicines for the millions of people in developing countries dying of HIV/AIDS. Patented AIDS medicines were extremely expensive, and campaigners focused their attention on the refusal of patent-holding companies to offer significant discounts or permit the use of generic alternatives and on the limits that the TRIPS Agreement and the trade policies of countries such as the United States put on countries' abilities to override patents.

Activists adopted confrontational tactics and achieved rapid results. In April 2000, they forced thirty-nine multinational drug companies to abandon a high-profile lawsuit challenging a South African law designed to reduce the price of medicines there. Another important victory came in 2001 at a WTO ministerial meeting in Qatar.

As the Wall Street Journal reported, "unlike in 1993, when intellectual-property protections were first negotiated as part of the initial WTO pact, this time the [drug company] lobbyists were matched by AIDS activists who proved to be a well-coordinated group of opponents." The result was the "Doha Declaration on the TRIPS Agreement and Public Health," which states that the TRIPS Agreement "can and should be interpreted and implemented in a manner supportive of WTO Members' right to protect public health and, in particular, to promote access to medicines for all."

The Doha Declaration also set in motion a process that resulted in the first ever amendment to TRIPS (or indeed, to any core WTO agreement), which gives developing countries marginally more flexibility to use generic medicines. But perhaps the most significant measure of the success of the campaign has been the drastic fall in the price of antiretroviral medicines.

In a few years, the world-best price of first-line triple-combination HIV/AIDS therapy purchased from originator companies dropped by ninety-five per cent, and generics became available in many developing countries at a discount of ninety-nine per cent. In 2003, two Swarthmore students, Nelson Pavlovsky and Luke Smith, founded the "Swarthmore Coalition for the Digital Commons." The group met regularly to discuss what they called issues of the "intellectual commons," including "drug patents, music downloading, and Monsanto's crop patents."

It was just a discussion group, until Pavlovsky and Smith decided to post internal memos from a company called Diebold on the Internet. The memos described systemic flaws in Diebold's voting machines, and Diebold was keen to keep them out of the public eye. The company sent the students (and many others like them) cease-and-desist letters asserting that the posting was an act of copyright infringement.

The students found some powerful allies, Diebold retreated, and Smith and Pavlovsky ultimately won a copyright abuse suit against the company. They then set their sights on a new challenge: creating a "Free Culture" movement on college campuses around the world, in order to "reverse the recent radical expansion of intellectual property" and "build a technological and cultural movement to defend the digital commons."

Despite its recent beginnings, it now has more than thirty stages in the United States and some germinating in other countries too. As a profile in the New York Times recently

remarked, members of the Free Culture movement “talk about the group’s goals with something like the reverence that earlier generations displayed in talking about social or racial equality.”

The Free Culture movement is in many ways the child of the Creative Commons movement, which was begun in 2001 by law professor Lawrence Lessig, with the support of colleagues such as James Boyle and Eric Eldred. The Creative Commons project seeks to “build a layer of reasonable copyright on top of the extremes that now reign.” It does this by offering creators a series of copyright licenses that give users more rights than they would have under the default rules of copyright law, for example the right to make derivative works or to reproduce covered material for non-commercial purposes. Use of the licenses has grown exponentially in the last few years, and today they govern an estimated sixty million copyrighted works around the world.

Recent years have also seen the emergence and extraordinary success of free and open-source software paradigms. Such software is created by volunteers, and sustained by an open-licensing scheme that guarantees users’ rights to share and modify the software source code. Free and open-source software programmes such as Linux have become major components in the worldwide software market, and have been adopted by leading corporations such as IBM, HP, and Dell. Successful new enterprises have also emerged to sell services and support to users of free software, demonstrating that new business models can be built around informational products that are not governed by rules of exclusivity.

Because free and open-source software has been both wildly successful and contradicts “our longstanding perceptions of how people behave and how economic growth occurs,” it has also been central to the theorization of a new mode of production that is characteristic of the digital networked age: the “commons-based peer production model,” which “relies on decentralized information gathering and exchange.” Increasingly, programmers have moved beyond bottom-up private-ordering strategies to advocate top-down legal change. They have focused particularly on software patents, which many believe present a significant threat to the collaborative processes that produce free and open-source software.

Thus, programmers—supported by corporations that have put open-source software at the center of their business models—recently launched a vocal campaign against an E.U. patent directive that would have ensured the availability of software patents throughout Europe. By mobilizing hundreds of demonstrators and lobbyists, these programmers helped persuade the European Parliament to abandon that directive. A German parliamentarian who led the attempt to pass the law gave the following pithy account of its demise: ““They produced a whole movement....

Industry was sleeping.” More and more programmers seem to be warming to the activist role. In France, computer enthusiasts recently staged street demonstrations and demanded to be arrested for violating digital rights management systems that, for example, prevent an individual from playing iTunes music files on a portable player

other than an iPod. In the past several years, there have also been attempts by key architects of this new politics to build something more sustained and interconnected. The recent surge of advocacy around intellectual property issues has always oscillated between local and global, often within very short timeframes and in the same campaign. But attempts to conjoin all of these groups under the rubric of the “access to knowledge movement” and to try to create a common framework through which to articulate their concerns came only very recently.

Perhaps the most significant flashpoint was in 2004, when Brazil and Argentina, seeking to capitalize on the growing discontent among public interest groups and developing-country governments regarding international IP law, proposed that WIPO revisit its mandate and adopt what they called a “development agenda.”

The aim was to secure a new commitment within WIPO to the concerns and needs of developing countries and a willingness to explore the potential of non-IP based models of innovation, such as open-source software and open genomics. In conjunction with this proposal, advocates organized a meeting in Geneva that brought policymakers and business representatives together with participants in the access-to-medicines, free-software, Creative Commons, open-science, and open-publishing campaigns. The meeting endorsed the proposal for a development agenda, and produced a three-page document entitled the “*Geneva Declaration on the Future of the World Intellectual Property Organization*.” Signed by more than five hundred individuals, this declaration reads like nothing so much as a manifesto for a new movement.

“Humanity faces a global crisis in the governance of knowledge, technology, and culture,” the Geneva Declaration proclaims, and cites as evidence widespread lack of access to medicines, global inequalities in access to education and technology, growing obstacles to follow-on innovation, misappropriation of the public domain, and increasing concentration and anticompetitive practices in the knowledge industries.

It goes on to emphasize the success of new modes of knowledge production such as open-source software and Wikipedia, and insists that “humanity stands at a crossroads—a fork in our moral code and a test of our ability to adapt and grow.” It poses the choice as one between existing policies, which it calls “intellectually weak, ideologically rigid, and sometimes brutally unfair and inefficient,” and new models to produce and govern informational goods. It calls for a “moratorium on new treaties and harmonization of standards that expand and strengthen monopolies and further restrict access to knowledge,” urges the negotiation of a “Treaty on Access to Knowledge and Technology,” and demands fundamental procedural reforms in WIPO to render it more responsive to the needs of developing countries and more open to participation from public interest groups.

The WIPO membership voted to create a committee to consider the adoption of a development agenda, and NGOs and activists took advantage of the opening by convening further international meetings to define that agenda. One result was a document that might be thought of as a preliminary campaign platform for this new

mobilization: a draft Access to Knowledge Treaty. Two themes unite many of the proposals in the text: the idea that “restrictions on access ought to be the exception, not the other way around,” and that “both subject matter exclusions from, and exceptions and limitations to, intellectual property protection standards are mandatory rather than permissive.” Notably, the treaty addresses not only users’ rights, but also rights and structures intended to benefit open-source models of innovation.

The Access to Knowledge Treaty is a prototype more than a completed proposal, but intraissue collaborations have since spread to other fora. In 2005, leading figures in the free software movement, the farmers’ rights movement, the open genomics movement, the Creative Commons, librarian organizations, and the access-to-medicines movement, along with the Minister of Culture of Brazil, drafted the “Adelphi Charter,” a set of principles that they contend governments and international agencies should respect when modifying IP laws.

Low-level conferences to generate dialogue between groups such as software programmers and farmers’ rights advocates are also being organized. Knowledge-rights activists are also increasingly issuing calls for greater cooperation and sharing between them. An editorial recently published by the farmers’ rights organization GRAIN, for example, urged that if those “working on free software, no-patents-on-life, access to generic drugs, traditional medicine, digital rights, peer-to-peer networking and ‘fair use’ came together and formulated one common platform to rein in the IPR system, the effect could be explosive.” New umbrella organizations such as “IP-Watch” and “IP Justice” have recently been created to contribute to and to report on the activities of these groups.

These overt attempts to build interissue platforms are also mirrored by a wave of everyday cross-references and collaborations. For example, when asked why the Brazilian government was migrating its computers to open-source software, Brazil’s top technology official, Sergio Amadeu, explained that the shift would reduce licensing fees, support a national effort to increase computer access, and promote the development of local technological industries. Then he added a fourth reason: “Free software is like generic drugs.” Creative Commons explains that it drew inspiration for its licenses from those that govern free software. Richard Stallman, one of the founders of the free-software movement, recently published a letter in an open content journal declaring that free software and open publishing are “based on the same fundamental principle: knowledge contributes to society when it can be shared and developed by communities.”

A prominent farmer’s rights organization encourages visitors to its Web site to learn about “Software Freedom Day.” Scientist Richard Jefferson, who recently invented an important new genomics tool that is governed by an open-source license (also modeled on open-source software licenses), states that “everything that open source has been fighting in software is exactly where we find ourselves now with biotechnology.”

NGOs and activist coalitions that emerged independently of one another to contest the contours of IP rights in seeds, medicines, software, genetic material, and cultural

goods are thus beginning to build links to one another. The structure of this emerging mobilization is more akin to a network than a pyramid. Like many networks, this one is characterized not only by horizontal connections between actors engaged in advocacy around IP, but also by “hubs,” or groups and individuals who are prominent within the network for the density and intensity of their connections. These key participants are seeking to create a set of shared principles, arguments, and identities between groups that are otherwise divided by their substantive or regional focus.

These groups have also begun to have, jointly and severally, a significant impact on IP law. Some such effects: the successes of the access-to-medicines campaign in obtaining an amendment to TRIPS and bringing down the prices of HIV/AIDS medicines; the success of free-software programmers in preventing the codification of software patents in Europe; and the expansive growth of the private ordering schemes introduced by proponents of free software and the Creative Commons. Some of the most significant changes that these groups have inaugurated are those underway at WIPO. The political dynamic triggered by the development-agenda process has created a significant challenge to the agency’s mandate. Negotiations over new treaties creating broadcasting rights and further harmonizing substantive patent law very recently fell apart, with developed countries disagreeing with one another, developing countries objecting, and A2K groups in active opposition.

AUK advocates, in concert with supportive developing country governments, successfully pressed WIPO to create a new standing committee to discuss the impact of IP on development, to make the agency’s technical assistance programmes better serve the interests of developing countries, and to “promote norm-setting activities related to IP that support a robust public domain in WIPO’s Member States.” Developing countries have also urged WIPO to take up proposals to articulate minimum exceptions and limitations in areas such as copyright, pointing to the possibility that WIPO could be used to create ceilings, and not just floors, for IP rights.

In the United States, significant changes in patent law are also underway. Major reports from both the Federal Trade Commission and the National Academy of Sciences have expressed concern that patent law has become overgrown and have recommended major reform. After many years of implicitly ceding the realm to the Federal Circuit, the Supreme Court has stepped decisively back into the arena. The past few years have seen “the highest level of patent activity at the Court in forty years.” The result has been a series of decisions that have diminished the power of patents. Precisely how much awaits determination in the lower courts, but these cases have sent a clear signal of concern that the U.S. patent system has become too strong and may be stifling rather than promoting innovation.

A new bipartisan agreement on trade policy that was recently negotiated in Congress has ratcheted back the U.S. Trade Representative’s (USTR’s) mandate to increase IP protection through free trade agreements, particularly with developing country trading partners.

As of this writing, a bipartisan patent reform bill is also pending that would make meaningful changes in the patent system in order to address perceived problems with patent quality and the expense of patent litigation. The process that has led to this reform effort has also turned up significant rifts in the coalition of IP industries. As one newspaper editorial page characterized it, patent reform negotiations have brought on a fight between two usually like-minded allies, biotech and computer technology. Other groups—financial services, universities, research firms, and small inventors—are taking sides too.... In broad terms, large tech firms want patents harder to get, easier to challenge and worth less when infringement occurs.

Big players, such as Cisco or Intel, claim “patent trolls” target their products with longshot claims over a small part in a router or computer chip.... Biotech has a different gripe. A pill built around a lab-created molecule or protein fold may need only a few patents. Biotech firms spend millions of dollars to develop silver-bullet drugs that need years of sales to pay back the investment. These firms want the nuclear option that tough patent defences bring: Infringe on our product, and we’ll destroy you in court. All of these signals point to a significant shift in the political valence of the field of IP in a few short years. A once strong industry coalition is beginning to fray. Courts, legislators, and international agencies are increasingly receptive to arguments that IP rights have become too restrictive.

Calls for a reorientation in the field are increasingly audible, and public interest groups and developing countries are enjoying increasing success in asserting their agenda at local and international levels. Groups that have emerged to contest IP norms in different subject areas are now forming alliances across their differences. How should we understand these events?

The predominant account in IP scholarship of the recent expansion in IP law draws on public choice theory. William Landes and Richard Posner, for example, point to the “inherent asymmetry between the value that creators of intellectual property place on having property rights and the value that would-be copiers place on the freedom to copy without having to obtain a license” that results from the fact that exclusive rights “can shower economic rents on the holder of that right, but copiers can hope to obtain only a competitive return.” These pressures, they suggest, account for some portion of the recent and rapid growth in copyright law.

Over the years, many scholars have called attention to the same issue, often drawing upon Jessica Litman’s important early work demonstrating that the drafting of U.S. copyright law had been effectively delegated to competing interest groups. Today, a wide range of IP scholars have come to agree that rights holders have had a theoretically and practically disproportionate influence on IP lawmaking, and to use public choice theories to explain this fact. The boldest articulations of the argument suggest that public choice pressures will exist wherever there are exclusive rights in information. Yochai Benker, for example, writes: Our legislative process demonstrates a systematic imbalance in favour of the expansion and deepening of exclusive rights to information

at the expense of the public domain. The imbalance exists because the benefits of such rights are clearly seen by, and expressed by, well-defined interest holders that exist at the time the legislation is passed. In contrast, most of the social costs—which are economic, social, political, and moral—are diffuse and likely to be experienced in the future by parties not yet aware of the fact that they will be affected by the extension of rights. He further contends that “it is never the case that the diffuse and future users [of information] will band together to expand fair use. Even if they were to band together, it is impossible that copyright owners would remain unaware of the initiative and fail to offer substantial opposition in the legislative process.”

Those who have sought to provide remedies have invoked a familiar public choice solution to the problem: courts. Benkler has argued, for example, that “it is the role of courts to prevent the systematic and excessive expansion of exclusive rights by serving as a backstop” against the powerful lobbying advantages that exclusive-rights-based industries enjoy. He proposes that courts apply intermediate scrutiny to all copyright legislation. Neil Netanel has contended similarly that “rigorous, albeit ‘intermediate,’ scrutiny is warranted in [copyright] cases in part because speech entitlement allocations give rise to a suspicion of successful rent seeking by the highly organized interests to whom the entitlements are granted.”

Such arguments were in fact explicitly (and unsuccessfully) presented to the Supreme Court as grounds for striking down a retroactive extension of the copyright term in *Eldred v. Ashcroft*. These arguments, in their strongest form, suggest that public mobilization against increasingly strong IP law is improbable or even impossible. And yet, such a mobilization appears to be emerging and changing the political valence of IP law. Paradoxically, the *Eldred* case may have played a role in this, by providing a locus for mobilization of groups outside the court. At the same time, an industry coalition that once appeared invincible now shows signs of division.

Existing public choice accounts in the field did not predict this. Public choice theory presumes that social actors have fixed interests, and that they do not need to make complex judgements to determine how their interests can be advanced. It also does not explain how non-material motivations and resources affect collective action. The frame-analytic perspective can help us understand the socially mediated process through which preferences and collective-action strategies are developed. It can therefore offer us richer accounts of both the A2K mobilization and the mobilization of IP industries.

FRAME MOBILIZATION IN IP INDUSTRIES

Can the recent increase in IP protection both locally and globally really be explained as the direct product of the rents associated with exclusive rights regimes, as the broadest public choice accounts suggest? A closer examination indicates not. In their account of the recent strengthening of copyright law, William Landes and Richard Posner point to a gap in the logic of public choice theories: if IP laws invariably create rent-seeking pressures, the public choice problems associated with IP law are timeless. Why, then,

have they suddenly come to have such a significant effect? Economic models suggest that information has become more valuable, but this has been a much more gradual process than the recent shifts in the law. To explain the inflection point that seems to have occurred about thirty years ago, Landes and Posner turn to influences such as political context and “ideological currents.”

They cite, for example, the “free-market ideology” that came to prominence in the late 1970s and argue that “given the historically and functionally close relation between markets and property rights, it was natural for free-market ideologists to favour an expansion of intellectual property rights.” They refer to such forces as the decline of U.S. competitiveness internationally and the decreased hostility of new Supreme Court nominees and executive agencies to strong IP rights. Their conclusion: “political forces and ideological currents... abetted by interest-group pressures that favour originators of intellectual property over copiers, may explain the increases” in copyright protection that we have seen.

One of the most recent and sophisticated public choice accounts in the field thus ultimately treats ideology and context, and not material interests, as the fulcrum of change. Other scholars have also pointed to the influence of the conceptual realm when they seek to understand the timing and velocity of recent changes in IP law. Mark Lemley and William Fisher, for example, argue that the term “intellectual property” did not come into widespread use in the United States until the 1960s.

What we today call IP was once more commonly called “industrial property” or “monopoly,” and many scholars contend that this shift in terminology itself has helped to legitimate and effectuate the recent expansion of IP law. Most such arguments focus on the purported impact of the word property, or the “proprertization” of intellectual property. They point out that it is not obvious that patents, copyright, and trademarks should be thought of as a species of property. They might instead be treated as a branch of tort law (for example, through rubrics of unfair competition or misappropriation), or as a kind of government regulation or subsidy. And of course patent, copyright, and trademark law are quite unlike property law in many ways. Others contend that what has changed is less the use of the word “property” in relation to immaterial goods, but the “use of the combination of the words ‘intellectual’ and ‘property’ as a catch-all phrase to denote a large variety of disparate rights—in other words, the ‘intellectualizing’ of property.”

Scholars have thus pointed to several areas in which ideas and acts of interpretation may affect the structure of IP law, but there has been as yet no systematic attempt to link these insights together and incorporate them into our accounts of the political economy of the field. The insights of framing theory are rarely applied to corporate actors, perhaps because they are usually excluded by definition from the social movements that framing theorists usually study. We apply the paradigm here not to suggest that the IP industries constitute a social movement, but to call attention to the processes of interpretation that industry actors must engage in before they act, and

particularly before they act collectively. Even corporate interests are not invariably fixed and given. As importantly, corporate actors' theories about how such interests can be advanced (diagnoses and prognoses, in frame-analytic terms) are subject to the usual set of interpretive difficulties that framing theorists describe.

That does not mean that the assumptions of public choice theory—that actors have fixed interests and are able to ascertain with a high degree of certainty how to advance them—are not useful in some settings and at some times. In fact, these assumptions frequently may be adequate for describing institutionalized actors in settled times; framing processes may be in turn more salient and important at times of ideational and environmental uncertainty. The digital networked economy has evolved at breakneck pace, and IP law alongside it. This may be a good example of the kind of unsettled context in which interests and interpretations are unpredictable, even in the most institutionalized and materially oriented sectors. Material interests and aims are not just properties of social actors; rather, they arise out of social processes mediated by collective understandings.

Consider an example. Businesses have available to them many different theories of how they might profit from the production of information. IP rights are one such system; publicly financed prizes and grants are another. An IP strategy requires significant up-front investments and unpredictability at the back end and raises the price of informational inputs as well as outputs.

IP rights can also be very expensive to enforce; the government “trough” might in some circumstances be viewed as a more secure source of funds than the market, and of course is also a possible target of rent seeking. One classic criticism of provisioning programmes derives precisely from the possibility of rent-seeking and capture that they generate. The point is not that IP is not the best way for a particular industry to obtain the greatest rents, but that whether this is so may at times be difficult to tell objectively, and contingent on contextual factors. It will depend not only on complex judgements about economics, but also on the likelihood that one frame or another will resonate with allies and bystanders.

The broadest public choice claim, that IP rents themselves will invariably lead to strong demands for more IP, is perhaps best understood as a powerful descriptive account of a particular historical moment, one that has been conditioned by many more influences than opportunities for rents. As Landes and Posner point out, IP—unlike strategies of patronage or prize funds—was a good fit with the “ideological currents” of the 1980s. A self-interested business in the 1980s might well have made assumptions about the state, markets, and property rights—operating with the “free market” frame, for example—in ways that affected not only what it could achieve through the political process, but also what it could understand as its own interests. In other words, the imperative of interpretation affects the corporate sector too, and corporate collective action is thus necessarily shaped by interpretive choices that are influenced by pre-existing discursive opportunities and structures.

Such everyday interpretative frames become collective-action frames in the sociological sense when they are used to foster, sustain, or legitimate collective action. The term “intellectual property” has been very much “in vogue” in recent years. One reason may be that the term provides a collective-action frame for industry groups, one that unifies a disparate set of industries and at the same time capitalizes on the positive associations that come with the concept of “property.” The popularity of the term “intellectual property,” that is, may not simply affect policymakers and judges, or be “evidence” that such decision makers are treating IP more like property. It may also be evidence of a process of dialogic framing.

The term “intellectual property” appears to do several different kinds of framing work. As scholars such as Lemley and Fisher have emphasized, the term conceptually links regimes like copyright and patent law with the strong rights of exclusion and cultural legitimacy associated with real property law. Justin Hughes has recently demonstrated, however, that regular references to copyright as “property” or “literary property,” and to patents as “industrial property,” reach back to the nineteenth century, and that the word “property” figured in the original title and text of the Statute of Anne.

“Even if ‘intellectual property’ was a recent concept,” he contends, “no one has provided a serious explanation of how ‘intellectual property’ leads to the propertization of copyright in a way that ‘property’ and ‘literary property’ did not in the eighteenth, nineteenth, and twentieth centuries.” Framing theory may provide us with such an explanation. For one, concepts of property have of course themselves changed over time, providing different resonance today than in earlier years. A second explanation turns precisely on the “intellectualization” of property, a trend that Hughes does not dispute. The term “intellectual property,” that is, constructs a diverse group of industries as having common interests.

Framing theory also suggests that the term facilitates alliances between these groups, and helps them appeal to a broader cohort of contiguous groups and bystanders. The frame builds a bridge between patent and copyright, perhaps permitting copyright industries to draft off of the arguments that the pharmaceutical industry makes about the importance of exclusive rights to innovation. Similarly, it may permit patent-based industries to benefit from arguments about piracy and the breakdown of law that copyright owners make. (Of course, each member of this newly framed group is also rendered vulnerable to the attacks made on their new allies.)

The importance of collective-action frames for industry actors is particularly apparent in the international realm. The TRIPS agreement is widely attributed to the efforts of a very small number of industry leaders in the United States who came together to articulate a common interest and persuade the legislative and executive branches that IP protection was crucial to the balance of trade in the United States. But as analysts of the process have shown, the alliance itself was not a foregone conclusion; industry groups had to create a new trade committee in order to create a common agenda that would unite Hollywood producers, publishing interests, the software sector, industrial manufacturers,

and the chemical and pharmaceutical industries. Participants initially understood themselves as “strange bedfellows” and had significant disagreements about strategy. Part of how they united and gained the support of policymakers was by forging a common identity as intellectual property industries, and by framing the use of their products without permission as “theft.” The industry lobby was “particularly effective in translating their private interests into a matter of public interest,” for example by “packaging its demands as a solution to America’s trade woes” and “appealing to America’s long-standing free trade ethos.” It made the case that TRIPS was not only good for American business, but also good for global innovation, and for developing countries specifically.

Although there were undoubtedly many factors that worked to produce the acquiescence of developing countries, those who have studied it have concluded that the success of TRIPS required not just pressure and transfer payments, but also interventions in the realm of ideas. Some might contend that frames are merely epiphenomenal expressions of material conditions. But as the preceding examples indicate, framing processes can themselves affect material circumstances and outcomes. As Landes and Posner show, framing processes may sometimes define “tipping points” and thus affect outcomes in a political contest. Frames can thus themselves be thought of as contextual resources. But they are also resources that are not fully in the control of those who seek to use them. Rather, they set up chains of argument and counterargument that are difficult to predict *a priori*.

For example, when a coalition of industry groups promoted TRIPS as an agreement that would promote development, they may have paved the way for the agreement’s adoption, but they also paved the way for a critique of that argument (and thus of TRIPS) by developing countries and A2K advocates. Similarly, the industry attack on “pirates” who steal intellectual property has been inverted into an attack on industry “biopirates” who “steal” traditional knowledge and genetic resources.

Arguments about the importance of copyright to innovation have been contested by showing that an extension to an already long copyright term adds only a miniscule amount to the incentive effect of the original term. The argument that exclusive rights are essential to sustaining innovation in software has been undermined by scholars who chronicle the success of free and open-source software. The frames that are chosen to advance a particular political claim can thus be challenged in ways that can help turn back the gains of a group. Frames can also shape the framer’s response to a counterattack. Consider the co-evolution of TRIPS and the access-to-medicines campaign. When seeking to legitimate TRIPS, representatives of the pharmaceutical industry argued that the agreement, and strong patent rights, was good for developing countries. When the access-to-medicines campaign drew attention to the price implications of patents in developing countries and the extremity of the AIDS crisis, the industry responded in several ways, all of which continued to operate within the initial frame that insisted on the importance of medicines to developing countries.

They first offered limited drug donation programmes and argued that lack of medical infrastructure was a much more important barrier to access than patents. This strategy largely failed. Companies then lowered prices more significantly, to what they said were “no profit” prices in some countries. These offers were also criticized, with advocates claiming that the reductions were not steep enough and the programmes were too bureaucratic and limited. Eventually, companies began issuing limited licenses to generic suppliers—the very same move that a few years earlier they had adamantly resisted. Around the same time, the access-to-medicines campaign took an interesting turn, one perhaps influenced, and at least facilitated, by the frames set forth by their opponents. In order to counter companies’ arguments that strong patents are necessary for R&D in poor countries, medicines campaigners collected evidence that roughly ninety per cent of the world’s R&D funds went to diseases that cause only ten per cent of the world’s disease burden, and that this state of affairs was a predictable result of the small market share that the world’s poor represent. They also created new organizations specifically to stimulate R&D for neglected diseases.

Encountering and seeking to discredit the companies’ “innovation frame” thus seems to have helped direct the access-to-medicines campaign towards new initiatives and advocacy around R&D. In response, drug companies that had historically spent very little on neglected diseases began to invest much more in charitable attempts to develop medicines for neglected diseases. Paradoxically, pharmaceutical companies may have set the stage for a new moment of global consciousness about the medical needs of developing nations when they framed their arguments for strong global patent rights in terms that trumpeted the importance of access to medicines for all people.

As the access-to-medicines campaign unfurled, the process of argument and counterargument followed a path enabled (though not determined) by the arguments made by companies. The claims of activists forced companies to reduce prices, grant licenses, and allocate resources to R&D to bolster their claim that strong patent rights were not inconsistent with access to medicines and R&D for the poor. Had the conditions of the political culture of the time permitted companies to defend TRIPS with regard to a different claim—say, as an agreement that concentrated R&D resources in the wealthiest countries and that extracted maximum resources from developing countries—then different responses would have been possible and different material outcomes would have resulted.

FRAME MOBILIZATION IN A2K

The existence and success of the A2K mobilization likewise requires us to understand not just material interests, but how people collectively construct their sense of interests and opportunities, and how acts of framing can help groups to build support, recruit allies, and exert political leverage. Many of those involved in the A2K mobilization have acute material stakes in calling for changes in IP laws. Arcane aspects of patent law can be understood to affect the lives of people with AIDS in South Africa in a

profoundly immediate way, for example. But even this fact, the frame-analytic perspective tells us, requires an act of interpretation before it can be the basis of political mobilization. In South Africa, for example, high government officials responded to the AIDS crisis and the high price of patented medicines not in an anti-IP frame, but in an “AIDS denialist” frame. Other interpretations are possible, including the one offered by drug companies: that AIDS drugs are sophisticated and that their high cost represents their “true” price, when the massive expenditures and risk required for their development are considered. Why then did so many AIDS patients and access-to-medicines campaigners focus on IP as a problem and on international trade rules and companies as the proper register in which to respond? Posing this question helps to elucidate a broader point: the recent politicization of IP was not inevitable, even for those most directly and materially affected by strong IP laws.

Rather, it is the result of movement-building dynamics that have their roots in intersections between expanding IP law and emergent frameworks of antiglobalization, human rights, environmentalism, and cyberutopianism. The access-to-medicines campaign, for example, both understood and built claims against strong patent laws through frameworks of international human rights discourse and corporate malfeasance. Farmers’ rights advocates tapped into environmental frameworks and the antiglobalization movement. The open-source and Creative Commons movements have drawn upon conceptions of the freedom of cyberspace and the abundance of the digital age. All of these frameworks both helped orient these groups towards an understanding of IP as a problem and build bridges to others who might support their cause.

Access-to-medicines campaigners could use the human rights frame to create connections with human rights organizations and institutions in Geneva and New York. Farmers’ rights campaigners’ arguments about sustainable development linked them to environmental groups. Claims for protection of traditional knowledge were framed in a way that drew connections to indigenous rights claims. Thus, each of these groups mobilized frames that made certain alliances and political arguments possible.

But these acts of framing also made other alliances more difficult—including some of the alliances that are beginning to emerge under the rubric of A2K. AIDS activists, for example, have drawn on humanitarian and human rights frames to argue that medicines are “essential” and categorically more important than cultural goods like “Barbie dolls or CDs.” Farmers’ rights groups often take strong stances against genetically modified organisms, in conflict with those who argue for open-source biotechnology on the grounds that it will allow scientists to engineer new products for poor farmers. To the extent that groups concerned with farmers’ rights and traditional knowledge draw upon antitechnological discourses, this creates obvious possibilities for conflict with those advocates of free and open-source software who see in new digital technologies a revolution in the making.

How and why, then, did groups working with such different frames build bridges to one another? As the next Part argues, these groups are being brought together through

an encounter with IP law itself, operating here both as a form of constraint and as a field of meaning. Law has provided these groups with a frame—that of “intellectual property”—that encourages them to understand themselves as having similar problems and interests, and that facilitates joint counter mobilization against the “IP industries.” This is, of course, somewhat ironic, given the purported power of the concept of IP to promote the agenda of rights holders. The paradoxical implication here is that the consolidation of the concept of intellectual property also created a scaffold for the creation of a broad countermovement.

This comports with one of the central insights of dialogic framing theory, that groups borrow discourses from their opponents. It also suggests that successful acts of framing can help to consolidate ground for opposition as well as advance the framer’s cause. IP is the broad frame that is bringing these groups together. But these groups are also making use of narrower frames to create common arguments criticizing the existing IP system and to create a sense of themselves as related. At a recent conference, for example, a key actor in the A2K mobilization, James Love (of the NGO Knowledge Ecology International), offered the following list of terms that align the groups in question and that distinguish them from the IP industries:

A2K groups are actively and self-consciously creating new concepts in order to construct their interests as related. Frames of the “information commons” and the “public domain,” for example, are at the heart of the A2K mobilization. The most common definition of the public domain is the realm of “IP-free resources,” unprotected either because they were ineligible for protection in the first place or because they have been “freed” by invalidation or expiry of the relevant IP right. The first scholarly call to recognize something called the public domain came in 1981, but the term did not come into broader academic usage until much more recently.

It has since become a central frame for the A2K mobilization (although not always uncontroversially). James Boyle explained why: “We need a change in the way that these [IP] issues are understood, a change that transforms even our perceptions of self-interest, making possible coalitions where none existed before.” This could be done, he suggested, through the development of “affirmative arguments for the public domain” and the “use of the language of the commons to defend the possibility of distributed methods of non-proprietary production.” “Like the environment,” Boyle contended, “the public domain must be ‘invented’ before it is saved.” The concept of the “commons” had to be invented in much the same way, drawing on the notion of the commons as a historical concept in real property law, and adapting it to the non-rival domain of information in order to combat the frame of the “tragedy of the commons.” As Boyle describes it:

Increasingly, intellectual property scholars and information economists have turned to the theorists of the commons in trying to understand innovation.... In the debates over intellectual property policy, we have been familiar with a conceptual scheme that portrays “intellectual property” as a monopoly, and “the public domain,” as its conceptual

opposite—a realm of vaguely defined “freedom.” In contrast, the commons literature gives us a conceptual scheme in which property, seen as a regime of individual, legal, market-based control is juxtaposed to its conceptual opposite—the well-run commons, a realm of collective, and sometimes informal, controls that avoids the tragedy of the commons without a need for single party ownership.

The former juxtaposes monopolies against freedom, the latter juxtaposes individual formal controls against collective, and often informal, ones. Both give us a realm of property and a realm in which its opposite, or alternative, are offered. Boyle thus illustrates the power of collective-action frames to combat, and possibly overcome, what might at first appear to be insurmountable public choice problems. Acts of framing can create a sense of commonality between people who previously understood themselves as unrelated. They can also render interests that are diffuse suddenly salient, particularly once we incorporate non-material interests into our theories of action.

Savvy acts of framing can also help groups recruit support for their cause. Love’s Tables illustrate the framing struggle between IP and A2K activists, with IP proponents mobilizing terms such as “piracy” and “theft,” and the A2K mobilization countering with arguments about “sharing,” “freedom,” and “access.” Thus, we can also see how A2K activists are seeking to build collective frames that resonate across their different issue areas and that supplement their issue-specific frames of, for example, environmentalism and peer-to-peer production. In other words, the most pessimistic public choice accounts of the tectonics of this area of law do not reckon with the mediating power of frames.

As framing theorists point out, the “passage from condition to [political] action is a contingent and open process mediated by a number of conjectural and structural factors.” The A2K and IP mobilizations are both illustrations of this fact. (This is not to say that the A2K movement will necessarily continue to enjoy success, or even that it will continue to mobilize through the same rubric and frames. Nor is it to suggest that industry groups’ significant material advantages will not give them disproportionate influence in the political arena. Rather, it is to draw attention to the mediating role of framing processes and to the contingent acts of meaning making that have facilitated the emergence of both mobilizations.)

We can also find in the A2K mobilization many illustrations of the fact that frames are not simply epiphenomenal. Concepts of framing, for example, help explain the rapid proliferation of open-licensing schemes across the A2K mobilization. As critics point out, there may be vast differences in the material realms in which these licenses are being employed, and it is not obvious that they will work in the realm of copyright or biotechnology as they do in the realm of software. But as groups have been drawn together through their encounter with law into an “A2K” frame, they are trying these strategies, perhaps as a form of solidarity or because they believe that their problems are related. Frames also matter because they can help determine who occupies and is linked to a particular mobilization. The A2K mobilization has seized on concepts of the

“public domain” and the “commons” as themes that unify groups working in different areas of IP and as anchors for arguments about the damage that strong IP does to public welfare. But this creates tension with some groups that have contributed to the growing critique of IP—most evidently, those that argue for protection of traditional knowledge.

There are, of course, many different ways of conceptualizing the claim for protection of traditional knowledge, some more and some less in tension with a call for open access and the protection of a public domain. But as participants in the effort to build an A2K movement have pointed out, tensions between advocates of openness and advocates for protection of traditional knowledge may create a cleavage in what is currently a fragile coalition. This fissure would, in a very material sense, be attributable to framing choices made by movement actors.

EXPANSION IN NATURE AND SCOPE OF INTELLECTUAL PROPERTY LAWS

Other criticism of intellectual property law concerns the expansion of intellectual property, both in duration and in scope. In addition, as scientific knowledge has expanded and allowed new industries to arise in fields such as biotechnology and nanotechnology, originators of technology have sought IP protection for the new technologies. Patents have been granted for living organisms, (and in the US, certain living organisms have been patentable for over a century)

The increase in terms of protection is particularly seen in relation to copyright, which has recently been the subject of serial extensions in the United States and in Europe. With no need for registration or copyright notices, this is thought to have led to an increase in orphan works (copyrighted works for which the copyright owner cannot be contacted), a problem that has been noticed and addressed by governmental bodies around the world. Also with respect to copyright, the American film industry helped to change the social construct of intellectual property via its trade organization, the Motion Picture Association of America. In amicus briefs in important cases, in lobbying before Congress, and in its statements to the public, the MPAA has advocated strong protection of intellectual-property rights. In framing its presentations, the association has claimed that people are entitled to the property that is produced by their labour.

Additionally Congress’s awareness of the position of the United States as the world’s largest producer of films has made it convenient to expand the conception of intellectual property. These doctrinal reforms have further strengthened the industry, lending the MPAA even more power and authority. The growth of the Internet, and particularly distributed search engines like Kazaa and Gnutella, have represented a challenge for copyright policy. The Recording Industry Association of America, in particular, has been on the front lines of the fight against copyright infringement, which the industry calls “piracy”. The industry has had victories against some services, including a highly publicized case against the file-sharing company Napster, and some people have been

prosecuted for sharing files in violation of copyright. The electronic age has seen an increase in the attempt to use software-based digital rights management tools to restrict the copying and use of digitally based works. Laws such as the Digital Millennium Copyright Act have been enacted, that use criminal law to prevent any circumvention of software used to enforce digital rights management systems.

Equivalent provisions, to prevent circumvention of copyright protection have existed in EU for some time, and are being expanded in, for example, Article 6 and 7 the Copyright Directive. Other examples are Article 7 of the Software Directive of 1991 (91/250/EEC), and the Conditional Access Directive of 1998 (98/84/EEC). This can hinder legal uses, affecting public domain works, limitations and exceptions to copyright, or uses allowed by the copyright holder. Some copyleft licenses, like GNU GPL 3, are designed to counter that. Laws may permit circumvention under specific conditions like when it is necessary to achieve interoperability with the circumventor's programme, or for accessibility reasons; however, distribution of circumvention tools or instructions may be illegal.

In the context of trademarks, this expansion has been driven by international efforts to harmonise the definition of "trademark", as exemplified by the Agreement on Trade-Related Aspects of Intellectual Property Rights ratified in 1994, which formalized regulations for IP rights that had been handled by common law, or not at all, in member states. Pursuant to TRIPs, any sign which is "capable of distinguishing" the products or services of one business from the products or services of another business is capable of constituting a trademark.

Patents, Trademarks and Copyrights

PATENTS

WHAT IS A PATENT?

A patent is a document, issued by the federal government, that grants to its owner a legally enforceable right to exclude others from practicing the invention described and claimed in the document. Congress allows this right, for a term ending twenty years from the date of filing of an application for patent, to encourage the public disclosure of technical advances and as an incentive for investing in their commercialization. Thus, the overall progress of technical innovation is favoured, while at the same time inventors are rewarded for their specific contributions. Like other forms of property, the rights symbolized by a patent can be inherited, sold, rented, mortgaged and even taxed. When a patent expires, or is held invalid, the right to exclude the others ceases. The public is the ultimate beneficiary of the technical advance.

Under What Conditions Is A Patent Granted?

Congress has specified that a patent will be granted if the inventor files a timely application which adequately describes a new, useful and unobvious invention of proper subject matter. To be timely, an application must be filed within one year of certain acts (by the inventor or others) which place the invention in the hands of the public *i.e.*, patented or published anywhere in the world, on sale or in public use in this country. This one-year grace period, however, is not available in most foreign countries.

A U.S. inventor who wants to obtain corresponding foreign patents must first file an application in the U.S. before any divulgation, whether in written or oral form, of the invention to the public. The description of the invention in the application must be complete enough to enable others to practice the invention. Moreover, the application must describe the best manner (“best mode”) known to the inventor of carrying out the invention. The described invention must be new *i.e.*, not invented first by another or identically known or used by others in this country or patented or published anywhere in the world before the actual invention date (not the application filing date). The invention also must be useful *i.e.*, serve some disclosed or generally known purpose.

The requirement of unobviousness means that the differences between the invention and the prior public knowledge in its technical field must be such that a person having ordinary skill in this field would not have found the invention obvious at the time it was made. The proper subject matter of a patent is any product, process, apparatus or composition, including living matter such as genetically engineered bacteria or plants. Special provisions also permit patents directed to certain distinct and new varieties of plants (Plant Patent) and new original and ornamental designs for articles of manufacture (Design Patent). Purely mental processes, newly discovered laws of nature and methods of doing business are not proper subjects for a patent.

Why Obtain A Patent?

Most inventors seek a patent to obtain the actual or potential commercial advantages that go along with the right to exclude others. Given the high cost of research and development, the opportunity to recoup these costs through commercial exploitation of the invention may be the primary justification for undertaking research in the first place.

Patent rights can be commercially exploited in two basic ways:

1. Directly, by the inventor’s practice of the invention to obtain an exclusive marketplace advantage (as where the patented technology results in a better product or produces an old product less expensively) and/or
2. Indirectly, by receiving income from the sale or licensing of the patent.

It is important to note that a patent (*i.e.*, the right to exclude others) does not give the inventor the right to practice the invention. The inventor can practice his invention only if by so doing he does not also practice the invention of an earlier unexpired patent. While only one patent can be granted on a particular invention, it is easy to see how more than one patent could be infringed by making a single product. For example, consider that A has a patent on a new type of door and B invents an improved door of this type with a special lock. B could not sell the improved locking door since A’s patent broadly covers all doors of this type.

On the other hand, A could not incorporate the improved lock in his basic door since B’s patent covers this combination. In these circumstances both A and B can be free to practice the best technology (locking door) only if each grants a patent license to the other. The indirect exploitation of a patent may be exclusive, *e.g.*, by selling all rights

in the patent or granting an exclusive license. Licenses can be non-exclusive, allowing many parties, including the inventor, to practice the invention simultaneously. A patent may also provide commercial advantages in addition to the potential for an exclusive market position or licensing income. A patent often lends business credibility to start up ventures and can open doors to both technical assistance and financing necessary to bring a new product to market. An improvement patent may also provide the barter necessary to cross license any basic patents held by others which block the path to market.

How to Obtain A Patent

Patents are obtained through a complex administrative proceeding in the United States Patent and Trademark Office. Since the legal rules that govern this proceeding are quite extensive and often complicated, it is strongly recommended that an inventor seek the assistance of an experienced patent attorney before beginning this process. Before actually applying to the Patent and Trademark Office there are several important preliminary steps that should be followed to prevent possible loss or damage to future patent rights. One of the most important of these preliminary steps is proper record keeping.

Since United States patents are granted to the first inventor, it may become necessary to prove when the invention was made. This is best accomplished by making a complete record of the invention from the first idea right up through development of commercial products. The invention record should clearly describe the invention with words and pictures (photographs, sketches, drawings, *etc.*) and should explain fully how it operates or is used. Each page of invention record should be signed and dated in ink by the inventor. The record should also be reviewed as it is made by at least one other trustworthy person who is capable of understanding the invention, who should sign and date the record under the notation read and understood by....

Another important preliminary step is the determination of whether the invention is likely to be considered patentable by the Patent and Trademark Office, and if so, whether a patent which might be granted would be broad enough in its coverage to be worthwhile in a commercial sense. Such a preliminary evaluation of patentability should be made by a patent attorney, based in part on the prior patents and other materials located in a search of relevant records in the Patent and Trademark Office. While the attorney's opinion that the invention should be patentable is not a guarantee that the patent will be granted, if he finds that the invention probably is not patentable or economically worthwhile, the considerable cost and effort of going forward with the process can be avoided.

The next step in the process of obtaining a patent is the preparation of a patent application. A patent application is a legal document, which must fully describe the invention with words and, where appropriate, drawings, and which includes claims which define the legal boundaries of the invention. It is essential to the validity of the

patent, and its ability to adequately protect the invention, that the invention be described and claimed completely and precisely. Accordingly, the inventor should tell the patent attorney everything about the invention, including what problems it solves and what difficulties were overcome to make it work. Particularly important is the duty to tell the attorney about prior patents or other prior inventions of which the inventor is aware, so this information can be disclosed to the Patent and Trademark Office.

The patent application will also contain a Declaration and Power of Attorney form which the inventor must sign indicating that he has read and understood the application and affirming that he is the first inventor. The application and a filing fee are then sent to the Patent and Trademark Office to effect a filing. Congress has recently authorized a new form of preliminary patent application known as a Provisional Application, which can be filed at a lower cost and without claims and certain other formalities. This Provisional Application is not examined, but must be replaced by a conventional application within a year. The benefit of this new form of application is that it does not count in determining the expiration date of the patent, which is measured from the date of filing of the conventional application.

The filing of an application for a patent does not create any enforceable rights since the courts will only stop an infringer after the patent is granted. Nevertheless, marking a device Patent Pending or Patent Applied For may discourage potential infringers since it puts them on notice that they may have to stop production once the patent is granted. It is unlawful to use such a notice unless an application for patent is actually pending in the U.S. Patent and Trademark Office. After the patent has issued, it is also good practice to mark the products sold under the patent with the patent number because it gives the inventor certain additional legal rights.

In the Patent and Trademark Office the application undergoes a process called Examination. After an initial processing stage (which may take 6-9 months or more) a Patent Examiner will review the application and write a letter (called an Office Action) commenting on it. The First Office Action often is a refusal to grant the patent, and the applicant then has an opportunity to modify the application to overcome the Examiner's objections. With the inventor's help, the patent attorney will reply in writing to the Office Action, usually making some changes and arguing that others are not necessary. Typically, at least two such exchanges between Patent Examiner and attorney are necessary to resolve all the legal and technical issues. In general, it now takes an average of about 18-22 months from filing to complete the examination process. During this period the application is kept secret, *i.e.*, only Government personnel and persons authorized by the inventor are permitted to examine the file.

When the Examiner is satisfied that the application is in proper form and its claims are allowable, the applicant is notified that a patent will be granted upon payment of final government fees. In order to keep the patent in force until it expires it also is necessary to pay progressively higher maintenance fees at 3 1/2, 7 1/2 and 11 1/2 years after the original grant. While the patent grant makes the information in the application

available to the public, the inventor has the right to prevent others from making, using or selling what is claimed for as long as the patent remains in force.

THE INDIAN PATENTS AND DESIGN ACT OF 1911 AND ITS AMENDMENTS

The Indian Patents and Design Act of 1911 was the operative legislation when India gained independence from the British in 1947. But soon after independence, legislators came to believe that the laws framed by the British were inappropriate and that the 1911 act was at odds with the aspirations and goals of a young, newly independent nation. With that in mind, two committees, the Patents Enquiry Committee, also known as the Tek Chand committee, and the committee on the Revision of the Patents Law, also known as the Ayyangar committee, were appointed to review the adequacy of the Indian patent system and to adapt it to conform with national goals. The resulting legislation was the Patents Act of 1970. The patent system in India is currently regulated by the Patents Act of 1970, which replaces the Indian Patents and Designs Act of 1911. Between 1947 and 1970, debate raged over whether or not to weaken the patent regime.

Commentators agreed that it was necessary to have a patent regime that protected inventors. N. R. Ayyangar (1959) argued that a system that had been universally adopted and, for more than a century, the primary means of rewarding inventors and encouraging innovation could not be discarded without the “weightiest of reasons,” and he saw “none such.” But the primary question was, How could the regime benefit domestic industries, as opposed to multinational companies (MNCs)? Both committees believed that MNCs were taking undue advantage of the patent system and continuing the process of exploitation to which the country had been subjected throughout British rule. The committees generally agreed that a new patent regime had to be put in place that reduced the monopoly and exploitative position of MNCs, and that gave a boost to indigenous industries. As a result, patent laws were significantly weakened. The Indian Patents and Design Act of 1911 in relation to the Patent Act of 1970. We also discuss the reasons for the change from the 1911 act to the 1970 act.

COVERAGE

The first area of concern for planners was the apparent scope of the Indian Patents and Designs Act. The Tek Chand committee report pointed out that the 1911 act did not contain any provision indicating categorically what could be patented, but instead merely stated that for a product to be patentable it was necessary for it to relate a “manner of manufacture”.

The report concluded that the law was wide open to varying interpretations, given its breadth and its failure to define what could be patented. The interpretation taken by the Indian Patent Office at the time was that a process, regardless of whether it was old

or new, was eligible for a patent. A new drug (a product) was patentable so long as the process for manufacturing it was described in full in the patent application. The process itself, however, did not necessarily have to be new. Under the 1911 act, third parties (in this case domestic industries and firms) were legally prevented from manufacturing new drugs introduced by the MNCs during the life of the patent. That was detrimental to domestic firms, Ayyangar argued, because in a patent application for a new drug, the patentee could describe all known possible processes, thereby effectively denying any third party the opportunity to create a similar drug using a different process.

As experience showed, the MNCs did exercise the legal rights given to them by the 1911 act. Indigenous firms could not use an old process, as specified by the MNCs, for at least sixteen years. Domestic firms were unable to circumvent these restrictions either by processing a patented drug into formulations or by importing it into the country for sale. A third party could, at least in theory, manufacture a new drug if it could do so by using a process not mentioned in the patent application. But determined MNCs were able to prevent third parties from producing patented drugs via different processes. For example, the Haffkine Institute, a public sector firm, produced tolbutamide from locally available raw materials.

The Haffkine Institute patented the process and granted Unichem Laboratories, a domestic firm, a license to produce the drug. Hoechst, an MNC, filed suit, claiming that the patent was invalid because tolbutamide had been manufactured on the basis of one of the formulas mentioned in a 1956 patent application by Hoechst. The Bombay High Court closely examined the Hoechst patent application and found that the process developed by the Haffkine Institute and used by Unichem was not specifically mentioned. Nonetheless, the Bombay High Court found in favour of Hoechst because the description in Hoechst's patent application was sufficiently vague as to be open to the interpretation of the court. Ayyangar argued that MNCs took undue advantage of the breadth of the law and that the loopholes in the definition of what could or could not be patented were detrimental to the national economy.

On the basis of his recommendations, the 1970 act was written to specify that “an ‘invention’ means any new and useful:

- Art, process, method or manner of manufacture.
- Machine, apparatus or other article.
- Substance produced by manufacture and includes any new and useful improvement of any of them”.

This new definition considerably narrowed the scope of protection and cleared up the former ambiguity about what was patentable. But Ayyangar did not stop there. After studying the patent systems of various countries, including the United States, Germany, and the United Kingdom, Ayyangar concluded that the weakened patent protection laws in Germany had given that nation's chemicals industry a boost. Further, he stated that because of the strategic and public policy importance of food and medicines, it was necessary for India to weaken patent protection for products and processes in those

areas. In response to those observations, the 1970 act lessened the protection for drugs and food by stating that in the case of inventions, “claiming substances intended for use, or capable of being used as food or as medicine or drug... no patent shall be granted in respect of claims for the substances themselves but claims for the methods or processes of manufacture shall be patentable”. Thus, under the 1970 act, product patents for drugs and food could no longer be claimed; only process patents were allowed.

TRADEMARKS

Brand names, a synonym for trademarks and service marks, are important intellectual properties upon which the public learns to rely to identify a source and a standard of quality of the products or services it purchases. Trademarks and service marks are words, phrases, designs, sounds or symbols. They are used on or in association with goods and in association with services to be performed. The public learns, through purchasing experience, that the goods or services bearing a mark come from a single (but possibly anonymous) source and will meet an expectation of a standard of quality that the goods or services sold under the mark have met in the past. This predictability of a level of quality and reassurance of a known source provides the owner or user of the mark with the benefit of good will held by the public for the product or service that he offers. This goodwill is often the cornerstone of the owner’s business.

Trademark rights arise in one of two ways: either by filing a mark with the United States Patent and Trademark Office based on a bona fide intent to make use of the mark on a product or in association with a service that will soon be offered to the public, or by actually using the mark in commerce on a product or in association with a service. Under the federal law, an application for registration may be filed based on actual use of the mark or a bona fide intent to use the mark in commerce. The filing date of an application with the United States Patent and Trademark Office to register a trademark or service mark which the applicant intends to use on a product or in association with a service establishes a date of constructive use of the mark.

The act of filing the application, therefore, through constructive use, can create a nationwide priority of rights in the mark against any other person who subsequently adopts the same or a confusingly similar mark. The intent to use the mark must be a bona fide intention or the application may be invalid and the applicant may be subject to certain penalties for filing a fraudulent claim of intent. The constructive use provided by the filing of the application claiming an intent to use the mark is contingent upon the mark ultimately issuing as a federal registration.

The federal registration will be issued only after the applicant has made actual use of the mark on the goods or in association with the service that it claims to have intent to use when the filing takes place. Actual use of the mark must be made within six months of the time that the Trademark Office issues a notice of allowance of the mark after the application has been filed, although that period may be extended up to thirty months in appropriate circumstances. When use is made and the federal registration issues, the

owner's rights in the mark are superior to all who have adopted the same or a similar mark for the same or a similar product or service subsequent to the filing date of the owner's application. Obtaining a federal registration based on the bona fide intent to use the mark or based on the actual use of the mark provides significant advantages.

Among these advantages are the following:

1. Federal registration confers the benefit of nationwide constructive use and, thus, a right of priority, over all subsequent users or filers, as of the application filing date.
2. Registration is constructive notice; it prevents acquisition of common law rights by innocent adoption and use of the same or a similar mark by another.
3. A certificate of registration creates a legal presumption and is sometimes conclusive evidence of the registrant's right to exclusive use of the mark and the validity of the mark's registration.
4. Registration permits the use of the “ “ symbol or other statutory designations of registration.
5. Goods bearing a mark which infringes on a registered mark may be excluded from entry into the U.S. by recordation with the U.S. Customs Service.
6. Registration furnishes the basis for foreign registration under a treaty known as the Paris Convention.

Those who make use of trademarks or service marks without filing for federal registration can obtain exclusive rights in those marks in the market in which the marks are used, if the use is prior to the filing date of any federal registrant or any other user in that market of a confusingly similar mark. This market may be small, *i.e.*, a town or part of a town, or large, *i.e.*, the entire United States. A wider area, referred to as the “zone of natural expansion,” may also be granted to the prior user if it can be shown that there were plans to expand before a junior user entered the market.

An application for a federal registration of a trademark or service mark filed in the United States Patent and Trademark Office is examined by government attorneys to ensure that the mark is not likely, when used on or in connection with the goods or services of the applicant, to cause confusion, or to cause mistake, or to deceive the consuming public that the goods or services bearing the mark have the same source, origin or approval as those sold under a previously registered mark which has not been abandoned or canceled. The Examiner also makes a determination that the mark is not inherently incapable of performing the identification function of a mark because, for example, it is the generic word for the product or service with which it is used or because it is descriptive of the products or service with which it is used.

Federal trademark registrations are valid for ten years and can be renewed for like periods, provided the mark is constantly used. Failure to use a mark can cause the rights in the mark to be lost. Such abandonment of the mark causes the mark to lose its significance to the public as an identification of a single source and a standard of quality. The use of the registration symbol with a mark indicates that it is federally registered.

The symbol TM is often used with unregistered trademarks to give notice to the public that the user is staking out a claim in the symbol as a trademark, but such use is optional. The symbol SM may similarly be used to designate a unregistered service mark.

Because trademarks and service marks can be licensed, they are a primary basis for the franchising industry in the United States. The public comes to learn to expect a certain quality standard from every facility, a fast food restaurant chain for example, that bears the same name or service mark. As long as the mark's owner maintains the same quality standards for each of the facilities bearing its name, it can license the use of the name to many people. Finally, there are special types of marks known as collective marks or membership marks which identify organizations and their members and certification marks which are used to identify the fact that a product or service meets a certain acceptable standard, *e.g.*, Good housekeeping seal.

COPYRIGHTS

Copyright is a statutory property right which grants to creators (authors) certain exclusive rights in their creations for a limited duration. Its purpose, as expressed in the Constitution, is to promote the Progress of Science and useful Arts by providing economic incentive for creative activity. Copyright protects intangible original works of authorship which are fixed in a tangible medium of expression. Put another way, copyright protects the expression of ideas, but not the ideas themselves. Copyright protects works such as books, pictorial, graphic and sculptural works, music, photographs, movies and computer programmes.

Copyright is often described as a bundle of rights, which include the right:

1. To reproduce the copyrighted work;
2. To make derivative works (such as a movie from a novel);
3. To distribute copyrighted works to the public
4. To perform, publicly, certain works (such as music); and,
5. To display, publicly, certain works (such as paintings).

Copyright arises upon creation and, under current law, endures for the life of the author plus 50 years. (The term of copyright for works created before 1978 is 75 years.) Copyright applies to both unpublished and published works. Registration of a copyrighted work with the Copyright Office in Washington, D.C. is not required for existence of the copyright; however, it is a prerequisite to a lawsuit for copyright infringement and to certain legal remedies.

Copyright registration is simple and inexpensive. The registrant (who may be the author, including an employer or other person for whom a work made for hire was created, or a person or entity which has acquired the copyright) completes a printed form provided by the Copyright Office and returns it to the Office with a small fee and copies of the work. Transfers of copyright ownership (including exclusive licenses) must be in writing to be valid. The familiar copyright notice is no longer required on copies of works published after March 1, 1989. However, it is still in The Copyright

Owner's interest to place a copyright notice on publicly distributed copies of published works. The notice should include the copyright symbol () or the word "Copyright" or its abbreviation (Copr.), followed by the year of first publication of the work and the name of the Copyright Owner.

In summary, it is important to realize that patents, trademarks, and copyrights constitute the basis on which the underlying intellectual property may be protected in law. It is therefore important that a great degree of skill be exercised in drafting the documents and following the procedures necessary for obtaining this protection. By relying on a specialist in the field who has good standing and recognized ability in his profession, the inventor or artist can be assured that the intellectual property will be adequately protected.

JUSTIFICATION

The usual justification of copyright is to enable creators of intellectual wealth to financially support themselves and give them a motive to continue publishing their creations. Without copyright, photographers, authors, artists, and journalists or anyone else who creates non-material economic wealth would have to find alternative means (other than licensing) to support themselves, like for example publishing a small subset of their creations and then requesting payment before they published more (also see the street performer protocol). In a capitalist economic system, food and house rent cost money so photographers, film directors, authors, painters, poets and other intellectual creators would have to find other jobs to support themselves if they couldn't get compensated for their intellectual work.

With copyright in place, the author of a book or the photographer of a photograph can charge users who want to get a copy of their creations and thus support themselves. Before copyright, authors generally requested a large-sum one-off payment from the printer of their book before publishing it. With copyright in place, and assuming efficient enforcement, authors, photographers and other intellectual workers can publish their creations immediately and wait for licensing requests from people who want to use or re-publish their works (an example of this model for the funding of photography is Corbis, Getty Images, Alamy and other stock photography image banks). Anti-copyright critics claim copyright law protects corporate interests while criminalizing legitimate use, while proponents argue the law is fair and just, protecting the interest of the creator.

HISTORY

Copyright came about with the invention of the printing press and with wider public literacy. As a legal concept, its origins in Britain were from a reaction to printers' monopolies at the beginning of the 18th century. Charles II of England was concerned by the unregulated copying of books and passed the Licensing of the Press Act 1662 by Act of Parliament, which established a register of licensed books and required a copy to be deposited with the Stationers' Company, essentially continuing the licensing of

material that had long been in effect. The British Statute of Anne (1710) further alluded to individual rights of the artist. It began, “Whereas Printers, Booksellers, and other Persons, have of late frequently taken the Liberty of Printing... Books, and other Writings, without the Consent of the Authors... to their very great Detriment, and too often to the Ruin of them and their Families:” A right to benefit financially from the work is articulated, and court rulings and legislation have recognized a right to control the work, such as ensuring that the integrity of it is preserved. An irrevocable right to be recognized as the work’s creator appears in some countries’ copyright laws.

Copyright laws allows products of creative human activities, such as literary and artistic production, to be owned and thus incentivized. Different cultural attitudes, social organizations, economic models and legal frameworks are seen to account for why copyright emerged in Europe and not, for example, in Asia. In the Middle Ages in Europe, there was generally a lack of any concept of literary property due to the general relations of production, the specific organization of literary production and the role of culture in society.

The latter refers to the tendency of oral societies, such as that of Europe in the medieval period, to view knowledge as the product and expression of the collective, rather than to see it as individual property. However, with copyright laws, intellectual production comes to be seen as a product of an individual, with attendant rights. The most significant point is that patent, and copyright laws support in fundamental and thoroughgoing ways the expansion of the range of creative human activities that can be commodified. This parallels the ways in which capitalism led to the commodification of many aspects of social life that hitherto had no monetary or economic value *per se*.

The Statute of Anne was the first real copyright act, and gave the publishers rights for a fixed period, after which the copyright expired. Copyright has grown from a legal concept regulating copying rights in the publishing of books and maps to one with a significant effect on nearly every modern industry, covering such items as sound recordings, films, photographs, software, and architectural works.

Prior to the passage of the United States Constitution, several States passed their own various copyright laws between 1783 and 1787, the first state being Connecticut. Contemporary scholars and patriots such as Noah Webster, John Trumbull, and Joel Barlow were instrumental in securing the passage of these statutes.

The Copyright Clause of the United States Constitution (1787) authorized copyright legislation: “To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.” That is, by guaranteeing them a period of time in which they alone could *profit* from their works, they would be enabled and encouraged to invest the time required to create them, and this would be good for society as a whole. A right to profit from the work has been the philosophical underpinning for much legislation extending the duration of copyright, to the life of the creator and beyond, to his heirs. The original length of

copyright in the United States was 14 years, and it had to be explicitly applied for. If the author wished, he could apply for a second 14-year monopoly grant, but after that the work entered the public domain, so it could be used and built upon by others. Thomas Jefferson, who strongly advocated the ability of the public to share and build upon the works of others, proposed as part of the Bill of Rights that a short timespan be protected: Art. 9. Monopolies may be allowed to persons for their own productions in literature and their own inventions in the arts for a term not exceeding – years but for no longer term and no other purpose.

The 1886 Berne Convention first established recognition of copyrights among sovereign nations, rather than merely bilaterally. Under the Berne Convention, copyrights for creative works do not have to be asserted or declared, as they are automatically in force at creation: an author need not “register” or “apply for” a copyright in countries adhering to the Berne Convention.

As soon as a work is “fixed”, that is, written or recorded on some physical medium, its author is automatically entitled to all copyrights in the work, and to any derivative works unless and until the author explicitly disclaims them, or until the copyright expires. The Berne Convention also resulted in foreign authors being treated equivalently to domestic authors, in any country signed onto the Convention. The UK signed the Berne Convention in 1887 but did not implement large parts of it until 100 years later with the passage of the *Copyright, Designs and Patents Act of 1988*. The United States did not sign the Berne Convention until 1989.

The United States and most Latin American countries instead entered into the Buenos Aires Convention in 1910, which required a copyright notice (such as *all rights reserved*) on the work, and permitted signatory nations to limit the duration of copyrights to shorter and renewable terms. The Universal Copyright Convention was drafted in 1952 as another less demanding alternative to the Berne Convention, and ratified by nations such as the Soviet Union and developing nations. The regulations of the Berne Convention are incorporated into the World Trade Organization’s TRIPS agreement (1995), thus giving the Berne Convention effectively near-global application. The 2002 WIPO Copyright Treaty enacted greater restrictions on the use of technology to copy works in the nations that ratified it.

SCOPE OF COPYRIGHT

Copyright may apply to a wide range of creative, intellectual, or artistic forms, or “works”. Specifics vary by jurisdiction, but these can include poems, theses, plays and other literary works, motion pictures, choreography, musical compositions, sound recordings, paintings, drawings, sculptures, photographs, computer software, radio and television broadcasts, and industrial designs. Graphic designs and industrial designs may have separate or overlapping laws applied to them in some jurisdictions.

Copyright does not cover ideas and information themselves, only the form or manner in which they are expressed. For example, the copyright to a Mickey Mouse cartoon

restricts others from making copies of the cartoon or creating derivative works based on Disney's particular anthropomorphic mouse, but does not prohibit the creation of other works about anthropomorphic mice in general, so long as they are different enough to not be judged copies of Disney's.

In many jurisdictions, copyright law makes exceptions to these restrictions when the work is copied for the purpose of commentary or other related uses. Meanwhile, other laws may impose additional restrictions that copyright does not – such as trademarks and patents. Copyright laws are standardized somewhat through international conventions such as the Berne Convention and Universal Copyright Convention. These multilateral treaties have been ratified by nearly all countries, and international organizations such as the European Union or World Trade Organization require their member states to comply with them.

OBTAINING AND ENFORCING COPYRIGHT

Typically, a work must meet minimal standards of originality in order to qualify for copyright, and the copyright expires after a set period of time (some jurisdictions may allow this to be extended). Different countries impose different tests, although generally the requirements are low; in the United Kingdom there has to be some 'skill, labour and judgement' that has gone into it. In Australia and the United Kingdom it has been held that a single word is insufficient to comprise a copyright work. However, single words or a short string of words can sometimes be registered as a trademark instead. Copyright law recognises the right of an author based on whether the work actually is an original creation, rather than based on whether it is unique; two authors may own copyright on two substantially identical works, if it is determined that the duplication was coincidental, and neither was copied from the other.

In all countries where the Berne Convention standards apply, copyright is automatic, and need not be obtained through official registration with any government office. Once an idea has been reduced to tangible form, for example by securing it in a fixed medium (such as a drawing, sheet music, photograph, a videotape, or a computer file), the copyright holder is entitled to enforce his or her exclusive rights. However, while registration isn't needed to exercise copyright, in jurisdictions where the laws provide for registration, it serves as *prima facie* evidence of a valid copyright and enables the copyright holder to seek statutory damages and attorney's fees. (In the USA, registering *after* an infringement only enables one to receive actual damages and lost profits.)

The original holder of the copyright may be the *employer* of the author rather than the author himself, if the work is a "work for hire". For example, in English law the *Copyright, Designs and Patents Act 1988* provides that if a copyrighted work is made by an employee in the course of that employment, the copyright is automatically owned by the employer which would be a "Work for Hire."

Copyrights are generally enforced by the holder in a civil law court, but there are also criminal infringement statutes in some jurisdictions. While central registries are kept in

some countries which aid in proving claims of ownership, registering does not necessarily prove ownership, nor does the fact of copying (even without permission) necessarily prove that copyright was infringed. Criminal sanctions are generally aimed at serious counterfeiting activity, but are now becoming more commonplace as copyright collectives such as the RIAA are increasingly targeting the file sharing home Internet user. Thus far, however, most such cases against file sharers have been settled out of court.

Cost of Enforcing Copyright

In most jurisdictions the copyright holder must bear the cost of enforcing copyright. This will usually involve engaging legal representation, administrative and or court costs. These costs, including time, should be taken into consideration when evaluating the benefits of enforcing copyright. In light of this, many copyright disputes are settled by a direct approach to the infringing party in order to settle the dispute out of court.

Copyright Notices in the United States

Prior to 1989, use of a copyright notice – consisting of the copyright symbol (©, the letter C inside a circle), the abbreviation “Copr.”, or the word “Copyright”, followed by the year of the first publication of the work and the name of the copyright holder – was part of U. S. statutory requirements. Several years may be noted if the work has gone through substantial revisions. The proper copyright notice for sound recordings of musical or other audio works is a sound recording copyright symbol (℗, the letter p inside a circle), which indicates a sound recording copyright, with the letter p indicating a “phonorecord”.

Similarly, the phrase *All rights reserved* was once required to assert copyright. In 1989 the United States enacted the Berne Convention Implementation Act, amending the 1976 Copyright Act to conform to most of the provisions of the Berne Convention. As a result, the use of copyright notices has become optional to claim copyright, because the Berne Convention makes copyright automatic. However, the lack of notice of copyright using these marks may have consequences in terms of reduced damages in an infringement lawsuit – using notices of this form may reduce the likelihood of a defence of “innocent infringement” being successful.

“Poor Man’s Copyright”

A widely circulated strategy to avoid the cost of copyright registration is referred to as the “poor man’s copyright”. It proposes that the creator send the work to himself in a sealed envelope by registered mail, using the postmark to establish the date. This technique has not been recognized in any published opinions of the United States courts. The United States Copyright Office makes it clear that the technique is no substitute for actual registration. The United Kingdom Intellectual Property Office discusses the technique and notes that the technique (as well as commercial registries) does not constitute dispositive proof that the work is original nor who the creator of the work is.

EXCLUSIVE RIGHTS

Several exclusive rights typically attach to the holder of a copyright:

- To produce copies or reproductions of the work and to sell those copies (including, typically, electronic copies)
- To import or export the work
- To create derivative works (works that adapt the original work)
- To perform or display the work publicly
- To sell or assign these rights to others
- To transmit or display by radio or video.

The phrase “exclusive right” means that only the copyright holder is free to exercise those rights, and others are prohibited from using the work without the holder’s permission. Copyright is sometimes called a “negative right”, as it serves to *prohibit* certain people (*e.g.*, readers, viewers, or listeners, and primarily publishers and would be publishers) from doing something they would otherwise be able to do, rather than *permitting* people (*e.g.*, authors) to do something they would otherwise be unable to do. In this way it is similar to the unregistered design right in English law and European law. The rights of the copyright holder also permit him/her to *not* use or exploit their copyright, for some or all of the term. There is, however, a critique which rejects this assertion as being based on a philosophical interpretation of copyright law that is not universally shared. There is also debate on whether copyright should be considered a property right or a moral right.

Useful Articles

If a pictorial, graphic or sculptural work is a useful article, it is copyrighted only if its aesthetic features are separable from its utilitarian features. A useful article is an article having an intrinsic utilitarian function that is not merely to portray the appearance of the article or to convey information. They must be separable from the functional aspect to be copyrighted.

LIMITATIONS AND EXCEPTIONS TO COPYRIGHT

Idea-expression Dichotomy and the Merger Doctrine

The idea-expression divide differentiates between ideas and expression, and states that copyright protects only the original expression of ideas, and not the ideas themselves. This principle, first clarified in the 1879 case of *Baker v. Selden*, has since been codified by the Copyright Act of 1976 at 17 U.S.C. § 102(b).

The First-sale Doctrine and Exhaustion of Rights

Copyright law does *not* restrict the owner of a copy from reselling legitimately obtained copies of copyrighted works, provided that those copies were originally produced by or with the permission of the copyright holder. It is therefore legal, for

example, to resell a copyrighted book or CD. In the United States this is known as the first-sale doctrine, and was established by the courts to clarify the legality of reselling books in second-hand bookstores.

Some countries may have parallel importation restrictions that allow the copyright holder to control the aftermarket. This may mean for example that a copy of a book that does not infringe copyright in the country where it was printed does infringe copyright in a country into which it is imported for retailing. The first-sale doctrine is known as exhaustion of rights in other countries and is a principle which also applies, though somewhat differently, to patent and trademark rights. It is important to note that the first-sale doctrine permits the transfer of the particular legitimate copy involved. It does not permit making or distributing additional copies. In addition, copyright, in most cases, does not prohibit one from acts such as modifying, defacing, or destroying his or her own legitimately obtained copy of a copyrighted work, so long as duplication is not involved. However, in countries that implement moral rights, a copyright holder can in some cases successfully prevent the mutilation or destruction of a work that is publicly visible.

Fair use and Fair Dealing

Copyright does not prohibit all copying or replication. In the United States, the fair use doctrine, codified by the Copyright Act of 1976 as 17 U.S.C. Section 107, permits some copying and distribution without permission of the copyright holder or payment to same. The statute does not clearly define fair use, but instead gives four non-exclusive factors to consider in a fair use analysis.

Those factors are:

- The purpose and character of your use
- The nature of the copyrighted work
- What amount and proportion of the whole work was taken, and
- The effect of the use upon the potential market for or value of the copyrighted work.

In the United Kingdom and many other Commonwealth countries, a similar notion of fair dealing was established by the courts or through legislation. The concept is sometimes not well defined; however in Canada, private copying for personal use has been expressly permitted by statute since 1999. In *Alberta (Education) v. Canadian Copyright Licensing Agency (Access Copyright)*, 2012 SCC 37, the Supreme Court of Canada concluded that limited copying for educational purposes could also be justified under the fair dealing exemption.

In Australia, the fair dealing exceptions under the *Copyright Act 1968* (Cth) are a limited set of circumstances under which copyrighted material can be legally copied or adapted without the copyright holder's consent. Fair dealing uses are research and study; review and critique; news reportage and the giving of professional advice (*i.e.* legal advice). Under current Australian law it is still a breach of copyright to copy, reproduce

or adapt copyright material for personal or private use without permission from the copyright owner. Other technical exemptions from infringement may also apply, such as the temporary reproduction of a work in machine readable form for a computer.

In the United States the AHRA (Audio Home Recording Act Codified in Section 10, 1992) prohibits action against consumers making non-commercial recordings of music, in return for royalties on both media and devices plus mandatory copy-control mechanisms on recorders.

Section 1008. Prohibition on certain infringement actions:

No action may be brought under this title alleging infringement of copyright based on the manufacture, importation, or distribution of a digital audio recording device, a digital audio recording medium, an analog recording device, or an analog recording medium, or based on the non-commercial use by a consumer of such a device or medium for making digital musical recordings or analog musical recordings.

Later acts amended US Copyright law so that for certain purposes making 10 copies or more is construed to be commercial, but there is no general rule permitting such copying. Indeed making one complete copy of a work, or in many cases using a portion of it, for commercial purposes will not be considered fair use. The Digital Millennium Copyright Act prohibits the manufacture, importation, or distribution of devices whose intended use, or only significant commercial use, is to bypass an access or copy control put in place by a copyright owner. An appellate court has held that fair use is not a defence to engaging in such distribution.

Accessible Copies

It is legal in several countries including the United Kingdom and the United States to produce alternative versions (for example, in large print or braille) of a copyrighted work to provide improved access to a work for blind and visually impaired persons without permission from the copyright holder.

TRANSFER AND LICENSING, AND ASSIGNMENT

A copyright, or aspects of it, may be assigned or transferred from one party to another. For example, a musician who records an album will often sign an agreement with a record company in which the musician agrees to transfer all copyright in the recordings in exchange for royalties and other considerations.

The creator (and original copyright holder) benefits, or expects to, from production and marketing capabilities far beyond those of the author. In the digital age of music, music may be copied and distributed at minimal cost through the Internet, however the record industry attempts to provide promotion and marketing for the artist and his or her work so it can reach a much larger audience. A copyright holder need not transfer all rights completely, though many publishers will insist. Some of the rights may be transferred, or else the copyright holder may grant another party a non-exclusive license to copy and/or distribute the work in a particular region or for a specified period of

time. A transfer or licence may have to meet particular formal requirements in order to be effective, for example under the Australian Copyright Act 1968 the copyright itself must be expressly transferred in writing. Under the U.S. Copyright Act, a transfer of ownership in copyright must be memorialized in a writing signed by the transferor. For that purpose, ownership in copyright includes exclusive licenses of rights. Thus exclusive licenses, to be effective, must be granted in a written instrument signed by the grantor. No special form of transfer or grant is required.

A simple document that identifies the work involved and the rights being granted is sufficient. Non-exclusive grants (often called non-exclusive licenses) need not be in writing under U.S. law. They can be oral or even implied by the behaviour of the parties. Transfers of copyright ownership, including exclusive licenses, may and should be recorded in the U.S. Copyright Office. (Information on recording transfers is available on the Office's web site.) While recording is not required to make the grant effective, it offers important benefits, much like those obtained by recording a deed in a real estate transaction.

Copyright may also be licensed. Some jurisdictions may provide that certain classes of copyrighted works be made available under a prescribed statutory license (*e.g.* musical works in the United States used for radio broadcast or performance). This is also called a compulsory license, because under this scheme, anyone who wishes to copy a covered work does not need the permission of the copyright holder, but instead merely files the proper notice and pays a set fee established by statute (or by an agency decision under statutory guidance) for every copy made. Failure to follow the proper procedures would place the copier at risk of an infringement suit. Because of the difficulty of following every individual work, copyright collectives or collecting societies and performing rights organizations (such as ASCAP, BMI, and SESAC) have been formed to collect royalties for hundreds (thousands and more) works at once. Though this market solution bypasses the statutory license, the availability of the statutory fee still helps dictate the price per work collective rights organizations charge, driving it down to what avoidance of procedural hassle would justify.

Creative Commons

Founded in 2001, Creative Commons (CC) is a non-profit organization which aims to facilitate the legal sharing of creative works. To this end the organization provides a number of copyright license options to the public, free of charge. These licenses allow copyright holders to define conditions under which others may use a work and to specify what types of use are acceptable. Terms of use have traditionally been negotiated on an individual basis between copyright holder and potential licensee.

Therefore, a general CC license outlining which rights the copyright holder is willing to waive enables the general public to use such works more freely. Six general types of CC licenses are available. These are based upon copyright holder stipulations such as whether he or she is willing to allow modifications to the work, whether he or she

permits the creation of derivative works and whether he or she is willing to permit commercial use of the work. As of 2009 approximately 130 million individuals had received such licenses.

DURATION

Copyright Term

Copyright subsists for a variety of lengths in different jurisdictions. The length of the term can depend on several factors, including the type of work (*e.g.*, musical composition, novel), whether the work has been published, and whether the work was created by an individual or a corporation. In most of the world, the default length of copyright is the life of the author plus either 50 or 70 years. In the United States, the term for most existing works is a fixed number of years after the date of creation or publication. Under most countries' laws (for example, the United States and the United Kingdom), copyrights expire at the end of the calendar year in question.

The length and requirements for copyright duration are subject to change by legislation, and since the early 20th century there have been a number of adjustments made in various countries, which can make determining the duration of a given copyright somewhat difficult. For example, the United States used to require copyrights to be renewed after 28 years to stay in force, and formerly required a copyright notice upon first publication to gain coverage. In Italy and France, there were post-war-time extensions that could increase the term by approximately 6 years in Italy and up to about 14 in France. Many countries have extended the length of their copyright terms (sometimes retroactively). International treaties establish minimum terms for copyrights, but individual countries may enforce longer terms than those.

In the United States, all books and other works published before 1923 have expired copyrights and are in the public domain. In addition, works published before 1964 that did not have their copyrights renewed 28 years after first publication year also are in the public domain, except that books originally published outside the US by non-Americans are exempt from this requirement, if they are still under copyright in their home country.

But if the intended exploitation of the work includes publication (or distribution of derivative work, such as a film based on a book protected by copyright) outside the U.S., the terms of copyright around the world must be considered. If the author has been dead more than 70 years, the work is in the public domain in most, but not all, countries. Some works are covered by copyright in Spain for 80 years after the author's death. In 1998 the length of a copyright in the United States was increased by 20 years under the Copyright Term Extension Act. This legislation was strongly promoted by corporations which had valuable copyrights which otherwise would have expired, and has been the subject of substantial criticism on this point. As a curiosity, the famous work *Peter Pan, or The Boy Who Wouldn't Grow Up* has a complex – and disputed – story of copyright expiry.

Public Domain

Copyright, like other intellectual property rights, is subject to a statutorily determined term. Once the term of a copyright has expired, the formerly copyrighted work enters the public domain and may be freely used or exploited by anyone. Courts in the United States and the United Kingdom have rejected the doctrine of a common law copyright. Public domain works should not be confused with works that are publicly available. Works posted in the internet for example, are publicly available, but are not generally in the public domain. Copying such works may therefore violate the author's copyright.

COPYRIGHT INFRINGEMENT

Piracy is considered to be the illegitimate use of materials held by copyright. For a work to be considered pirated, its illegitimate use must have occurred in a nation that has domestic copyright laws and/or adheres to a bilateral treaty or established international convention such as the Berne Convention or WIPO Copyright Treaty. Improper use of materials outside of this legislation is deemed "unauthorized edition", not piracy. Piracy primarily targets software, film and music. However, the illegal copying of books and other text works remains common, especially for educational reasons. Statistics regarding the effects of piracy are difficult to determine. Studies have attempted to estimate a monetary loss for industries affected by piracy by predicting what portion of pirated works would have been formally purchased if they had not been freely available. Estimates in 2007 stated 18.2 billion potential losses in consumer dollars lost as a result of piracy activities in the United States. International estimates suggest losses in the billions throughout the last decade. However other reports indicate that piracy does not have an adverse effect on the entertainment industry.

The Evolution of Property Rights in Cyberspace

The institutions of property — the network of primary and secondary rules regarding obligations to respect entitlements regarding access to, use of, and transfer procedures for assets — determine how resources will be brought into use. Resources will be used, however, no matter what the property rights system is. Even the lack of specified property rights is a property rights system: it implies that a resource can be used by anyone strong enough to claim access.

Indeed, since property rights are an important determinant of the allocation of resources, and therefore at least to a degree, the distribution of wealth, individuals have incentives to develop property rights arrangements to help them achieve their personal objectives. Creation of effective property rights requires persuading or inducing others to recognize rules (accept obligations) to respect the rights, of course.

After all, rules are generally not necessary if there are no conflicts to resolve. Furthermore, as David Hume (1751) emphasized over 250 years ago, the primary source of conflict between individuals is scarcity. Scarcity becomes apparent when a conflict over use arises, and the institutions of property evolve as such conflicts are resolved by individuals attempting to find ways to expand personal well-being or “wealth” in the face of this scarcity, whether the resolution is achieved through cooperative procedures such as negotiation, or through violence or threats of violence (coercion). Thus, in a positive sense, property rights are a matter of economic value rather than of legal definition or moral philosophy.

THE SECURITY OF PROPERTY RIGHTS IN CYBERSPACE

In the past, individual had little choice but to accept messages from all sources, inspect them, and reject data only after it was received, but the internet appears to be transforming into a system where address holders can make affirmative decisions to accept messages only from identifiable sources. In today's environment each individual can decide, or contract with someone else [e.g., an ISP, an employer, a specialist in screening e-mail] to decide when and with whom a connection to an address will be allowed. The trend, for individuals or organizations who consider the cost of spam, viruses, adware, spyware, and other threats to their computers and their time to be higher than the benefits of leaving their in-boxes unlocked, clearly is toward a situation in which verification from others will be required. A message will be allowed to enter only if the other person is identified as someone who is trustworthy.

Efforts to secure information that being sent from one party to another are also developing, particularly with regard to financial transactions. For instance, customers of several credit card providers (e.g., Citibank, MBNA, Discover) can request a temporary account number for use in on-line purchases. While these numbers are linked to the customer's account, they expire after one use, or after a customer-specified spending limit is reached. Depending on the provider, these numbers can be obtained over the phone, from the company's web site, or by downloading software that generates such numbers, either upon request or when the software detects that the cardholder is at an on-line retailer's checkout page. While these temporary numbers limit the potential for identity theft with spyware and hacking, they also can be used to prevent retailers from renewing purchases such as magazine subscriptions without sending a reminder to the customer. Furthermore, if a company stores the credit card number in a database, and hacker breaks into the file to steal numbers, the temporary number will be useless.

Encryption is another method of protecting information. It insures that information stored in accessible computers or sent over the internet cannot be read by anyone who does not have the key to the encryption. Identification and authentication provide means of privatizing rights to exclude others from an individual's in-box, but they simultaneously raise concerns about another right: the right to privacy and anonymity.

There appears to be a solution to this dilemma, however. Encryption along with public keys allows information to be coded and for its source to be verified while the individual's off-line identity can remain anonymous. The process sounds cumbersome and even infeasible to those of us used to dealing in real space, but in cyberspace it is virtually costless and very fast. It involves a pair of "keys" (mathematical algorithms) that are inverses of each other. One is a private key retained by the sender, and the other is a "public key" used to decode the message. It is not a true "public" key in the sense that anyone and everyone has it, however. It is a decoding key that

the private key holder provides to selected other individuals. Indeed, only a person who has a public key is able to decode the message, so the sender must provide the intended recipient with the public key (technically, the public key can be sent along with the message to the recipient).

The transmission is private, since it is coded, and only the person who has the private key can send the message, so sender's identity is verified. The technology is not sufficient, by itself, to establish the reliability of the sender, of course. The recipient may not know who sent a message with an attached public key. Thus, an assurance problem may stand in the way of such "verifiable" transactions without some additional reason to trust the sender. This problem also can be overcome, however, as can other types of assurance problems.

FACTORS INFLUENCING THE EVOLUTION OF PROPERTY RIGHTS IN CYBERSPACE

The economic model of property rights suggests that as a consequence of the rising external costs that we are witnessing in cyberspace, we should be seeing increasing investments in the creation and security of private property rights, and this is clearly occurring. One way of establishing property rights is through laws that threaten punishment for trespassers, thieves and others who violate claims. This deterrence approach requires that the threats of punishment be credible, however, and it is clear that while such threats are being made by some governments (*e.g.*, the U.S. Can Spam Act), the abilities of violators to locate outside those governments' jurisdictions and to cloak their identities make this an ineffective method for creating anything close to secure rights. The alternative is to protect individual pieces of property through watching, walling, and wariness. Much as the invention of barbed wire lowered the cost of defining property boundaries and excluding other people's cattle from land in the American West, new technological developments are lowering the cost of sorting (watching) and excluding (walling) unwanted data from IP addresses and personal computers, and for requiring authentication from senders (wariness).

Other factors are also important, such as innovations in the methods of establishing reputations in cyber commerce, as well as in contractual arrangements and other organizational methods of institutionalizing incentives, including developments in dispute resolution techniques. This sub-subsection will focus on technological innovations which are facilitating the ability of individuals to exclude others from using their addresses and computers, while most of the institutional changes.

Access to a person's computer and/or IP address can be limited if the benefits of doing so in the minds of the computer owner exceed the benefits. Indeed, there are growing numbers of technological innovations available for screening and filtering content in order to prevent entry of unwanted data. One area of considerable development in this regard involves efforts by parents to monitor and restrict children's access to

content. For instance, consider the growing use of instant messaging and online chat that allow internet users to communicate virtually instantaneously.

Some parents are increasingly concerned about who their children are communicating with, but physical methods of monitoring such activities are very costly and often ineffective. Technological methods of monitoring are increasingly available, however. Instant messaging and chat rooms are offered as part of many paid online services such as AOL and MSN, for instance, and these services offer control tools to parents that can constrain children's use of these options (they can also limit access to Web sites that are considered by parents to be inappropriate).

AOL subscribers are able to create restricted accounts, for instance, that limit chat and instant messaging opportunities as well as web browsing and e-mail opportunities. The AOL "Kids Only" default category blocks all instant messaging, for example, while the "Young Teen" category does not allow the exchange of images, files, voice, or videos. A "Mature Teen" category is also available that limits chat and Web browsing but not instant messaging. All of these categories can also be customized by individual parents, who can even control how long children can use AOL and when they can do so. Similarly, MSN parental control categories include Teen, Preteen, and Young Child with varying degrees of limitations as well as the ability of parents to customize settings. Parents can block all use of Microsoft Messenger, or restrict access to people on a specified contact list approved by the parents. MSN also allows parents to block file downloads, and will e-mail reports to parents about their children's online activities.

Many other internet access providers (ISPs) have similar options for parents. Of course, children can use browsers not provided by their ISP to circumvent such controls, and even though AOL and other providers offer some options for controlling these activities as well, there is more that parents can do as individuals if they remain concerned. In particular, there are a number of parental control software packages that can be purchased and installed. CyberPatrol can block instant messaging, for instance, and its Chat Guard feature filters specified strings of characters such as objectionable words, phone numbers, and names. Cybersitter fulfills similar functions, and also allows parents to record the text of instant messaging exchanges.

It also cannot be disabled without a password (unlike recording features in Yahoo and MSN messaging software, for instance). Spectorsoft records all e-mail, instant-messaging, and chat conversations and keeps parents informed of their children's online activities by e-mail. Of course, children can always find ways around constraints if they try hard enough, whether the constraints are imposed in geographic or cyber space. However, the fact that access to a child's computer address can be significantly limited, even for children who do not want such limits, suggests that individuals who do not want to receive certain kinds of content themselves can also dramatically limit access to their addresses. Software to filter incoming messages is developing rapidly, and it is being applied both by individuals (.e.g, Anti-Virus programmes purchased

from Norton, McAfee, and other firms) and through contractual arrangements. Indeed, the same ISPs that offer parents methods for controlling access to their children's computers, offer various methods for preventing access by spammers, Adware popups, viruses, and so on. AOL reported blocking 500 billion spam e-mails intended for their subscribers during 2003, for instance, or roughly 15,000 spams per AOL member. Casual observation of television advertising by such ISPs suggests that these filtering features are very attractive to consumers, and therefore, that they are becoming an increasing important focus of competition.

E-mail services like Yahoo Mail, Hotmail, Eudora, and Outlook also are providing spam filters. Hotmail reports catching 3.2 billion spam messages per day, while Eudora version 6.0's filter is said to have stop 97 percent of the spam directed at its users after it was released in September of 2003. Such filters are not perfect, of course.

As Barzel (1989: 5) stresses, property rights are never likely to be completely and perfectly delineated, given the costs of doing so relative to the benefits. Indeed, the Eudora filter is less accurate than it was when it was first released because spammers have altered their messages to avoid detection. This filter, probably one of the most effective, uses a method called Bayesian scoring. It looks at words in an e-mail message and assigns each word a probability of being part of a spam.

It then calculates a cumulative probability to estimate the total likelihood that the message is spam. The user can set the level of probability that he or she is comfortable with, and the programme will divert all messages that have a higher probability than the one chosen. Like other filters, the Bayesian scoring method does not filter all spam, and it can divert messages that are not spam. Furthermore, over time, spammers have adjusted by filling messages with words that the filter does not place high probabilities on. A spam message may have a large number of what appear to be random words at the bottom, for instance, or the ad might be surrounded by poetry. As a result, the number of words recognized as likely to be in spam are swamped by large numbers of words that the programme does not consider to be likely spam content.

Thus, the filters have to change over time as the spam does. One way that this happens is that the individual users of the filters can adjust the filter. If spam gets through, the user can designate it as such and future messages of that kind will be diverted. Diverted messages can also be sent to a "junk" mailbox that the user can then check occasionally to make sure that desired messages are not being diverted. If they are, the user can tag such messages as valid, so they will not be diverted in the future. In addition, of course, filter providers continue to experiment in an effort to refine the filtering process, just as spammers experiment to undermine it. In the process, such filtering software has become very effective at catching a huge majority of the spam being sent before it ever gets into e-mailboxes.

Others ways to exclude unwanted e-mail are also available. "Challenge-response" systems are used by some e-mail account providers, for instance, including Earthlink and Mailblocks. This technology sends an automatic challenge from an e-mail recipient

to the sender, asking for a response of some sort (*e.g.*, supply a generated password, or answer some question) in order to verification that the sender is a real person. Since most spam is sent in bulk by computers, spam mail is not verified by responding to the challenge, and therefore excluded. These systems have proven to be very effective at eliminating spam. Thus, Microsoft reported, in 2004, that it was experimenting with this technology as a possible added protection for the more than 100,000 million users of its Hotmail and MSN mail servers.

Indeed, Microsoft has purchased a number of other software firms specializing in security software and is reportedly spending about \$2 billion per year on computer security research and development, no doubt in part at least because of growing consumer demands for security and the increasing level of competition offering more secure e-mail and browsing systems. The increasing use of the Firefox browser apparently induced Microsoft to rush Explorer 7.0, with its new anti-spyware technology and improved anti-virus features, onto the market, for instance. E-mail checking services are also being offered. A user of PermitMail provides the service with a list of permitted addresses for senders, and if an incoming e-mail is not on the list, it is held. The sender receives a message indicating that the mail is being held, and that he or she can fill out a form asking to be put on the user's permit list. The sender must provide a name, and can write a message explaining why he or she should be added to the list.

Other kinds of identification and authentication processes are also developing rapidly. For instance, businesses like banks, brokerage firms, travel sites, and other on-line merchants, have started requiring authentication in an effort to distinguish between their legitimate customers and spam, and to protect their transactions from potential identity theft. One way to do this is by issuing customers a "hardware token," a small device (small enough to attach to a key chain) that displays a six digit number which changes once a minute. When customers access their accounts they type in their user names and passwords, but they also enter the numbers displayed on their tokens.

Several banks in Europe and Australia are already using tokens, and mandating that their online customers have one. A number of U.S. banks have issued tokens to their corporate customers and to their own employees. These banks are all considering the technology for their retail banking services, and U.S. Bancorp is planning an experimental test of the system, while E Trade Financial is offering this type of device to its customers as an option this year. In fact, virtually every bank in the U.S. apparently is seriously considering the technology.

A number of other technologies are also being developed that can provide similar authentication, such as smart cards (plastic cards with embedded microprocessor chips), biometrics (identification of people by their voice, physical characteristics, or fingerprints), and shared secrets (customers must answer a question that, at least in theory, only they know the answer to). In a similar vein, several ISPs have begun to "stamp" messages leaving their customers' addresses with a "digital signature" of the

customers' domain names, using strong cryptography to prevent alteration or counterfeiting of the signature. Recipients can then look for such a signature on incoming mail using DomainKeys software. This identification process is being supported by an organization of ISPs who have agreed to prevent spammers from operating through their ISPs.

Yet another option that is being considered involves pricing of e-mail. An Australian firm, CashRamSpam.com already is offering a payment system. Customers of the service can set a fee for incoming e-mail from unknown sources, and the firm will collect the fee, keep 10 percent and pass the rest to the customer. Refusal to pay means that the e-mail will be deleted, so a high fee is expected to eliminate all unwanted e-mail, while a lower fee might be paid by some advertisers, providing the customer with a source of income. Whether this system will catch on is unclear, but Microsoft Research is considering an alternative involving payment in time rather than money.

The system forces a sender to use the sending computer's resources to solve a computational puzzle devised for each message in order for the message to be "stamped": "The puzzle uses an intricate design involving the way a computer gains access to memory and resists a quick solution by speedy processors, requiring about ten seconds. It is not so long that you'd notice it for the occasional outgoing message, but if you have eight million Viagra messages queued up, good luck in getting each one stamped." An individual can then set his or her e-mail programme to filter all incoming messages from unfamiliar senders by examining them for their digital stamp (proof of having completed an assigned problem). Exclusion of non-payers is an issue with another kind of property as well. The free and easy sharing of files can significantly reduce the value of so-called intellectual property rights, particularly to music and films, resulting in substantial pressure by firms in these industries for governments to protect such rights. Legislatures are being lobbied and court cases are being pursued.

Yet, the solution to these intellectual property rights issues also is apparently coming through technological innovations rather than through government. This should not be surprising. After all, as Cox and Crews (2005: 2) explain, "The online intellectual property problems faced by copyright owners do not exist in isolation. They are rooted in the Internet's (perceived) anonymity [and free access] - but so are spam, cybersecurity, spyware, and authentication of transactions.

Indeed, to a large extent, we don't have an intellectual property problem; we have an anonymity/authentication[/free-access] problem." In this context, "digital rights management" (DRM) technology now provides a method of protecting copyrights to digital content distributed on line. A DRM system involves a technological container format for "packaging" the product. Thus, for instance, a music album will contain the music and track titles, along with a set of rules that the software and hardware music players must follow in order to play the material. This can include paying a price for each play of a song, or for a set number of plays. Alternatively, a consumer might subscribe to a service, perhaps for rentals or demos, and chosen song titles can

be valid as long as the subscription is maintained. More significantly, DRM address issues of redistribution, as encrypted controls can limit how users make and distribute copies of the digital files. As with the pricing arrangements, DRM is capable of a tremendous number of options regarding copying. For instance, a different price can be charged for a movie or music album that can be downloaded and/or copied than for the same film or music that cannot be reproduced. Thus, sellers can contract with consumers to meet whatever preferences they may have regarding transferability, but consumers will pay for the ability to reproduce. Formats supporting DRM include Windows Media and Advanced Audio Coding (AAC).

Apple Computer introduced its Internet Music Store in April, 2003 using AAC compression technology and a “light-handed” DRM system called FairPlay (Einhorn and Rosenblatt 2005: 4), for instance. Buyers of music can transfer songs to Apple iPod players, and burn unlimited numbers of CDs, but they can only transmit downloads to three other hard drives. The next version of Music Store will have a number of additional features. Real Network’s Rhapsody allows “all-you-can-eat” streaming for a \$9.95 per month fee, but charges 79 cents for individual burns.

There are several other copyright-respecting music services, including MusicMatch, with 99 cent downloads as well as a subscription service that permits on-demand steaming and playlist sharing with friends (non-subscriber friends can play the first 20 tracks on each playlist they receive, but only up to three times before paying for individual downloads or subscribing), and the re-launched Napster which charges 99 cents per track to download and burn individual songs and along with other subscription services.

DRM provides a way of contracting between suppliers of digital music and film products and their consumers, to give consumers specific use rights while preventing uncompensated redistribution. Norms regarding “fair use” (i.e., “reasonable usage”) of downloading and transferring are evolving through the competitive market process, as different firms are offering various alternatives: those that survive and prosper will be the firms that meet the expectations of sufficient groups of consumers while compensating the producers of music and movies for their products.

EXTERNALITIES AND THE EVOLUTION OF PROPERTY RIGHTS

Coase (1960) stresses the reciprocal nature of externalities, explaining that they arise when two or more individuals attempt to use the same asset or resource for conflicting purposes. The primary reasons for divergent expectations regarding access to and use of the resource are that the relevant property rights either are not clearly assigned or they are not effectively protected (i.e., rules of obligation either are not recognized or not respected). Therefore, a solution to the conflict may be achieved if one party is persuaded/induced to recognize that the other party has the relevant

property rights and/or to respect those rights (follow rules of obligation). To avoid the harm to one party, then, the other party must be harmed. The result of allocation and recognition of property rights is that one party will be able to benefit from using the resource and the other party will have bear costs if he or she wants to use it [*i.e.*, bargain to purchase it, or face the possibility of prosecution and liability for damages due to trespass or transfer without bargaining (theft)].

Because the allocation and enforcement of property rights determines the distribution of costs and benefits, the effected individuals obviously will have different opinions about how the rights should be assigned and protected. Thus, transactions costs arise as individuals attempt to create or protect property rights, because other individuals must either be persuaded or induced to accept obligations to respect claimed rights. Effective property rights are likely to arise when the benefits of creating them exceed the associated transactions costs. In fact, as Demsetz explains, "Property rights develop to internalize externalities when the gains from internalization become larger than the cost of internalization. Increased internalization, in the main, results from changes in economic values, changes which stem from.... changes in technology and relative prices."

Suppose that a sufficiently large number of people want to use a resource so that the resource is scarce. That is, the use by some individual or individuals, A, has a negative impact on the well being of (imposes a negative externality on) another individual (or individuals), B. Under these circumstances, each individual has incentives to access the resource and use it before other users do the same. Assume that B's use is prevented altogether if A is successful in using the resource first, for instance, and *visa versa*. In this case, each individual has incentives to claim the resource before others do. The potential means of successfully asserting that claim. The point here simply is that when A's use of a scarce resource precludes B's use and *visa versa*, the conflict becomes immediately apparent, as are the incentives to claim property rights. Things are not quite this simple, however.

It is costly to measure an asset's attributes, however, and the cost of increasingly fine delineation of an asset's attributes rise: "Because transacting is costly, as an economic matter property rights are never fully delineated". Individuals have incentives to delineate and develop rights to those attributes of an asset that are valuable relative to the cost of delineation and rights establishment. However, when costs mean that attributes are less than fully delineated and claimed, some attributes remain accessible to other individuals, *e.g.*, B, other than the person or persons, A, who has claimed rights to the delineated attributes. Whether intentionally or not, uses of unclaimed attributes of the resource by B can negatively affect the well being of A. Thus, externalities persist. As Barzel (1989: 5) explains, for instance,

The rights to receive the income flow generated by an asset are a part of the property rights over that asset. The greater is others' inclination to affect the income flow from someone's asset without bearing the full costs of their actions, the lower the value

of the asset. The maximization of the net value of an asset, then, involves that ownership or ownership pattern that can most effectively constrain uncompensated exploitation. The argument is more general than just income, however. Interference with the ability to obtain any type of subjective benefit will create incentives to look for ways to constrain such interference, and if cost effective means of doing so are discovered, more attributes will be delineated and property rights will evolve. The bundle of rights associated with a particular resource can be quite large, then, as access, use, and transferability rights can apply to and differ over many different attributes.

In some cases, many people can use a resource without actually precluding others from using the same resource, but in the process each person's use, or benefits from use, may be impaired in some way. That is, each person's use reduces the satisfaction that others obtain from their use. In the case of a free-access commons, for instance, crowding or congestion arises and the resource deteriorates in quality (each individual's use impairs every other individuals' uses to a degree) as the result of over use.

This has been called the tragedy of the commons, of course. The tragedy of the commons is not the inevitable outcome of free access, however, because the externalities create incentives to develop alternative property rights. In fact it is not the likely outcome unless the transactions costs of at least partial privatization (*e.g.*, quota rights like fishing licenses or pollution permits) are so high that such privatization is not warranted.

In this context, Johnson and Libecap (1982) contend that there are three types of costs that can be relevant:

1. Exclusion costs;
2. Internal governance costs when exclusive rights are shared by a group; and
3. Imposition of punishment for violating an open access constraint imposed by a strong coercive authority (*e.g.*, the state).

If these costs are not prohibitive, property rights should evolve when externalities become significant. For example, Demsetz (1967) uses his theory to explain the evolution of property rights among a community of Indian hunters of the Eastern part of Canada who, in the early 18th century, developed exclusive rights to take beaver furs. Prior to the arrival of European fur traders, beaver were so abundant relative to the native population's demand for beaver furs or meat, that if one individual harvested beaver from the commons, it had no noticeable impact on other individuals' ability to harvest beaver.

There was no reason to claim exclusive hunting rights. However, Europeans considered beaver to be very valuable. Therefore, when European fur traders began trading things that the Indians valued highly (European manufactured goods such as metal knives, axes, pots and pans, woolen blankets, guns and ammunition, *etc.*) for beaver, the Indians' incentives to hunt beaver increased dramatically. This created a common pool problem. Rather than allowing the destruction of the beaver population, Indians began claiming and recognizing exclusive hunting rights. Individual Indian

households claimed their own beaver populations along streams, and the community of Indians (related bands) recognized those claims. As a result, the population of beaver leveled off and was maintained, rather than being destroyed. Demsetz also argued that the Indians of the American Southwest failed to develop similar property rights because of the relatively high costs and low benefits from establishing exclusive hunting rights.

Factors Influencing the Evolution of Property Rights

Demsetz' (1967) property-rights-in-beaver example emphasizes that a change in relative values (*e.g.*, prices) can lead to a change in property rights. Rights can also evolve because of a technological change that reduces the cost of measuring attributes and/or establishing rights, however. For instance, Anderson and Hill (1975) explain that property rights in range land in the American West evolved from open range to private range with the introduction of barbed wire, and Ellickson (1993: 1330) explains that transferability of private use rights developed, at least in part because written language reduced the cost of record keeping.

Property rights can also evolve with changes in the institutions of policing and dispute resolution. Similarly, discoveries of new information that lead individuals to recognize that the resource in question is more valuable than previously believed, can also induce increased measurement and rights delineation. In addition, property rights can change because of changes in the relative opportunity costs of violence used to induce recognition of rights, or the relative costs of negotiation used to persuade potentially cooperative individuals to recognize each others' rights, perhaps due to technological changes that reduce the costs of violence or negotiation.

Thus, many things can change the value of property rights, and this in turn leads to changes in the rights structure. As Barzel (1989: 65) emphasizes in echoing Demsetz (1967), "People acquire, maintain, and relinquish rights as a matter of choice.... As conditions change... something that has been considered not worthwhile to own may be newly perceived as worthwhile." Such changes need not be exogenous, however. In fact, they are not likely to be. When substantial externalities arise, the desire to internalize them creates incentives to carry out research in an effort to develop new technologies and/or new information that will lower the transactions costs of rights creation. Thus, the technological advances and other increases in knowledge that allow property rights to evolve often are endogenous.

THE SECURITY OF PROPERTY RIGHTS

Alchain and Allen (1969: 158) propose a revealing definition of ownership rights: "the expectations a person has that his decision about the use of certain resources will be effective." This very non-legal and non-philosophical sounding definition stresses the fact that, in practice, rights (or expectations) are never absolute, and not just because of Barzel's (1989) point that delineation and development of rights to attributes

is costly. The strength or security of property rights (and therefore expectations) is a function of efforts made to protect or enforce rights claims, and the offsetting efforts to take or attenuate those claims.

There are many ways that others might attempt to capture control of resources, such as through warfare or theft, or through the political process by lobbying the legislature to change the rules. Given such threats, people incur costs as they attempt to enhance the security of the rights over the assets that they wish to control. Sherman (1983) discusses individual investments in watching, walling, and wariness, for instance. These activities can include investments of time (*e.g.*, watching, lobbying to prevent political transfers), or in the purchase of labour services (*e.g.*, guards, hired lobbyists), technology (*e.g.*, monitoring equipment for watching, better fences and locks as part of walling), and in the sacrifice of desirable activities that might increase the vulnerability of property (*e.g.*, avoiding certain dangerous areas, not going out at night). Technological changes can lower such costs by improving the productivity of individuals' time, or in labour and capital services, so incentives to perform research and development regarding security methods are high when rights are insecure.

Some of the costs of watching, walling, and wariness activities for individuals can also be reduced significantly through organizational innovations. A community of individuals might be formed, for instance, and voluntarily agree to respect each others' rights. In this context, Vanberg and Buchanan (1990: 18) define "trust rules" to be rules of behaviour toward others which individuals have positive incentives to voluntarily recognize, and explain that:

By his compliance or non-compliance with trust rules, a person selectively affects specific other persons. Because compliance and non-compliance with trust rules are thus "targeted," the possibility exists of forming cooperative clusters.... Even in an otherwise totally dishonest world, any two individuals who start to deal with each other - by keeping promises, respecting property, and so on - would fare better than their fellows because of the gains from cooperation that they would be able to realize.

Wealth can be enhanced for everyone involved in such trust relationships as property rights are made relatively more secure and relatively more private. As Hayek (1973: 107) explains, "The understanding that 'good fences make good neighbours', that is, that man can use their own knowledge in the pursuit of their own ends without colliding with each other only if clear boundaries can be drawn between their respective domains of free action, is the basis on which all known civilization has grown. Property, in the wide sense in which it is used to include not only material things... is the only solution men have yet discovered to the problem of reconciling individual freedom with the absence of conflict." Indeed, the absence of conflict may be the primary objective of some agreements, as suggested above, given the relatively low probability of winning in violent confrontations. Such implicit agreements are even seen between animal as property claims in the form of hunting ranges are delineated and recognized, thereby reducing conflict (Hayek 1973: 75). In other words, the

primary goal of some agreements may be to obtain non-material wealth in the form of “peace” or security, but of course, the production of more material wealth is also likely to be enhanced for everyone involved in such trust relationships. If property rights are made relatively more secure and relatively more private, after all, time horizons lengthen and incentives to use the property for production, rather than immediate consumption, increase. Thus, whatever the objective, all organizations function to a substantial degree by delineating various rights that individuals are expected to respect.

Members of such communities may also agree to cooperate in joint production of property protection against outside threats (*e.g.*, neighbourhood watch, pooling funds to employ security or policing services for an area, walling and gating an entire community, traveling in a group). But a significant source of transactions cost may stand in the way of adopting such rules voluntarily through an explicit or implicit contract: the assurance problem. Will individuals live up to their promises to cooperate, or will they attempt to free ride on the efforts of others in the group? Therefore, consideration of the nature of the cooperative process that can underlay a system of voluntarily recognized and protected property rights requires an examination of potential solutions to the assurance problem. These consideration will simultaneously provide an analysis of the spontaneous evolution of the rules of and potential for contracting to voluntarily transfer property rights (*e.g.*, trade), since the same assurance problem stands in the way of all contracts, whether they involve trading commitments to respect each others’ property rights and to cooperate in watching, walling and wariness, or they involve trading goods and services.

ASSURANCE PROBLEMS AND SOURCES OF CREDIBILITY IN CONTRACTING

There would be no assurance problem if everyone had full knowledge, but such perfect knowledge does not exist anywhere except in some economists’ mathematical models. Information is so scarce in the real world that trust or recourse generally must substitute for full knowledge in order to make promises credible. Trust, a willingness to make oneself vulnerable to another even in the absence of external constraints, certainly can evolve to support cooperative interaction such as contracting to create protection arrangements or to trade, but it takes time, and under some circumstances it can be limited to relatively small communities.

If individuals are going to be willing to deal with others that they do not trust, recourse in the form of credibly threatened sanctions against breaches of promises (perhaps supported by third party dispute resolution) is a necessary substitute. While trust and recourse are both substitutes for knowledge, however, they are not perfect substitutes for each other. Tradeoffs in transactions costs mean that under some circumstances trust provides a superior solution to assurance problems, while recourse may be more desirable under other conditions. Furthermore, there are alternative

institutional mechanisms for the provision of recourse, and they also are imperfect substitutes. In order to illustrate this, let us consider some of the differences between alternative institutional sources of trust and recourse.

Building Trust

Many of the concepts from game theory are useful in demonstrating the gains from cooperating and defecting in various contexts, and therefore, in thinking about determinants of trust. Game theory demonstrates that cooperation can arise through repeated interactions, for instance. Perhaps, for instance, at some point, different individuals decide to claim adjoining properties. Facing the likelihood of repeated interaction, they form relatively tentative bilateral relationships based on reciprocity incentives, implicitly promising to recognize each others' territorial claims.

Because the long-term reciprocal response is uncertain, a repeated-game situation does not guarantee unconditional cooperation even with the tit-for-tat threats to reinforce the positive incentives associated with remaining on good terms with the other party(ies) (*e.g.*, relatively secure property rights, the potential to focus resources to produce wealth rather than violence). The dominant strategy still depends on expected payoffs, frequency of interaction, time horizons, and other considerations. Furthermore, in emerging economic and social activities, repeated-dealing arrangements must be established. For instance, McMillan and Woodruff (1998), in their study of emerging trade in Vietnam explain that an entrepreneur tends to be very cautious when considering a potential trading partner. This can take time, of course, and that is obviously one of the drawbacks of exclusively relying on trust relationships.

Individuals may be able to gain the trust of others relatively quickly by offering some sort of bond or hostage. When an unknown party posts a bond with some trusted third party (*e.g.*, a reputable bank) as a guarantee that his promises are credible, for instance, he may be able to overcome a lack of trust. Similarly, individuals can invest in signals that demonstrate a commitment to high quality. As an example, consider Nelson's (1974) explanation of advertising of experience goods.

He notes that such advertising serves two primary functions for the rational buyer, and neither of these functions focus on the provision of direct information about the experience quality of commodities that are advertised: first, "advertising relates brand to function" and provides information about the general uses of the product, but second and more important, the volume of advertising is a signal to buyers that shows the extent of committed investment by the seller. What matters most to a rational buyer is not what advertising says about quality, but simply that it is a recognizable investment in non-salvageable capital: brand name.

Investments in other non-salvageable assets (*e.g.*, elaborate store fronts, charitable contributions, community service) can serve the same function. Essentially, investments in non-salvageable assets are offered as a bond to insure credibility. Information transmission is a key to the success of such a business strategy, as buyers must be aware

of such commitments. If they are, then as Klein and Leffler (1981) explain, the marginal cost to buyers of measuring such specialized or non-salvageable investments must also be less than the prospective gains: “If the consumer estimate of the initial sunk expenditure made by the firm is greater than the consumer estimate of the firm’s possible short-run cheating gain” he or she will tend to trust the seller.

When effective recourse is not available or is relatively costly, individuals who want to enter into cooperative relationships such as trade have very strong incentives to make such investments. Time is required to build reputations, of course, so new entrants may have to suffer through a considerable period of losses before they can expect to see investments in reputation building pay off. Indeed, since the payoff to such investments are delayed and very uncertain, incentives to make them tend relatively weak, and the emergence of cooperation based on such sources of recourse also can be quite slow. Some individuals may have reputations that they have developed in other activities, however, that can be transported into the new situation. Firms with international reputations may enter an emerging market and become established very quickly, for instance. There is another alternative as well.

Markets for reputation can also develop. For instance, a firm or other organization might develop a reputation for honestly assessing the quality of other firms’ products or services. Then when someone wants to enter a market and quickly establish a reputation, he or she can pay to have a product or service performance inspected/tested and “certified” by this reputable assessment organization. Moodys, Standard and Poors, Underwriters Laboratory and the Good Housekeeping Seal of Approval come to mind. Similarly, the American Automobile Association inspects and rates motels. Reputation can be purchased in other ways as well. One possibility is to contractually affiliate with an organization that is recognized to require that all affiliates meet specified standards for their products or services. Best Western motels is an example. Locally owned franchises of regional or national chains of motels, restaurants, retail outlets, and so on provide other examples. All such arrangements involve non-salvageable investments, of course, since failure to maintain quality will result in loss of the reputation signal that has been purchased.

From Trust to Reputation and Recourse

Most arguments about the inability of private parties to cooperate without the backing of a coercive power are explicitly or implicit prisoners’ dilemma arguments. However, the one-shot prisoners’ dilemma analogy does not characterize many kinds of interactions. When repeated dealing arrangements are valuable, each individual has implicit threat of punishment if the other party fails to live up to a promise, commits fraud or behaves opportunistically: the tit-for-tat response.

As more bilateral relationships are formed in recognition of the benefits from cooperation, and a loose knit group with intermeshing reciprocal relationships often begins to develop, however, tit-for-tat becomes a less significant threat. An exit threat

becomes credible when each individual is involved in several different games with different players, in part because the same benefits of cooperation may be available from alternative (competitive) sources.

And the fact is that individuals are generally involved in several “communities” as described by Taylor (1982: 26-30), wherein “the relations between members are direct and... many-sided”. When competitive alternatives within a community of transacting individuals make the exit option viable, Vanberg and Congleton (1992: 421) suggest that one strategy that can be adopted is unconditional cooperation until or unless uncooperative behaviour is confronted, and then imposition of some form of explicit punishment of the non-cooperative player as exit occurs. They label such a strategy as “retributive morality,” and the “blood-feuds” of primitive and medieval societies provide examples of such behaviour. This practice of retributive morality strengthens the threat to non-cooperative behaviour. However, the fact is that retributive morality or the blood feud played a much less significant role in primitive and medieval societies than is popularly perceived. After all, such violence is risky, and there is an even better alternative.

When individuals cooperatively engage in successful bilateral relationships within an evolving web of such relationships, others are likely to notice their cooperative behaviour and attempt to initiate mutually beneficial relationships with them. In other words, when information about cooperation spreads, such behaviour in one relationship can serve as investment in building a reputation for fair dealing, and this reputation can attract more opportunities. Importantly, however, when information spreads about non-cooperative behaviour, all of the beneficial relationships that the non-cooperative individual enjoys within the community can be put in jeopardy.

All members of the community have an exit option, and therefore they may cut off all relationships with someone who has proven to be untrustworthy in dealings with anyone else in the group. This means that there is a low cost option to retributive morality: unconditional cooperation whenever an individual chooses to enter into some form of interaction, along with a refusal to interact with any individual who is known to have adopted non-cooperative behaviour with anyone in the group and the spread of information about untrustworthy people. Vanberg and Congleton (1992) refer to this response as “prudent morality,” and given that reputation information spreads quickly within a group, the consequences of retributive and prudent morality become quite similar.

If everyone spontaneously responds to information the result is social ostracism, a very significant punishment, even though it is not explicitly imposed by a single retributive individual. Essentially, each individual’s reputation is “held hostage” by every other individual in the evolving group, *à la* Williamson (1983), and reputation is an ideal hostage. It is highly valued by the individual who has invested in building it, so a credible threat of destroying it can be a significant deterrent, and the threat is also credible because the reputation hostage has no value to the hostage holder and

the cost of destroying it (spreading truthful information) is low. That is, it is a non-salvageable asset (an asset that might be built relatively quickly by offering bonds or investing in even more non-salvageable assets, thus increasing the value of the hostage). Such an ostracism threat can be a very powerful source of recourse, where the “third party” providing the threatened sanction is the community of individuals who receive and respond to the information about non-cooperative behaviour. The spontaneous development of social ostracism illustrates another point.

As an informal community evolves from a web of bilateral trust relationships, group-wide norms also evolve. Note, in this context, that it is not the existence of “close-knit” communities that generates group-wide norms, as some have contended. Instead, norms and communities can evolve simultaneously as each affects the other: the evolution of norms of cooperation lead to the development of a web of interrelationships that can become a “close-knit” community, and the development and extension of such a community in turn facilitates the evolution of more effective norms. Thus, as Vanberg and Congleton (1992: 429) conclude, perceptions “of what is moral vary with relevant differences in exit costs.

At the high-cost end of the spectrum, moral justification for tit-for-tat and retributive behaviour seems to be fairly common, whereas prudent morality gains in importance as we move to the low-cost end.” Many group-wide norms are simply commonly adopted trust rules that apply for all interactions in an informal web of relationships. As this web of relationship begins to become recognized as a community, other rules can arise. Called “solidarity rules” by Vanberg and Buchanan, these are expected to be followed by all members of the group, because individual sacrifices associated with obeying solidarity rules produces shared benefits within the group.

Solidarity rules are things like “do not behave recklessly and put others at risk.” However, they can also involve rules about individuals’ obligations in cooperative production of rule-enforcement functions. Rules like “inform your neighbours about individuals who violate trust rules,” and “do not cooperate with individuals who behave in a non-cooperative fashion with someone else,” are solidarity rules in the sense that production of information and ostracism create benefits for everyone in the group by deterring non-cooperative behaviour. Significant limits on abilities to reason and to absorb knowledge means that individuals are not able to use conscious reason to evaluate every particular option in the array of alternatives that are available.

Therefore, rational individuals will often find it beneficial to voluntarily conform to a community’s rules in an almost unthinking way. And in this context, as Ridley (1996: 132) notes, “Moral sentiments... are problem-solving devices [that evolve]... to make highly social creatures effective at using social relations [by]... settling the conflict between short-term expediency and long-term prudence in favour of the latter.”

People conform to all sorts of faddish and ritualistic behaviours, of course, and even though they may appear to have nothing to do with evolving moral sentiments, they

actually may have similar functions: facilitating cooperation. After all, while individuals want to identify and exclude non-cooperative players, they also have strong incentives to identify themselves as cooperative. Outward conformity to a group's fads and rituals can serve as a signal of willingness to cooperate in order to be in a position to reap the rewards from participation in joint production and other forms of interaction within the evolving group. As Ridley (1996: 188) explains, "We are designed not to sacrifice ourselves for the group but to exploit the group for ourselves."

Incomplete knowledge, scarcity, and transactions costs mean that someone alleged to have violated a trust or solidarity rule may not be guilty, so "disputes" over guilt or innocence arise. Indeed, confrontations can arise under two different conditions in an expanding or increasingly dynamic group. In addition to disputes regarding allegations of violations of a norms, disputes can arise when property rights that are not clearly defined become valuable and conflicting claims to those rights are asserted. Some (and in a close-knit community, most) disputes can be solved by direct bargaining, but transactions costs can prevent successful bargaining in some cases.

Therefore, third-party dispute-resolution institutions are desirable in order to reduce the chances of violent confrontation (a dispute resolution process that can create considerable costs for other members of a community), and to increase the changes that the community can survive so its members can prosper from the mutually beneficial interactions it supports. Public courts are one source of such dispute resolution, but there are many other options as well. Contracting parties can specify some sort of alternative dispute resolution (ADR) for instance, be it mediation, arbitration, or some combination of the two. Individuals may also choose ADR after a dispute arises, even if they have not specified the option in a formal contract, possibly because they want to maintain a good relationship or because social pressures are brought to bare by other members of the community.

Third Party Dispute Resolution

Voluntary acceptance of ADR means that the selected third party must be acceptable to both disputants, so "fairness" is embodied in the dispute-resolution process. There are a wide variety of potential sources of ADR. Specialists can be selected from organizations like the International Chamber of Commerce, the American Arbitration Association, or any number of other private dispute resolution providers, including private ADR firms and for-profit "courts." Mechanisms for ADR (*e.g.*, arbitrator and mediator) selection actually vary widely, but they all are designed to guarantee the selection of an unbiased third party who will apply the norms that the parties share within their relevant community.

In general, the choice of an ADR provider is made without requiring explicit agreement by the two parties while still allowing for prescreening, and possibly more than one level of screening. For example, one common selection method involves a pre-approved list of professional mediators or arbitrators determined by contracting

parties, so if a dispute arises, a person is chosen from the list by some preset mechanism (*e.g.*, random selection, rotating selection, selection by a third party such as a governing board of a trade association).

Empirical evidence indicates that selection of arbitrators for a pre-approved list is based on the reputation of the arbitrators for impartiality and expertise in contractual matters that might arise. Another common arbitrator selection system gives the parties to a dispute the resumes of an odd numbered list of arbitrators from a larger pre-selected group (*e.g.*, pre-selected by a community such as a trade association, or provided by an organization like the ICC or the AAA), with each party having the power to successively veto names until one remains. Thus, a second level of screening is added at the time of the dispute, contributing “to the legitimacy of the arbitrator and his award in the eyes of the parties”. Since the parties are given the arbitrator’s resumes, they have information about experience, training, the nature of awards given in the past, and so on.

A similar practice provides the parties with a list and resumes of an odd number of potential arbitrators from a pre-approved list, with each disputant having the power to veto one less than half and rank the others, and the arbitrator who is not vetoed by either party and has the highest combined rank is chosen.

Both sides of the dispute may also provide a list of a fixed number of mediators or arbitrators with each being able veto any or all of the names on the other party’s list; if all names are vetoed each provides another list and the process is repeated (clearly, this procedure requires that both parties want to arbitrate, so they do not continue to provide unacceptable names). All such systems are intended to guarantee the appointment of a third party without requiring explicit agreement by the two parties while still allowing for prescreening, and possibly more than one level of screening, of the potential mediators or arbitrators.

Biased rulings are not likely in such a competitive environment where potential arbitrators are chosen before hand by the trading community (*e.g.*, as in the diamond traders associations) or where both parties have the power to reject judges proposed by the other party. Furthermore, successful arbitrators will be those who consistently apply the norms that members of the relevant community expect to be applied. Indeed, by choosing an arbitrator/mediator attempting to build a reputation of trustworthiness strong incentives are created for those aspiring to be chosen as third-parties in dispute resolution to avoid the appearance of bias.

The chosen arbitrator/mediator must convince individuals in the group that a judgement should be accepted, after all, since he has no coercive power to enforce it. More importantly, an appearance of bias will damage the individual’s reputation. The ruling can therefore be backed by an implicit threat of ostracism, although in general, dispute resolutions are likely to be accepted because individuals recognize the benefits of behaving in accordance with community members’ expectations, not because they fear ostracism.

Recourse through Contractual Associations

Both commitments and threats can be made more credible, and some uncertainty can be eliminated, if individuals with mutual interests in long-term interactions form “contractual” groups or organizations rather than waiting for trust or reputation institutions to evolve more slowly into informal communities. Potential contractual arrangements are numerous, including the implicit contracts of family bonds and ethnic or religious networks, clubs and other social organizations, and in the commercial area, arrangements such as indirect equity ties through pyramidal ownership structures, direct equity ties, interlocking directorates, and trade associations. As Khanna and Rivkin (2000) explain, for instance, business groups are actually “ubiquitous in emerging economies” (as evidence, they cite a large number of studies about groups such as *grupos* in Latin America, business houses in India, and *chaebol* in Korea). Many of these associations may form for reasons other than the development and enforcement of norms, of course, but once they develop, the cost of adding such functions is relatively low. In addition to creating strong bonds that facilitate interaction, an affiliation with such a group can be information generating in that it can imply a credible signal of reputable behaviour.

A contractual organization can provide a formal mechanism to overcome frictions in communication, insuring that information about any individual’s non-cooperative behaviour will be transmitted to others in the contractual community. Then group membership can include a contractual obligation to boycott anyone who fails to follow the group’s contractually accepted rules: specifically, any non-cooperative party will be automatically expelled from the organization. Such automatic ostracism penalties make the reputation threat much more credible.

These groups can also lower transactions costs by establishing their own unbiased dispute resolution arrangements. After all, allegations of non-cooperative behaviour are not necessarily true, so they may have to be verified. Within some organizations a single mediator or arbitrator or panel of mediators or arbitrators is chosen for a set period to arbitrate all disputes between members. Thus, prescreening by the formal group occurs as ADR providers are chosen from a competitive pool by the association through its membership approved selection process. For instance, in the diamond industry, arbitrators are elected from the organization’s membership. In many religious organizations, the religious leadership (e.g., priests, elders) provides this function. Those selected are likely to have considerable standing (reputation) within the community, and they have strong incentives to maintain their own reputation for fairness, so they are not likely to be biased or corruptible. These services do not have to be produced internally, however, as the group may contract with external mediation and/or arbitration specialists.

When a dispute involves new and unanticipated issues, an ADR supplier may be required to determine what rule should be applied to the situation. In this context, for instance, Lew explains that “Owing no allegiance to any sovereign State, international

commercial arbitration has a special responsibility to develop and apply the law of international trade.” The “law” that dominates international trade has, for the most part, evolved through contracting and the use of arbitration.

Customary Law

In modern societies the most visible types of rules are the “laws” designed and imposed by those with authority in nation-states, there are other rules (e.g., habits, conventions, norms, customs, traditions, or standard practices) that are actually much more important determinants of behaviour in many aspects of human activity. A key distinguishing characteristic of such rules is that they are initiated by an individual’s decision to behave in particular ways under particular circumstances. As Hayek emphasizes, adopting a behavioural pattern creates expectations for others who observe it and this create an obligation to live up to those expectations.

Furthermore, as Mises (1957: 192) explains, when individuals who interact with one another observe each others’ behavioural patterns they often emulate those that appear desirable so such behaviour and accompanying obligations spread. In other words, these rules evolve spontaneously from the bottom up rather than being intentionally designed by a legislator, and they are voluntarily accepted rather than being imposed. No central “authority” with coercive powers is necessary to produce the resulting cooperative social order, as obligations are largely self-enforcing: it pays for each party to behave as expected in order to be able to expand wealth over the long run through mutually beneficial interaction.

Pospisil (1971, 1978) distinguishes between “legal” arrangements that evolve from the top down through command and coercion, which he calls “authoritarian law,” and systems of trust and solidarity rules that evolve from the bottom up through voluntary interaction, which he refers to as “customary law”. Such a norm-based cooperative arrangement often can be characterized as a “legal system” following Hart’s (1961) definition of law, since, it has primary rules of obligation (e.g., recognized norms), and it can be backed by secondary rules or institutions of recognition (e.g., reciprocities, mechanisms to spread information about reputations, ostracism, mutual insurance, cooperative policing), adjudication (e.g., negotiation, arbitration, mediation), and change [e.g., innovations in behaviour followed by observation, emulation and conformity, contracting, dispute resolution].

For an obligation to achieve the status of a “customary law” it must be recognized and accepted by the individuals in the affected group. In other words, a strong consensus rule applies, and as a result, customary law tends to be quite conservative in the sense that it guards against mistakes. Nonetheless, flexibility and change often characterize customary law systems. For instance, if conditions change and a set of individuals decide that, for their purposes, behaviour that was attractive in the past has ceased to be useful, they can voluntarily devise a new contract stipulating a new behavioural rule. Thus, an existing norm (custom) can be quickly replaced by a new

rule of obligation toward certain other individuals without prior consent of, or simultaneous recognition by, everyone in the group. Individuals entering into contracts with these parties learn about the contractual innovation, however, and/or others outside the contract observe its results, so if it provides a more desirable behaviour rule than older custom, it can be rapidly emulated. Contracting may actually be the most important source of new rules in a dynamic system of customary law. For instance, many innovations in commercial law have been initiated in contracts and dispersed quickly through the relevant business community.

Alternatively, as conditions change, the inadequacy of existing customary rules can be revealed when a dispute arises. Negotiation is probably the primary means of dispute resolution for members of a customary law community, reinforcing the contention that contracting is a primary mechanism for initiating rapid change in customary law. If direct negotiation fails, however, the parties to a dispute often turn to an arbitrator or a mediator. Since a dispute can suggest that existing rules are unclear or insufficient, new customary rules can be and often are initiated through third-party dispute resolution. Unlike public court precedent, such a decision only applies to the parties in the dispute, but if it effectively facilitates desirable interactions the implied behaviour can spread rapidly through the community, becoming a new rule.

No community evolves in complete isolation. Anthropological and historical evidence suggests that intra-group conflict has been an almost ubiquitous characteristic of human history, of course. Since a key function of customary-law communities is to establish and secure private property rights, and such rights are insecure if outsiders are able to “invade” and take the property, one joint product of a cooperative group is likely to be mutual defence. In fact, an external enemy can strengthen group cohesion, leading some to actually suggest that norms are important because they enable groups to be sufficiently united to deter their enemies, not because they allow people to create order (coordination) within their groups. Clearly, norms that support the production of mutual defence evolve, and an important part of an individual’s belief system will be a “communitarian” one (*e.g.*, tribalism and “a concept of them and us” where individuals are expected to aid in the defence of the “community”). In fact, however, these communitarian norms simply evolve along with intra-group norms of cooperation as part of the overall objective of creating an environment conducive to the pursuit of subjective well-being.

Polycentric Governance

Inter-group competition does not have to involve violence. Voluntary associations imply both the ability to voluntarily join a group, given acceptance by existing members, and to voluntarily withdraw, for instance, so inter-group movements are a distinct possibility. Indeed, individuals have incentives to “migrate” to the group which best facilitates pursuit of their objectives. Furthermore, individuals in groups that gain cooperative members enjoy more opportunities for mutually beneficial interaction so

they have incentives to compete to attract or hold membership (the basis of customary law groups may be geographic proximity, but it also may be kinship, functional proximity as in a trade association or the “business community”, religion, or any of a number of factors that create repeated dealings and/or reputation effects). As a result, members of customary legal systems have incentives to imitate desirable institutions and rules developed elsewhere.

Competition and emulation lead to standardization of many rules and institutions across similarly functioning groups, although differences may remain, reflecting preferences of various groups’ members. A group does not necessarily have to expand to expand opportunities for beneficial interaction. Indeed, if individuals want to interact, but only on some dimensions, or if they want to maintain different sets of rules for different dimensions of interaction, then parallel “localized” mutual support groups may be maintained while a “second order of clustering” is established, facilitating a relatively limited scope for interaction.

A group whose members insist on strictly imposing their own morality and penalties on outsiders would probably be unable to initiate beneficial inter-group interaction. Thus, if people wish to simultaneously facilitate inter-group interaction and impose laws that differ substantially from the norm in other groups, they have strong incentives to inform outsiders of the differences in order to avoid conflict and minimize the difficulty of maintaining non-standard laws. Part of the reciprocal agreements with other groups may be the explicit recognition of differences in laws and procedures for treating conflicts. This in turn implies that as inter-group interactions expand, a hierarchical jurisdictional arrangement may become necessary.

For example, each localized group may have jurisdiction over rules for and disputes between its members. Rules for inter-group interactions can differ from both groups’ internal rules, although they certainly do not have to, and disputes between members of a confederation of different groups are settled by some “higher” confederation level adjudication process. Note that these are not “higher courts” where disputes can be appealed from within-group dispute resolution mechanisms (*e.g.*, they are not a supreme authority). Rather, this is a jurisdictional hierarchy defining the role of each adjudication process and allowing for increasingly more distant interactions. This allows for differences between the law applied within groups and between groups; a monopoly in law is not required.

A judgement involving an inter-group dispute will have to be considered to be a fair one by members of both groups, of course. Thus, an equal number of individuals from each group might serve as an arbitration board, or a mutually acceptable third party (*i.e.*, an arbitrator or mediator with a reputation for good judgement) might be chosen. This provides another reason for the tendency toward standardization of rules across parallel groups with similar functions, at least for those functions carried out in the process of inter-group interactions. Some individual members of each group must recognize the potential benefits of inter-group interaction and be willing to bear

the cost of initiating institutional innovations. Furthermore, the resulting innovations must involve more than just dispute resolution, because such interaction faces an assurance problem.

Each individual must feel confident that someone from the other group will not be able to renege on a promise and then escape to the protection of that other group. After all, at least initially, repeated game and reputation affects are localized within each group, and there is limited potential for a boycott sanction. Thus, for second order clustering to develop, some sort of inter-group insurance or bonding arrangement becomes desirable, along with an apparatus for inter-group dispute resolution. For instance, as inter-group interaction develops the mutual support group can become a surety group as well. Membership in a group then serves as a signal of reputable behaviour, and if a member of a group cannot or will not pay off a debt to someone from the other group, the debtor's group will. The individual then owes his own group members so the boycott threat comes into play once again.

Limits to the extent of an inter-group network of cooperation are determined by the relative costs and benefits of information about other groups and their legal systems. The costs of establishing inter-group legal arrangement depend in part on how "distant" the groups are from one another, where distance can be in terms of geographic space, or in terms of the behavioural norms that are relevant to the groups. Thus, extensive interaction between starkly different groups may not arise.

However, these limits are stretched as individuals become members of several groups. After all, as Mises (1957: 257) explains, "Man is not the member of one group only and does not appear on the scene of human affairs solely in the role of a member of one definite group.... The conflict of groups is not a conflict between neatly integrated herds of men. It is a conflict between various concerns in the minds of individuals."

Thus, the relatively limited jurisdictions of some customary communities are not as constraining as they might appear to be. A person may simultaneously belong to many groups that have well established customs (and be subjects to the commands of several rule-making authorities, *e.g.*, as in a formal federalist system of government), so being in one community does not preclude dealing with people in all other communities. A person may belong to a trade association, a homeowners association, a religious group, a fraternal organization, and so on, for instance, each with its own rules and governance institutions. The membership of all of these communities can differ, although considerable overlap may also occur, so individuals may deal with other individuals on some dimensions but not on all dimensions.

Indeed, in any complex society, there are many distinguishable systems of rules and institutions, and yet people from many of these different systems interact regularly without having to call upon any legal authority. Thus, inter-group cooperation appears to be the norm rather than the exception, and it appears to be quite widespread. And with good reason: as Gluckman (1955: 20) suggests, "multiple membership of diverse

groups and in diverse relationships is... the basis of internal cohesion in any society.” An all inclusive legal system would eliminate the benefits of competition and emulation and undermine the incentives for innovation, as Berman (1983: 10) explains:

It is this plurality of jurisdictions and legal systems that makes the supremacy of law both necessary and possible.... The very complexity of a common legal order containing diverse legal systems contributes to legal sophistication. Which court has jurisdiction? Which law is applicable? How are legal differences to be reconciled?.... The pluralism of... law... has been, or once was, a source of development, or growth — legal growth as well as political and economic growth. It also has been, or once was, a source of freedom.

The competitive/cooperative relationship between consensual customary legal systems is driven by the desire to facilitate voluntary mutually-beneficial interactions rather than a desire for legal sovereignty. Thus, many different customary systems can co-exist and interact. An understanding of customary law requires that individuals and their organizations be the points of reference rather than “society” as a whole: “there may then be found utterly and radically different bodies of “law” prevailing among these small units, and generalization concerning what happens in ‘the’ family or in ‘this type of association’ made on the society’s level will have its dangers. The total picture of law-stuff in any society includes along with the Great Law-stuff of the Whole, the sublaw-stuff or bylaw-stuff of the lesser working units”.

Customary law can be geographically extensive and functionally decentralized (i.e., specialized), in contrast to the law of geographically defined states which tends to be functionally centralized and geographically constrained. Thus, customary law can have different sized jurisdictions for different functions. In some areas of law, economies may be considerably more limited than any state, for instance, so existing political entities are too large geographically or functionally. In other areas of law, such as international commerce, some of these economies appear to be greater in geographic scope than any existing nation can encompass, although many also are narrow in functional scope, as international trade associations may be the most efficient source of rules and governance for many groups of traders.

EVOLVING PROPERTY RIGHTS IN CYBERSPACE

The internet originated in the 1960s with ARPANet, a decentralized computer-based communications network set up by the United States Department of Defence and the National Science Foundation. It has developed into a global technological system of networked computer networks with an estimated 900 million users. In addition, the types of communication that flow over the internet have exploded as new applications have been developed (*e.g.*, Telnet, FTP, Gopher, LISTSERV, Usenet, EDI, E-mail, and so on). One of the most influential developments, the World Wide Web, is only 15 years old (it was created in 1990 by Tim Berners-Lee at CERN). Thus, many uses of the internet are relatively recent, and are clearly evolving. For instance,

Lucking-Riley and Spulber (2001: 55) report various estimates and projections of the magnitude of business-to-business (B2B) e-commerce. A 1999 estimate by the Gartner group put the level of B2B e-transactions at \$90 billion, while a 2000 estimate by Jupiter Communications concluded that these transactions amounted to \$336 billion (out of a total of \$11.5 trillion).

The accuracy of such estimates are suspect, of course, because e-commerce is a global phenomenon and much of it probably is not reported to any government authority. Nonetheless, they do reveal that B2B e-commerce is substantial, and probably growing. Business-to-consumer (B2C) trading is much more visible, and this component of e-commerce has attracted much more attention from governments and the media (and probably from academia).

Yet, the Gartner Group estimate of B2C e-transactions was \$16.7 billion in 1999, including both retail sales and brokerage fees for on-line financial transactions, compared to the \$90 billion in B2B transactions. Growth is also evident in B2C e-trading, of course, as Bakos (2001: 69) reports that such trade reached \$45 billion for 2000, for instance, and Tedeschi (2004a) cites a Shop.org report that estimated 2003 retail sales for U.S. retailers (not including sales of travel services) on line to be \$72 billion (5.4 percent of total U.S. retail receipts for the year).

Nonetheless, B2B e-commerce is likely to continue to be larger and less apparent than B2C e-commerce for institutional reasons. Before turning to such developments, and more importantly, their institutional underpinnings, however, cyberspace itself must be briefly described. After all, as Crews (2003: 1) notes, “the Internet wasn’t originally designed to be a mass commercial and consumer medium that it is today. If one were to design a commercial network today from the bottom up, it would probably” look very different. The technological configuration of the internet has raised the cost of some forms of institutional development, but it does not preclude all such innovations.

The internet is not simply a technological system. It also is an implicit agreement, or perhaps more accurately, “a loosely-organized international collaboration of autonomous, interconnected networks” (Internet Engineering Taskforce, RFC 2026) that allows bits to flow among computers using a particular language or “protocol.” In may be useful consider the result in terms of layers.

At the bottom there is a physical infrastructure (*e.g.*, cable, satellites, routers, DSL, WiFi). The next layer up consists of applications (*e.g.*, browsers, e-mail, VoIP), and at the top there is a content layer (text, music, speech, images). The bottom two layers separate mechanical transportation of bits from the protocol.

The protocol divides the bits into packages that can be reassembled after transport. Internet hosts (computers that store data that is included in the internet) have domain names that are translated into IP addresses. The TCP/IP protocol allows very heterogeneous infrastructure networks to transport data from one IP address to another. Routers are computers that link the component networks. They contain routing tables,

and they mechanically look up parts of the address in a data package in order to send the package on to another router that is closer to the final destination. Thus, the technical part of data transmission is very mechanical.

The “free access” characteristic of the internet applies to the two lower levels. For instance, the logical layer is allowed to run across all (or at least enough of) of the infrastructure layer to reach any IP address in the network of networks. That is, the installers of cables and wireless connections and routers are expected to mechanically and indiscriminately allow TCP/IP to work. Furthermore, the logical layer presumably does not discriminate between applications, so anyone can send any kind of information over the internet. The early tradition of free access to the internet also presumed that, like the infrastructure and the logical layers, the applications layer would not discriminate against particular kinds of content. Thus, free access also presumably extended to that layer in that anyone could send anything to everyone else. Importantly, however, the top two layers are not actually global at all.

They are developed and installed by individuals and organizations such as firms and governments, in order to use the lower layers. If developers of an application choose not to freely allow any and all content to be deposited into an address or addresses in their part of the network of networks, the internet can still function as a global communications system. Everyone is still free to send whatever content they want to send, but some individuals or organizations might refuse to accept delivery of some kinds of content. Therefore, even though the internet does require free access to the bottom two layers to function as a global communications system, given its current technological configuration, it does not require free access into all IP addresses by all types of applications of all types of content.

Externalities in Cyberspace and the Evolution of Property Rights

Recall that externalities arise when two or more parties want to use the same scarce resource for conflicting purposes. Externalities are an inevitable result of common access to the internet, a system comprised of scarce resources. Indeed, as more and more uses of this resource are discovered, and more and more people enter the commons, it is not at all surprising to find increasingly costly externality problems arising. People want to use this resource to lower their costs of buying and/or selling goods and services, to transmit and store data, to communicate with friends and colleagues, and for entertainment, but some people also get pleasure from causing harm to others. The net is sufficiently large to support tremendous numbers of activities, but crowding (*i.e.*, conflicting uses) is becoming an increasingly significant problem.

An obvious example is spam. The “bulk-mailing” of large numbers of e-mails advertising viagra and other prescription drugs, pornography sites, and the huge array of other goods and services, crowd the internet, slowing traffic, and more importantly at this point, they crowd e-mail inboxes, raising the cost to recipients who do not want to deal with such “junk mail.” As Stross (2005: 1) puts it, recipients of e-mail bear

the costs: "It is nominally free, of course but it arrives in polluted form. Cleaning out the stuff once it reaches our in-box, or our Internet service provider's, is irritating beyond words... This muck ... is a bane of modern life."

A December 2004 survey suggested that internet users were spending an average of 10 working days per year dealing with spam, and at least some industry analysts estimate that the 2005 cost of spam to business due to lost productivity and additional network maintenance costs will be around \$50 billion for the year. This is not surprising since spam accounts for an estimated 80 percent of all e-mail traffic. Spammers go beyond just filling in-boxes, and slowing internet traffic, however.

They also "commandeer personal computers as zombie spam transmitters," by using a virus to install programmes on other people's computers so they will unknowingly act as free relays while simultaneously making it more difficult for others to break through the "cloak of anonymity" to find out the true source of the spam. Some estimates suggest that 50 percent or more of the spam flowing through the internet is handled by "hijacked machines". As a consequence, "The overlapping and truly global networks of spam-friendly merchants, e-mail list resellers, virus-writers and bulk e-mailing services have made identifying targets for prosecution a daunting process. Merchants whose links actually appear in junk e-mail are often dozens of steps and numerous deals removed from the spammers, Mr. Jennings said, and providing culpability 'is just insanely difficult.'"

Most spam involves efforts to convince consumers to purchase various products or services. While spammers probably consider such bulk mailings to be legitimate business practice, as an effective and low cost form of advertising which clearly results in enough sales to cover their costs, they obviously do not consider the external costs they generate for other web users, and especially for recipients who do not want to receive the spam. After all, despite laws passed by various governments that make some of these activities illegal, they are not likely to be held liable for these costs, which, are becoming very high.

Not all spam consists of annoying advertising, however. Phishing was a hacker-coined term which originally referred to the act of stealing AOL accounts, but phishers are expanding their targets. June of 2004 saw 492 different mass e-mailings attempting to convince customers of Citibank that they should provide confidential financial information to them (*e.g.*, account numbers), along with 285 mass mailings to eBay users. Since then, similar mailings have falsely represented themselves as coming from many other financial institutions. Spam of this kind does more than simply crowd e-mail inboxes. It is an intentional effort to harm (impose costs on) others. Viruses and worms are similar in this regard.

A 2002 survey conducted by the FBI and the Computer Security Institute reports that virus attacks caused losses of \$49.97 million over a twelve month period during 2001 and 2002, and other surveys put the cost much higher, ranging up to \$12 billion. Estimates suggest that the "love bug" virus of 2000 cost the global economy \$2.62

billion alone, after all, and the Coordination Center of the Software Engineering Institute reports that attacks on business and government computers doubled in 2001 over the 2000 level, with an addition 26,829 attacks during the first quarter of 2002 (another doubling over the previous year).

Under reporting of such attacks is likely since many businesses and government agencies do not want to reveal their vulnerability. 2003 has been called “the year of the worm” by some computer experts. The year started with the Slammer worm infecting almost 75,000 servers in just ten minutes in January. Among the most spectacular of the infestations that followed was the Blaster worm which infected hundreds of thousands of computers, and Sobig. F, which, at one point, resulted in one out of every 17 e-mail messages being transmitted carrying a copy of Sobig.F. One estimate put the global cost to business of 2003’s computer viruses at \$55 billion.

Mydoom. A followed in January of 2004, however, and spread even faster than Sobig.F (at its peak, one in five e-mail messages contained a copy of Mydoom.A). The worldwide cost of these worms, for cleanup and lost productivity, clearly is many billions of dollars. The people who intentionally create such attacks obviously do not consider the tremendous costs they are imposing on others, or perhaps more accurately, they do not consider these consequences to be costs at all, since they apparently get personal pleasure from their actions. In fact, such attacks often have a specific target that the programmer apparently has a personal grudge against, but they impose tremendous costs of millions of innocent bystanders in the process. The Blaster worm was developed to attack Microsoft, for instance (it exploited flaws in Windows and attempted to bombard a Microsoft web site with data), while Mydoom.A reprogrammed computers to attack the Web site of another software firm, SCO.

Like viruses, Adware and Spyware are programmes that are actually installed on a person’s computer by someone else, but their purpose is not to disrupt or destroy. Those who deploy these programmes want users to continue their internet activities, as they are secretly monitored. These programmes create very real external costs, however. For one thing, they tend to overburden PC’s, making them operate slower as they respond to the processes prompted by hidden programmes. In addition, Adware monitors online activity in order to display pop-up ads to users who show interest in particular kinds of services or products. Large numbers of popup ads create considerable cost themselves, as they typically must be deleted before the user can continue whatever he or she is doing. The extent of the spread of Adware is difficult to determine, because users generally do not know that it has been installed. In all likelihood, however, such programmes are becoming ubiquitous. For instance, one Adware vendor, Claria, reportedly had 29 million users running its Adware products on their computers in 2004, up from 1.5 million in 2000. The company’s 2003 revenues from firms using popup ads were 90.5 million, which produced a profit of \$35 million.

Spyware is even more “parasitic.” These programmes are installed on computer hard drives by piggybacking on software programmes that people intentionally download,

or by being sent through security gaps in Web browsers when users visit certain Web sites. Spyware automatically records and discloses everything the person does on line. It can be used to learn passwords, as well as information facilitating identity theft, such as account and credit card numbers, social security numbers, and other information a user keys into his or her computer.

Indeed, identity theft has become an issue of considerable concern, whether the thefts are accomplished through low tech activities like phishing, or high tech activities such as distributing spyware and intentionally hacking into targeted databases containing information such as credit card and social security numbers. The Federal Trade Commission reports that an estimated 27.3 million Americans suffered the theft of their identities over the 5-year period from April 1998 to April 2003, and that over a third of these thefts occurred during the final 12 months of the period (O'Brien 2004: 2). A substantial portion of these identity thefts was perpetrated over the internet, and the costs of these thefts was tremendous.

The final 12 month period referred to above saw an estimated \$48 billion in losses for financial institutions and businesses as a result of these identity thefts, for instance, on top of the uncountable costs to the direct victims of the thefts (an estimated \$5 billion in out-of-pocket expenses were born by victims in order to reestablish their financial identities, but the psychological and time costs were probably much higher). Clearly, the internet has made property rights to many non-cyber assets (*e.g.*, assets other than IP addresses and computers) relatively insecure as well, because those assets (*e.g.*, individual credit and payment arrangements) are used in on-line transactions.

Intellectual Property Rights and Agreement

THE ORIGINS OF INTELLECTUAL PROPERTY RIGHTS

The public willingness to bestow private property rights to creative expressions, designs, and innovations can be traced back to ancient times. Copyright protection was first observed in the fifteenth century after the invention of the printing press. During the sixteenth and seventeenth centuries, it was common in most countries for the sovereign to grant exclusive publishing rights for a fee.

Unauthorized publications were prohibited because they reduced the sovereign's revenues. In 1710, the first statute recognizing the author as the primary recipient of copyright protection was passed in England. Subsequent court decisions in England confirmed the rights of authors to exclude others from publishing their works without permission as a common law right, rather than a statutory right.

Many civil law countries during the eighteenth and nineteenth centuries began enacting statutes, such as the 1793 French statute protecting the rights of authors. In the United States, the Constitution contains a provision calling for federal copyright protection. Trademarks evolved from the early practices of marking property to indicate ownership. The practice of marking property appeared during the fifth and sixth centuries B.C. where Greek vases bore the marking of the potter. Modern trademark protection started in the trade guilds where craftsmen were required to place production marks on

their goods. While an individual's mark on the goods served for identification purposes, the concept of trademark and unfair competition evolved from the premise that it was unfair to present one's goods as the goods of another.

Systematic legal protection of trademarks began to evolve in the early nineteenth century. Courts in both the United States and England created the tort action of "passing off," where "the defendant had used plaintiff's mark to deceive consumers into thinking that plaintiff was the source of defendant's goods." Eventually, courts began categorizing the stealing of another's mark as trademark infringement, rather than "passing off" or unfair competition cases. Today, the mark not only can be placed on goods, but also can serve to identify services, associations, and certifications. Very little is known about the development of trade secrets because of the secrecy of the facts surrounding them. The legal protection of trade secrets can be traced back to early Roman times. For instance, a master could bring a legal action and obtain a monetary remedy against a third party who enticed the master's slave to reveal the master's secrets.

During the Middle Ages in Europe, the craft guilds successfully provided protection for craft secrets by keeping the secrets of the method of production within the guilds. In nineteenth century Britain, protection of trade secrets was provided through the apprenticeship system, where tricks of the trade were passed from the principal to the apprentice. In Asia, family businesses and the sense of lifetime employment with a particular employer minimized the loss of secrets via employee turnover. Today, international competition and the high cost of technological development have intensified the importance of trade secret protection.

During the Renaissance, the Italian states were among the first Western governments to grant patents. For example, during the fifteenth century, the Venetians codified a general patent statute in order to spur the introduction of new technologies. The practice of granting patent rights eventually spread throughout Europe in the centuries that followed, where often royalty had the power to grant patents.

In the seventeenth century, the English government adopted the Venetian model, whereby the English government, not the royalty, granted inventors limited monopolies on their inventions and ideas. This English model became known as the Statute of Monopolies. In the late eighteenth century, both France and the United States enacted comprehensive patent laws with the understanding that patent rights were individual rights, rather than a royal prerogative. By the late nineteenth century, many countries around the world, including Japan, had adopted comprehensive patent statutes. However, many former colonies, such as those of the British Empire, adopted less comprehensive patent systems.

IPR PROTECTION SCORE ANALYSIS

Correlations between the IPR protection scores and Rapp and Rozek's patent scale scores, Sherwood's IPR scores and Seyoum's three new IPR measures were calculated for the countries in the IPR protection data set. The results show that Rapp and Rozek's

scale score correlates inconsistently across the three types of IP laws and protection. The highest correlations are with patent laws and protection, which correlate at .56 and higher. The lowest correlations were with trademark laws and copyright protection. While patents are moderately correlated with Rapp and Rozek, the other two forms of IPR protection do not correlate as highly. The copyright and trademark variables' low correlation with Rapp and Rozek's patent variable is further indication of the theoretical importance of treating intellectual property law components separately. One can not be used as a proxy for the other because nations do not assign equal importance and attention to them.

Seyoum's measures reflected moderate positive correlation with the IPR protection scores, with the highest correlations in the area of patent protection and the lowest correlations in trademark protection. There was no significant correlation between any IPR protection scores and Sherwood's measures, which may be due in part to the low number of observations available in Sherwood's data set. The correlations among the other variables were also mixed. The strongest correlations were between Seyoum's patent measure and Rapp and Rozek's measure (.883), indicating a close relationship between the two. There was moderate correlation between Seyoum's measures for trademarks and copyrights and Rapp and Rozek's patent measure. Sherwood's measure had a low level of correlation with Rapp and Rozek's measure.

However, the low and insignificant correlations of Sherwood's measure with Seyoum's measure cannot be assigned much importance given the low number of cases the two data sets had in common. Additionally, the low correlations among the IPR protection scores and the other scores may be a product of the differences in the methodology used to create these data. The enforcement component in the IPR protection score has the capacity to enhance or to detract from the strength of the laws.

It is possible that nations may have weak laws but strong enforcement of those laws or strong laws but weak enforcement of them. Measures that do not capture this distinction will correlate modestly with measures that do. This stage has attempted to accomplish two goals. The first was to call attention to the need for additional quantitative research in the area of IPR protection. Given the scarcity of such research, more will be needed before we can begin to address how IPR protection affects processes such as investment and capital movement decisions. The second goal was to contribute to the literature on measuring IPR laws. The importance of an enforcement component in assessing the strength of IPR protection in countries should not be overlooked. Measures that do not take enforcement into account may unintentionally overestimate the protection afforded IPR in a particular country.

While the approach taken here is an improvement, additional information on enforcement practices from government sources and private foundations will continue to improve these measures. Two veins of work in the measurement of IP protection are also needed in light of the new TRIPS agreement. Future research in such measures should be directed at measuring how nations are complying with the agreement. Research

in this direction becomes important because when nations come into compliance with the agreement, the distinguishing factor, theoretically, should be in how nations are enforcing those laws. Additionally, research into the nature of the enforcement component is needed. Enforcement of IPR laws is a two dimensional concept that involves the institutional capacity together with the institutional will to enforce laws. The first element addresses the actual institutions needed to enforce the laws (*i.e.* the statutes, the nature of the judiciary, technical expertise, and policing organizations).

The second addresses whether those institutions actually carry out the enforcement of the laws. The second dimension is a behavioural component as opposed to an institutional and resources component. The institutional dimension is a necessary condition for enforcement, but it alone is not sufficient to constitute enforcement; the institutional will to do so is needed.

Research in this direction will contribute much to the discussion concerning the role of IPR protection in business and national development. The intellectual property regime is part of society's innovation system, and its intent is to provide incentives to innovate by allowing innovators to restrict the use of the knowledge they produce by allowing the imposition of charges on the use of that knowledge, thereby obtaining a return on their investment. But it is important when thinking about intellectual property rights (IPR)—which include patents, copyrights, and various other parts of the intellectual property system—to realise that there are many other parts of society's innovation system. There are other ways of financing and producing research—for instance, through universities and government-supported research labs.

In fact, would argue, perhaps immodestly, that the most important ideas are those that are generated in universities, and many of the most important intellectual advances are not covered at all by the patent system. Look at the basic idea underlying the computer, Alan Turing's "Turing Machine": it was not protected by the patent system. Ideas like asymmetric information are not covered by intellectual property.

Another example of important innovations not driven by IPR is the open source movement, which has been particularly successful in software. Even when research is financed by firms, there are other ways of providing returns on knowledge instead of using patents, such as trade secrets and advantages that come naturally to the first entrants in a market economy. There are also other ways of providing incentives; one of them will discuss is a prize system, which has actually been a part of the innovation system for a couple hundred years. Will argue that, from a societal perspective, prizes have marked advantages over the patent system.

Our innovation system rests on foundations of basic research, and most basic research occurs within academia and government-sponsored research laboratories. Monetary returns are only a small part of what motivates these researchers. Thus, the basic framework of what induces people to engage in research is really not reflected in the intellectual property regime. Obviously, research has to be financed. It takes resources, so the question is not just how we motivate research but also how we finance it. Financing

research through monopoly profits may be neither the most efficient nor the most equitable way of doing so. The key issue is the role of the patent system, or the intellectual property regime more broadly, within the economy's innovation system. To address the next issue, the design of the patent or IPR system more generally, one has to understand the limited (though still important) role of the IPR regime to address the next issue: the design of the patent or IPR regime.

There are a host of questions, many of which absorb much of the energy of the lawyers involved in obtaining and fighting patents. These issues include what can be patented, the breadth of the patent, the standard of novelty, the procedures for granting and challenging patents, the rules for patent enforcement, and the notion of responsibilities as well as rights. For instance, to get a patent you have to disclose enough information that somebody could replicate what is being patented (though often firms try to get away with disclosing as little as possible). Disclosure has long been an important part of the patent and intellectual property regime; it is one of the reasons why IPR can enhance innovation: people can build on that knowledge.

Knowledge is the most important input into knowledge. Interestingly, in some of the more recent intellectual property disputes, the notion of disclosure has been contested. Microsoft has, by most accounts, not wanted to disclose even its protocols (required for interoperability with applications). The European Union has insisted that it at least disclose specifications for its protocols and Microsoft has tried everything not to comply with the European Union's requirement, even to the point of risking millions of dollars in fines. Furthermore, as with any property right, there are restrictions on the use of intellectual property.

The fact that you have a property right does not mean you can do anything that you want with it. Owning a bat does not give you the right to hit someone over the head with it. In the case of intellectual property, one of the restrictions is that you cannot engage in abusive, anticompetitive behaviour. The rights of governments to issue compulsory licenses form another important part of the patent regime. One of the responses to the abusive, anticompetitive practices has been to restrict the use of patents, effectively insisting on compulsory licensing, sometimes through forming patent pools.

In the consent decree in the case of the antitrust action against AT&T in the 1950s, AT&T had to make its patents available to anybody that wanted to use them. In my Tunney filing in the Microsoft case, we argued that one way of dealing with that company's anticompetitive behaviour was to limit its intellectual property rights. Such restrictions would, suggested, enhance innovation in our economy. More generally, the design of the patent system can affect the efficiency of the economy and its innovativeness.

The current patent system imposes large costs on the economy, and one of the questions is whether there are reforms that could increase the economy's efficiency and innovativeness. Believe there are. To understand these issues one has to look at the economic foundations of knowledge.

STATIC INEFFICIENCY

The important basic idea is that knowledge is a public good. Economists use the concept of a public good as a technical term; Paul Samuelson defined it precisely more than fifty years ago. A public good is a good whose consumption is non-rivalrous. By contrast, private goods can only be consumed by one person. For instance, only one person can sit in a chair. The same goes for food: if eat the hamburger, you cannot eat it. Knowledge, however, is different. We have just shared with you some of the things we know, but sharing this knowledge with you has not taken away from what we know. Thus, knowledge has the quality of non-rivalrous consumption.

Another way of putting it is that there is no marginal cost associated with the use of knowledge. Thomas Jefferson described this much more poetically. Jefferson said that knowledge was like a candle: when one candle lights another it does not diminish the light of the first candle. Understanding this concept is at the core of understanding efficiency in the use of knowledge. It is more efficient to distribute knowledge freely to everybody than to restrict its use by charging for it. Free distribution, however, could cause problems for creating incentives for the production of innovation, and that is the dynamic issue. Before turning to the question of dynamic efficiency, though, we want to emphasize that efficiency in use means knowledge should be freely available.

The problem is that intellectual property rights circumscribe the use of knowledge and thus, almost necessarily, cause inefficiency. Not only does intellectual property create a distortion by restricting the use of knowledge, but it also does something even worse: it creates monopoly power. Monopoly leads not just to inequities but also to major distortions of resource allocations; limiting monopoly power and its abuses is the focus of anti-trust policy. There is a quandary. We not only tolerate this distortion and inefficiency by restricting the use of knowledge, which creates monopoly power, but we sanction it: it is part of our legal framework because we hope it will promote innovation.

If we do not design this legal framework properly, however, it may actually impede innovation. That is where as suggested, eventually take this argument. Before getting there, though, want to point out that the social cost of the distortion of the patent system is particularly high. (By contrast, Schumpeter suggested that it would be lower than for other forms of monopolization.)

- *Access to Medicine*: One of the reasons that the costs are so high is that the patent system impedes access to lifesaving drugs for billions. Opposed TRIPS (the so-called Agreement on Trade-Related Aspects of Intellectual Property Rights, part of the Uruguay WTO agreement signed in 1994) so strongly in part because of these effects. Indeed, believe one of the main reasons the pharmaceutical industry was pushing for TRIPS was that they wanted to reduce access to generic medicines. These are so disliked by the drug companies for the same reason that they are so liked by everybody else: the prices of generic drugs are very low. The low price means that people who could not afford the

brand name drugs still can buy the generic drugs, and the competition with the generics drives down the price of the brand name drugs. The loss of sales to generics as well as the lower prices in turn lowers the profits of the large (brand name) pharmaceutical companies; it is understandable why they have pushed so hard (and contributed so much to campaigns) for IPR.

One example that shows the magnitude of what is at stake (and that has received a lot of attention), are the AIDS drugs. One year's treatment of the brand name drugs, not the most recent ones, but the older ones which are less expensive, costs \$10,000. In a poor developing country, where the per capita income is \$300, or even \$3,000, a person with AIDS is not going to be able to afford \$10,000.

They might be able to afford the generic medicines, which sell for less than \$200. When the trade ministers signed the TRIPS agreement in Marrakesh in the spring of 1994, they were in effect signing the death warrants on thousands of people in sub-Saharan Africa and elsewhere in the developing countries. This is one of the reasons that TRIPS has generated such immense concern.

- *Leveraging Monopoly Power and Other Abuses of IPR:* The efficiency costs of IPR go beyond the direct effects associated with monopolization of the particular product covered by the patent: Intellectual Property Rights generate monopoly power that can be used to leverage further monopoly power.

The most obvious example is Microsoft, which has leveraged its monopoly power in operating systems to obtain a dominant position in applications like word processing (Microsoft Office) and Internet browsers (Internet Explorer). The courts and regulators in the United States, in the European Union, and in South Korea have all ruled against Microsoft. There is little disagreement about the fact that Microsoft has engaged in abusive, anticompetitive practices. The only debate is what to do about it; because Microsoft has so much monopoly power and has obtained such a dominant position, it is not easy to figure out how to deal with the problem.

There is actually a long history of abusive uses of intellectual property rights. Going back to the beginning of the last century, the automobile and the airplane were two of the most important innovations, and the development and success of both were impeded by IPR. In the case of the automobile, a patent was granted, but it was an excessively generic patent for a four-wheeled, self-propelled vehicle.

The person who received the patent had no intention of developing the automobile; instead, he used it to form a cartel. Often, the best, or at least the easiest, way of making money is not to come up with a better idea but to form a monopoly or a cartel and restrict competition. For those seeking easy profits, competitive marketplaces are very bad because they drive down prices and erase profits. IPR can be an important way of maintaining a cartel.

Most of the automobile companies went along with the idea of the patent-driven cartel, except for one. There was one innovator who had a different conception of what the automobile was about, and that was Henry Ford. He had the idea of a people's car,

at a low price; that idea was totally inconsistent with the cartel's view of a high-priced, restricted-use vehicle. Ford challenged the patent. Had he not had the resources and the determination, he would not have succeeded in this legal battle; however, he eventually did prevail. Had he failed, the development of the automobile as we know it would have been delayed for a long time.

The automobile patent illustrates two other problems with the patent system: The first is the difficulty of determining the "boundaries" of property rights. In the case of "real" property (land), we can define the property right precisely by the geographical boundaries. We may have to further specify whether the property right extends to mineral rights and air rights, and whether there are public rights of way. But the limitations are easy to specify and are well understood. In the case of intellectual property, there are no such natural borders. Should the intellectual property right have extended to all self-propelled vehicles, or only to those that are propelled in a particular way?

These boundary issues are related to another issue: it is widely agreed that if patents are to promote innovation, they should be given for, and only for, an idea that is "novel." If the original patent is overly broad, and encompasses all four-wheeled, self-propelled vehicles, then a truly novel way of self-propelling may not be granted a patent or may have to pay the owner of the original patent so much as to attenuate incentives for innovation.

Drug companies and others have tried to extend the effective life of a patent by making incremental improvements on existing drugs. In a recent case, Indian courts ruled against Novartis' attempt to do so. Whether a product is novel depends on the state of knowledge throughout the world. It is often difficult for the patent office to determine whether someone in the rest of the world might have had the idea before this patent was granted; in the case of the automobile, it is arguable that others in Europe had the idea before the granting of the 1895 patent.

The United States has often been somewhat provincial in its approach to knowledge within the patent system. For instance, the United States gave a patent for basmati rice. Indians had been consuming basmati rice for a thousand or more years, but those in the patent office in Washington responsible for reviewing the patent application obviously had not had the pleasure of eating basmati rice. They thought it was a wonderful innovation and granted a patent to it. Had India implemented and enforced this patent, every time anybody in India ate basmati rice they would have to send a check to Texas to pay for this idea which they thought was theirs in the first place.

Ordinarily, property rights are argued for as a means of achieving economic efficiency; intellectual property rights, by contrast, result in a static inefficiency which can only be justified by the dynamic incentives. These examples suggest that the static inefficiencies may be greater than is often thought. Later, will argue that the dynamic benefits may be less. Of course, any method of raising funds for innovation has a social cost. In the case of a monopoly, the way you raise funds is through the disparity between the price and the marginal cost. The patent system, however, is not an optimal way of raising money

because it is not an optimal tax; it creates a particularly large set of distortions, which is one of the reasons why it is particularly objectionable. Later in this talk, as suggested, discuss the issue of financing research more broadly.

- *Schumpeterian Competition*: Standard competitive equilibrium theory has paid very little attention to innovation. The only rigorous proof of the efficiency of competitive markets is provided by the Arrow-Debreu model, and that model assumes that technology is fixed. One might think this is strange; how could economic theory pay any attention to models that assume technology is fixed in a dynamic economy?

That is a question that sociologists ought to address, but the Arrow-Debreu competitive model is the standard, reigning paradigm, and sadly, it ignores innovation. Most first-year graduate courses in economics, and most introductory textbooks (until my textbook came out), simply did not talk about innovation in any systematic way. But there was a strand of thought associated with Joseph Schumpeter that focused on innovation and argued that this competition for innovation resulted in temporary monopolies. One monopoly followed after another; new firms tried to displace the existing monopolist. In this sense, there could be intense competition. This kind of competition was referred to as Schumpeterian competition.

Some of my earlier research, however, pointed out that Schumpeter was wrong about the temporary nature of monopoly. Monopoly power, once established, can easily be perpetuated. Not only is it possible to perpetuate monopoly power, in fact, there is an incentive to do so. This is particularly evident in the case of network externalities and in situations where there are important switching costs, such as those that arise from learning. That is one of the reasons why Microsoft is so difficult to deal with. The monopoly power persists even after the bad practices which enabled it to create that power have been terminated.

- *Further Costs to the Patent System*: The transactions costs associated with our IPR system are a further problem, although whether you view them as costs or benefits may depend on your position in the economy. For lawyers, transaction costs are a benefit, because they are a source of their income. But from a social perspective, these administrative costs are a social cost. Litigation over patents also introduces uncertainty. Of course, there is always an intrinsic uncertainty of research, but it is compounded by the risk of patent infringement and the associated risk of litigation.

DURATION OF INTELLECTUAL PROPERTY RIGHTS

The duration of a patent term within TRIPs has been standardized to a minimum of twenty years from the date of filing; copyrights are standardized at fifty years on sound recordings and at least fifty years on motion pictures and other works. Trademark

protection is established for seven years and is indefinitely renewable as long as the trademark is in use. TRIPs requires contracting states to expedite administrative procedures for the granting of patents as much as possible to avoid a *de facto* reduction in the effective life of a patent. To the extent that this provision proves effective, it will be a boon for U.S. businesses, which have continually been frustrated by the long backlogs in granting intellectual property rights in developing countries.

SCOPE OF INTELLECTUAL PROPERTY RIGHTS

Throughout the developing world, the list of products declared ineligible for intellectual property protection is long, and it includes software, pharmaceuticals, and the products of biotechnology. TRIPs expands the existing scope of protection for trade secrets, computer programmes, and databases. In the case of pharmaceuticals, although TRIPs extends the scope of what is patentable, it excludes natural or artificially produced biological material, animals, and plant varieties. The exclusions clearly represent a blow to the U.S. biotechnology industry, in which the United States has a distinct comparative advantage. TRIPs also confers on patentees the following exclusive rights: the exclusion of third parties from the acts of producing, distributing, selling, or importing a patented product or process; full protection of the product obtained through a patented process; and the flexibility to sell or license patent rights. Although protection for information-intensive products is not airtight, TRIPs increased the previous level of protection.

LICENSING

This is an area where international disputes are common. TRIPs has failed to resolve this conflict, because it allows nations to grant compulsory licenses for “adequate remuneration” after considering each case “on its individual merits” and after attempting to negotiate “reasonable commercial terms” with the holder of the intellectual property right. TRIPs specifies that compulsory licenses must be non-exclusive and of limited duration, but then it puts the fox in charge of the henhouse by mandating that appeals of compulsory licenses be subject to the jurisdiction of the nation imposing the license. The result is that developing countries can continue to require compulsory licenses.

TRANSITION PERIOD FOR LESS DEVELOPED COUNTRIES

TRIPs allows a grace or transition period of ten years for developing countries to extend intellectual property rights to those industries not previously subject to intellectual property protection. The TRIPs provisions do not require “pipeline” or retroactive protection for patents that are covered under the new proposed laws but were not protected under the old laws. Both the grace period and the lack of pipeline protection are particularly troubling issues for U.S. investors in the pharmaceutical sectors of

developing countries in which an industry exists that produces pirated pharmaceuticals and where displacement of workers and capital is an economic and social concern.

DISPUTE SETTLEMENT MECHANISMS

Violations of the TRIPs Agreement can now be enforced through arbitration panels. Moreover, consequential international penalties can usually be applied to countries that infringe patent and copyright provisions. Within this framework, the piracy of intellectual property rights abroad can be penalized in accordance with the WTO-TRIPs rules. As a way to penalize the infringement of U.S.-held intellectual property rights, the United States may, after a favourable arbitration ruling, totally or partially bar U.S. exports to countries found in violation of TRIPs provisions. As a result, developing countries could no longer simply disregard international complaints. This WTO legal framework is expected to enhance the enforcement of patent and copyright laws throughout the developing world.

The TRIPs Agreement attempts to cure the problems occurring in international market transactions of high-technology goods and services. It also benefits developing countries that have had limited access to high technologies because of their lack of technological capabilities. By providing clear and enforceable rules plus dispute-settlement mechanisms, the TRIPs Agreement promotes the institutional capacity of many developing nations to absorb complex technologies. In short, the principles enunciated in the TRIPs Agreement are an example of international legal convergence.

His stage poses three questions:

- How does India's existing legislation differ from the TRIPs Agreement, and what changes must India make to its intellectual property laws to be in compliance with TRIPs?
- Why has India actually *weakened* its intellectual property regime since the 1970s?
- How will TRIPs-mandated changes in patent legislation affect the Indian pharmaceutical industry? He argues that India's Drug Price Control Order, "more than patents, is responsible for the low prices of drugs and pharmaceuticals, and that any increase in prices has occurred, and will continue to occur, mainly because of deregulation and decontrol in the prices of drugs." He concludes that the Indian pharmaceutical industry should embrace stronger patent protection as set out in TRIPs.

International treaties notwithstanding, the legal international convergence towards U.S. standards of protection can be extremely slow. The process of intellectual property reform must be carried out by the developing countries themselves.

TRIPs contains a provision granting developing countries a ten-year transition period in which to come into compliance with increased protection for intellectual property rights. As a result, the challenge facing the developed world is to persuade developing countries to implement the obligations of TRIPs as quickly as possible. Responses to

pressures to enact intellectual property law harmonization can be accelerated by incentives operating on developing-country governments through U.S. foreign policy channels. The speed with which developing countries adopt standards of intellectual property protection in compliance with the TRIPs Agreement, however, will depend largely on local domestic conditions and political pressures. Intellectual property protection in Latin America, Edgardo Buscaglia analyses the play of some of those domestic conditions and political pressures in Latin America.

“Because much intellectual property is produced only after considerable financial investment, the actual, perceived, and expected losses of U.S. firms that are attributable to inadequate intellectual property protection influence the willingness of firms to transfer technology” to the developing world. From the point of view of developing-country leaders, however, the feasibility of reforming intellectual property rights regimes sooner rather than later depends on the leaders’ assessment of costs and benefits.

Raising the standards of intellectual property protection presents long-term benefits but short-term costs. Buscaglia argues that politicians in developing countries see the enforcement of intellectual property rights as hampering their chances for reelection and, among other things, reducing campaign contributions made by lobbies representing local pirate industries. Furthermore, the benefits of defining and enforcing intellectual property rights are more distant and less tangible. Buscaglia concludes that when governments are unstable and when politicians do not expect to gain the longterm benefits of legal reforms, politicians do not have the incentive to make those reforms.

With the globalization of the world economy, developing nations are finding that maintaining competitiveness is a critical factor in development. Intellectual property protection has increasingly become a factor in attracting foreign and domestic investment and in maximizing a country’s economic growth potential. As policymakers in developing countries struggle to attract investment and world-class technologies to their shores, they are beginning to pay more attention to their standards of intellectual property protection, realizing, in the words of Pogo, that “we have seen the enemy and it is us.”

Global governance in intellectual property (IP) has changed dramatically since the 1980s. What was once principally an instrument of national policy, where countries could tailor their IP regimes in accordance with national economic and social conditions, is now increasingly a policy area subject to international disciplines, where countries that fail to conform to new global standards risk retaliation in the form of losing trade privileges.

The centerpiece of the new international regime is the inclusion of the Agreement on Trade Related Aspects of Intellectual Property Rights (TRIPs) in the new World Trade Organization (WTO). Is TRIPs a “friend or a foe” of international development? Observers from across the ideological and methodological spectrum tend to criticize TRIPs, assail its implications for development, and regret the inclusion of IP in the WTO. The World Bank and the United Nations Development Programme (organizations that are not frequently in agreement on development issues) express concern that full

implementation of TRIPs will generate significant resource transfers from poorer countries that primarily use IP to wealthier countries where most IP owners are based.

Development economists argue that TRIPs encourages poorer and middle-income countries to adopt IP systems that are inappropriate for their levels of scientific and technological development. Liberal trade economists complain that TRIPs introduces protectionism into an institution geared towards dismantling barriers to the flow of goods and services, and those contentious debates over IP clutter the multilateral negotiating agenda and the WTO's dispute settlement system. That scholars who disagree on so much agree on this issue suggests that trade and TRIPs and development are indeed foes, and it is difficult to argue otherwise. We explore the possibilities that TRIPs, for all its warts, may become a positive asset for the WTO, which in turn could benefit developing countries. We do so by emphasizing how conflicts over IP and TRIPs can have system-strengthening effects.

Global conflicts over IP that occur outside the WTO can spark countries' enthusiasm for improving the multilateral trading regime. And rather than muddle the agenda and divert attention, difficulties with TRIPs can help developing and developed countries gain sharper focus on the workings of the WTO and potentially contribute to measures to improve the procedural mechanisms of the institution.

Before discussing the contemporary politics of IP and the possibilities of system strengthening, a brief remark on the IP-trade relationship and the origins of TRIPs is in order. IP may be "trade related" in the sense that changes in trade flows correspond to changes in how countries regulate IP, but the decision to embed IP in the international trade regime is separate and distinct from the technical, economic rationale for or against doing so.

That is, it was the intense activism and immense power of the biotech, chemical, pharmaceutical, seed, software, and entertainment industries in the 1980s and 1990s that created the link between IP and trade, and thus made IP "trade related." These sectors shared a common interest in fortifying the protection of intellectual property rights (IPRs) worldwide, because they depend for their profits on the economic rents that accrue from copyright and patent protection, and they had the political muscle to place IP prominently on the international trade agenda.

Hence, IPRs are not part of the trade regime for the sake of trade (*i.e.*, because weak IPRs were a fetter on global commerce), but, to the contrary, IPRs are part of the trade regime for the sake of IPRs. Making countries' ability to access export markets conditional on changing national practices in IPRs was regarded by key business constituents and policymakers as an effective mechanism for ratcheting up standards across the board. Indeed, the name of the subsequent agreement is misleading, for TRIPs does not limit itself to "trade related" aspects of countries' IP regimes, such as how state regulations affect importing or exporting firms, but rather is a broad-based agreement that affects virtually all aspects of how countries go about establishing and protecting IP.

More than a decade after the introduction of TRIPs, the governance of IP remains one of the most charged issues on the international agenda. TRIPs is regarded as unsatisfactory by most developing and developed countries. Many developing countries resent and resist the imposition of new standards for the establishment and protection of IP that go far beyond what countries have typically offered at similar stages of development. They seek to retain the ability to use IP management as a tool for achieving broader development goals—for example, by relaxing standards of IP protection in strategic sectors to facilitate access to foreign technology or to lower the prices of technologically intensive consumer goods such as essential medicines.

Most developed countries, in contrast, regard the standards for IP protection established by TRIPs as too weak and too easily circumvented. They seek to harmonize IP protection at a higher level and thereby guarantee high returns for their own innovative firms that benefit from IP. The differences in the perspectives of “developed” versus “developing” countries are most accentuated in the realm of patents. In contrast to other realms of IP, such as copyright, where many developing countries are not just users but also generators of IP, patterns of global patenting activities demonstrate extraordinary asymmetry. During the period 1997-2004, for example, firms and organizations from ten developed countries accounted for more than 90 per cent of all patents granted by the United States Patent and Trademark Office (USPTO), with the United States, Japan, and Germany alone accounting for nearly 80 per cent.

In contrast, firms and organizations from the top ten developing and transition economies accounted for less than 7 per cent of the patents granted, with greater than 5 per cent coming from Taiwan and South Korea. Another way of capturing the asymmetry is to measure patent applications cross-nationally in terms of applicant’s residency. Data from the World Bank indicate that developing countries are generally importers and users of foreign knowledge: more than 97 per cent of patent applications in middle-income countries come from abroad, while in low-income countries, foreign applications account for all but one-fifth of 1 per cent of the total.

The contemporary “North-South” politics of IP are defined by these different perspectives and interests. While developing countries have fought to retain and consolidate the flexibilities they have under TRIPs, developed countries have pushed for heightened IP protection through regional and bilateral trade agreements (RBTA) that offer market access above and beyond what is available in the WTO in exchange for IP practices that are above and beyond what is required under TRIPs. Indeed, the contrast between TRIPs and the IP provisions in most RBTA is striking. Consider the case of patents, which confer limited monopoly rights to the owners of new, non-obvious, and industrially useful ideas.

The patentees’ rights are limited in three ways:

1. The rights are non-automatic, in that the owner must apply for a patent and the state must formally grant private ownership rights;
2. The rights are non-absolute, in that the third parties have some automatic

rights of use and the state has rights to regulate how patentees use their private rights; and

3. The rights are temporally bounded, in that when patents expire, what is treated as private property enters into the public domain.

The important point to underscore here is that international agreements on IP set standards for how countries set the boundaries of these limitations—how easy or difficult it is to obtain a patent, the extent to which the patentee can exclude others from freely using the idea and operate independently of state regulations, and how long the exclusive rights last.

And if one compares the WTO with RBTAs, it is evident that the IP provisions in the latter place significantly tighter restrictions on how countries establish these boundaries: establishing private rights becomes more automatic (states have limited ability to declare certain types of inventions “non-patentable”), the private rights are more absolute (third-party use and the state’s regulatory discretion are both significantly diminished), and the private rights are longer (patent terms are more easily renewable and extensions are at times automatic).

Does the cloud have a silver lining? While the IP-trade and IP-development relationships are both problematic, perhaps the three-way relationship between TRIPs, WTO, and development can be made more positive. Consider three ways that conflicts over IP can strengthen and improve the WTO and thus make it work better for developing countries: reinvigourating national commitments to the multilateral trading system, clarifying international trade rules, and inspiring procedural reform to make the WTO more user friendly.

REINVIGOURATING MULTILATERALISM

The fact that IPRs have become part and parcel of regional and bilateral trade agreements increases the interest of many countries in the multilateral setting. Although developing countries have reason to be alarmed by the IP obligations entailed in RBTAs, such agreements have continued to spread like wildfire.

Once one country gains preferential access to the US market, for example, its neighbours (and at times competitors for export shares and foreign investment) feel compelled to do the same, and so on. And this process is fueled by the United States, which actively plays developing countries off each other and induces this competition. Bit by bit, more and more countries find themselves agreeing to even more restrictive and problematic standards for granting and protecting IP. The best way to dampen the spread of RBTAs is to preserve and strengthen the WTO, because a WTO that works can decrease the attractiveness of bilateral and regional accords and give developing countries reliable alternatives to the United States.

For example, expanding market access in the WTO, by increasing the sectoral scope of trade liberalization and reducing most-favoured-nation tariffs, dilutes the preferential access that comes with RBTAs and thus will lead to diminished enthusiasm for such

agreements. But developing countries will not get better market access for nothing. Thus, to the extent they have an interest in addressing the underlying dynamics driving the proliferation of RBTAs, they have an interest in serious bargaining and negotiations at the multilateral level. That means building coalitions, actively participating in negotiations in multiple arenas, brokering compromises among themselves and with developed countries, and making concessions. Developing countries will not be able to dictate subsequent agreements, of course, but the realization that an unstable WTO contributes to the proliferation of RBTAs and to an even more threatening global IP environment could focus energies on making the WTO work better.

THE TRIPS AGREEMENT ON INTERNATIONAL TRADE

International trade opens up a country's market and stimulates economic growth. While intellectual property protection can influence trade flows, the net impact of higher levels of intellectual property protection on international trade remains uncertain. A study conducted by Keith E. Maskus and Mohan Penubarti, concluded that a higher level of intellectual property protection can have different effects on the imports of otherwise identical countries.

First, even though increased intellectual property protection can increase the holder's market power on his or her intellectual property fights, this results in a decrease in demand for the protected goods because countries are unwilling to pay the royalty. Thus, there is a decrease in trade. Second, greater intellectual property protection could possibly increase demand for the protected product because people are unable to get the product on the black market. Depending on which scenario occurs, the trade results will differ. The net trade results will depend on which effect dominates.

For example, "if the market-power effect is more substantial than the market-expansion effect, trade flows may decrease in the aftermath of the reform. If the opposite occurs, strengthened IPR protection will lead to trade expansion." Thus, the final answer regarding whether a higher level of protection for intellectual property fights diminishes international trade will be based on the particular situation and the figures involved.

The winners under the TRIPs Agreement are clearly the firms in developed countries that invest heavily in research and development and export their technology and goods to other countries. Firms exporting technology abroad now have an enforceable legal remedy against foreign infringers of their intellectual property rights. In particular, pharmaceutical and agricultural chemical industries are the strongest winners because of the "supernatural" property rights they receive under the TRIPS agreement.

Indirectly, developed countries benefit through the increased financial and welfare gain of their own citizens and firms. At first glance, it appears that the biggest losers are the NICs, who previously depended on imitating foreign technology as a tool for industrializing and growing economically. With the implementation of the TRIPs

Agreement and the increased limits on these countries' abilities to imitate foreign technology free of cost, these firms are forced to either develop their own technology or to obtain licenses from the patent owners.

However, studies have demonstrated that a stronger intellectual property protection system can possibly work to the advantage of developing and less developed countries. The TRIPs Agreement will have varying welfare and economic effects on developing countries depending on the economy and characteristics of each country. Even though "[t]here remains considerable uncertainty concerning the impact of the TRIPs Agreement on global economic development," not all developing and less developed countries will be losers from the TRIPs Agreement.

Multinational Corporations are profoundly disquieted by the Charter of Economic Rights and Duties, particularly those provisions dealing with expropriation and compensation. They are also troubled by the provisions for technology transfer, which essentially mean surrendering the heretofore exclusive patents and trade secrets of advanced technology to competitors in the Third World.... Much of this apprehension derives from a novel but intensely held perspective; the popular image of the world is no longer an infinitely expanding organism but rather a claustrophobic spaceship with limited and increasingly overtaxed resources, distributed hereafter to the winners of zero-sum competitions (your win is my loss).

With the revolutionary changes in the world economy, developed countries like the United States and Britain have been trying to maintain their dominant positions in the world economic order by encouraging the development of new technological innovation.

In the meantime, some developing countries have been adopting some of these new technological innovations without reimbursement to the developed countries. In response, developed countries have formulated national laws to protect against infringement of intellectual property rights. However, these national laws have not been effective in combating the problem of infringement of intellectual property rights. As a result, nations adopted both the Paris Convention for the Protection of Industrial Property and the Berne Convention for the Protection of Literary and Artistic Works to establish an international system of protection for intellectual property rights.

However, both the Paris Convention and Berne Convention failed to establish the strong international system needed for policing against infringements of intellectual property rights. As a result, the developed countries moved for the strengthening of intellectual property rights under the Uruguay Round of GATT. Negotiations between developing and developed countries during the Uruguay Round led to the adoption of the TRIPs Agreement. The TRIPs Agreement offered a more comprehensive and higher level of protection for intellectual property rights than found in either the Paris or Berne Conventions. Developing countries argue that the provisions set forth in the TRIPs Agreement hinder their economic growth because intellectual property rights serve to stop the transfer of technology. Many of the developing countries are economically disadvantaged in comparison to other nations.

These countries were originally subject to Western colonialism and imperialism during the nineteenth and twentieth centuries. Their corporations are small and undercapitalized, their educational and technological systems are underfunded, and their political systems are fragmented. In addition, least developed countries maintain that they will face a formidable cost in enforcing the TRIPs Agreement.

Several recent economic studies have shown that higher levels of intellectual property protection do not necessarily result in the stopping or slowing down of the transfer of technology. These studies have discovered that strengthening protection for intellectual property rights result in possible economic development and growth. In a report entitled “The TRIPs Agreement and the Developing Countries,” commissioned by the WIPO, the United Nations Conference on Trade and Development outlines the financial and other implications of the TRIPs Agreement on developing countries. The objective of the report is “to assist developing countries in collaboration with the WIPO and the WTO to identify opportunities provided by the TRIPs Agreement, including for attracting investment and new technologies.”

The report finds that by strengthening protection on the intellectual property rights, there may be a positive impact on developing countries through increases in local innovation, foreign direct investment, and technology transfers. In addition, the report notes that depending on the country’s economy, technological development, and existing system of protection for intellectual property rights, the impact of the TRIPs Agreement will differ significantly among countries. Newly industrialized countries with strong industrial and technological bases will most likely benefit from the TRIPs Agreement.

COUNTERING VIEWS ON THE EFFECTS OF THE TRIPS AGREEMENT

Developed and developing countries are split over several issues regarding the benefits and burdens of the TRIPs Agreement. The main argument between developed and developing countries arises over whether the TRIPs Agreement is focused more to benefit the industries of developed countries rather than the developing countries. During the negotiations of the Uruguay Round, concerns were raised regarding whether the TRIPs Agreement damages the economic growth of developing countries. The member nations realised that the TRIPs agreement would potentially impact the economic interest of the developing countries because most developing countries do not currently have the legal infrastructure necessary to protect intellectual property rights and often depend on foreign technology to economically develop.

Differing views are advanced regarding the positive and negative impacts of the TRIPs Agreement and the extent of these impacts on developing countries and their economic growth. Some countries, particularly developed ones, maintain that a stronger intellectual property rights system is in the best interest of the developing countries. However, developing countries contend that they have little to gain from the TRIPs

Agreement. There are several arguments supporting the benefits of the TRIPs Agreement for developing countries and their economic growth. Some theorists argue that a stronger intellectual property rights system will encourage foreign direct investment and the transfer of technology from developed countries to developing countries. These theorists believe that foreign firms are currently reluctant to invest in countries that do not have a strong legal system to protect intellectual property rights because these firms fear that their technology will be stolen and sold to third parties.

Thus, with an adequate legal infrastructure to protect intellectual property rights, foreign firms will be more willing to transfer technology to developing countries, therefore, increasing foreign direct investment. Another argument is “if developing countries do not adopt high levels of [intellectual property rights], their scientists and other innovators will leave” because the national laws are not sufficient to protect innovations and ideas. Thus, a strong national system to protect intellectual property rights will encourage domestic scientists to remain in their countries and will also encourage local research and development and domestic enterprises.

In the long run, developing countries will gain the international reputation of being a competitive supplier of technology and innovations and become less dependent on other countries for technology. The citizens of developing countries would own more intellectual property rights and the disparity in the number of intellectual property rights owned between developed countries and developing countries would decrease.

Studies and past experiences have shown that the key to continual economic success lies in the technological development within a country. On the other hand, many argue that the TRIPs Agreement is counterproductive. Developing countries have denounced the developed countries’ interest in TRIPs as being “self-serving and hypocritical.”

Because developing countries are not major producers of intellectual property products, developing countries see no real benefits deriving from strengthening the system protecting against infringement of intellectual property rights. To many developing countries, an intellectual property protection mechanism translates into a blockade of the transfer of technology and an impediment to economic development.

Most developing countries believe that intellectual property products should be available at minimal costs. If the TRIPs Agreement is enforced, the developing countries would be faced with the formidable cost associated with enforcing intellectual property rights. Monitoring is expensive and most developing countries will have to create a civil justice system to hear intellectual property disputes. These member countries will have to set up copyright, trademark, and patent offices and staff them with trained personnel. Although developing countries initially resisted the TRIPs Agreement, it was eventually accepted for a variety of reasons. First, the developing countries’ acceptance of the TRIPs Agreement was to prevent the United States from imposing further unilateral trade sanctions on violating countries.

Under the TRIPs Agreement, complaining parties, such as the United States, will be forced to comply with the WTO’s dispute settlement system. Second, Articles 65 and

66 of the TRIPs Agreement provide substantial concessions to developing countries. Developing and less developed countries have an extended period of time to comply with the TRIPs Agreement. Even though the developing and less developed countries have signed onto the TRIPs Agreement, the question remains whether the TRIPs Agreement adversely affects the economic growth of developing countries and widens the developmental gap between the developed and developing nations.

ON THE ECONOMIES OF DEVELOPING COUNTRIES

In the past, there have been very few studies examining the direct relationship between intellectual property protection and economic development in the setting of the developing countries. Older studies demonstrate that increased protection of intellectual property rights does not necessarily affect the transfer of technology in developing countries. For example, a United Nations study did not find a correlation between developing countries with strong intellectual property protection and increased foreign direct investment. Countries with the highest levels of foreign direct investment, such as Argentina, Brazil, China, and Thailand, have the lowest levels of intellectual property Protection.

On the other hand, countries with high levels of intellectual property protection, such as Nigeria, have not attracted higher levels of foreign direct investment than other similarly situated countries. With the increasing academic and policy-oriented interest in intellectual property protection, several new measures and studies concerning the effect of higher levels of intellectual property protection on economic development and growth have emerged. Numerous studies have been conducted to determine the effects of strengthening the protection of intellectual property rights through the TRIPs Agreement on economic growth, foreign direct investment, and technology transfer. The empirical data gathered in these new studies comes from either surveys of investors or from econometric works evaluating the impact of the TRIPs Agreements on developing countries. These newer studies show a positive correlation between stronger intellectual property protection and increased foreign trade and foreign direct investment.

EMPIRICAL STUDIES

Many theorists have argued that firms in developed countries are more likely to invest in countries with higher levels of intellectual property protection. Without a high level of protection for intellectual property, firms are reluctant to invest in production that entails “a significant transfer of proprietary knowledge, such as R&D and technology-intensive manufacturing processes” due to the fear that domestic firms will steal the protected technology and produce and sell the protected product to third parties at a lower price. Thus, foreign corporations are unwilling to enter into joint ventures with local firms in countries which do not have a strong system of protection for

intellectual property rights. In order to attract the foreign direct investment needed to create the technological basis for growth, developing countries need to implement a stronger system of protection for intellectual property rights. One of the leading economic analyses regarding the impact of increased protection for intellectual property rights on foreign direct investment was conducted by Edwin Mansfield.

Based on surveys from executives of major U.S. corporations and patent attorneys, Mansfield found that there was a strong positive correlation between higher levels of intellectual property protection and foreign direct investment. Moreover, Mansfield discovered that although a higher level of intellectual property protection increased foreign direct investment, this positive correlation did not exist in all types of foreign direct investments and in all industrial sectors. For example, a stronger positive correlation between intellectual property protection and foreign direct investment existed in investment associated with research and development facilities than in investment related to sales and distribution outlets. Mansfield also found that when a country enacted a stronger intellectual property protection system, firms in the chemical, pharmaceutical, machinery, and electrical equipment industries increased the amount of foreign joint venture and foreign direct investment, while firms in the transportation equipment, metals, and food industries did not.

In an attempt to explain the relation between the U.S. foreign direct investment flow into sixteen developing countries and newly industrialized countries and the level of protection for intellectual property rights, Jeong-Yeon Lee and Mansfield compiled an index of intellectual property protection for each of these sixteen countries by using a multivariate regression analysis. Lee and Mansfield discovered that countries with stronger intellectual property protection attracted significantly higher levels of foreign direct investment.

For instance, a survey confined to the chemical industry demonstrated that the percentage of a firm's investment devoted to research and development was directly related to the perceived strength of the intellectual property protection. However, Lee and Mansfield's study of intellectual property protection has been criticized for having two main weaknesses. Critics argue that the index is too subjective in character because the surveys include the firms' personal perception of what elements influence investment.

Moreover, critics point out that the sample countries chosen for the study all have some technological capabilities and have a larger number of intellectual property disputes than the average countries. Thus, the results of the study are more likely to overstate the positive correlation between intellectual property protection and foreign direct investment. Robert Sherwood also conducted a comprehensive economic study that ranked national intellectual property rights regimes. Sherwood's data is based on a combination of a survey-based index and an "on the books" analysis of eighteen developing countries. Sherwood believes that because the research and development of new technology bears higher financial risk than any other commercial activity, stronger intellectual property protection helps reduce this risk and stimulates higher levels of

investment in the developing countries' technology. As a result, the creation of new technology "fosters considerable economic growth and enhances social welfare." Sherwood's analysis starts each country with a maximum score of 100. Points are then subtracted from the maximum score for weaknesses in one of eight major headings: enforceability; administration; treaties; and substantive laws for copyright, patents, trademarks, trade secrets, and life forms.

The index then adds up to three points for a country's strong general commitment to intellectual property protection. Sherwood finds that while exact answers to the question of whether higher levels of intellectual property protection stimulates foreign direct investment are unavailable, overall trends and effects are complete and helpful. Sherwood also discovers that "the impact of the TRIPs Agreement on most developing countries is likely to be slightly negative in the short run (one to two years) and increasingly favourable as local firms and individuals begin to realise the potential benefits for their activities." Furthermore, Sherwood notes that while intellectual property protection provided by the TRIPs Agreement is sufficiently strong enough to stimulate international trade flows and foreign direct investment, it falls short of encouraging domestic innovation, such as domestic research and development of innovative technology.

In other words, local scientists may still leave the developing countries for countries with stronger intellectual property protection. Furthermore, Sherwood finds that depending on the economy of the country and the country's current technological state, the impact of the TRIPs Agreement will differ among countries. For example, "[e]xtremely poor, weakly endowed countries differ from what are now called 'semi-industrialized' countries in that they are less likely to have the ability to 'pirate' the intellectual property of others for private gains, and therefore have less to lose in moving to robust protection levels." As with Lee and Mansfield's analysis, Sherwood's index is impaired by its subjective character.

In both Sherwood's and Mansfield and Lee's studies, a positive correlation between strengthening of intellectual property protection and the increase of foreign direct investment generally exists. By providing higher levels of protection for intellectual property rights in general, the TRIPs Agreement encourages the foreign direct investment that developing countries need to create and develop technology.

INTERNATIONAL AGREEMENT ON PROTECTION OF INTELLECTUAL PROPERTY RIGHTS

With the globalization of the economic market and the increase in international trade, industries in developed countries have experienced an increased number of infringements on intellectual property rights. National intellectual property laws have been ineffective in combating problems associated with intellectual property rights infringement. With the increase in trade, industries have begun to lobby for an international system of protection for intellectual property rights. During the past century,

nations attempting to harmonize intellectual property rights have entered into a number of international agreements. Even though each of these international agreements have covered different types of intellectual property rights and have varied in terms of their objectives, the “ultimate goal has been movement towards internationally recognized and enforceable intellectual property rights.” The two most important multilateral agreements providing international protection for intellectual property rights are the Paris Convention for the Protection of Industrial Property, which protects against trademark and patent infringement, and the Berne Convention for the Protection of Literary and Artistic Works, which protects against copyright infringement.

The Paris Convention for the Protection of Industrial Property (“Paris Convention”), signed on March 20, 1883, is one of the oldest international agreements on the protection of intellectual property rights. Signatories to the Paris Convention include the United States, the United Kingdom, Mexico, and over 100 other countries. The objective of the Paris Convention is to provide “protection of industrial property.... The protection of industrial property has as its object patents, utility models, industrial designs, trademarks, service marks, trade names, indications of source or appellations of origin, and the repression of unfair competition.”

The Paris Convention grants national treatment of industrial property within member nations; whereby, member nations have to treat its own citizens and citizens of other member nations similarly under its national law protecting intellectual property rights. The Paris Convention, however, contains two major loopholes. First, members of the Paris Convention are not required to abide by any substantive standards of patent protection. As a result, a member nation has the individual choice to provide as little or as much intellectual property protection as it decides, so long as it treats both domestic and foreign claimants similarly under its national intellectual property law. Second, under the Paris Convention, disputes between member nations are to be settled by the International Court of Justice (“ICJ”).

Many member nations, however, do not recognize the ICJ and believe it has no jurisdiction within their countries. Furthermore, ICJ rulings are often ignored, even by member nations that recognize its jurisdiction, leaving the disputing party without a legal remedy for infringements. The Berne Convention for the Protection of Literary and Artistic Works (“Berne Convention”), signed on September 9, 1886 and effective on December 5, 1887, was the first multilateral copyright treaty. The Berne Convention strives to provide protection for literary, scientific, and artistic works.

The Berne Convention has at least 121 signatories, including the United States, France, Mexico, and the United Kingdom. Like the Paris Convention, the Berne Convention sets forth national treatment, providing that “[p]rotection in the country of origin is governed by domestic law. However, when the author is not a national of the country of origin of the work for which he is protected under this Convention, he shall enjoy in that country the same rights as national authors.” Unlike the Paris Convention, the Berne Convention includes a minimum standard of protection, whereby exclusive

rights must be granted to an individual for his or her work over a duration of time. The Berne Convention, however, fails to outline clear legal remedies by which copyright holders may enforce their rights against infringers. Specifically, there are no provisions within the Berne Convention that provide for sanctions against member states that do not uphold their obligations under the convention. Moreover, under the Berne Convention, producers of sound recordings have neither protection nor legal remedies against infringement.

Established in 1967 under the WIPO convention, the World Intellectual Property Organization (“WIPO”) is an intergovernmental organization with headquarters in Geneva, Switzerland. The objective of the WIPO is “the promotion of the protection of intellectual property throughout the world through cooperation among States, and for the administration of various multilateral treaties dealing with the legal and administrative aspects of intellectual property.”

As one of the sixteen specialized divisions of the United Nations, the WIPO is responsible for the administration and enforcement of the Paris Convention and the Berne Convention. The infrastructure of the WIPO is divided into two major divisions—industrial property and copyright. A substantial part of the activities and resources of the WIPO are devoted to the establishment of world-wide intellectual property protection by promoting cooperation among nations and facilitating the transfer of technology from developed countries to developing countries.

As of August 2000, there were over 171 countries that were members of the WIPO, including the United States, Argentina, Brazil, Mexico, the United Kingdom, and Japan. The WIPO, however, has its share of weaknesses. The WIPO has both a weak enforcement mechanism and a weak dispute settlement mechanism. The WIPO is unable to sanction violators and the provisions covering legal remedies are vague. The WIPO dispute settlement mechanism has been called “effectively worthless” because it relies on the voluntary cooperation of the party receiving the unfavourable ruling in enforcing a dispute resolution. In an attempt to give the WIPO more power in enforcing the Berne and Paris Conventions, the United States and other developed countries have attempted to change and improve the provisions governing the enforcement and dispute mechanism. They wanted to give the WIPO more specific tools for legal enforcement, such as the ability to impose sanctions.

Many developing countries, however, argued that WIPO had too much power in administering the Berne and Paris Conventions. Because the WIPO is dominated by “U.N.-style voting blocs,” the result has been a stalemate; therefore, no changes have been made to the WIPO’s ability to enforce the provisions set forth in the Conventions. In summary, the multilateral attempt to protect industries and individuals globally against infringement on their intellectual property rights have largely failed because the Paris Convention, the Berne Convention, and the WIPO do not offer a comprehensive and effective international system of protection against infringements on intellectual property rights.

THE TRIPS AGREEMENT

Due to the inadequacy of the Paris and Berne Conventions in protecting against infringement of intellectual property rights, the United States, Japan, and the European Community pushed for international protection of intellectual property rights to be added to the agenda of the Uruguay Round of General Agreement on Tariffs and Trade (“GATT”) in 1994. GATT earlier addressed the issue of protection for intellectual property rights in 1979 during the Tokyo Round of GATT.

At the Tokyo Round, however, the member nations rejected the Anti-Counterfeit Code, which would have addressed trademark counterfeiting. On April 15, 1994 as part of the negotiations and agreements of the Uruguay Round of GATT, the Trade-Related Aspects of Intellectual Property Rights (“TRIPs Agreement”) was adopted in Marrakesh, Morocco. The TRIPs Agreement did not actually become effective until January 1, 1995. Because the TRIPs Agreement is categorized as a multilateral agreement under the WTO, any country that wishes to join the WTO must also agree to abide by the TRIPs Agreement. Currently, there are 138 countries participating in the WTO and the TRIPs Agreement. Current signatories include the United States, Japan, Mexico, the United Kingdom, and France.

The purpose of the TRIPs Agreement is to provide adequate and effective protection for intellectual property rights in order to reduce impediments to international trade and promote global competition. The TRIPs Agreement covers eight forms of intellectual property, including: copyright and related rights; trademarks; geographical indications; industrial design; patents; layout-designs of integrated circuits; confidential information or trade secrets; and contractual licenses. The TRIPs Agreement is not considered a new concept in international law, rather it supplements the existing international agreements governing the protection of intellectual property rights on a global scale. The TRIPs Agreement outlines the minimum standards of substantive protections for each form of intellectual property that each member nation must provide in their national law. Member nations then have the freedom to determine the appropriate method of implementing the TRIPs Agreement within their own legal system.

The TRIPs Agreement is divided into three main sections—standards, enforcement, and dispute settlement. Parts I and II of the TRIPs Agreement highlight the substantive standards of intellectual property protection and outline the agreement’s three core commitments—minimum standards, national treatment, and most-favoured nation treatment. The section on minimum standards defines the subject matter to be protected, the intellectual property rights conferred, and the minimum duration of protection. While the minimum substantive standards in the Agreement are partially based on the principles and provisions set forth in the Berne and Paris Conventions, the TRIPs Agreement adds a substantial number of additional obligations on matters where the other conventions are either silent or are seen as inadequate.

The national treatment provisions of the WIPO govern the TRIPs Agreement, where “[e]ach Member shall accord to the nationals of other Members treatment no less

favourable than it accords to its own nationals with regard to the protection of intellectual property....” The idea of national treatment flows into the concept of most-favoured nation (“MFN”).

Article 4 of Part I states “with regard to the protection of intellectual property, any advantage, favour, privilege or immunity granted by a Member to the nationals of any other country shall be accorded immediately and unconditionally to the nationals of all other Members.”

Part III of the TRIPs Agreement sets forth general obligations governing the enforcement procedures for the protection of intellectual property rights. The TRIPs Agreement uses both internal and external enforcement mechanisms. The internal enforcement mechanism primarily concerns private actions by requiring member countries to ensure that enforcement procedures, as specified in the Agreement, are made available under national laws. Section 2 of Part III of the TRIPs Agreement, entitled Civil and Administrative Procedures and Remedies, outlines the standards of evidence, injunctions, damages, and other remedies required under national law.

External enforcement mechanisms permit the use of trade sanctions to force compliance with the TRIPs Agreement. The success of using trade sanctions to enforce compliance will depend on the legitimacy of the WTO dispute resolution process. Unlike the Berne and Paris Convention, the TRIPs Agreement adopted the detailed Dispute Settlement Understanding (“DSU”), established by the WTO. Unlike the old DSU system, the new DSU mechanism guarantees a right to adopt the dispute panel’s decision unless there is a consensus to reject the panel’s decision.

Appellate review of the dispute panel’s decision on the violation is also guaranteed. The offending member must then either change the inconsistent measure or negotiate a compensatory settlement by a set deadline. If the offending member fails to change the inconsistent measure or compensate the injured party, the WTO may choose to retaliate by withholding trade benefits in the same sector. Articles 65 and 66 of Part VI of the TRIPs Agreement contain the transitional arrangements. Developed countries have had to comply with all the provisions of the TRIPs agreement since January 1, 1996. Developing countries and countries in transition from a centrally-planned economy to a market economy had until January 1, 2000 to comply with all requirements of the Agreement—with the exception of previously uncovered patent protection, where the deadline is January 1, 2005.

Less developed countries (“LDCs”) have until January 1, 2006 to completely conform to the Agreement. Although developing countries and LDCs have had longer transitional periods, they, along with developed countries, have had to comply with Articles 3, 4, and 5 of the TRIPs Agreement since January 1, 1996. Articles 3, 4, and 5 outline the national treatment and most favoured nation treatment obligations.

The TRIPs Agreement has had a higher success rate in the protection of international intellectual property rights than the WIPO. The combination of a minimum standard for the protection of intellectual property rights with a detailed dispute settlement

mechanism has made it easier for countries to determine their intellectual property rights and obligations. However, the TRIPs Agreement still has its own share of problems. While most developed countries have clearly taken steps to change and adopt national laws that are in compliance with the TRIPs Agreement, there are some developed countries that have not changed their national laws to completely comply with the TRIPs Agreement.

For instance, the new European Community regulation, which requires the use of a single trademark for a product within the European Community, possibly violates Article 20 of the TRIPs Agreement because it imposes an unjustified encumbrance on the trademark holder. Even though most developing countries have argued against the benefits of the TRIPs Agreement, developing countries have started to change their national laws to comply with the TRIPs Agreement.

For example, developing countries in Southeast Asia have begun to change their national laws and set forth the administrative structure in order to comply with the TRIPs Agreement. Even though most developing countries likely will not meet the compliance deadline and the problem of ineffective enforcement will still exist, these developing countries, in the long run, will certainly offer foreign claimants better intellectual property protection against possible infringements.

In addition, the DSU system appears to be effective in bringing violating countries into compliance, as demonstrated by the case against India for patent infringement. As a developing country, India has failed to implement both the “mailbox” rule under Article 70.8 and the “pipeline” provision under Article 70.9 by the set deadline of January 1, 1995. After an unsuccessful consultation with India under the WTO DSU system, a panel was formed by the Dispute Settlement Body. The dispute panel ruled that “the lack of legal security in the operation of the mailbox system in India is such that the system cannot adequately achieve the object and purpose of Article 70.8 and protect legitimate expectations contained therein for inventors of pharmaceutical and agricultural chemical products.” With the threat of withholding of trade benefits by the WTO, India has recently begun to change its laws to comply with this aspect of the TRIPs Agreement.

VIEWS ON THE EFFECT OF THE TRIPS AGREEMENT

THE ECONOMIC STATUS OF DEVELOPING COUNTRIES

With the increase in international trade and the expansion of the global market starting in the 1970s, one recognizes a dramatic change in the international economic structure. Large multinational corporations with far-flung corporate networks and global factories dominate this new economic order. Many nations, like the United States and four of the Newly Industrialized Countries (“NICs”), have experienced unprecedented levels of

economic growth and development. However, not all nations have enjoyed this unprecedented level of economic growth and development. Economic growth and development in the world have always been geographically uneven. Since the end of World War II, many of the developing countries have been attempting to catch up economically with the developed countries by promoting industrialization and using trade policy to cope with the disparate domestic economic development.

Economists and international organizations have largely focused their attention on the development gap between the countries of the Northern Hemisphere and the countries of the Southern Hemisphere. They specifically attempt to understand the disturbing problem of why, despite economic growth and some improvement in living conditions in the Southern Hemisphere, the divide between rich and poor nations has not diminished.

For instance, between 1950 and 1980, the growth in real per capita income in the Southern Hemisphere amounted to \$81, in comparison to the \$5,807 improvement in the Northern Hemisphere. The difference in social and physical well-being also exemplifies the gap between the two hemispheres. The average life expectancy in the Northern Hemisphere in 1989 was seventy-six years, but the average life expectancy in the Southern Hemisphere was only sixty-three years. In addition, in 1989, the average diet in the Northern Hemisphere consisted of over twenty-five per cent more calories than the average diet in the Southern Hemisphere. In 1990, the World Bank Report estimated that a majority of the 950 million people of the world suffering from chronic malnutrition lived in the South. Moreover, the differences in the medical resources available to the citizens of the Northern and Southern Hemispheres are striking.

In 1984, Northern Hemisphere countries averaged one doctor for every 450 people, while the Southern Hemisphere averaged one doctor for every 4,990 people. The economic structures of the Southern and Northern Hemispheres also differ dramatically. Agriculture accounts for about 19 per cent of the total Gross Domestic Product (“GDP”) in the South while agriculture accounts for only 3 per cent of the GDP in the North. In the 1980s, the average prices for agricultural goods decreased, causing a balance of payment problem in developing countries.

THEORIES ON THE KEY TO ECONOMIC GROWTH

Traditionally, the economies of developing countries depend on the exporting of agricultural goods. However, economists have noted that developing countries cannot economically develop by simply increasing current production because of the phenomenon of “immizerizing growth”. Immizerizing growth is defined as a situation in which the increase in production of goods causes a decrease in economic growth. For example, an increase in agricultural goods may hurt market equilibrium where market supply outweighs market demand.

As a result, agricultural prices are driven down, resulting in a net welfare loss. In an attempt to help developing countries grow economically, neo-classical economists during the past decade have analysed and formulated economic growth theories. The Solow

model, formulated by Robert Solow, is one of the leading growth theories in the area of developmental economics.

In his model, Solow argues that there are three factors of production—labour, capital, and technical progress. Previous growth models account for only two factors of production—labour and capital. Solow recognized that technical progress acts as an “enlarger” of the number of hours worked. For example, four working hours in the nineteenth century is equivalent to one hour or less in the twentieth century due to technology. Thus, technology gives poorer countries an opportunity to catch up with wealthier countries.

In summary, developing countries should expand their technological bases to increase growth and attain the levels found within industrialized countries, such as the United States and Britain. Thus, governments in developing countries have to promote industrialization through the economic policy of protection of infant industries through the use of tariffs, import substitution, and subsidies. While some third world countries have successfully created and accumulated a large supply of technology, most are unable to create and maintain high levels of capital and technology. The “Two Gap” Theory recognizes two main constraints limiting a developing country’s ability to gain technology. First, developing countries are often unable to save enough capital to create and maintain a technological base in order to promote economic growth. Second, a balance of payment problem typically plagues developing countries, where the cost of industrial imports far exceeds agricultural export revenues. As a result, the governments of developing countries often owe money to other nations, and, consequently, do not have the capital to invest in technology.

One solution to the problems listed in the Two Gap Theory is foreign direct investment. Developing countries often look towards foreign direct investment to supplement the lack of domestic capital to invest in technology. For developing countries, a popular solution has been to encourage the free flow of technology from industrialized countries to developing countries. Developing countries often depend on the new technology originating from laboratories and universities in the developed countries. Recently, developing countries have increasingly demanded access to Western technology. These demands have gone as far as to call for an absolute abolition of the international intellectual and industrial property regimes and for the creation of the free flow of technology across national boundaries.

Developing countries argue that the goal of international protection of intellectual property rights reflects the interests of those with dominant economies in the world, specifically the United States, the European Union, and Japan. The protection of intellectual property rights does not reflect the interest of developing countries. Moreover, developing countries maintain that under the doctrine of “uneven development,” the developed countries became wealthy at the expense of the developing countries because the developed countries were first to industrialize and have not given developing countries a chance to advance. The developed countries now actually owe the developing

countries a share of this technology because it was earned at the expense of developing countries. In addition, developing countries perceive intellectual property rights as a method of transferring money from developing countries to developed countries through the form of rent transfer. In short, developing countries perceive the existing international economic order and legal structure as unfair because the developed countries receive far more benefits under this system.

Believing that the current international economic order is unfair, governments of developing countries have pushed the developed countries to change their policies and create a new international economic order. The Declaration on the Establishment of a New International Economic Order sets forth the principle of “giving to the developing countries access to the achievements of modern science and technology, and promoting the transfer of technology and the creation of indigenous technology for the benefit of the developing countries in forms and in accordance with procedures which are suited to their economies.”

This principle was further expanded in the Charter of Economic Rights and Duties of States, which states that:

- Every State has the right to benefit from the advances and developments in science and technology for the acceleration of its economic and social development.
- All States should promote international scientific and technological co-operation and the transfer of technology, with proper regard for all legitimate interests including, *inter alia*, the rights and duties of holders, suppliers and recipients of technology.

Even though developing countries call for a new economic order and depend on the transfer of technology to fuel their economic growth, the TRIPs Agreement increases protection of intellectual property rights and restricts the transfer of technology. As a result, developing countries are obligated to alter their national intellectual property laws to comply with the TRIPs Agreement. In the end, the TRIPs Agreement limits the available range of trade and industrial policy options available to developing countries.

THE IDEOLOGY OF PRIVATE RIGHTS IN INTELLECTUAL PROPERTY

The thinkers whose ideas we have outlined in the previous part never had much to say about intellectual property as such. Their chief concern was to justify material property including land. The question we must pose here then is, how well do the theories stand up when they are applied to intellectual property? Justin Hughes has argued that Locke’s labour theory approach applies fairly well. His position is based on an essentially neo-Platonic interpretation of ideas as existing in an immaterial “commons”. When these idea-forms are mixed with individual labour they become effective in the material world.

For Hughes intellectual laborers are entitled to property rights in a manner analogous to those who toil on the natural commons and thereby create value-added. Hughes, in fact, goes as far as to argue that the Lockean approach applies with greater force to intellectual property than it does to physical property. This is because the essence of the intellectual commons renders unnecessary the “Lockean proviso” and takings leave “enough and as good” for others. While in a developed market economy where scarcities prevail the proviso might be an important qualification, an important characteristic of intellectual goods is non-rivalry of use.

Hence, any taking from the Platonic ether of potential ideas does nothing to diminish the ability of anyone else from doing likewise. In fact, the development of ideas has a greater potential to expand the intellectual commons inasmuch as the development of a particular idea often gives rise to additional intellectual creation. The first difficulty with Hughes application of the Lockean theory is his argument that ideas are the result of the labour expended by individuals. We find this a dubious proposition both as a matter of how new ideas emerge and as a valid justificatory argument for private property. Ideas as such are the result of social rather than individual creation.

This does not mean that individuals do not make particular contributions to the stock of intellectual and cultural capital. Clearly, however, such contributions must necessarily work on previously existing ideas that are themselves the product of a social process. Even the most novel of individual human insights, for example, require the use of language—a social product for which no one person may take credit.

The notion that ideas already have a prior existence in a “common” only waiting to be discovered by individuals is an absurdity. The human race is not somehow endowed by their Creator with intellectual raw materials out of which new and useful ideas and concepts are developed. Rather these “raw materials” are themselves entirely the result of a social process and, therefore, a social product. Granting that individuals do make contributions to knowledge in ways that are useful to others, could it be argued that they deserve some reward for their effort? Locke’s labour theory certainly argues that they do. As Hughes points out Locke’s thinking was grounded in the presumption that labour itself was something that people would ordinarily seek to avoid in the absence of adequate reward.

Property then is justified as the proper reward for such labour. How applicable is this sort of argument in the context of intellectual property? Are scientists, inventors, and other creative people well-described by this view of human nature? Neither do believe that the creative work that is usually associated with intellectual property to be an anathema, or even unpleasant, to the people who typically engage in it. Quite to the contrary, those who engage in intellectual and creative endeavors derive a large amount of satisfaction in the creative process itself and find ultimate reward in discovery and creation. This view doesn’t require that we take an especially Pollyannaish attitude about human nature. Intellectual and creative people are not all indifferent to worldly comforts and material goods. We do not believe, however, that such motivations figure

as the primary incentives that explain their works. Most would regard these rather as the coincidental benefits of their success. Marginal increases or decreases in material rewards would not then have an appreciable effect on the rate of creative or inventive activity.

What about the status of the “Lockean proviso” as it applies to intellectual property? Does the appropriation of intellectual goods leave enough and as good for others? As Hughes points out it is in consideration of this question that the differences separating intellectual property from other kinds of property become most apparent. The appropriation and use of most physical property by individuals obviously, at least temporarily, deprives others from its simultaneous possession and/or use.

The creation of private property right legitimizes this exclusionary relationship. In the case of intellectual property right, on the other hand, the point is to create the basis for and legitimize such a relationship where none would otherwise exist. That is, given the public goods characteristics of intellectual goods, the creation of private property in such goods involves two types of issues. The first is, as noted, non-rivalry in use. Use of an idea in no way impedes anyone else’s full use of the same idea. Moreover, the marginal cost of use of an idea is zero. Once an idea exists it costs nothing for as many people who wish to make use of it to do so. The second issue is the difficulty of exclusion. How can those who are not inclined to pay for the use of an idea be excluded from its use? This is the clear purpose of intellectual property right.

In a sense then the Lockean proviso is both well met as well as irrelevant. It is well met in the sense that the appropriation of intellectual property certainly does leave enough and as good for others. At the same time, however, the very purpose of copyrights, patents, and the other instruments of institutionalized rights in such property is to prevent free access to such “enough and as good.” In contrast to Hughes then, we argue that this public goods aspect of intellectual production renders it unfit as private property. Hughes also argues that because new ideas provide the basis for additional intellectual insights that protection is called for. In this intellectual property protection adds to the extensiveness of the “commons” and thereby creates a basis for an expansion of generalized human welfare.

Here it is evident that Hughes is appealing to an essentially utilitarian, rather than rights based, argument for private property protection. As before, we have our doubts about the essential assumption that private property rights are a necessary or appropriate incentive for the creation and dissemination of intellectual goods and services.

Why shouldn’t a fixed monetary amount be sufficient as a reward or public recognition and approbation? Other issues relevant to the utilitarian perspective will be examined below. Another qualification to place on the legitimacy of private property in Locke’s theory is that it not result in waste.

In Locke’s theory this applies with obvious force to the waste that may result if physical property is accumulated beyond the ability of its possessor to make adequate (efficient) use of. If land is accumulated, for example, but remains fallow because its

owner lacks the time or other complementary inputs to employ it efficiently, then such accumulation would lack legitimacy, particularly if it deprived another of its efficient use. For Locke this qualification did not amount to an effective limit on the accumulation of private property in a money economy since property could always be accumulated in a fungible and non-perishable form—namely, money capital.

The money economy, recall according to Locke, emerges as a matter of the common assent of humankind. Again believe this aspect of Locke's theory as it applies to intellectual property is suspect. The relevant question is, To what extent does the monopolization of new and valuable ideas via patents and copyrights involve waste? It would seem a considerable loss to limit the benefits of discovery and innovation when they involve zero marginal cost of production and very little or no cost of dissemination.

Hughes makes the argument that the waste qualification is irrelevant to intellectual property inasmuch as ideas are not perishable. This is also a very questionable assertion. The history of patents is rife with examples of cases where new ideas and innovations have been delayed or permanently lost by interests who used patents to prolong their existing intellectual and physical capital stock. Finally, contrary to Locke, it is fairly common knowledge that the historical emergence of the money economy involved something more dramatic (and violence-prone) than the common assent of humankind.

Interestingly Hughes also seems to suggest that the concept of intellectual property rights is of limited applicability. He identifies certain categories of ideas that are subject to becoming "depropertized." These include common, "everyday ideas", *e.g.* walking your dog on a leash, and extraordinary ideas, *e.g.* the human genome map.

The latter category itself is decomposable into:

- Ideas that reveal extraordinary facts about the world, *e.g.* the theory of electromagnetism.
- Those that have already established widespread use, *e.g.* the use of architectural columns. Hughes cites particular case law that have liberated certain ideas into the sphere of the public domain. The general principle he derives is expressed as follows:

In short, some ideas become "depropertized." Originally, they could have been subject to private ownership (unlike the first kind of extraordinary ideas), but the pressure to keep them in the common increases as the ideas become increasingly important to the society. As an idea becomes extraordinary, it is clear the common will not have "enough and as good" if the rights to the idea continue to be privately held. Here again, we believe that Hughes' position should be greeted with skepticism. While he asserts that some ideas become "depropertized", he offers no explanation as to how, when, and if this is likely to happen in particular cases. The most problematical case involves what he refers to as "extraordinary ideas."

In the current climate of rapid technological change and scientific discovery there exists many candidates that may qualify as falling into this category ranging from biotechnology to a new understanding of the creation of the cosmos.

Many of these discoveries have the potential to radically modify our understanding of ourselves and our potential. Who is to decide, and under what conditions, what constitutes sufficiently important and fundamental ideas as to qualify it as a public idea? Would, for example, private control over a new drug capable of ridding the world of the AIDS virus leave “enough and as good” for those not wealthy enough to afford its purchase price? Obviously the answer is “no.”

The Lockean proviso then is clearly violated in such a case. It would seem, in fact, that for an extraordinary idea to become depropertized, a necessary, and perhaps even sufficient, condition would be that established property rights in it would have to be difficult to enforce leading to their wholesale violation. The typical rejoinder to this argument, of course, is that to the extent that private property rights are not respected or effectively enforced, then the incentive for their creation and development in a market economy is undermined. At this point the justificatory ground has shifted again from a Lockean basis to a utilitarian one. Before we do, however, we would like first to take up possibilities offered by the Hegelian, “personality”—based justification as it might apply to intellectual property.

IP AND PERSONALITY

Recall that Hegel’s primary concern is his concept of freedom which requires the ability of individuals to overcome their narrow subjectivity and “externalize” itself as part of a higher and larger embodiment—to wit, an ethical community. Property is logically necessary towards this end insofar as it permits people to develop personality. For Hegel the creative act that results in simultaneously property and personality was viewed primarily in terms of the transformation of a material reality. There seems no reason, however, why his theory could not accommodate intellectual property as well. Certainly the creation and expression of an idea is often a powerful means of extending or externalizing subjectivity.

As Hughes notes this is especially true of artistic work. Yet Hughes goes on to argue that the personality argument might even preclude certain, less artistic types of intellectual property. He writes, Poetry seems to lend itself to personality better than trade secrets, symphonies better than microchip masks. Should poetry as a category receive more protection than microchip masks? Should some categories receive no protection at all from the personality justification? In this interpretation believe that Hughes is taking a narrower understanding of personality than that intended by Hegel. Hughes seems to view property merely as the receptacle of personality rather than the means of development of personality as well.

It is true, as Hughes himself argues, that the creation of some types of intellectual property is subject to greater constraints than others. For example, ideas that serve utilitarian objectives, *i.e.* satisfy needs, must often also satisfy efficiency requirements. To the extent that such constraints exist, Hughes claims production of ideas reduces the opportunity for personality expression. This interpretation, however, misses the more

general focus of Hegel's views on personality development as a means to overcome the narrowness of individual subjectivity.

It would seem since by Hegel's requirement that such overcoming join the individual to an ethical community, scientific discovery would contribute as much to personality development as conventionally understood artistic expression. Moreover, discoveries in science and the practical arts certainly provide their discoverers with social recognition, a requirement for personality explicitly recognized by Hegel. Hughes argues that this recognition follows more forcefully from money payments made to the holders of intellectual property rights. He writes: "Recognition" involves more than lip service. If say "this forest is your property" and then proceed to flagrantly trespass, cut your timber, and hunt your deer, we have not recognized your property rights.

Similarly, verbal recognition of an intellectual property claim is not equal to the recognition implicit in a payment. Purchasers of a copyright work or licensees of a patent form a circle of people recognizing the creator as a person.

There are several problems with this interpretation of Hegel's recognition motivation for private property. First of all, why is monetary payment an appropriate and sufficient means of providing the necessary recognition? Most scholarly work, such as that routinely published in academic journals for example, provides no direct compensation to their creators at all. It does oftentimes however provide substantial professional recognition and even, in some cases, wider public acclaim. Second, even if it is admitted that some intellectual property is deserving of a monetary reward type of recognition, what is the appropriate amount of such reward? Hughes apparently is prepared to grant full rights of monopoly exploitation inherent in copyright and patent protection.

How do we know if such rights are appropriate or excessive? In cases where the original creator has alienated his property rights to a third party, what are the appropriate rewards to this new possessor? Should they be the same as those of the original creator? Whatever we decide on the answers to these latter two questions it would certainly seem that we are fully outside the purview of anything related to Hegel's recognition requirement. Clearly it is private property rights in ideas themselves rather than for personality as described by Hegel. Indeed it is hard to understand how monopoly exploitation of intellectual property can aid in the externalization of the individual into a ethical community, particularly as the uninhibited use of such property in no way deprives the creator of the same use. To the contrary, we can cite many cases where the free dissemination of ideas provides the opportunity of personality development of creators as well as that of the beneficiaries of their creations. The argument that creators need incentive for their creations, once again, takes us into the philosophical ground tilled by the utilitarian school and we turn to a consideration of these arguments next.

THE UTILITARIAN DEFENCE

Utilitarian arguments are by far the most popular defences of private property and it is not surprising that they find their way into discussions of intellectual property. As

we've seen in the present discussion other arguments based on right and personality sometimes appeals to utilitarian considerations to buttress weak spots in their own justificatory edifices.

The most commonly encountered version of the utilitarian argument cites private property rights and their economic exploitation as providing the necessary incentive for the production of useful knowledge. In this sense the production of intellectual products is not regarded any differently than any other form of capitalist production. That is, the intellectual capitalist is motivated by the prospect for high returns, and in their absence will turn his or her talents, time, and effort elsewhere in the economy. This reallocation of intellectual resources, it is presumed, will result in a Pareto inferior state compared to that in which intellectual property rights are established and respected.

As noted earlier, we believe there is reason to doubt that the producers of ideas can realistically be characterized as intellectual capitalists who are first and foremost motivated by the prospect of substantial economic profit. Even if we grant, however, that some firms exist who employ creative thinkers for this very reason, the utilitarian justification runs up against a rather formidable and obvious logical difficulty.

Clearly the goal of utility maximization is best served by the widest possible dissemination of new knowledge and the products and services derived from that knowledge. The purpose of intellectual property rights, however, is precisely to limit this dissemination contingent on the willingness and ability to pay. Admitting that idea-related production might suffer in the absence of some required reward, the important utilitarian question is, What is the minimum required reward necessary to procure the maximum dissemination of new and useful knowledge? In many cases this minimum required reward will be provided by the opportunity inherent in first-to-market production and marketing advantages.

These can be considerable when the new goods are produced under conditions of imperfect competition owing to scale economies, control of necessary inputs including human capital, or other factors, and combined, as they often are, with substantial advertising. Do not believe, however, that any regime of private patent rights in ideas or intellectually derived goods and services can be presumed to efficiently answer this question. Patents create monopoly power that is inconsistent with social welfare maximization. Moreover, when private, profit-maximizing firms are given the necessary protection to induce them to produce and market intellectual property and its derivatives, there can be no presumption that such firms will be responsive to human need that is not effectively represented in the market by purchasing power. These facts constitute then a double incidence of market failure.

The neo-classical economic response to the problem of market failure is frequently given expression in what is known as the Coase Theorem. According to this result economic efficiency is maximized when agents are free to bargain and property rights are clearly specified. It matters little that the owners of such property are private individuals or a single public authority.

The Theorem is often applied to the problem of common public goods. In the absence of assignable property rights economic agents will have little incentive to engage in the efficient use of such resources often leading to their overexploitation. There are several reasons, however, why the Coase Theorem has doubtful application as a defence of private property rights in ideas. First, the need for assignable property rights does not definitively establish an argument for private property rights. There is no reason in principle why government might not efficiently allocate the use of common public goods according to some democratically chosen objective. Second, ideas are not the type of “common public goods” that the Theorem has in mind. Air, water, and in a bygone era, common pasture lands, are “public goods” in the sense that they share the characteristic of non-exclusiveness of use.

The need to institutionalize exclusive use is the motivation behind assigning property rights. What such goods do not share with ideas, however, is non-rivalry of use. One particular use by a particular individual(s) of “common public goods” such as air, water, and land may indeed preclude other uses by other people. This cannot be said of intellectual “goods and services” since one person’s use of an idea in no way conflicts with another’s use of it.

In fact, where intellectual property is concerned it probably makes much more sense to speak of “the tragedy of the anti-commons” in the manner of Heller and Eisenberg (1998). In this scenario, a resource (new knowledge) is subject to underuse when multiple owners of “upstream knowledge” have the right to exclude others leaving no one with an effective privilege of “downstream” use. This is perhaps best illustrated with the recent widely published announcements surrounding the mapping of the human genome. Private profit-seeking companies have taken out patents on thousands of DNA sequences that preclude their study by other scientists.

There is a huge speculative aspect to this behaviour since the patent-seeking companies are betting that by gaining control over a sequence of DNA, they will eventually come to possess the entire gene of which it is a constituent part. The gene in turn could be the key to an important (and profitable) diagnostic or treatment procedure or drug. The tragedy, however, is that this sort of speculation impedes the process of scientific discovery. More typically, market failure calls out for public policy relief in the form of market regulation or the direct provision of public goods. What is clear then is that some sort of market intervention seems inevitable where intellectually derived goods and services are concerned. The important question is whether this intervention takes the form of state-established monopoly rights to private firms to produce and distribute, or something else, *e.g.* direct state provision, market regulation of intellectual goods and services.

It is also clear that to this point much more has been accomplished in terms of providing intellectual property rights with international application than any other imaginable formula for treating the complex of questions surrounding knowledge based production. There is nothing in the recently established TRIPS agreement that addresses

the “tragedy of the anti-commons” problem, for example. The reasons for this preference are to be found in a consideration of international political economy rather than philosophically-based justificatory arguments. The main arguments of this paper may be summarized as follows: traditional justificatory defences of exclusive, private property rights apply very weakly, if at all, to intellectual property. Lockean and Hegelian (personality) justifications are found ultimately to appeal to essentially utilitarian arguments. Utilitarian arguments themselves, however, fail insofar as exclusionary property rights cannot be presumed to promote social welfare particularly in a world comprised of manifest market imperfections and inequality in the distribution of income and wealth.

It is evident that several important questions raised in this document require appeal to empirical evidence for their answers. Do the producers of intellectual goods, for example, respond to material incentives? Does the flow of innovation change as the legal regime governing intellectual property change? Economists and others have investigated these and related questions but have not arrived, to my knowledge, to any definitive results. Needless to say, however, the lack of widespread agreement on the empirical evidence has not served as a deterrent to the establishment of the TRIPS to protect intellectual property. The recent establishment of the TRIPS was largely the result of efforts made by the US Trade Representative on behalf of such patent-intensive US multinational companies as Pfizer Chemical and IBM.

These and other companies launched an educational campaign under the aegis of the GATT’s Advisory Committee on Trade Policy and Negotiations (ACTPN). This effort was directed at the USTR as well as to members of Congress to get them to understand the benefits that would follow from a stronger IPR regime. That the campaign was successful is evidenced both by the fact that the USTR made acceptance of the TRIPS a condition for US Participation in the Uruguay Round of the GATT talks as well as threatened recalcitrant alleged violators of US IPRs with unilateral trade sanctions. It borders on the trivial to note that justificatory arguments of the sort we have considered in this document are typically ideological rationalizations for the real political economic power wielded by groups whose interests they serve. It should not for this reason be concluded, however, that the arguments are themselves trivial.

THE STRUCTURE OF THE TRIPS AGREEMENT

There are seventy-three articles in the text, divided into seven different parts. Part I covers general provisions and basic principles. Part II pertains to standards for specific intellectual property rights and covers copyrights and related rights (also known as neighbouring rights), trademarks, geographical indications, industrial designs, patents, the layout designs of integrated circuits (mask works), and the protection of undisclosed information (trade secrets). There is also a section on the control of anticompetitive practices in contractual licenses. Part III addresses the enforcement of intellectual property rights, part IV deals with the acquisition and maintenance of intellectual property

rights, and part V covers dispute prevention and settlement. Part VI applies to transitional arrangements, and part VII addresses institutional arrangements.

BASIC PRINCIPLES AND OBLIGATIONS OF THE TRIPS AGREEMENT

Article 3 of the TRIPs Agreement states that each “Member shall accord to the nationals of other Members treatment no less favourable than that it accords to its own nationals” with respect to the protection of intellectual property, a condition known as national treatment.” The TRIPs Agreement also requires that every member be given most-favoured-nation (MFN) treatment with regard to intellectual property. It establishes that “any advantage, favour, privilege or immunity granted by a Member to the nationals of any other country shall be accorded immediately and unconditionally to the nationals of all other Members” (Article 4).

There are, however, exceptions to these obligations. With respect to national treatment, exceptions identified in international conventions — such as the Paris Convention, the Berne Convention, the Rome Convention, and the Washington Treaty — are recognized by the TRIPs Agreement. These exceptions include the suspension of privileges to rightsholders in the event of a national emergency. Each member country has to implement and enforce standards pertaining to all seven areas of intellectual property.

STANDARDS

This section encompasses copyrights, trademarks, other standards, and patents.

COPYRIGHTS

Copyright law protects a work of authorship from the time that it is created. It covers the original expression of an idea rather than the idea itself. Related rights (or neighbouring rights, as they are sometimes called) in turn protect the work of performers, phonogram (sound recording) producers, and broadcasters.

Article 9 of the TRIPs Agreement states that copyrights will protect works covered by Articles 1-21 of the Berne Convention. That excludes Article 6bis of the Berne Convention, which deals with “moral rights” — an author’s inalienable right to protect the integrity of his or her work and to object to changes that would be prejudicial to his or her honour or reputation.

Even though the TRIPs Agreement does not require the recognition of moral rights, India’s Copyright Act of 1957 recognized moral rights after it was amended in 1994. Prior to this round of amendments, the Copyright Act did not recognize Article 6bis of the Berne Convention, despite the fact that India was a member of the Berne Convention. The 1994 amendment to India’s Copyright Act establishes standards exceeding those laid down by the TRIPs Agreement. A few remaining areas have yet to be amended, but they will require only minor changes.

Article 9 of the TRIPs Agreement also requires members to recognize that “copyright protection shall extend to expressions and not to ideas, procedures, methods of operation or mathematical concepts as such.” Thus, copyright protection can be given only to the articulation and exposition of ideas, and not to the ideas themselves. For example, a mathematical formula cannot be copyrighted, though a product that results from that formula can be.

Indian legislation is not explicit in that respect, although established case law sets precedents in that direction. Indian courts have shown by their judgements that originality does not necessarily mean a new idea. Originality could imply a different form or a unique method of presentation. What is protected by the legislation is the *form* in which the ideas are expressed, not the idea itself.

Computer programmes and compilations of data are protected as literary works in Article 10 of the TRIPs Agreement. In that area, the Indian Copyright Act is satisfactory and protects computer software and compilations of data. Further, TRIPs stipulates that the term of protection extends for the life of the author of a work plus fifty years. Indian legislation goes a step further and protects the work for the author’s lifetime plus sixty years.

Article 14 of the TRIPs Agreement relates to the protection of the rights of performers, producers of phonograms, and broadcasting organizations. The TRIPs Agreement states that performers have the authority to prevent the “fixation of their unfixed performance and the reproduction of such fixation.” Performers also have the authority to prohibit public broadcasts of their live performances without due authorization from them. Producers of phonograms are awarded the right to “authorize or prohibit the direct or indirect reproduction of their phonograms.”

Broadcasters have the same rights as performers regarding the fixation of performances and the reproduction and broadcast of these performances. The term of protection for performers and producers of phonograms is fifty years from the date of the performance or the creation of the phonogram. Broadcasters are given protection for up to twenty years from the end of the calendar year in which the broadcast took place. Indian legislation covering “neighbouring rights” is similar in all but a few areas.

Section 37 of the Indian Copyright Act gives broadcasters a twenty-five-year term of protection, as opposed to the twenty years stipulated by the TRIPs Agreement. For performers’ rights, also, the Copyright Act provides twenty-five years of protection, in contrast with the fifty years stipulated by TRIPs. Rental rights are also covered by the disciplines of the TRIPs Agreement. They provide that, “at least” for computer programmes and cinematographic works, the title-holder should be allowed “to authorize or prohibit the commercial rental to the public of originals or copies of their copyright works” (Articles 11 and 12).

That obligation, however, applies to cinematographic works only when widespread copying linked to rental of these works is impairing the economic rights of the title-

holder. No parallel to rental rights exists in Indian legislation. Since there is no intrinsic opposition to that clause in the agreement, future amendments to the law will probably be made to accommodate the stipulations of the TRIPs Agreement. The discussion indicates that Indian legislation with regard to copyrights is relatively uncontroversial. In some areas, the standards are more stringent and explicit than in the TRIPs Agreement. Since the amendment of the Copyright Act in 1994, Indian legislation is almost in line with the provisions of the TRIPs Agreement.

The strength of copyright legislation reflects the interests of domestic industries affected by the lack of strong intellectual property rights. For many years the domestic film industry, one of the largest and most influential industries in the country, has been clamoring for stronger protection of its intellectual property. Rampant piracy has taken a toll on the profitability of that industry. Similarly, the stronger copyright law also benefits the burgeoning software industry based in the Bangalore, the “Silicon Valley” of India. The software industry has become one of the fastest growing industries, with an international reputation.

TRADEMARKS

Trademarks are commercial symbols used to identify goods and services or their producers. The guidelines defining what constitutes a trademark, as laid down by the TRIPs Agreement, do not conflict seriously with the Indian counterpart of trademark protection, the Indian Trade and Merchandise Marks Act of 1958, or with existing Indian case law. By virtue of an application for a trademark, an enterprise selling its goods under a particular mark acquires a limited exclusive right to use that mark for those products. In India, registration of a trademark is essential to claim protection under the existing Trade and Merchandise Marks Act of 1958. Proprietors of unregistered trademarks may also sue infringers for misrepresenting infringed goods as legitimate.

Article 15 of the TRIPs Agreement provides for registration of trademarks and of service marks. The Indian counterpart of service marks does not exist. To remedy that deficiency, a new bill was introduced to provide for the protection of service marks and “well known” marks. After a comprehensive review of the Trade and Merchandise Marks Act of 1958, the Trade Marks (Amendment) Bill was introduced in the Parliament in 1993. It was intended to repeal and replace the 1958 act.

The bill proposed to introduce protection for service marks. The bill also provided for a greater degree of protection for trademarks and a greater degree of punishment for infringement. Among other proposals, the Trade Marks (Amendment) bill also extended the period of protection from seven to ten years. The bill was examined by the Department Related Parliamentary Committee for Industry, and a modified bill was passed by the *Lok Sabha* (lower house of Parliament) in 1995. But in the *Rajya Sabha* (upper house) it was referred to a Select Committee of its members. The bill has since lapsed. As will be discussed later, the Trade and Merchandise Marks Bill has faced the same problems

as the Patents Bill. The TRIPs Agreement states that members may allow registrability to depend on use. In contrast, the Indian Trade and Merchandise Marks Act of 1958 stipulates that the registrar cannot deny an application for registration of a trademark, even in light of a registrant's apparent intentions not to use the trademark, so long as the registrar is satisfied that a company is about to be formed.

Article 19(1) of TRIPs states that registration of a trademark may be canceled only after an uninterrupted period of three years of non-use, unless valid reasons based on the existence of obstacles to such use are shown by the trademark owner. Clear guidelines are set forth in section 46.1 of the Indian Trade and Merchandise Marks Act of 1958, under which trademarks can be taken off the register. These include the intention not to use the trademark or the fact that, one month after application for renewal of the trademark, it had not been used for a period of five years or longer (as opposed to three years, as provided by TRIPs).

In India, courts have taken the view that obstacles to non-use may include import restrictions and prohibitive tariffs. As for the term of protection, Article 18 of the TRIPs Agreement defines it as a period of seven years, renewable indefinitely. The Indian legislation agrees with TRIPs in that respect and covers trademarks for the same period. Finally, consistent with Article 21 of the TRIPs Agreement, Indian legislation does not provide for compulsory licensing for trademarks.

The changes that should be made to trademark legislation in India are not very radical. Yet despite the fact that, legally, the TRIPs Agreement and Indian legislation are mostly congruent, for many years foreign trademarks were difficult to protect in India. The Foreign Exchange Regulation Act (FERA) of 1973 placed stringent restrictions on the use of foreign brand names in India by prohibiting any type of payment (direct or indirect) for their use in India. That meant that foreign exchange regulations made it difficult for a company using a foreign trademark to make payment for the use of this trademark to the parent company. So foreign trademarks were unprotected by default — not because of legislation *per se* but because of Indian financial regulations. But in 1991 all that changed, with the relaxing of controls and the general liberalization of the economy. In 1992, foreign brand names were allowed into the country by virtue of a nonlegislative policy change made by the government of India.

A trickier hurdle remained, however. The common law rule in India was that a trademark owner could prove its goodwill in a mark only by proving actual sales of the relevant branded goods in India. International reputation alone could not protect a mark in India. That policy created an interesting paradox. Since, by definition, foreign branded goods could not be sold within the country, there was no way of proving actual sales of a particular branded good in India. And since international reputation was not enough to prove goodwill in a mark, domestic firms could sell products under a foreign brand name and claim it as their own. But some recent court decisions have changed that aspect of Indian common law, bringing trademark law up to date with the TRIPs Agreement.

OTHER STANDARDS

With regard to *industrial designs*, the main difference between the Indian legislation and the TRIPs Agreement involves the period of protection. Although Article 26(3) of the TRIPs Agreement stipulates a ten-year term of protection for industrial designs, the Indian Patents and Design Act provides for a period of five years. *Geographical indications* are not explicitly mentioned in Indian law. The TRIPs Agreement does not require any special statute to be set up as long as legislation provides legal means for redress. India's Trade and Merchandise Marks Act allows redress, and case law establishes the recognition of geographical indications in Indian courts.

Until recently, the Indian government believed that the Trade and Merchandise Marks Act were adequate to address issues concerning geographical indications, and that new legislation was not required. But in reality, there is a strong need for concrete legislation covering geographical indications. That need was highlighted by the furor over the patenting of a new strain of rice by an American company, RiceTec Inc.

There was outrage when that new strain was named *basmati*. India and Pakistan have been arguing that the term *basmati* is used for a variety of rice that is unique to the Indian subcontinent and that any such misrepresentation would be detrimental to basmati rice exports from the two countries. RiceTec Inc., which applied for a patent in July 1994 and received it in September 1997, is already marketing its product globally under brand names like Kasmati and Texmati. The company claims its product is a "basmati-type" rice. The issue, therefore, is not one of opposition to the patent itself but to calling the patented brand of rice *basmati*. To reflect that concern regarding the limited scope of the TRIPs Agreement with regard to geographical indications, a recent communication to the World Trade Organization General Council proposed some implementation issues it wanted addressed at the Seattle ministerial meeting. It proposed that "additional protection for geographical indications shall be extended for products other than wines and spirits."

India does not have legislation that corresponds with Article 35 of the TRIPs Agreement, which protects the layout designs (topographies) of integrated circuits. But no new legislation is required as long as means of redress for infringement exist and the term of protection is at least ten years. That should not be a problem, because India is a signatory of the Washington Treaty on Integrated Circuits. Layout designs can also be protected through special legislation or through laws on copyrights, utility models, or industrial designs.

India similarly lacks a counterpart for the protection of undisclosed information (Article 39), and the legislature will have to create a statute to protect undisclosed information or trade secrets before the end of the ten-year transition period. On the whole, most components of TRIPs are provided for by Indian legislation. Only minor changes will be needed for India to comply fully with the TRIPs Agreement. These are unlikely to pose a major problem, because there is generally little opposition within the Indian polity to most of the TRIPs provisions.

Those regarding copyrights, trademarks, undisclosed information, geographical indications, layout designs, and industrial designs have been accepted without much resistance. That acceptance reflects India's self-interest. For example, copyright protection is stronger under certain Indian laws than it is in the TRIPs Agreement, and Indian courts in recent times have enforced decisions on copyright disputes with criminal procedures and fines. The importance given to copyright protection probably reflects the demands of the Indian film industry, which is the largest in the world, and the burgeoning software industry based in Bangalore. Similarly, Indian courts reflect the increasing need to attract foreign investment by ensuring trademark protection for international brands. The interests of an increasingly competitive domestic industry are also satisfied by greater protection of domestic industry trademarks.

In general, most of the provisions of the TRIPs Agreement coincide with India's self-interest. But one area elicits little or no consensus within the country, and controversy about it seems never to wane; that is patent protection. It remains a difficult area for India because the interests of the country do not coincide with the provisions of the TRIPs Agreement.

PATENTS

The area of patents has been the most controversial element of the TRIPs Agreement. Until a few years ago, few lay people knew much about patents. The subject of patents and patent protection belonged to the arcane world of intellectual property lawyers. But the issue has caused diverse people, from industry barons to university students to farmers to activists, to stream onto the streets to protest against the patent provisions of TRIPs. What does TRIPs stipulate and how does it differ from Indian legislation in this area? Patents protect inventions that are novel, non-obvious, and useful. A patentee has the right to exclude others from using, making, or selling the patented invention for a limited period of time. Article 27 of the TRIPs Agreement requires that all signatory countries put in place an effective patent system for essentially all branches of technology.

The TRIPs Agreement also requires that patents run for twenty years from the date the patent application is filed (Article 33). Patent rights are clearly presented and include the right to prevent third parties from making, using, offering for sale, selling, or importing the product without prior consent from the patent holder (Article 28).

Remedies for infringement include the availability of damages (Article 45) and injunctive relief (Article 44). TRIPs shifts the burden of proof, thus making process patents easier to enforce than they would be under a regime following the ordinary rules about burden of proof. The TRIPs Agreement affords protection to all fields of technology. But a controversial aspect of Article 27 is paragraph 3 and especially subparagraphs 3(b). Article 27 excludes the granting of patents for "diagnostic, therapeutic and surgical methods for the treatment of humans or animals... and plants and animals other than micro-organisms, and essentially biological processes for the production of plants or animals other than non-biological and microbiological processes."

The method of protection is left up to the members, but it must be “effective.” Article 27 makes clear that plant varieties must be protected. Indian legislation, however, does not provide for the patenting of life forms of any kind — not even plant varieties. The issue of protecting plant varieties and, generally, the patenting of life forms has been an emotional one. Some people argue that if plant varieties are protected, the large multinational firms that usually hold these patents will raise the price of seeds, making them unaffordable for farmers.

The issue of plant varieties, a highly contentious one, has not been dealt with in this stage. It is properly the subject of an entire stage. My focus will be on patents as they relate to pharmaceuticals. There are five main differences between existing Indian patent legislation and the TRIPs Agreement. The first difference concerns the coverage of patents. Section 5 of the Indian Patents Act of 1970 states the following:

In the case of invention,:

- Claiming substances intended for use, or capable of being used, as food or as medicine or drug,
- Relating to substances prepared or produced by chemical processes (including alloys, optical glass, semi-conductors and inter-metallic compounds), no patent shall be granted in respect of claims for the substances themselves, but claims for the methods or processes of manufacture shall be patentable.

Thus, the Indian Patents Act of 1970 did not allow product patents in the areas of food, medicines, drugs, and chemicals. *Only processes could be patented.* That notion conflicts with TRIPs, which clearly stipulates that members must provide patents, whether product or process, for all fields of technology.

The second discrepancy between the Indian Patents Law and that of the TRIPs Agreement concerns the duration of protection for patents. Section 53(1) of the Patents Act establishes that process patents in the areas of food, medicines, and drugs have a term of five to seven years, and any other patent has a fourteen-year term. TRIPs, however, states that patent duration uniformly must be for twenty years (Article 33). Working a patent, or commercially producing the subject matter of the patent, is the third main difference between Indian law and the TRIPs Agreement. Under existing Indian legislation, importing a product is not regarded as equivalent to working a patent.

Indian law makes clear that patents are granted to encourage production of inventions in India, not merely to act as vehicles of monopoly power for the patent holders importing the product. TRIPs, however, mandates that importation of a patented product is tantamount to working the patent: importing a patented product entitles the patent holder to the same privileges and protection as would producing the product in the country. In short, there can be no distinction between imported and domestic products with regard to patent protection.

The fourth and probably most contentious area for India is the issue of compulsory licensing. To prevent the patent holder from misusing or abusing the temporary monopoly awarded, Indian law allows compulsory licenses to be awarded to third parties.

Except in cases of national emergency, like war, TRIPs makes the grant of a compulsory license subject to a number of conditions. While TRIPs does not rule out compulsory licensing altogether, it does make such a practice difficult. The fifth problematic issue is the reversal of the burden of proof. Ironically, while other portions of the TRIPs patent regime have attracted a great deal of negative publicity, reversing the burden of proof has been ignored.

Article 34 states: If the subject matter of a patent is a process for obtaining a product, the judicial authorities shall have the authority to order the defendant to prove that the process to obtain an identical product is different from the patented process. Therefore, Members shall provide, *in at least one of the following circumstances*, that any identical product when produced without the consent of the patent owner shall, in the absence of proof to the contrary, be deemed to have been obtained by the patented process:

- (a) If the product obtained by the patented process is new;
- (b) If there is a substantial likelihood that the identical product was made by the process and the owner of the patent has been unable through reasonable efforts to determine the process actually used.

Any Member shall be free to provide that the burden of proof indicated in paragraph we shall be on the alleged infringer only if the condition referred to in subparagraph (a) is fulfilled or only if the condition referred to in subparagraph (b) is fulfilled.

TRIPs stipulates, pursuant to clause (b), that when a plaintiff holding a process patent believes the patent is being infringed and has made “reasonable” but unsuccessful efforts to determine the process used by the alleged infringer, the burden of proof will be shifted to the defendant. The rationale behind this provision is that, when a plaintiff holding a process patent discovers a product that he thinks has been produced using his patented process, it may not be possible to prove infringement because the plaintiff would not have access to the technology used by the alleged infringer. Under such circumstances, only the alleged infringer has access to proof of whether or not infringement is occurring. The alleged infringer must thus prove that the product is non-infringing, that is, that it has been produced using a different process. That provision of the agreement has obvious applications to the pharmaceutical industry and, in particular, to biotechnological processes.

Clause (a), however, implies that even when a totally new product is produced, the defendant must prove that an alternative and unique process has been used. Governments are bound to implement either clause (a) or clause (b), but not necessarily both. It is likely that the Indian government will choose clause (b) over clause (a).

Although India has no explicit law regarding reversal of the burden of proof, there are precedents to indicate that the idea is not totally alien to Indian law. For example, section 78 of the Trade and Merchandise Marks Act of 1958 states, “Any person who... (c) makes, disposes of, or has in his possession, any die, bloc, machine, plate or other instrument for the purpose of falsifying, or being used for falsifying, a trade mark... shall, unless he proves that he acted without intent to defraud, be punishable” (emphasis

added). That example does not relate to patents in particular, although the principle is similar. India has not implemented that aspect of TRIPs and will have to bring the provision explicitly within the purview of patents, particularly process patents. Though Indian legislation with regard to patents is not in compliance with the TRIPs Agreement at present, that is not for lack of trying. After the conclusion of the Uruguay Round, there were very few stipulations in the Marrakech Agreement that required immediate action by developing countries. But two articles in the TRIPs Agreement, Articles 70.8 and 70.9, did require immediate action.

Developing countries did not need to provide patent protection for pharmaceutical and agricultural chemical products until the year 2000. Article 70.8 stipulates that those countries must create a system by which applications for those types of products can be accepted and placed in a “mailbox” for examination after the year 2000. And Article 70.9 stipulates that if those claimants have been given patents in other WTO-member countries, then “exclusive marketing rights” (EMRs) must be awarded to the patent holders for a period of five years in the country where there is no immediate protection. If India failed to allow for those two provisions, it would jeopardize its place in the WTO and its right to utilize the transition period.

At the time, the minority government of P. V. Narashima Rao of the Congress Party knew full well that it would not be able to get any amendments to the Patents Act of 1970 approved. They realised that support would not be readily forthcoming from the Bharatiya Janata Party (BJP) and the Left parties. To get around that problem, the government issued an ordinance at the end of December 1994, immediately *after* the end of the parliamentary session. Ordinances have the same force as an Act of Parliament, and are considered as law. But those ordinances must be accepted by both houses within six weeks of the beginning of a new session of Parliament.

Presidential ordinances were issued for copyrights and patents, pending approval from Parliament. The patent ordinance included all the amendments needed to comply with the provisions of the TRIPs that came into force immediately. Early in 1995 the patents ordinance was accepted by the *Lok Sabha* of Parliament, where the government had a small working majority, but was stalled in the *Rajya Sabha*, where it had no majority.

Ultimately, the ordinance collapsed. The BJP government promulgated another ordinance on January 8, 1999, with a view to fulfilling India’s obligations under Articles 70.8 and 70.9. An identical Patents (Amendment) Bill was passed by the *Rajya Sabha* in its last session in 1998, though the bill is pending consideration in the *Lok Sabha*.

To compensate for the failure to meet its WTO obligations, the Indian government instituted an administrative order, and has been accepting applications ever since. The United States has long been demanding implementation of Articles 70.8 and 70.9 of the TRIPs Agreement and was not satisfied with the administrative order, arguing that it failed to fulfill the conditions set forth in TRIPs. In December 1997, the WTO Dispute Settlement Body (WTO-DSB) issued an adverse ruling against India, which was adopted

in the third week of January 1998. The WTO-DSB, in turn, held that, since the Indian Patents and Design Act does not recognize product patents and merely provides for process patents, an administrative order could not override the provisions of a parliamentary act.

India sought an extension of time to alter the law, owing to special circumstances such as a “government in transition” and the existing political impasse. Soon after the ruling, negotiations began to take place between India and the United States, as stipulated by the dispute settlement rules, to work out a time frame. Eventually, India decided to invoke the more formal method of arbitration to gain time. In effect, Indian legislation with regard to patents is now exactly where it was before TRIPs. It has become increasingly difficult to get new patent legislation passed in the current climate of political uncertainty. So far we have discussed the TRIPs Agreement and how Indian legislation differs with respect to intellectual property rights. Most of the provisions of TRIPs have parallels in Indian legislation, but the most contentious area remains the issue of patents. There is much disagreement on the subject within the country, and opposition still remains strong against the patent component of the TRIPs Agreement. But the opposition to stronger patent laws is nothing new.

THE WEAKENING OF THE PATENT REGIME

Contrary to popular perception, India has had a long tradition of intellectual property protection, a legacy handed down by the British. British India enacted its first law regarding domestic patent rights in 1856, granting exclusive privileges to inventors of new products for a period of fourteen years. That act was renewed in 1859, with revisions based on the British Patent Act of 1852. Subsequently, the Indian Patents and Design Act of 1911 replaced all previous legislation.

Cyber Security as Science

A science of cyber security has to deal with a combination of peculiar features that are shared by no other area of study. First, the background on which events occur is almost completely created by humans and is digital. That is, people built all the pieces. One might have thought that computers, their software, and networks were therefore completely understandable. The truth is that the cyber-universe is complex well beyond anyone's understanding and exhibits behaviour that no one predicted, and sometimes can't even be explained well. On the positive side, the cyber-universe can be thought of as reduced to the 0s and 1s of binary data. Actions in this universe consist of sequences of changes to binary data, interleaved in time, and having some sort of locations in space. One can speculate as to why mathematics is so effective in explaining physics, but the cyber-world is inherently mathematical. Mathematics is a natural way for reasoning about it and this point of view is a theme that appears repeatedly in this report. Second, cyber security has good guys and bad guys. It is a field that has developed because people have discovered how to do things that other people disapprove of, and that break what is thought to be an agreed-upon social contract in the material world. That is, in cyber security there are adversaries, and the adversaries are purposeful and intelligent.

Thus, the reasoning in cyber security has to be both about the constructed universe and the actions and reactions of the adversaries. Definitions and concepts will shift over time in cyber security, as in all sciences. A successfully chosen name is a bridge between scientific knowledge and common sense, between new experience and old habits. The conceptual foundation of any science consists of a complicated network of

names of things, names of ideas, and names of names. It evolves itself, and its projections on reality changes. This quote echoes the observations of a presentation to JASON by Peter Gallison of Harvard in which he compared the type of science encountered in cyber security to a “Manichaeian science” or science in the presence of adversaries. Particularly important for this type of science (and indeed any branch of science) was the development of a common language so that precise arguments could be formulated that addressed directly the problems of interest. We emphasize this point because the definition of cyber security itself is imprecise. The subject includes correctness properties like confidentiality and integrity, and liveness properties such as are associated with denial-of-service attacks, together with a growing set of unpleasant surprises.

SECURITY DEGRADES OVER TIME

We observed earlier that success in cyber security requires constant attention. Not only do the threats and attacks change, but the old techniques tend to become less effective. Technology advances, and the properties of the security system become better understood.

UNIX PASSWORDS

The original UNIX passwords were up to 8 ASCII non-control characters. The passwords aren’t stored directly. Rather hashes (some one-way function) are stored. In modern systems these hashes aren’t easily read, but if a machine is compromised, then the hashes can be taken away, and one can attempt to break passwords by making up likely candidates, applying the one-way function, and seeing if the result matches any of the hashes. There are about 2 original UNIX passwords, which seemed like quite a lot at the time. Nowadays, one could imagine trying about 32 million guesses (2) per second on a single CPU core, so testing the entire space would only take about 3 days on 1,000 cores. That’s not a particularly large computer, but the calculation is also embarrassingly parallel, so the 1,000 cores could be 1,000 machines in a (small) botnet. Or one could use 9,000 machines, still not a large botnet, and do the whole calculation in one night. For completeness we note that there has been a lot of work on making it harder to guess passwords; for instance, by using much slower one-way functions and much larger spaces. Another approach is to force people to use more complex passwords, but that has other sociological ramifications which are unfortunately very difficult to deal with in terms of any science. An extreme version turned up in the recent Russian spy case, where “Ricci said the steganographic programme was activated by pressing control-alt-E and then typing in a 27-character password, which the FBI found written down on a piece of paper during one of its searches.”

Lock Bumping

A second example of security decreasing over time comes from the common locks people use on doors. A simple technique called “lock bumping” makes all these locks

quite easy to open with just a little practice. The locks were, of course, vulnerable, but it didn't much matter until the technique was popularized starting about 2005. There are plenty of explanatory videos online. The implication for cyber-security is that the nature of the attack can change over time. It will not in general be possible to anticipate all attacks. For this reason it becomes necessary to constantly survey the extant modes of attack to understand the nature of current technology as well as continually attack one's own systems in the hopes of finding heretofore undiscovered attacks.

The Role of Secrecy

All security regimes include some secrets. The TSA doesn't explain everything it is looking for; the credit card processors don't fully explain their criteria, and so on. In the computer science community, security by obscurity is generally considered bad, probably based on examples of awed cryptography. There seems to be little theoretical work on this subject. We mention a few examples of keeping (and not keeping) secrets. Commercial anti-virus programmes, of the sort that run on your home computer, reveal some of the weaknesses of totally public defences. In principal, a malware author could just buy these, and adjust the malware until none of them detect it. The market has made this easier; there are online services that for a fee will run uploaded malware through a large number of commercial programmes, and report the result.

Game theory and economics teach us that keeping our private valuations secret sometimes adds value. Much effort has gone into auction mechanisms that get bidders to reveal their private valuations. On the other hand, under various circumstances, this "asymmetric information" can lead to lemon markets, where either no market exists, or only people who are forced to will trade in the market. Secrecy might also be required by the rules. For instance, the DoD might use logs of network traffic to tune its network defences. These logs might easily contain material that should not be made public. For these reasons, there may be some benefit in exploring the role an enhanced role for secrecy and obfuscation, particularly for DoD systems. Finally, one might ask if keeping the source code secret adds much to security.

First, there may be other reasons for keeping source code secret, such as the trade secrets incorporated in it. Certainly attackers with source code are likely to proceed differently, as they have a better chance of finding attacks based on logical errors, and this would be especially valuable for targeted attacks. On the other hand, there seems to be no particular shortage of effective attacks when source code is unavailable. But the code owner may have no realistic alternatives to making the source available. As an example consider the following announcement: "The agreement that we signed with the FSB is an extension of Microsoft's Government Security Programme (GSP)," Microsoft said in a statement on Friday. "The purpose of the GSP is to increase trust with national governments. In the case of the Russian agreement, GSP participation will facilitate the development of the next generation of secured solutions for Russian government agencies based on the latest Microsoft technologies and Russian cryptography."

CYBER SECURITY: A PROBLEM OF NATIONAL IMPORTANCE

TRUSTING SYSTEMS IN A DANGEROUS WORLD

The Nation's information technology (IT) infrastructure, still evolving from U.S., technological innovations such as the personal computer and the Internet, today is a vast fabric of computers – from supercomputers to handheld devices – and interconnected networks enabling high-speed communications, information access, advanced computation, transactions, and automated processes relied upon in every sector of society. Because much of this infrastructure connects one way or another to the Internet, it embodies the Internet's original structural attributes of openness, inventiveness, and the assumption of goodwill. These signature attributes have made the U.S., IT infrastructure an irresistible target for vandals and criminals worldwide. The PITAC believes that terrorists will inevitably follow suit, taking advantage of vulnerabilities including some that the Nation has not yet clearly recognised or addressed. The computers that manage critical U.S., facilities, infrastructures, and essential services can be targeted to set off system wide failures, and these computers frequently are accessible from virtually anywhere in the world via the Internet.

THE INFORMATION TECHNOLOGY INFRASTRUCTURE IS 'CRITICAL'

Most Americans see and use the components of the IT infrastructure – mainly desktop computers connected to the Internet – that enable e-mail, instant messaging, exchange and downloading of sound and images, online shopping, information searches, interactive games, and even telephony. Americans also work with the information technologies that drive day-to-day operations in industry and government and are relied upon by organisations large and small for a range of functions including design, manufacturing, inventory, sales, payroll, information storage and retrieval, education and training, and research and development. In fact, economists credit successful applications of information technologies throughout the economy for the spectacular gains in U.S., productivity over the last decade.

Less visible, and certainly less well understood, is the fact that these technologies – computers, mass storage devices, high-speed networks and network components such as routers and switches, systems and applications software, embedded and wireless devices, and the Internet itself – are now also essential to virtually all of the Nation's critical infrastructures. Computing systems control the management of power plants, dams, the North American power grid, air traffic control systems, food and energy distribution, and the financial system, to name only some.

The reliance of these sensitive physical installations and processes on the IT infrastructure makes that infrastructure itself critical and in the national interest to safeguard. The electric power generation industry, for example, relies on a range of IT

systems and capabilities. As in other industries, power companies implement business management systems for administrative and information services. But the power industry uses much more information technology.

It relies on supervisory control and data acquisition (SCADA) systems to collect information about system operation, help regulate and control power generation, optimise power production, respond to changing power demands and system parameters, control distribution, and coordinate among the various generation and storage facilities within a power company system. Increasingly, SCADA systems are also used to integrate electric companies into regional or national power grids to optimise power production, minimize production and distribution costs, and provide backup services. This requires a private network that often includes links to the Internet. A cyber attack that disables key Internet nodes could disrupt the power network's communications. And if an entity within the private network is compromised, an attacker could gain direct control of the SCADA systems and their data and operation.

Today, the Internet also is used to manage essential services provided by business and government, such as electronic financial transactions, law enforcement dispatch and support, emergency response and community alerts, and military communications. Banks, for example, rely on extensive distributed Internet and information services, both for customer interaction and in inter-bank operations. To assure reliability and security of its most sensitive systems, the banking industry, like the power industry, uses private networks and is vulnerable to cyber attacks that cripple Internet nodes and/or result in unauthorised access to data and services. Such shared Internet links, for example, enabled the "Slammer" worm to disable a major bank's ATM system and an airline's computer system, even though they were not directly connected to the Internet.

During a national emergency, it is imperative that the Nation's communications infrastructure be available for emergency response coordination. Today, that vital infrastructure is vulnerable to a variety of denial of service attacks, including the release of simple viruses and worms that can disrupt Internet communications as well as more sophisticated attacks in which modems from compromised servers are used to flood key parts of the telephone network. The latter example demonstrates how vulnerability in one system (*e.g.*, the Internet) can be exploited to attack a totally separate system (*e.g.*, the telephone network). These examples illustrate how computing and computer communications have become integral to virtually every domain of activity in the U.S., today. Those systems are interconnected and interdependent in highly complex ways, which are often surprisingly fragile.

UBIQUITOUS INTERCONNECTIVITY = WIDESPREAD VULNERABILITY

The Internet – now a global network of networks linking more than 300 million computers worldwide – was designed in a spirit of trust. Neither the protocols for neither network communication nor the software governing computing systems (nodes)

connected to the network were architected to operate in an environment in which they are under attack. Indeed, the protocols used by the Internet today are derived from the protocols that were developed in the 1960s for the Federal government's experimental ARPANET. Only a few researchers used ARPANET and they were trusted to do no harm. The civilian networks, such as NSFNET, that developed from ARPANET into the Internet likewise did not incorporate security technologies at the system software or network protocol levels

Ubiquitous interconnectedness – first exhibited by the Internet and further extended in local area networks, wide area networks, and wireless and hybrid networks – has generated whole new industries, rejuvenated productivity in older ones, and opened new avenues for discourse and education and an unprecedented era of collaborative science and engineering discovery worldwide. That is indeed good news. The bad news is that ubiquitous interconnectivity provides the primary conduit for exploiting vulnerabilities on a widespread basis. Despite efforts in recent years to add security components to computing systems, networks, and software, the acts of a hostile party – whether a terrorist, an adversary nation, organised crime, or a mischievous hacker – can propagate far and wide, with damaging effects on a national or international scale.

For example:

- In the past several years, worms such as Code Red, which defaces World Wide Web sites and/or launches distributed denial of service (DDoS) attacks, and Slammer, which severely degraded the Bank of America's ATM network in January 2003, have caused damage estimated in the billions of dollars.
- The Department of Defence responded to the Code Red worm by disconnecting its unclassified network (NIPRnet) from the Internet to protect it from infection. This protective measure disabled the Army Corps of Engineers' control of the locks on the Mississippi River, since, the NIPRnet was used to transmit commands to the locks through the Internet.
- By using a laptop computer and radio transmitter, a former contractor for an overseas wastewater system was able to assume command of hundreds of control systems that manage sewage and drinking water. Over a period of two months, hundreds of thousands of gallons of putrid sludge were intentionally released from the wastewater system.
- Many businesses are now being attacked by cyber extortionists who demand payment in return for not attacking the businesses' Web presence. Seventeen percent of the 100 companies surveyed in a 2004 poll by Carnegie Mellon University-*Information Week* reported being the target of some form of cyber extortion.
- Identity theft is a rapidly increasing problem for Internet users. One of the simplest methods of stealing a user's identity is known as "phishing," a technique that uses fake e-mail messages and fraudulent Web sites to fool recipients into divulging personal financial data. Consumers Union estimates

that 1 per cent of U.S., households fell victim to such attacks at a cost of \$400 million in the first half of 2004.

SOFTWARE IS A MAJOR VULNERABILITY

Network connectivity provides “door-to-door” transportation for attackers, but vulnerabilities in the software residing in computers substantially compound the cyber security problem. As the PITAC noted in a 1999 report, the software development methods that have been the norm fail to provide the high-quality, reliable, and secure software that the IT infrastructure requires. Software development is not yet a science or a rigorous discipline, and the development process by and large is not controlled to minimize the vulnerabilities that attackers exploit. Today, as with cancer, vulnerable software can be invaded and modified to cause damage to previously healthy software, and infected software can replicate itself and be carried across networks to cause damage in other systems.

Like cancer, these damaging processes may be invisible to the lay person even though experts recognise that their threat is growing. And as in cancer, both preventive actions and research are critical, the former to minimize damage today and the latter to establish a foundation of knowledge and capabilities that will assist the cyber security professionals of tomorrow reduce risk and minimize damage for the long term. Vulnerabilities in software that are introduced by mistake or poor practices are a serious problem today. In the future, the Nation may face an even more challenging problem as adversaries – both foreign and domestic – become increasingly sophisticated in their ability to insert malicious code into critical software.

ATTACKS AND VULNERABILITIES ARE GROWING RAPIDLY

Today, the threat clearly is growing. Most indicators and studies of the frequency, impact, scope, and cost of cyber security incidents – among both organisations and individuals – point to continuously increasing levels and varieties of attacks. The data show that the total number of attacks – including viruses, worms, cyber fraud, and insider attacks in corporations – is rising by over 20 percent annually, with many types of attacks doubling in number.

For example, 83 percent of financial service organisations experienced compromised systems in 2003, more than double the percentage in 2001. Moreover, the reported level of security incidents almost certainly understates the actual level. There are few incentives – but strong disincentives – for large organisations to report incidents in a public forum. Targets of cyber attacks typically are concerned that widespread disclosure of their victimisation could shake public confidence in their operations, not to mention attract other attackers. Technology-oriented indicators clearly reflect the rapid growth in the rate of cyber attacks. For example, ICSA Labs reports that the monthly percentage of personal computers infected by a virus has grown from 1 percent in 1996 to over 10 percent in 2003.

From January to June of 2004, the rate at which new hosts were compromised and incorporated into “bot armies” rose from well under 2,000 a day to more than 30,000 a day. When compromised hosts are incorporated into bot armies, they can be used as platforms for launching denial of service attacks against a given target or to distribute “spam” e-mail without the knowledge or consent of the owners or operators. Trends within large organisations are also disturbing. For example, the percentage of organisations that experienced virus disasters (defined as those with more than 25 simultaneous infections or with major impact from infection) has grown nearly every year over the last decade, with 92 percent of organisations reporting such incidents during 2003. Symantec reports that 40 percent of the networks controlled by the Fortune 100 companies were exploited to originate hostile worm traffic, despite the fact that these companies have taken a variety of protective measures.

The cost, downtime, and days to recover from significant virus events have also trended upward for each of the past nine years. Meanwhile, the number of identified system and network vulnerabilities has also risen. The Computer Emergency Response Team Coordination Center (CERT/CC) at Carnegie Mellon University reports that 3,780 new electronic vulnerabilities were published in 2004, more than a 20-fold increase from 1995.

Once published, most of the details of vulnerabilities are available for miscreants to begin attacking or developing attack tools and techniques, thus forcing users and organisations to scramble to assure that their defences are adequate. The Symantec security threat report notes that in the first half of 2004, for example, the average time between the public disclosure of vulnerability and the release of an associated exploit was 5.8 days, substantially reduced from the months estimated in prior years. In fact, many IT system designs continue to incorporate characteristics that make these systems vulnerable to attack. In some instances, system designs may be pushing the state of the art, so their vulnerabilities may not be understood until they are deployed.

In other instances, vulnerabilities may be designed into systems because the developers lack technical knowledge or fail to execute best practices. In this brief report, PITAC can point to only a few examples to provide the reader with a sense of the vulnerabilities of IT systems. But it is clear that without action, IT vulnerabilities will become more severe, as computers, cell phones, and embedded systems proliferate globally and as expanding “always-on” high-speed or broadband connections enable attacks to propagate more rapidly and with more force than the occasionally connected low-bandwidth modems that were the norm until recently.

ENDLESS PATCHING IS NOT THE ANSWER

A broad consensus among computer scientists is emerging that the approach of patching and retrofitting networks, computing systems, and software to “add” security and reliability may be necessary in the short run but is inadequate for addressing the Nation’s cyber security needs.

As computer security expert and PITAC member Eugene Spafford testified before the House Science Committee:

- Security cannot be easily or adequately added on after the fact and this greatly complicates our overall mission. The software and hardware being deployed today have been designed by individuals with little or no security training, using unsafe methods, and then poorly tested. This is being added to the fault-ridden infrastructure already in place and operated by personnel with insufficient awareness of the risks. Therefore, none of us should be surprised if we continue to see a rise in break-ins, defacements, and viruses in the years to come.

Granted, our IT infrastructure may be less secure right now than it could be if all known security best practices were applied everywhere. But Professor Spafford's comment suggests that, even if all best practices were fully in place, in the absence of any fundamental new approaches we would still endlessly be patching and "plugging holes in the dike."

FUNDAMENTALLY NEW SECURITY MODELS, METHODS NEEDED

We urgently need to expand our focus on short-term patching to also include longer-term development of new methods for designing and engineering secure systems. Addressing cyber security for the longer term requires a vigorous ongoing programme of fundamental research to explore the science and develop the technologies necessary to design security into computing and networking systems and software from the ground up. Fundamental research is characterised by its potential for broad, rather than specific, application and includes farsighted, high-payoff research that provides the basis for technological progress. The vast majority of cyber security research conducted to date has been based on the concept of perimeter defence.

In this model, what is "inside" an information system or network is protected from an "outside" attacker who tries to penetrate it to gain access to or control its data and system resources. However, once the perimeter is breached (whether by virtue of a technical weakness such as a software vulnerability or an operational weakness such as an employee being bribed or tricked to reveal a password), the attacker has entirely free rein and can compromise every system connected in a network with not much more effort than is required to compromise only one.

This weakness of the perimeter defence strategy has become painfully clear. But it is not the only problem with the model. The distinction between "outside" and "inside" breaks down amid the proliferation of wireless and embedded technologies connected to networks and the increasing complexity of networked "systems of systems." One element of a more realistic model for cyber security may be a principle of mutual suspicion: Every component of a system or network is always suspicious of every other component and access to data and other resources must be constantly reauthorised. More generally, cyber security would be an integral part of the design process for any

large, complex system or network. Security add-ons will always be necessary to fix some security problems, but ultimately there is no substitute for systemwide end-to-end security that is minimally intrusive.

CENTRAL ROLE FOR FEDERAL R&D

“The National Strategy to Secure Cyberspace” states that the private sector has the most important role to play in cyber security. The PITAC agrees with this conclusion as it pertains to relatively short-term efforts to improve the security of today’s systems and networks. But the Federal government has a vital, irreplaceable role to play as well. As at earlier stages of the digital revolution, Federal investment in fundamental research is required to fill the pipeline with new concepts, technologies, infrastructure prototypes, and trained personnel needed for the private sector to accomplish its cyber security mission.

The Government can also promote technology transfer mechanisms that accelerate adoption of these new technologies by industry, in part by supporting the development of performance metrics, models, datasets, and test beds so that new products and best practices can be evaluated. Federally sponsored fundamental research is a unique national investment in the production of new knowledge that can be broadly used across all sectors of society for the common good. Such research takes place primarily in universities and national laboratories.

As for-profit entities, companies typically focus on short-term results or proprietary research that can provide near-term competitive advantage. It is the mission of research universities to take the long-term view of a problem. Unclassified research performed in universities and national laboratories has the added benefit of creating trained talent in the field, as university graduates obtain employment in industry, universities, and government. University graduates who pursue advanced degrees in IT become the new generation of research leaders; other graduates frequently become involved in start-up companies, which have historically played critical roles in the IT industry.

Research at universities also accelerates changes in the education of new college graduates, as researchers rapidly move new ideas into undergraduate courses and textbooks. Fundamental research focuses on problems of extraordinary difficulty and complexity that often require a number of years to solve. The Federal government has long played a central role in supporting fundamental research in information technology. The results of this research lie at the heart of many of today’s billion-dollar information technology industries – industries that are transforming our lives, driving our economy, and enhancing our security.

Fundamental research is a “public good” – hence, the role of the Federal government in supporting it. The result of a highly effective interplay of Federally supported fundamental research, industry-supported applied research, and industry product development: The United States today is the world leader in information technology. An expanded portfolio of Federal cyber security R&D efforts is required because today

we simply do not know how to model, design, and build systems incorporating integral security attributes such as mutual suspicion – or any other fundamental security innovations.

In addition, we face substantial new challenges from the constant stream of emerging technologies. For example, we do not fully understand the security ramifications of networks of embedded devices. In that context, a principle of mutual suspicion would have to consider controlled access to the sub-networks, the information stores, the devices that are interconnected, and the computing and communication resources of a given network. In our current methods of software development, security is simply one more incremental requirement further burdening an already cumbersome, slow, and expensive process.

The add-on approach will not address our fundamental need for far-sighted advances in systems and software technologies that provide innovative new approaches to the problem of security. We urge a rethinking of the Federal investment balance between military/intelligence and civilian cyber security R&D. In part, this is because the military and intelligence communities rely on the commercial Internet and commercial providers of computing systems and software for the bulk of their own operations.

It is only through fundamental research in civilian cyber security that we can hope to address the strategic and pervasive vulnerabilities of our national IT infrastructure. We also underscore the importance of technology transfer because new concepts do not appear in products automatically. For this to happen, IT vendors must build into their products and services new security functionalities. But vendors respond to what users demand, and it is only recently that most users – corporations, government agencies, and individual users – have begun to care about cyber security.

In the absence of significant demand for cyber security, IT vendors have mostly chosen to add new features for which customers are willing to pay. (Ironically, the addition of new features and added complexity often leads to the introduction of more security vulnerabilities.) This market-driven bias away from cyber security is the “valley of death” for cyber security noted by many analysts.

R&D may provide the knowledge and the proof of operational feasibility, but in the absence of customer demand for the security that may be provided, vendors have little incentive to include new security technologies in their products.

CYBER SECURITY THREATS IN COUNTER- ATTACKS

Many people have proposed methods of “counter-attack” or “active response” to deal with today’s cyber security threats. There is a common belief that traditional approaches to the problem are inadequate. Patching software vulnerabilities is labour intensive, so many people don’t do it until there’s an attack. The legal system is slow to

respond, courts lack the necessary technical expertise, and prosecutions do not occur often enough to deter attacks.

HISTORY OF CYBER SECURITY

The history of successful counter-attacks in the world of cyber security is a short one. There are an increasing number of papers and publications which make proposals, but concerns over legal issues as well as technical challenges have kept the list of successes small. In Sept. 1998, the Pentagon is reported to have counter-attacked some activists who were doing a Denial of Service attack of the Department of Defence's web sites, responding to their requests with an applet which ran on the attackers' machines and forced them to reboot. In 1999, a California ISP Conxion wrote a script which caused DoS packets sent to the World Trade Organization's web site to be sent back to the attackers. Of course, such an attack would not have worked if the attackers had used what's known as "IP address spoofing", which would have made the packets appear that they came from a different computer from where they did.

In 2001, several defences were created in response to the Code Red II worm. One, called CRclean, if it received a probe from the worm, would make use of a back door which the worm placed on the infected machine to load a neutralizing agent and halt the spread of the worm. But it would also install itself on that machine. Another, called CodeGreen, would actually scan the internet for computers with the IIS vulnerability that allowed CodeRed, then download the patch and place it on the machine, and clean up the back doors.

It is clear that there are serious legal and ethical issues with both of these worms, even the former, which is still a worm even though it only reacts to attacks. In 2003, the Welchia worm was released onto the Internet. Welchia would remove the Blaster worm and patch the vulnerability that Blaster used to spread. However, Welchia also caused severe network congestion. In late 2004, Lycos Europe got fed up with spammers and launched a popular campaign called "Make Love Not Spam" which attracted over 100,000 users. Users could install a screen saver on their desktop which would send requests to web sites which were known to advertise via spam. This effort was successful in causing some of the sites to change locations. (<http://www.makelovenotspam.com>) Lycos reportedly got around the illegality of DDoS attacks by claiming to only take 95 per cent of the target sites' bandwidth, though, David Dittrich finds this reasoning dubious at best. Lycos ended this campaign in December 2004. Some companies such as Symbiot Security are currently developing counter attack tools and devices, but there is a lot of skepticism about deployment of devices because of liability concerns.

Cyber Security Incident Response

Counter-attacks are part of the overall process of cyber security incident response. This consists of five stages:

1. Intrusion detection.

2. Identifying the attacker.
3. Defending against the attack.
4. Designing and deploying a counter-attack.
5. Post-conflict investigation and support.

Intrusion Detection: Intrusion detection first identifies if there is an intrusion or intrusion attempt in the network or machine connected to network. Monitoring can be done on the network or on end-hosts. Firewalls and virus scanners also contribute. There are various techniques, such as anomaly detection, and policy-based approaches.

Identifying the Attacker

This very important step towards counter-attack, basically knowing who your enemy is and where is he located. Identifying the attacker can be difficult, since an attacker may use one compromised host to attack another. Also, with today's network architecture it can be quite difficult to locate the real attacker. Packets can be sent with spoofed source addresses, and tracing them through multiple hops and different autonomous systems can be a complicated process. Finding out the specific server and location can be critical in determining what the counter-attack would be.

Defending against the Attack: Before an attack, software patches are used to fix vulnerabilities. More interesting are defences once an attack against a real vulnerability has happened. Diversions can be set forth to surprise an attacker. Or once the attack has been detected, countermeasures can be put in place to stop further damage. Also, diversion tactics like Honey Pots can be used to keep the rest of the network safe. The information from these diversions (or information gained during the delays caused by these tactics) can be harnessed to engineer a specific counter-attack.

Designing and Deploying a Counter-Attack: Counter-attacks take a surprising variety of forms. Designing a counter- attack should take into account the following factors: the method of attack; the support of the network infrastructure (*i.e.* are ISPs performing egress filtering); collateral damage; and costs.

Post-Counter-Attack Investigation and Support: In order to aid the post-counter-attack investigation a counter- attack should have inbuilt mechanism to log its trail and its operation history. Ideally, it will also provide a means of discovering the hacker's techniques and identity. The more risk hackers have of being discovered, the less likely they are to go hacking.

Anti-worms

A worm spreads itself by sending out packets which look for certain responses from computers. It will analyse those responses to deduce if it thinks the replying machine has the vulnerability that it exploits. These packets represent a worm's 'signature'. The machine the worm is targeting can detect the signature of the worm and thus know its mode of propagation. An anti-worm is generated, which exploits the same vulnerabilities as the original worm or uses the back doors opened by the original worm. The antiworm would

either fix the attacking host, neutralize the worm, or bring it down gracefully, potentially informing the owner. It may also install itself for further propagation.

In addition to the stated goal, an anti-worm adheres to following guidelines to minimize collateral damage:

- It is bug-free and does not leave other vulnerabilities that could be exploited further.
- It can be uninstalled gracefully when owner asks for it.
- It makes minimal changes to host that it operates on.
- It should not be able to be used for propagating malicious content.

The countermeasure, or 'Neutralizing Agent' used, can take several forms. For example, it could patch the vulnerability and remove the botnet or worm. Less invasive changes include blocking the port over which the worm sends messages, or using a mutex and rebooting the machine to lock the botnet from being able to execute.

Also, one would have to remove any backdoors left by the worm.

In addition, deployment strategy of the anti-worm will determine its effectiveness and collateral damage it may cause.

- *Active Deployment:* The anti-worm is itself a worm; it propagates across the network by scanning and seeking exploited vulnerabilities. This is the fastest way to remove the original worm from the network, but this also causes the most collateral damage by affecting the network traffic a lot.
- *Passive Deployment:* The anti-worm waits on a host till another host attacks it, and then the anti-worm cures only the attacking host. Collateral damage is bare minimum, but the anti-worm is left behind on the new host.
- *Controlled Deployment:* The anti-worm propagation is controlled via some statistic of number of infected/vulnerable host to number host it cured. This can be very effective and efficient.
- *IDS base Deployment:* The anti-worm uses the intrusion detection sensors which monitor traffic and may deploy an anti-worm against an attacker. As with passive deployment, collateral damage is limited to computers actively launching attacks, but this is less invasive because it does not leave a worm, it just neutralizes the machine.

Spam

Spam is unsolicited e-mail. A virus is a programme which, when executed, installs malware such as root-kits, key-stroke loggers, *etc.* These programmes are then used to send sensitive information to the attacker without the user's consent. Both spam and viruses spread through e-mail and require human interaction to activate them. E-mail servers as well as the hosts can detect spam with the help of spam detectors like Symantec's TurnTide. Once the spam is identified one counter attack is achieved by automatic flooding of the Sender Server with bounced e-mails and mail notification to server's administrator. (Even if sender addresses are spoofed, one may be able to identify the last mail server that forwarded a message.)

Botnet

An attacker establishes a botnet by first spreading a virus or worm that takes over the vulnerable hosts. Each worm or virus then calls Home to tell that it has successfully captured the Host. It also establishes a command channel such that the Attacker through the home can control the host. Botnets can be used for DDOS attacks, phishing *etc.*

Counter-Attack

- Counter-attack involves identifying the attacker by monitoring the compromised Host and intercepting its communication to the Home machine. The Home machine is probably the attacker's base or intermediate node for communication between the host and the attacker base machines. Once that machine is identified, you hack back into that machine bring it down. For hacking back, you can use same techniques as used by the hackers to take control of the attacking machine.
- Taking down the Home machine only removes the attacker for a short duration, the network of compromised hosts still exists. A more effective counter-attack would be to takeover or monitor the command channel for communication between the Home and other compromised hosts. If possible, identify other compromised hosts and fix them or take them down as preventive measure.

CYBER-SECURITY RISKS

New risk factors and challenges to data and communications networks seem to be evolving as rapidly as the spread of high-speed Internet infrastructure. Among these compelling risks are: Computer worms and viruses. Debilitating worms and computer viruses have demonstrated destructive capabilities for a number of years, as evidenced by the damage caused by such programmes as Sasser, Blaster, Netsky, Welchia and Code Red. Organized criminal activity. The booming growth of ICT-services has spawned new gray-and black-market opportunities that organized criminal elements are exploiting for huge financial advantage. Financial institutions, for example, have already experienced significant losses through “phishing” and “pharming” operations.

The more significant the volume of revenues that flow over ICT-based networks, the greater will be the incentive for organized criminal elements to corrupt or economically exploit high-value data resources. A global “black economy” can, for example, generate financing for terrorism, as well as “off-budget” funding for military, police, or national security agencies of nation states.

A NETWORK MODEL FOR CYBER SECURITY

A well-balanced cyber security policy framework is highly complex. Such a policy framework has no bounds or limits—geographical or jurisdictional. It necessarily encompasses a full spectrum of business, societal, and governmental interests. Although cyber security policy will inevitably address grave concerns about national and global

security, as well as well-organized criminal activity, it is fundamentally about creating the very underpinnings of stable economic growth and open, transparent, just, and vibrant societies. A “network model”—essentially, a communications network—illustrates how actors in the public and private sectors can work together to manage future cyber risks. Rather than focus on institutions and functions, the focus is on processes, procedures, and information flows within these sectors. In other words, the framework promotes thinking in terms of functions, rather than titles or national or international institutions.

The focus on information flows is intended to help policymakers and other stakeholders visualize the systemic, inter-related nature of cyber security and how the actions of individual entities are likely to impact the decisions and responses of other participants in the network. It should be noted that such a model may require significant changes in traditional roles of intelligence and (inter)national security agencies, requiring them to operate on a more collaborative, interinstitutional basis.

The network model is based on three types of nodes or groups of stakeholders (policymakers, policy implementers, and operational personnel), which exchange three types of information on cyber-security: assessments, responses, and policy. Information exchange occurs both among the nodes, as well as between individual nodes and their peers in outside organizations at all levels. The model operates in the same way at four different levels, as risk management generally has the same components, regardless of whether it is conducted at the level of a firm, industrial sector, nation, or international organization. First, it requires communications across “organizational” boundaries. Second, it requires information exchange between stakeholders at different levels in the same organization and with other organizations regarding threats that have been encountered and handled. Third, procedures for handling such incidents are required and, fourth, legal or law enforcement sanctions may need to be applied.

At the national level, the major network “nodes” are central coordinating bodies, telecommunications regulators and e-economy ministries, intelligence agencies, law enforcement bodies, and national and governmental Computer Emergency Response Teams (CERTs) and Information Sharing and Analysis Centres (ISACs).

The model works at the international level, where the top-level node is comprised of multinational firms, political and military alliances, which coordinate with such international bodies as the ITU, G8, and the United Nations and its specialized agencies. The implementation node in the international model is comprised of individual national ministries (telecommunications, intelligence, e-economy, and defence ministries, together with law enforcement bodies) and the operational level of national firms (including software and hardware vendors), CERTs, and national defence organizations.

The drivers of various information exchanges—the factors that shape and influence critical flows of information—include regulatory requirements, liability obligations, and market signals (*e.g.*, risk-related information through which the performance of a network may be influenced). Each national networked relationship may operate with its own unique set of additional drivers. Policymakers may further shape certain drivers

to influence node behaviour and performance. It may be possible, for example, to strengthen the influence of market-related drivers by using government procurement policies. Insurance and rating companies may use liability strategies to influence the cost of capital and the incentives for security-related investments.

PROMOTING AUTOMATION OF SECURITY

Several commenters to the NOI discussed how they use automated methods to detect potentially dangerous web behaviour in order to prevent users from exposing themselves to risk suggesting that others could be doing the same. These entities said that they also were providing incentives to the owners of bad websites to reform.

By some accounts, approximately 80 percent of successful online attacks are attributable to known vulnerabilities that can be addressed with implementation of widely agreed upon industry standards, proper configurations and patches. As more computing services are based in the cloud and move further away from centralized enterprises, automating security will likely become even more important than it is today. Enterprise and service delivery will need to address vulnerabilities easily and quickly in order to assure customers of security. In particular, the automated sharing of threats and related signature information among government agencies, among the private sector, and between public and private entities is becoming more commonplace.

With leadership from NIST, the National Security Agency (NSA), DHS and the U.S. CIO Council, the U.S. government leads efforts to automate configuration and vulnerability management. The private sector has also begun to adopt automation protocols such as Security Content Automation Protocol (SCAP) and Continuous Monitoring. These efforts offer enterprises of all sizes the ability to better update security compliance at potentially lower costs and pave the way for future automated protocols.

The security automation initiative is a public/private collaboration that spans multiple government agencies, more than 30 security tool vendors, and a host of end user organizations. The goal of the project is to enable the efficient and accurate collection, correlation, and sharing of security relevant information including software vulnerabilities, system configurations and network events across disparate systems in the enterprise. Security automation work currently supports reference data such as the National Vulnerability Database (NVD), which provides software vulnerability information to users worldwide, and system configurations such as the U.S. Government Configuration Baseline (USGCB) through the Technology Infrastructure Sub-committee of the CIO Council. Standardization of security information has created opportunities for innovation in the private sector and research into new information domains like network events and asset management is expected to foster additional innovation in those markets. Through procurement strategies, the U.S. government can continue to provide tools for leveraging security automation technologies, leveraging existing vendor investment while encouraging additional investment in support of new specifications and standards.

COMPUTER SECURITY

Computer security is the process of preventing and detecting unauthorized use of your computer. Prevention measures help you to stop unauthorized users (also known as “intruders”) from accessing any part of your computer system. Detection helps you to determine whether or not someone attempted to break into your system, if they were successful, and what they may have done.

Computer security is a branch of computer technology known as information security as applied to computers and networks. The objective of computer security includes protection of information and property from theft, corruption, or natural disaster, while allowing the information and property to remain accessible and productive to its intended users.

The term computer system security means the collective processes and mechanisms by which sensitive and valuable information and services are protected from publication, tampering or collapse by unauthorized activities or untrustworthy individuals and unplanned events respectively. The strategies and methodologies of computer security often differ from most other computer technologies because of its somewhat elusive objective of preventing unwanted computer behaviour instead of enabling wanted computer behaviour.

USE SECURITY SOFTWARE THAT UPDATES AUTOMATICALLY

The bad guys constantly develop new ways to attack your computer, so your security software must be up-to-date to protect against the latest threats. Most security software can update automatically; set yours to do so. You can find free security software from well-known companies. Also, set your operating system and web browser to update automatically. If you let your operating system, web browser, or security software get out-of-date, criminals could sneak their bad programmes – malware – onto your computer and use it to secretly break into other computers, send spam, or spy on your online activities. There are steps you can take to detect and get rid of malware. Don’t buy security software in response to unexpected pop-up messages or emails, especially messages that claim to have scanned your computer and found malware. Scammers send messages like these to try to get you to buy worthless software, or worse, to “break and enter” your computer.

TREAT YOUR PERSONAL INFORMATION LIKE CASH

Don’t hand it out to just anyone. Your Social Security number, credit card numbers, and bank and utility account numbers can be used to steal your money or open new accounts in your name. So every time you are asked for your personal information – whether in a web form, an e-mail, a text, or a phone message – think about whether you can really trust the request. In an effort to steal your information, scammers will do everything they can to appear trustworthy. Learn more about scammers who phish for your personal information.

CHECK OUT COMPANIES

When you're online, a little research can save you a lot of money. If you see an ad or an offer that looks good to you, take a moment to check out the company behind it. Type the company or product name into your favourite search engine with terms like "review," "complaint," or "scam." If you find bad reviews, you'll have to decide if the offer is worth the risk. If you can't find contact information for the company, take your business elsewhere. Don't assume that an ad you see on a reputable site is trustworthy. The fact that a site features an ad for another site doesn't mean that it endorses the advertised site, or is even familiar with it.

PERSONAL INFORMATION

If you're shopping or banking online, stick to sites that use encryption to protect your information as it travels from your computer to their server. To determine if a web site is encrypted, look for https at the beginning of the web address (the "s" is for secure). Some websites use encryption only on the sign-in page, but if any part of your session isn't encrypted, the entire account could be vulnerable. Look for https on every page of the site you're on, not just where you sign in.

PROTECT YOUR PASSWORDS

Here are a few principles for creating strong passwords and keeping them safe:

- The longer the password, the tougher it is to crack. Use at least 10 characters; 12 is ideal for most home users.
- Mix letters, numbers, and special characters. Try to be unpredictable – don't use your name, birthdate, or common words.
- Don't use the same password for many accounts. If it's stolen from you – or from one of the companies with which you do business – it can be used to take over all your accounts.
- Don't share passwords on the phone, in texts or by e-mail. Legitimate companies will not send you messages asking for your password. If you get such a message, it's probably a scam.
- Keep your passwords in a secure place, out of plain sight.

HARDWARE MECHANISMS THAT PROTECT COMPUTERS AND DATA

Hardware based or assisted computer security offers an alternative to software-only computer security. Devices such as dongles may be considered more secure due to the physical access required in order to be compromised.

COMPUTER SECURITY RISKS TO HOME USERS

Information security is concerned with three main areas:

- Confidentiality—information should be available only to those who rightfully have access to it.

- Integrity—information should be modified only by those who are authorized to do so
- Availability—information should be accessible to those who need it when they need it.

These concepts apply to home Internet users just as much as they would to any corporate or government network. You probably wouldn't let a stranger look through your important documents. In the same way, you may want to keep the tasks you perform on your computer confidential, whether it's tracking your investments or sending e-mail messages to family and friends.

Also, you should have some assurance that the information you enter into your computer remains intact and is available when you need it. Some security risks arise from the possibility of intentional misuse of your computer by intruders via the Internet. Others are risks that you would face even if you weren't connected to the Internet (*e.g.* hard disk failures, theft, power outages). The bad news is that you probably cannot plan for every possible risk. The good news is that you can take some simple steps to reduce the chance that you'll be affected by the most common threats—and some of those steps help with both the intentional and accidental risks you're likely to face. Before we get to what you can do to protect your computer or home network, let's take a closer look at some of these risks.

COMPUTER SECURITY BASICS

Making a computer secure requires a list of different actions for different reasons. There is a secondary rule that says security is an on going process. No matter how well a system is designed, if it is never changed that gives any potential infiltrator all the time in the world to examine the security for flaws. The information described here is neither detailed nor comprehensive. This should, however, serve as a good overview of the types of security measures sometimes taken. What measures are appropriate are best determined on a case by case basis.

Physical Security

Theft is the physical threat of most concern and rightfully so. Keeping rooms locked is a good idea, but not always feasible. Keeping computers locked to a wall or table is a good deterrent against a casual, shoplifting style, theft but it will not deter a professional with a shopping list. We have seen a thief use a crow bar to remove a computer along with a portion of the formica table top (they were then foolish enough to take it to a repair shop with the table top still attached). There are very loud alarms which sound when the power cable is unplugged. A combination of locks and alarms is an excellent theft prevention system for computer labs which must be publicly accessible, particularly at late hours.

Computer hardware is protected from fire damage by smoke detectors and sprinkler systems just like any other equipment. Computers are unique in that the most costly

damage is the loss of data which can be prevented by storing back up tapes in remote locations. Surge protectors and uninterruptable power supplies are a low cost investment that can save very costly equipment damage.

These are particularly important if the computer must be used continuously or if your region is prone to severe thunder storms or frequent power outages. Some surge protectors have the ability to protect the phone line going to a modem also. The modem and mother board can be more readily damaged by lightning hitting a phone line than by lightning hitting the power lines because the computer power supply provides a minimal amount of protection.

Data Integrity

Backing up data is the single most important step in preventing data loss. Entire companies have gone out of business due to losing valuable information. An enormous amount of man hours are spent every year reproducing information which was lost in some manner. Back ups can be on removable disks, tapes, paper printouts or other computer systems. It is important to periodically put copies of these back ups in remote physical locations to prevent losing the original and back up data through fire, *etc.* In today's world, virus protection is a necessity for any PC or Macintosh and viruses are starting to appear on UNIX systems also. No system is completely safe from viruses since manufacturers have inadvertently shipped new computers with viruses on the hard drive and minted CDs with viruses.

For very important data, RAID systems are used. RAID stands for "Redundant Array of Inexpensive Disks". A RAID system is a computer with eight or more hard drives and software for storing data on those drives. Every byte of data is spread across all of these drives along with a parity bit that tells if it was an odd or even byte. In the event that a disk fails, it's contents can be completely reconstructed from the data on the other seven disks. This is a good way to store critical data which could not be reproduced, but the expense may not be justified otherwise.

Data Security

The primary threat to data security is illegal computer hackers. Studies show that the largest percentage of hackers are young men motivated by status with other hackers, malicious intent or the excitement of a challenging game. There have also been even more harmful cases of corporate spying and embezzlement of funds. Accounts on both multiuser machines and micro computers can be protected by passwords. Passwords can be very effective or not effective at all. Insecure password include ones that are easily guessed, never changed, shared or written down somewhere.

Some systems, particularly UNIX, have password files which are encrypted but readable by all users. Hackers have developed automated programmes, such as "crack", to break the passwords in these files by raw brute force, trial and error techniques. Since it could take months to crack well chosen passwords, some systems use a password

aging system that requires all users to set new passwords periodically. There are also programmes to prevent users from setting easily guessed passwords such as words in the dictionary, common names or permutations on the account name. Systems holding data belonging to multiple users, such as UNIX or Windows NT, set an owner for each file and permissions defining who is allowed to read or write to it. Many hacker attacks are centered around finding flaws in the file permission system. There are ways to set default permissions and ways to control how much individual users can control their own file permissions.

Since most security attacks are now initiated from a remote location via the network, many organizations now separate their internal networks from the Internet with a firewall. A firewall is a piece of software running on a dedicated machine with two network boards. The software can filter which network traffic is allowed to pass between the internal and external networks. This is a very effective security measure, but there is an unfortunate tendency for organizations to make the firewall their only security measure making any breach of security across the firewall a breach for every machine in the whole organization. An even higher level of security can be achieved by not having any connection between the internal network and the internet or not even having an internal network.

Data encryption provides a second layer of security. Once someone gains access to data, that data is useless if it has been scrambled by an encryption programme which requires a second password to unscramble it. Passwords themselves should always be stored in an encrypted form.

Today's encryption systems are similar to military code systems but not as sophisticated as the systems used by the armed forces. Almost all encrypted data can be unencrypted without the password by the use of a very large amount of time on very powerful computers. Security is provided by making the encryption complex enough that no one would be likely to have enough computer power to break say a message about the merger next month in less than six months, at which time the message is no longer valuable.

There must always be someone able to fix a computer system by using a second password protected account called "system", "administrator", "root" or "superuser" which bypasses the file permission system. One of the most serious security attacks is one which gains the password to this account. As well as particularly stringent security for this account, the encryption systems, ensure that there is a second layer of protection against this type of attack. This also provides for a segmented internal security system, if such is necessary.

E-mail is particularly insecure. Mail messages are simple ascii files that travel across the network where no password is necessary to get to them. E-mail is easily forged and can be altered. Of course, no one would have any particular reason for tampering with many personal messages, but people conducting sensitive business transactions over E-mail would be wise to use some sort of E-mail encryption system, such as PGP. These

systems have several functions including encrypting the message itself, verifying who sent the message and verifying that it was not tampered with. Audit trails are a means for the system administrators to find out if security has been breached and how much damage was done. Audit trails are records made by various pieces of software to log who logged into a system, from where and what files were accessed.

How Hackers Get In

Here is the typical sequence of steps used to gain illegal entry into a computer system.

- *Learn about the system.* Trying to connect to a system using networking utilities like telnet and ftp will be unsuccessful without a password, but even unsuccessful logins will often still display the machine manufacturer, and version of the operating system.
- *Look for openings.* Try known security flaws on that particular machine and operating system. Unless the system administrator is very diligent about installing security patches, many machines have openings in the security just waiting to be found.
- *Try sniffing to get a password.* Sniffing is when a machine has software to watch all of the network traffic and saves the messages corresponding to a valid user entering their password from a remote location.
- *Try spoofing.* Many machines share disks with other machines that are classified as “trusted hosts”. In order to share the data on these disks the two machines must communicate without a password. Spoofing is when someone configures a third machine to use the network address of one of the trusted hosts to impersonate that machine. If the spoofing machine responds faster than the true trusted host, communications will be carried out with it unnoticed. Spoofing requires that the infiltrator have physical access to the network in a location that falls close to the target machine in the network topology, which usually means being physically close to the target machine.
- *Get into the system and cover tracks.* Once one of the above techniques is successful in gaining access to the system, the first order of business is to alter any records that would reveal the presence of an illegal entry to the system administrators.
- *Try to get superuser access.* Just as there are many ways to get into a user account, there are many ways to get into the root level account or get equivalent access to the machine.
- *Make back doors.* Once entry has been gained, that access can be used to intentionally install security breaches so that the hacker can still get back into the system if the original method of entry is cut off.
- *Use the system.* At this point, the hacker can steal data, destroy information, alter files, use CPU time, lock everyone else out of the system, *etc.*

How to Combat Illegal Entry

Here are a list of ways to make computers more secure and some minimal suggestions for when they should be used. For systems that are critical to operation, all of these and more may be warranted.

- *Physical security.* Keep doors locked if feasible. Install locks on accessible but attended machines. Install locks and alarms on machines left unattended.
- *Back up files.* This should be done on all computers.
- *Use a surge suppressor.* All computers.
- *Use an uninterruptable power supply.* Critical systems.
- *Periodic virus checking.* All PC and Macintosh computers. High volume or critical multiuser machines.
- *Continual memory resident virus checking.* PCs or Macs used by many people, such as in public labs. When data routinely comes from many sources.
- *Firewalls.* For organizations that can conduct business with limits on the internet services accessible from inside the organization. Where outside access to company data could do significant harm to the business.
- Having no internet connection or no internal network at all is done when data is particularly sensitive or reliability is of key importance. Bank record systems and air traffic control systems are some examples.
- Programmes to enforce the use of good passwords. Systems with a moderate to large number of users.
- *Password aging.* Systems which have a large number of users or are a likely target for illegal entry.
- *Remove old accounts.* Old, unused accounts are just that many more passwords for someone to find out. If it is not feasible to remove old accounts, the passwords can still be deleted. This is done by setting a null password for which no possible password will give access to the account.
- *Smart cards.* There are various varieties of smart cards to act as passwords electronically. One example is a card with a number that changes every ten seconds and has its internal clock synchronized to one in the central computer. This way, even if someone get the password, it is only good for ten seconds. This expense is only warranted when someone would have a clear motive for trying to break into a system.
- *Install security patches to the operating system.* Invisible security patches should be installed anytime systems are being upgraded. On systems with many users or that are likely targets for illegal entry, the system administrator should install new patches frequently or perhaps instantly when available. Many break ins occur within 24 hours of when a security flaw and patch is announced. This occurs when someone has targeted a particular machine and hopes to figure out how to take advantage of the flaw before the system administrators

upgrade the system. For this reason, many flaws are not announced until a patch or temporary work around can be announced with them. Networking patches and network software upgrades are particularly important.

- *Security checking software.* There are programmes, like Satan, which will test a system for many known security flaws. These programmes were created so that administrators can test the integrity of the system, but they are also a favourite tool for the first step in infiltrating a system. It is a good idea to do this periodically. The software can be set to check many machines on a network without interrupting the people using those machines. There are programmes to check the system from the inside as well as checking network vulnerabilities.
- *Break in detection software.* There are also pieces of software to alert the system administrators when security is being tested by a known technique. This is a good way to know of an attack before they have gained entry.
- Some level of audit trail should be kept on any multiuser system and any system with sensitive data. Some level of auditing is built into many multiuser operating systems. An audit trail has to be maintained before a break in occurs in order to do any good.
- Use software to prevent sniffing, such as Kerberos or secure shell. These software packages allow remote logins to be authenticated, without sending an unencrypted password over the network. We have seen an increase in sites using these systems, particularly where many users login to machines remotely. The difficulty is setting up a system which is secure and reliable as well as not inconveniencing the users.
- *Encryption of disk files.* Disk files should be kept encrypted when the data is particularly important. Passwords, social security numbers and credit card numbers should always be encrypted. Many accounting systems use encryption.
- Do not use your credit card over the web unless your browser (not their web page) identifies it as a secure server. Even at that it is advisable only to do so with reputable companies that you are familiar with. You should never need a credit card number to get something that is free.
- Encrypted E-mail software should be used when someone would have a reason to want to see, forge or alter E-mail messages.
- *Random manual monitoring.* For a few businesses that deal with very sensitive information and must use networks, the security administrators will occasionally manually look at the information being passed over the network, particularly through the firewall. This probably is not warranted unless security is important enough to be paying someone solely as a security manager.
- *Hiring tiger teams.* A tiger team is a group of honest expert hackers that are hired to break into your system in order to give you an analysis of your security. This is generally done by banks or others with extremely sensitive data.

NETWORK SECURITY

Recent interest in security was fuelled by the crime committed by Kevin Mitnick. Kevin Mitnick committed the largest computer-related crime in U.S. history. The losses were eighty million dollars in U.S. intellectual property and source code from a variety of companies. Since then, information security came into the spotlight. Public networks are being relied upon to deliver financial and personal information. Due to the evolution of information that is made available through the Internet, information security is also required to evolve. Due to Kevin Mitnick's offence, companies are emphasizing security for the intellectual property. Internet has been a driving force for data security improvement. Internet protocols in the past were not developed to secure themselves. Within the TCP/IP communication stack, security protocols are not implemented. This leaves the Internet open to attacks. Modern developments in the internet architecture have made communication more secure.

BRIEF HISTORY OF INTERNET

The birth of the Internet takes place in 1969 when Advanced Research Projects Agency Network (ARPANET) is commissioned by the department of defence (DOD) for research in networking. The ARPANET is a success from the very beginning. Although originally designed to allow scientists to share data and access remote computers, e-mail quickly becomes the most popular application. The ARPANET becomes a high-speed digital post office as people use it to collaborate on research projects and discuss topics of various interests. The InterNetworking Working Group becomes the first of several standards-setting entities to govern the growing network. Vinton Cerf is elected the first chairman of the INWG, and later becomes known as a "Father of the Internet."

In the 1980s, Bob Kahn and Vinton Cerf are key members of a team that create TCP/IP, the common language of all Internet computers. For the first time the loose collection of networks which made up the ARPANET is seen as an "Internet", and the Internet as we know it today is born. The mid-80s marks a boom in the personal computer and super-minicomputer industries. The combination of inexpensive desktop machines and powerful, network-ready servers allows many companies to join the Internet for the first time. Corporations begin to use the Internet to communicate with each other and with their customers. In the 1990s, the Internet began to become available to the public. The World Wide Web was born. Netscape and Microsoft were both competing on developing a browser for the Internet. Internet continues to grow and surfing the internet has become equivalent to TV viewing for many users.

SECURITY TIMELINE

Several key events contributed to the birth and evolution of computer and network security. The timeline can be started as far back as the 1930s. Polish cryptographers created an enigma machine in 1918 that converted plain messages to encrypted text. In

1930, Alan Turing, a brilliant mathematician broke the code for the Enigma. Securing communications was essential in World War II.

In the 1960s, the term “hacker” is coined by a couple of Massachusetts Institute of Technology (MIT) students. The Department of Defence began the ARPANET, which gains popularity as a conduit for the electronic exchange of data and information. This paves the way for the creation of the carrier network known today as the Internet. During the 1970s, the Telnet protocol was developed. This opened the door for public use of data networks that were originally restricted to government contractors and academic researchers.

During the 1980s, the hackers and crimes relating to computers were beginning to emerge. The 414 gang are raided by authorities after a nine-day cracking spree where they break into top-secret systems. The Computer Fraud and Abuse Act of 1986 was created because of Ian Murphy’s crime of stealing information from military computers. A graduate student, Robert Morris, was convicted for unleashing the Morris Worm to over 6,000 vulnerable computers connected to the Internet. Based on concerns that the Morris Worm ordeal could be replicated, the Computer Emergency Response Team (CERT) was created to alert computer users of network security issues.

In the 1990s, Internet became public and the security concerns increased tremendously. Approximately 950 million people use the internet today worldwide. On any day, there are approximately 225 major incidences of a security breach. These security breaches could also result in monetary losses of a large degree. Investment in proper security should be a priority for large organizations as well as common users.

SECURITY IN NETWORKS

Networks their design, development, and usage are critical to our style of computing. We interact with networks daily, when we perform banking transactions, make telephone calls, or ride trains and planes. The utility companies use networks to track electricity or water usage and bill for it. When we pay for groceries or gasoline, networks enable our credit or debit card transactions and billing. Life without networks would be considerably less convenient, and many activities would be impossible. Not surprisingly, then, computing networks are attackers’ targets of choice. Because of their actual and potential impact, network attacks attract the attention of journalists, managers, auditors, and the general public.

For example, when you read the daily newspapers, you are likely to find a story about a network-based attack at least every month. The coverage itself evokes a sense of evil, using terms such as hijacking, distributed denial of service, and our familiar friends viruses, worms, and Trojan horses. Because any large-scale attack is likely to put thousands of computing systems at risk, with potential losses well into the millions of dollars, network attacks make good copy.

The media coverage is more than hype; network attacks are critical problems. Fortunately, your bank, your utility company, and even your Internet service provider

take network security very seriously. Because they do, they are vigilant about applying the most current and most effective controls to their systems. Of equal importance, these organizations continually assess their risks and learn about the latest attack types and defence mechanisms so that they can maintain the protection of their networks.

In this chapter we describe what makes a network similar to and different from an application programme or an operating system. In investigating networks, you will learn how the concepts of confidentiality, integrity, and availability apply in networked settings. At the same time, you will see that the basic notions of identification and authentication, access control, accountability, and assurance are the basis for network security, just as they have been in other settings.

Networking is growing and changing perhaps even faster than other computing disciplines. Consequently, this chapter is unlikely to present you with the most current technology, the latest attack, or the newest defence mechanism; you can read about those in daily newspapers and at web sites. But the novelty and change build on what we know today: the fundamental concepts, threats, and controls for networks. By developing an understanding of the basics, you can absorb the most current news quickly and easily. More importantly, your understanding can assist you in building, protecting, and using networks.

NETWORK CONCEPTS

To study network threats and controls, we first must review some of the relevant networking terms and concepts. This review does not attempt to provide the depth of a classic networking reference. Our study of security focused on the individual pieces of a computing system, such as a single application, an operating system, or a database. Networks involve not only the pieces but also importantly the connections among them. Networks are both fragile and strong. To see why, think about the power, cable television, telephone, or water network that serves your home. If a falling tree branch breaks the power line to your home, you are without electricity until that line is repaired; you are vulnerable to what is called a single point of failure, because one cut to the network destroys electrical functionality for your entire home.

Similarly, there may be one telephone trunk line or water main that serves your home and those nearby; a failure can leave your building, street, or neighbourhood without service. But we have ways to keep the entire network from failing. If we trace back through the network from your home to the source of what flows through it, we are likely to see that several main distribution lines support an entire city or campus. That is, there is more than one way to get from the source to your neighbourhood, enabling engineers to redirect the flow along alternative paths. Redundancy makes it uncommon for an entire city to lose service from a single failure. For this reason, we say that such a network has resilience or fault tolerance.

Complex routing algorithms reroute the flow not just around failures but also around overloaded segments. The routing is usually done automatically; the control programme

is often supplemented by human supervision or intervention. Many types of networks have very high reliability by design, not by accident. But because there often is less redundancy near a network's endpoints than elsewhere, we say that the network has great strength in the middle and fragility at the perimeter.

From the user's perspective, a network is sometimes designed so that it looks like two endpoints with a single connection in the middle. For example, the municipal water supply may appear to be little more than a reservoir (the source), the pipes (the transmission or communication medium), and your water faucet (the destination). Although this simplistic view is functionally correct, it ignores the complex design, implementation, and management of the "pipes." In a similar way, we describe computer networks in this chapter in ways that focus on the security concepts but present the networks themselves in a simplistic way, to highlight the role of security and prevent the complexity of the networks from distracting our attention. Please keep in mind that our network descriptions are often abstractions of a more complex actuality.

The Network

A network in its simplest form, as two devices connected across some medium by hardware and software that enable the communication. In some cases, one device is a computer (sometimes called a "server") and the other is a simpler device (sometimes called a "client") enabled only with some means of input (such as a keyboard) and some means of output (such as a screen). For example, a powerful computer can be a server, but a handheld personal digital assistant (PDA) or a cell phone might be a network client. In fact, because more consumer devices are becoming network-enabled, network security issues will continue to grow.



Fig. Simple View of Network.

Although this model defines a basic network, the actual situation is frequently significantly more complicated.

- The simpler client device, employed for user-to-computer communication, is often a PC or workstation, so the client has considerable storage and processing capability.
- A network can be configured as just a single client connected to a single server. But more typically, many clients interact with many servers.
- The network's services are often provided by many computers. As a single user's communication travels back and forth from client to server, it may merely pass through some computers but pause at others for significant interactions.

- The end user is usually unaware of many of the communications and computations taking place in the network on the user's behalf.

The user at one of the lettered client machines may send a message to System 3, unaware that communication is actually passing through the active Systems 1 and 2. In fact, the user may be unaware that System 3 sometimes passes work to System 4. A single computing system in a network is often called a node, and its processor (computer) is called a host. A connection between two hosts is known as a link. Network computing consists of users, communications media, visible hosts, and systems not generally visible to end users. Systems 1 through 4 are nodes. In our figure the users are at the lettered client machines, perhaps interacting with Server F. Users communicate with networked systems by interacting directly with terminals, workstations, and computers. A workstation is an end-user computing device, usually designed for a single user at a time. Workstations often have powerful processors and good-sized memory and storage so that they can do sophisticated data manipulation (such as converting coded data to a graphical format and displaying the picture). A system is a collection of processors, perhaps including a mixture of workstations and independent processors, typically with more processing power and more storage capacity than a workstation.

Environment of Use

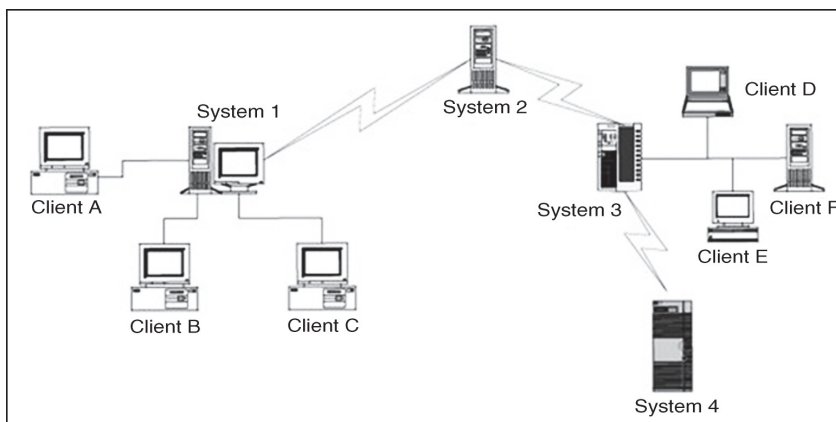


Fig. More Complex but More Typical View of Networks.

The biggest difference between a network and a stand-alone device is the environment in which each operates. Although some networks are located in protected spaces (for example, a local area network in a single laboratory or office), at least some portion of most networks is exposed, often to total strangers. The relatively simple network is a good example. Systems 2, 3, and 4 are remote from System 1, and they may be under different ownership or control.

Networks can be described by several typical characteristics:

- *Anonymity.* You may have seen the cartoon image that shows a dog typing at a workstation, and saying to another dog, "On the Internet, nobody knows

you're a dog." A network removes most of the clues, such as appearance, voice, or context, by which we recognize acquaintances.

- *Automation.* In some networks, one or both endpoints, as well as all intermediate points, involved in a given communication may be machines with only minimal human supervision.
- *Distance.* Many networks connect endpoints that are physically far apart. Although not all network connections involve distance, the speed of communication is fast enough that humans usually cannot tell whether a remote site is near or far.
- *Opacity.* Because the dimension of distance is hidden, users cannot tell whether a remote host is in the room next door or in a different country. In the same way, users cannot distinguish whether they are connected to a node in an office, school, home, or warehouse, or whether the node's computing system is large or small, modest or powerful. In fact, users cannot tell if the current communication involves the same host with which they communicated the last time.
- *Routing diversity.* To maintain or improve reliability and performance, routings between two endpoints are usually dynamic. That is, the same interaction may follow one path through the network the first time and a very different path the second time. In fact, a query may take a different path from the response that follows a few seconds later.

Shape and Size

The way a network is configured, in terms of nodes and connections, is called the network topology. You can think of the topology as the shape of the network. The topology ranges from very simple, such as two hosts connected by one path, to very complex, such as the Internet. These two extremes highlight three dimensions of networks that have particular bearing on a network's security.

- *Boundary.* The boundary distinguishes an element of the network from an element outside it. For a simple network, we can easily list all the components and draw an imaginary line around it to separate what is in the network from what is outside. But listing all the hosts connected to the Internet is practically impossible. For example, a line surrounding the Internet would have to surround the entire globe today, and Internet connections also pass through satellites in orbit around the earth. Moreover, as people and organizations choose to be connected or not, the number and type of hosts change almost second by second, with the number generally increasing over time.
- *Ownership.* It is often difficult to know who owns each host in a network. The network administrator's organization may own the network infrastructure, including the cable and network devices. However, certain hosts may be connected to a network for convenience, not necessarily implying ownership.

- *Control.* Finally, if ownership is uncertain, control must be, too. To see how, pick an arbitrary host. Is it part of network A? If yes, is it under the control of network A's administrator? Does that administrator establish access control policies for the network, or determine when its software must be upgraded and to what version? Indeed, does the administrator even know what version of software that host runs?

The truth is that, for many networks, it is difficult and at times impossible to tell which hosts are part of that network, who owns the hosts, and who controls them. Even for networks significantly smaller than the Internet, major corporate, university, or government networks are hard to understand and are not even well known by their system administrators. Although it seems contrary to common sense, many corporations today have no accurate picture of how their networks are configured. To understand why, consider a network of automated teller machines for a multinational bank. The bank may have agreements with other banks to enable customers to withdraw money anywhere in the world. The multinational bank may understand its own bank's network, but it may have no conception of how the connecting banks' networks are configured; no "big picture" shows how the combined networks look or operate. Similarly, a given host may be part of more than one network. In such a situation, suppose a host has two network interfaces. Whose rules does that host (and that host's administrator) have to follow? Depicting, configuring, and administering networks are not easy tasks.

Mode of Communication

A computer network implements communication between two endpoints. Data are communicated either in digital format (in which data items are expressed as discrete binary values) or analog (in which data items are expressed as points in a continuous range, using a medium like sound or electrical voltage). Computers typically store and process digital data, but some telephone and similar cable communications are in analog form (because telephones were originally designed to transmit voice). When the transmission medium expects to transfer analog data, the digital signals must be converted to analog for transmission and then back to digital for computation at the receiving end. Some mostly analog networks may even have some digital segments, so the analog signals are digitized more than once. These conversions are performed by a modem, which converts a digital data stream to tones and back again.

Media

Communication is enabled by several kinds of media. We can choose among several types, such as along copper wires or optical fibre or through the air, as with cellular phones. Let us look at each type in turn.

Cable

Because much of our computer communication has historically been done over telephone lines, the most common network communication medium today is wire. Inside

our homes and offices, we use a pair of insulated copper wires, called a twisted pair or unshielded twisted pair (UTP). Copper has good transmission properties at a relatively low cost.

The bandwidth of UTP is limited to under 10 megabits per second (Mbps), so engineers cannot transmit a large number of communications simultaneously on a single line. Moreover, the signal strength degrades as it travels through the copper wire, and it cannot travel long distances without a boost. Thus, for many networks, line lengths are limited to approximately 300 feet. Single twisted pair service is most often used locally, within a building or up to a local communications drop (that is, the point where the home or office service is connected to the larger network, such as the commercial telephone system). Although regular copper wire can transmit signals, the twisting reduces crossover (interference and signal transfer) between adjacent wires.

However, as speeds or capacities change, the basic ranking of two technologies tends to remain the same. Another choice for network communication is coaxial (coax) cable, the kind used for cable television. Coax cable is constructed with a single wire surrounded by an insulation jacket. The jacket is itself surrounded by a braided or spiral-wound wire. The inner wire carries the signal, and the outer braid acts as a ground. The most widely used computer communication coax cable is Ethernet, carrying up to 100 Mbps over distances of up to 1500 feet. Coax cable also suffers from degradation of signal quality over distance. Repeaters (for digital signals) or amplifiers (for analog signals) can be spaced periodically along the cable to pick up the signal, amplify it, remove spurious signals called “noise,” and retransmit it.

Optical Fibre

A newer form of cable is made of very thin strands of glass. Instead of carrying electrical energy, these fibres carry pulses of light. The bandwidth of optical fibre is up to 1000 Mbps, and the signal degrades less over fibre than over wire or coax; the fibre is good for a run of approximately 2.5 miles. Optical fibre involves less interference, less crossover between adjacent media, lower cost, and less weight than copper. Thus, optical fibre is generally a much better transmission medium than copper. Consequently, as copper ages, it is being replaced by optical fibre in most communication systems. In particular, most long distance communication lines are now fibre.

Wireless

Radio signals can also carry communications. Similar to pagers, wireless microphones, garage door openers, and portable telephones, wireless radio can be used in networks, following a protocol developed for short-range telecommunications, designated the 802.11 family of standards. The wireless medium is used for short distances; it is especially useful for networks in which the nodes are physically close together, such as in an office building or at home. Many 802.11 devices are becoming available for home and office wireless networks.

Microwave

Microwave is a form of radio transmission especially well suited for outdoor communication. Microwave has a channel capacity similar to coax cable; that is, it carries similar amounts of data. Its principal advantage is that the signal is strong from point of transmission to point of receipt. Therefore, microwave signals do not need to be regenerated with repeaters, as do signals on cable. However, a microwave signal travels in a straight line, presenting a problem because the earth curves. Microwave signals travel by line of sight: The transmitter and receiver must be in a straight line with one another, with no intervening obstacles, such as mountains. A straight microwave signal transmitted between towers of reasonable height can travel a distance of only about 30 miles because of the earth's curvature. Thus, microwave signals are "bounced" from receiver to receiver, spaced less than 30 miles apart, to cover a longer distance.

Infrared

Infrared communication carries signals for short distances (up to 9 miles) and also requires a clear line of sight. Because it does not require cabling, it is convenient for portable objects, such as laptop computers and connections to peripherals.

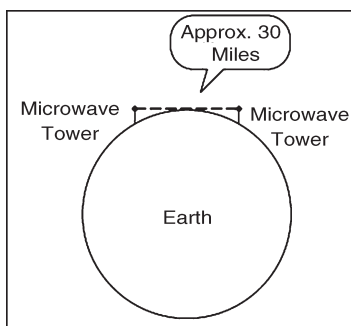


Fig. Microwave Transmission.

An infrared signal is difficult to intercept because it is a point-to-point signal. However, it is subject to "in the middle" attacks in which the interceptor functions like a repeater, receiving the signal, extracting any desired data, and retransmitting to the original destination the original signal or a modified version. Because of line-of-sight requirements and limited distance, infrared is typically used in a protected space, such as an office, in which in-the-middle attacks would be difficult to conceal.

Satellite

Many communications, such as international telephone calls, must travel around the earth. In the early days of telephone technology, telephone companies ran huge cables along the ocean's bottom, enabling calls to travel from one continent to another. Today, we have other alternatives. The communication companies place satellites in orbits that are synchronized with the rotation of the earth (called geosynchronous orbits), so the satellite appears to hover in a fixed position 22,300 miles above the earth. Although

the satellite can be expensive to launch, once in space it is essentially maintenance free. Furthermore, the quality of a satellite communication link is often better than an earthbound wire cable.

Satellites act as nave transponders: Whatever they receive they broadcast out again. Thus, satellites are really sophisticated receivers, in that their sole function is to receive and repeat signals. From the user's point of view, the signal essentially "bounces" off the satellite and back to earth. For example, a signal from North America travels 22,300 miles into the sky and the same distance back to a point in Europe.

We can project a signal to a satellite with reasonable accuracy, but the satellite is not expected to have the same level of accuracy when it sends the signal back to earth. To reduce complexity and eliminate beam focusing, satellites typically spread their transmissions over a very wide area. A rather narrow angle of dispersion from the satellite's transmitter produces a fairly broad pattern (called the footprint) on the surface of the earth because of the 22,300-mile distance from the satellite to earth. Thus, a typical satellite transmission can be received over a path several hundred miles wide; some cover the width of the entire continental United States in a single transmission. For some applications, such as satellite television, a broad footprint is desirable. But for secure communications, the smaller the footprint, the less the risk of interception.

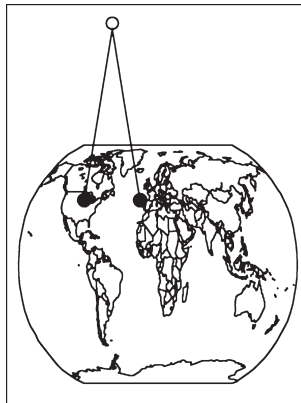


Fig. Satellite Communication.

Protocols

When we use a network, the communication media are usually transparent to us. That is, most of us do not know whether our communication is carried over copper wire, optical fibre, satellite, microwave, or some combination. In fact, the communication medium may change from one transmission to the next. This ambiguity is actually a positive feature of a network: its independence. That is, the communication is separated from the actual medium of communication. Independence is possible because we have defined protocols that allow a user to view the network at a high, abstract level of communication (viewing it in terms of user and data); the details of how the communication is accomplished are hidden within software and hardware at both ends.

The software and hardware enable us to implement a network according to a protocol stack, a layered architecture for communications. Each layer in the stack is much like a language for communicating information relevant at that layer. Two popular protocol stacks are used frequently for implementing networks: the Open Systems Interconnection (OSI) and the Transmission Control Protocol and Internet Protocol (TCP/IP) architecture. We examine each one in turn.

ISO/OSI Reference Model

The International Standards Organization (ISO)/Open Systems Interconnection model consists of layers by which a network communication occurs. How communication works across the different layers. We can think of the layers as creating an assembly line, in which each layer adds its own service to the communication. In concert, the layers represent the different activities that must be performed for actual transmission of a message. Separately, each layer serves a purpose; equivalent layers perform similar functions for the sender and receiver. For example, the sender's layer four affixes a header to a message, designating the sender, the receiver, and relevant sequence information. On the receiving end, layer four reads the header to verify that the message is for the intended recipient, and then removes this header.

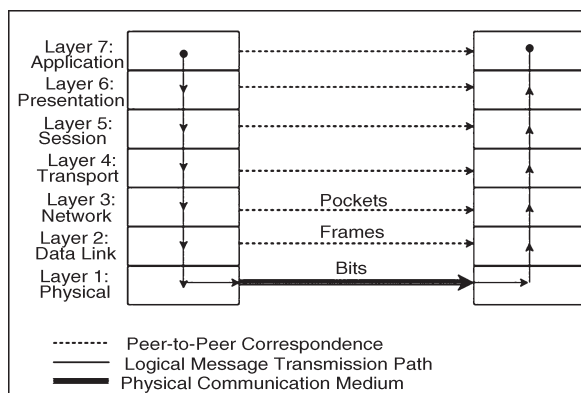


Fig. ISO/OSI Network Model.

Each layer passes data in three directions: above with a layer communicating more abstractly, parallel or across to the same layer in another host, and below with a layer handling less abstract (that is, more fundamental) data items. The communications above and below are actual interactions, while the parallel one is a virtual communication path. Parallel layers are called “peers.”

Let us look at a simple example of protocol transmission. Suppose that, to send e-mail to a friend, you run an application such as Eudora, Outlook, or Unix mail. You type a message, using the application's editor, and the application formats the message into two parts: a header that shows to whom the message is intended (as well as other things, such as sender and time sent), and a body that contains the text of your message. The application reformats your message into a standard format so that even if you and

your friend use different mail applications, you can still exchange e-mail. This transformation is shown in Figure below.

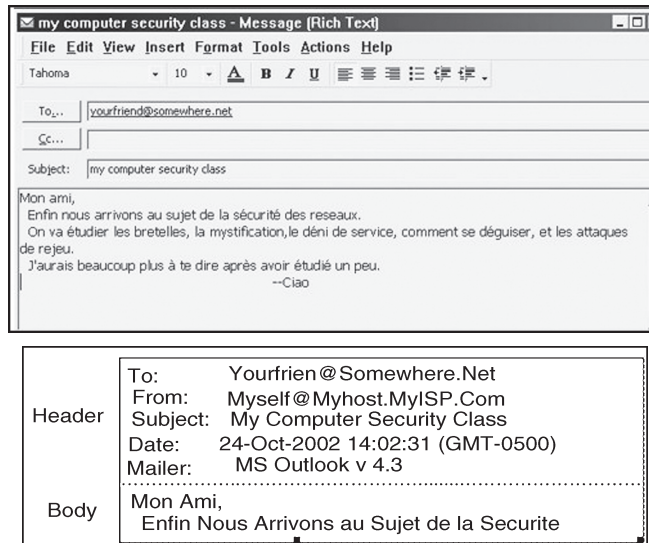


Fig. Transformation.

However, the message is not transmitted exactly as you typed it, as raw text. Raw text is a very inefficient coding, because an alphabet uses relatively few of the 255 possible characters for an 8-bit byte. Instead, the presentation layer is likely to change the raw text into something else. It may do compression, character conversions, and even some cryptography.

An e-mail message is a one-way transfer (from sender to receiver), so it is not initiating a session in which data fly back and forth between the two endpoints. Because the notion of a communication session is not directly relevant in this scenario, we ignore the session layer for now. Occasionally, spurious signals intrude in a communication channel, as when static rustles a telephone line or interference intrudes on a radio or television signal. To address this, the transport layer adds error detection and correction coding to filter out these spurious signals.

Addressing

Suppose your message is addressed to yourfriend@somewhere.net. This notation means that "somewhere.net" is the name of a destination host (or more accurately, a destination network). At the network layer, a hardware device called a router actually sends the message from your network to a router on the network somewhere.net. The network layer adds two headers to show your computer's address as the source and somewhere.net's address as the destination. Logically, your message is prepared to move from your machine to your router to your friend's router to your friend's computer. (In fact, between the two routers there may be many other routers in a path through the networks from you to your friend.) Together, the network layer structured with destination

address, source address, and data is called a packet. The basic network layer protocol transformation is shown in Figure below.

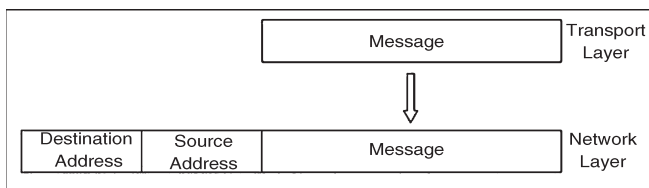


Fig. Network Layer Transformation.

The message must travel from your computer to your router. Every computer connected to a network has a network interface card (NIC) with a unique physical address, called a MAC address (for Media Access Control). At the data link level, two more headers are added, one for your computer's NIC address (the source MAC) and one for your router's NIC address. A data link layer structure with destination MAC, source MAC, and data is called a frame. Every NIC selects from the network those frames with its own address as a destination address. The data link layer adds the structure necessary for data to get from your computer to another computer (a router is just a dedicated computer) on your network.

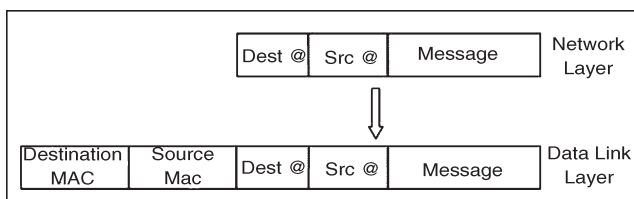


Fig. Data Link Layer Transformation.

Finally, the message is ready to be sent out as a string of bits. We noted earlier that analog transmissions communicate bits by using voltage or tone changes, and digital transmissions communicate them as discrete pulses. The physics and electronics of how bits are actually sent are handled at the physical layer. On the receiving (destination) side, this process is exercised in reverse: Analog or digital signals are converted to digital data. The NIC card receives frames destined for it. The recipient network layer checks that the packet is really addressed to it. Packets may not arrive in the order in which they were sent (because of network delays or differences in paths through the network), so the session layer may have to reorder packets. The presentation layer removes compression and sets the appearance appropriate for the destination computer. Finally, the application layer formats and delivers the data as an e-mail message to your friend.

The layering and coordinating are a lot of work, and each protocol layer does its own part. But the work is worth the effort because the different layers are what enable Outlook running on an IBM PC on an Ethernet network in Washington D.C. to communicate with a user running Eudora on an Apple computer via a dial-up connection in Prague.

Moreover, the separation by layers helps the network staff troubleshoot when something goes awry.

Layering

Each layer reformats the transmissions and exchanges information with its peer layer. Let us summarize what each layer contributes. A typical message that has been acted upon by the seven layers in preparation for transmission. Layer 6 breaks the original message data into blocks. At the session layer (5), a session header is added to show the sender, the receiver, and some sequencing information. Layer 4 adds information concerning the logical connection between the sender and receiver.

The network layer (3) adds routing information and divides the message into units called packets, the standard units of communication in a network. The data link layer (2) adds both a header and a trailer to ensure correct sequencing of the message blocks and to detect and correct transmission errors. The individual bits of the message and the control information are transmitted on the physical medium by level 1. All additions to the message are checked and removed by the corresponding layer on the receiving side.

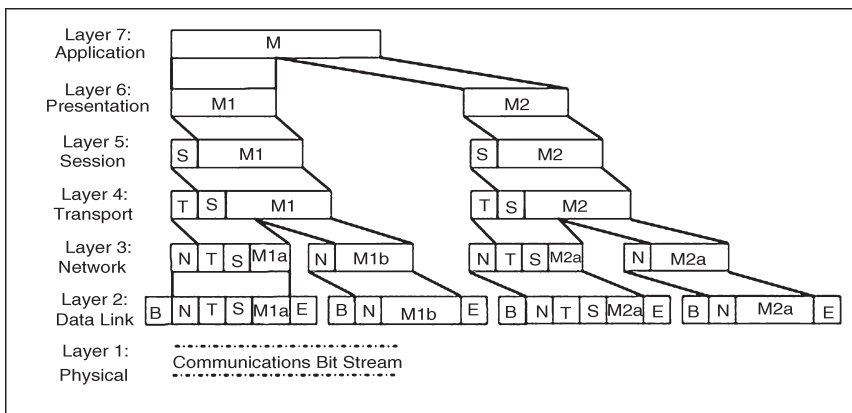


Fig. Message Prepared for Transmission.

The OSI model is one of several transmission models. Different network designers implement network activities in slightly different combinations, although there is always a clear delineation of responsibility. Some designers argue that the OSI model is overly complex; it has too many levels and so other models are typically shorter.

TCP/IP

The OSI model is a conceptual one; it shows the different activities required for sending a communication. However, full implementation of a seven-layer transmission carries too much overhead for megabit-per-second communications; the OSI protocol slows things down to unacceptable levels. For this reason, TCP/IP (Transmission Control Protocol/Internet Protocol) is the protocol stack used for most wide area network

communications. TCP/IP was invented for what became the Internet. TCP/IP is defined by protocols, not layers, but we can think of it in terms of four layers: application, host-to-host (end-to-end) transport, Internet, and physical. In particular, an application programme deals only with abstract data items meaningful to the application user. Although TCP/IP is often used as a single acronym, it really denotes two different protocols: TCP implements a connected communications session on top of the more basic IP transport protocol. In fact, a third protocol, UDP (user datagram protocol) is also an essential part of the suite.

The transport layer receives variable-length messages from the application layer; the transport layer breaks them down into units of manageable size, transferred in packets. The Internet layer transmits application layer packets in datagrams, passing them to different physical connections based on the data's destination (provided in an address accompanying the data). The physical layer consists of device drivers to perform the actual bit-by-bit data communication. How each layer contributes to the complete interaction.

The TCP protocol must ensure the correct sequencing of packets as well as the integrity (correct transmission) of data within packets. The protocol will put out-of-sequence packets in proper order, call for retransmitting a missing packet, and obtain a fresh copy of a damaged packet. In this way, TCP hands a stream of correct data in proper order to the invoking application. But this service comes at a price. Recording and checking sequence numbers, verifying integrity checks, and requesting and waiting for retransmissions of faulty or missing packets take time and induce overhead. Most applications expect a flawless stream of bits, but some applications can tolerate a less accurate stream of data if speed or efficiency is critical.

A TCP packet is a data structure that includes a sequence number, an acknowledgment number for connecting the packets of a communication session, flags, and source and destination portnumbers. A port is a number designating a particular application running on a computer. For example, if Jose and Walter begin a communication, they establish a unique channel number by which their computers can route their respective packets to each of them. The channel number is called a port. Each service uses a well-known port, such as port 80 for HTTP (web pages), 23 for Telnet (remote terminal connection), 25 for SMTP (e-mail), or 161 for SNMP(network management). More precisely, each of these services has a waiting process that monitors the specified port number and tries to perform its service on any data passed to the port. The UDP protocol does not provide the error-checking and correcting features of TCP, but it is a much smaller, faster protocol. For instance, a UDP datagram adds 8 bytes for control information, whereas the more complex TCP packet adds at least 24 bytes.

Addressing

Scheme for communication to occur, the bits have to be directed to somewhere. All networks use an addressing scheme so that data can be directed to the expected recipient.

Because it is the most common, we use the Internet addressing scheme known as IP addresses in our examples, since it is the addressing handled by the IP protocol. All network models implement an addressing scheme. An address is a unique identifier for a single point in the network. For obvious reasons, addressing in shared, wide area networks follows established rules, while addressing in local area networks is less constrained. Starting at the local area network, each node has a unique address, defined in hardware on the network connector device (such as a network interface card) or its software driver. A network administrator may choose network addresses to be easy to work with, such as 1001, 1002, 1003 for nodes on one LAN, and 2001, 2002, and so forth on another.

A host on a TCP/IP wide area network has a 32-bit address, called an IP address. An IP address is expressed as four 8-bit groups in decimal notation, separated by periods, such as 100.24.48.6. People prefer speaking in words or pseudowords, so network addresses are also known by domain names, such as ATT.COM or CAM. AC.UK. Addressing tables convert domain names to IP addresses.

The world's networks are running out of unique addresses. This 32-bit standard address is being increased to 128 bits in a scheme called IPv6. But because 32-bit addresses will remain for some time, we focus on the older version.

A domain name is parsed from right to left. The rightmost portion, such as .COM, .EDU, .NET, .ORG, or .GOV, or one of the two-letter country specific codes, such as .UK, .FR, .JP, or .DE, is called a top-level domain. A small set of organizations called the Internet Registrars controls these top-level domains; the registrars also control the registration of second-level domains, such as ATT in ATT.COM. Essentially, the registrars publish addresses of hosts that maintain tables of the second-level domains contained in the top-level domain.

A host connected to the Internet queries one of these tables to find the numeric IP address of ATT in the .COM domain. AT&T, the company owning the ATT Internet site, must maintain its own host to resolve addresses within its own domain, such as MAIL. ATT.COM. You may find that the first time you try to resolve a fully qualified domain name to its IP address, your system performs a lookup starting at the top; for subsequent attempts, your system maintains a cache of domain name records that lets it resolve addresses locally. Finally, a domain name is translated into a 32-bit, four-octet address, and that address is included in the IP packets destined for that address. (We return to name resolution later in this chapter because it can be used in network attacks.)

Routing Concepts

A host needs to know how to direct a packet from its own IP address. Each host knows to what other hosts it is directly connected, and hosts communicate their connections to their neighbours. For the example network of Figure above, System 1 would inform System 2 that it was one hop away from Clients A, B, and C. In turn, System 2 would inform its other neighbour, System 3, that it (System 2) was two hops

away from Clients A, B, and C. From System 3, System 2 would learn that System 3 was one hop away from Clients D and E, Server F, and System 4, which System 2 would then pass to System 1 as being a distance of two hops. The routing protocols are actually more complex than this description, but the concepts are the same; hosts advertise to their neighbours to describe to which hosts (addresses) they can route traffic and at what cost (number of hops). Each host routes traffic to a neighbour that offers a path at the cheapest cost.

THE CYBER SECURITY STRATEGY

The Federal Government aims at making a substantial contribution to a secure cyberspace, thus maintaining and promoting economic and social prosperity in Germany. Cyber security in Germany must be ensured at a level commensurate with the importance and protection required by interlinked information infrastructures, without hampering the opportunities and the utilization of the cyberspace. In this context the level of cyber security reached is the sum of all national and international measures taken to protect the availability of information and communications technology and the integrity, authenticity and confidentiality of data in cyberspace.

Cyber security must be based on a comprehensive approach. This requires even more intensive information sharing and coordination. The Cyber Security Strategy mainly focuses on civilian approaches and measures. They are complemented by measures taken by the Bundeswehr to protect its capabilities and measures based on mandates to make cyber security a part of Germany's preventive security strategy. Given the global nature of information and communications technology, international coordination and appropriate networks focusing on foreign and security policy aspects are indispensable. This includes cooperation not only in the United Nations, but also in the EU, the Council of Europe, NATO, the G8, the OSCE and other multinational organizations. The aim is to ensure the coherence and capabilities of the international community to protect cyberspace.

PROTECTION OF CRITICAL INFORMATION INFRASTRUCTURES

The protection of critical information infrastructures is the main priority of cyber security. They are a central component of nearly all critical infrastructures and become increasingly important. The public and the private sector must create an enhanced strategic and organizational basis for closer coordination based on intensified information sharing. To this end, cooperation established by the CIP implementation plan is systematically extended, and legal commitments to enhance the binding nature of the CIP implementation plan are examined.

With the participation of the National Cyber Security Council (cf. objective 5), the integration of additional sectors is examined and the introduction of new relevant technologies is considered to a greater extent. Whether and where protective measures have to be made mandatory and whether and where additional powers are required in

case of specific threats have to be clarified, too. Furthermore we will examine the necessity of harmonizing rules to maintain critical infrastructures during IT crises.

STRENGTHENING IT SECURITY IN THE PUBLIC ADMINISTRATION

The public administration will further enhance the protection of its IT systems. State authorities have to serve as role models for data security. We will create a common, uniform and secure network infrastructure in the federal administration (“federal networks”) as a basis for electronic audio and data communication. We will continue to press ahead with the implementation plan for the federal administration. Should the IT security situation get worse, this plan may be aligned accordingly. Effective IT security requires powerful structures in all federal authorities. For this reason resources must be deployed appropriately at central and local level.

To facilitate implementation through uniform action by authorities, joint investments into the Federal Government’s IT security will be made regularly in line with budgetary possibilities. Operational cooperation with the federal Länder, particularly with regard to CERTs (computer emergency response teams), will be further intensified by the IT planning council.

NATIONAL CYBER RESPONSE CENTRE

To optimize operational cooperation between all state authorities and improve the coordination of protection and response measures for IT incidents we will set up a National Cyber Response Centre. It will report to the Federal Office for Information Security (BSI) and cooperate directly with the Federal Office for the Protection of the Constitution (BfV) and the Federal Office of Civil Protection and Disaster Assistance (BBK). Cooperation in the National Cyber Response Centre will strictly observe the statutory tasks and powers of all authorities involved on the basis of cooperation agreements. The Federal Criminal Police Office (BKA), the Federal Police (BPOL), the Customs Criminological Office (ZKA), the Federal Intelligence Service (BND), the Bundeswehr and authorities supervising critical infrastructure operators all participate in this centre within the framework of their statutory tasks and powers.

Quick and close information sharing on weaknesses of IT products, vulnerabilities, forms of attacks and profiles of perpetrators enables the National Cyber Response Centre to analyse IT incidents and give consolidated recommendations for action. The interests of the private sector to protect itself against crime and espionage in cyberspace should also be adequately taken into account.

At the same time respective responsibilities must be observed. Every stakeholder takes the necessary measures in its remit on the basis of the jointly developed national cyber security assessment and coordinates them with the competent authorities as well as partners from industry and academia. Since security preparedness is best achieved by early warning and prevention, the Cyber Response Centre will submit recommendations to the National Cyber Security Council both on a regular basis and

for specific incidents. If the cyber security situation reaches the level of an imminent or already occurred crisis, the National Cyber Response Centre will directly inform the crisis management staff headed by the responsible State Secretary at the Federal Ministry of the Interior.

CYBER PROTECTION/CYBER SECURITY LIABILITY

The dawn of the internet provided many new avenues for doing business, and the protection of computer data has become a main concern in today's working environment. "Cyber Liability," as it is called, encompasses a gamut of new risks for businesses that not only conduct business over the internet, but that also may store private confidential information subject to several federal laws. To provide some insurance protection for these new risks, most insurance companies have developed cyber security liability policies which cover both first party "the insured" and third party liabilities which can arise because of attacks by outside parties or because of losing of sensitive data due to mistakes on the part of the insured. Up until a few years ago, these policies were available under most property and general liability policies. However, since 2005 the loss of intangible property has been excluded under standard property policies; likewise, bodily injury or property damage, previously covered under the general liability policy, is also excluded if it emanates from a cyber breach or error. Some examples of the claims that have occurred are as follows:

THEFT OF DIGITAL ASSETS

A regional retailer contracted with a third party service provider. A burglar stole two laptops from the service provider containing the data of over 800,000 clients of the retailer. Under applicable notification laws, the retailer-not the service provider-was required to notify affected individuals. The total expenses incurred for crisis management notification of customers cost nearly \$5 million. In a second example a home health care organization had backup data, laptops and disks containing social security numbers, clinical and demographic information.

In a small number of cases, patient financial data was stolen. In total, over 365,000 patient records were exposed. The organization settled with the State Attorney General and provided patients with free credit monitoring, credit restoration to patients that were victims of identity fraud, and reimbursement to patients for direct losses that resulted from the data breach.

Human Error

An employee of a private high school mistakenly distributed via e-mail the names, social security numbers, birth dates and medical information of students and faculty creating a privacy breach. Overall, 1,250 individuals' information was compromised.

Malicious Code

A juvenile released a computer worm that caused a launch of denial of service attack against a regional computing consulting and application outsourcing firm. The infection caused an 18 hour shut down of the entity's computer systems. The computer firm incurred extensive cost and expenses to repair and restore their systems, as well as business income expenses which totaled approximately \$875,000. The cyber liability policy is sufficiently flexible so that a business can buy either third party or first party coverages, or both. Here is an example of how the language in a carrier's insurance policy protects first party assets:

Loss of Digital Assets

"We will indemnify you for loss you incur, in excess of the deductible, as a result of damage, alteration, corruption, distortion, theft, misuse, or destruction of your digital assets directly caused by a covered cause of loss". In this case, digital assets mean electronic data and computer programmes that exist in a computer system. Please note that digital assets do not include computer hardware. In addition to this, protection can be provided for the business income loss that may result from any direct first party claims. Here is an example of the coverage grants available for third party protection under a typical insurance carrier's policy:

Network Security and Privacy Liability Coverage

The insurance company "will pay on your behalf those amounts, in excess of the applicable deductible, which you are legally obligated as damages on claim expenses arising from your acts, errors or omissions or from acts, errors or omissions for others for whom you are legally responsible, including outsourcers, or vendors provided such acts, errors or omissions follow a security breach or privacy breach". An example of this claim is the loss of a laptop containing sensitive information, which results in the public disclosure of a person's private information. Also, this insuring agreement would cover unauthorized access into your computer system, a denial of service attack against your computer system, or an infection of your computer system by malicious code. There are a number of other coverage protection grants available on most policies, which The Fedeli Group, after an assessment of your risks, can review with you. Please contact The Fedeli Group team to arrange for this type of risk assessment and possible valuable insurance protection.

THE FUTURE OF CYBER SECURITY

The product of human ingenuity and innovation, cyberspace now delivers a range of critical services to more citizens around the world than ever before. Yet, the online world as we know it stands at the threshold of unprecedented change. Being invited to speak at the EastWest Institute's Worldwide Security Conference in Brussels this week provided an opportunity to examine the needs faced by the global security community as we prepare to

meet the needs of the Internet's next billion users. The International Telecommunication Union (ITU) reported that the number of Internet users reached the two billion threshold in March of this year and according to a Boston Consulting Group report, another billion are expected to come online in the next four years, bringing the total number of Internet users worldwide to about three billion by 2015. Planning to ensure that our online world—cyberspace—is trustworthy, resilient and secure as we move into this uncertain future, policy leaders need to consider the fundamental changes that are occurring in cyberspace, and the policy issues that these changes will likely present and that will need to be addressed. Looking towards the future of three billion users, four factors will fundamentally change the future of cyberspace security: people, devices, data, and cloud services.

THE GLOBAL ONLINE POPULATION EXPANDS

The emergence of the next billion Internet users will impact security in two ways. Most important will be the impact of the demographic characteristics of these users. Consider that the next billion users will (1) be younger, (2) spend more time online, (3) be more mobile, (4) see the world through social media and apps, and (5) make greater use of natural interfaces. These five factors could hasten the onset of totally digital lifestyles making connectivity seemingly as essential as oxygen. The next billion could also help foster new innovation in the development and application of technology. Separately, however, this emerging user population also represents a greatly expanded “target rich” environment for cybercriminals that want to exploit their data, social networks, and devices via botnets or other means.

Internet of Things to Immersive Computing

These new users will require new devices. According to The Boston Consulting Group, the number of Internet-connected devices is predicted to exceed 15 billion—twice the world's population—by 2015, and to soar to 50 billion devices by 2020. “Devices” of course refers to more than smart phones, netbooks and tablets. It also systems such as smart grids, intelligent transportation, Health care monitoring, smart manufacturing, and environmental sensors. The advent of powerful wireless devices that both run infrastructure and deliver infrastructure services, including providing access to cloud services, means that cybercriminals and other threat actors need not merely target traditional, and increasingly protected, commercial software and consumer applications to execute attacks with significant consequences. Attackers may well target the embedded software, firmware and hardware in these devices to attack the infrastructure or seize control of the devices and turn them into sensors that can report status, collect personally identifiable information, or conduct other espionage.

Data: Rapid Increases in Understanding

The striking growth in the number of users and devices will also produce an exponential growth in the amount of data that is being generated, stored, analysed

and transformed into innovation and knowledge. Analysing large data sets—so-called big data—will become a key basis of competition, underpinning new waves of productivity growth, innovation, and consumer surplus, according to research by MGI and McKinsey's Business Technology Office. However, such data sets also represent attractive targets for organized cyber criminals and other threat actors. From a security standpoint, safeguarding these huge data sets, protecting privacy and integrity, will require concerted global effort requiring collaboration among governments, the private sector, and users.

Cloud Computing: The Information Society Enabler

With an exponentially growing community of increasingly mobile users, cloud computing will commensurately grow in importance. It will fundamentally change how businesses operate, how every manner of services are delivered, and even how lives are lived. On the positive side, the security best practices implemented by an effective cloud provider may rival or surpass the measures that cloud customers might themselves be able to provide, resulting in enhanced security. Yet there are global issues that will need to be addressed in terms of transparency and jurisdiction to enable cloud services that are both secure and scalable to service the needs of this expanded user community across multiple countries.

Reducing the Cyber-attack Surface

Reducing the cyber-attack surface can be achieved by industry and government working in partnership to make the ICT infrastructure less susceptible to attack and compromise. One important way to achieve this is through concerted action to address risks in the supply chain for information and communications technology products and services. Vendors and service providers need to build and maintain world-class approaches to secure software and hardware development methodologies. Microsoft began this journey over 10 years ago and has openly shared its Security Development Lifecycle. The non-profit alliance SAFECODE provides a platform for companies to share, both within the software development community and more broadly, information on secure software development techniques that have proven to be effective as well as those that have not. Industry needs to do more. For their part, governments need to understand the nature of ICT supply chain risk more clearly and work collaboratively with one another and with vendors to develop a common risk management framework rooted in core principles that both address supply chain integrity concerns and preserve the fruits of global free trade. Such a system should be risk-based, transparent, flexible, and should recognize the realities of reciprocal treatment in the global economic environment.

Improving Internet Health

Improving Internet health requires a global, collaborative approach to protecting people from the potential dangers of the Internet. Despite the best efforts at education

and protection, many consumer computers host malware and may be part of a “botnet,” unbeknownst to their legitimate owners. There is currently no concerted mechanism to shield users from or help them mitigate these risks. Such infected computers do not simply expose their owners’ valuable information and data; they place others at risk too. This threat to greater society makes it essential that the technology ecosystem take collective action to combat it. Work has been underway in industry circles to build cooperation among various stakeholder groups including ISPs, software vendors, and others; to leverage investments made in key regions of the world; and to create a future roadmap for an Internet health system. The active discussions of cyber security policy and legislation now underway in many nations afford a ripe opportunity to promote this Internet health model. As part of this discussion, it is important to focus on building a socially acceptable model. While the security benefits may be clear, it is important to achieve those benefits in a way that does not erode privacy or otherwise raise concern.

Jurisdiction and Sovereignty

Issues of jurisdiction and sovereignty have quickly come to the fore in the era of the Internet. The Internet does not tend to make geographical and jurisdictional boundaries clear, but Internet users remain in physical jurisdictions and are subject to laws independent of their presence on the Internet.

As such, a single transaction may involve the laws of at least three jurisdictions:

1. The laws of the state/nation in which the user resides,
2. The laws of the state/nation that apply where the server hosting the transaction is located, and
3. The laws of the state/nation which apply to the person or business with whom the transaction takes place.

So a user in one of the United States conducting a transaction with another user in Britain through a server in Canada could theoretically be subject to the laws of all three countries as they relate to the transaction at hand.

Jurisdiction is an aspect of state sovereignty and it refers to judicial, legislative and administrative competence. Although jurisdiction is an aspect of sovereignty, it is not coextensive with it. The laws of a nation may have extra-territorial impact extending the jurisdiction beyond the sovereign and territorial limits of that nation. This is particularly problematic as the medium of the Internet does not explicitly recognize sovereignty and territorial limitations. There is no uniform, international jurisdictional law of universal application, and such questions are generally a matter of conflict of laws, particularly private international law. An example would be where the contents of a web site are legal in one country and illegal in another. In the absence of a uniform

jurisdictional code, legal practitioners are generally left with a conflict of law issue. Another major problem of cyber law lies in whether to treat the Internet as if it were physical space (and thus subject to a given jurisdiction's laws) or to act as if the Internet is a world unto itself (and therefore free of such restraints). Those who favour the latter view often feel that government should leave the Internet community to self-regulate. John Perry Barlow, for example, has addressed the governments of the world and stated, "Where there are real conflicts, where there are wrongs, we will identify them and address them by our means."

We are forming our own Social Contract. This governance will arise according to the conditions of our world, not yours. Our world is different" (Barlow, *A Declaration of the Independence of Cyberspace*). A more balanced alternative is the Declaration of Cybersecession: "Human beings possess a mind, which they are absolutely free to inhabit with no legal constraints. Human civilization is developing its own (collective) mind. All we want is to be free to inhabit it with no legal constraints. Since you make sure we cannot harm you, you have no ethical right to intrude our lives. So stop intruding!". Other scholars argue for more of a compromise between the two notions, such as Lawrence Lessig's argument that "The problem for law is to work out how the norms of the two communities are to apply given that the subject to whom they apply may be in both places at once" (Lessig, *Code* 190).

Though rhetorically attractive, cybersecession initiatives have had little real impact on the Internet or the laws governing it. In practical terms, a user of the Internet is subject to the laws of the state or nation within which he or she goes online. Thus, in the U.S., Jake Baker faced criminal charges for his e-conduct, and numerous users of peer-to-peer file-sharing software were subject to civil lawsuits for copyright infringement. This system runs into conflicts, however, when these suits are international in nature. Simply put, legal conduct in one nation may be decidedly illegal in another.

In fact, even different standards concerning the burden of proof in a civil case can cause jurisdictional problems. For example, an American celebrity, claiming to be insulted by an online American magazine, faces a difficult task of winning a lawsuit against that magazine for libel. But if the celebrity has ties, economic or otherwise, to England, her or she can sue for libel in the British court system, where the standard of "libelous speech" is far lower. It's nice to know that there is such law to prevent "CYBER CRIME".

NET NEUTRALITY

Another major area of interest is net neutrality, which affects the regulation of the infrastructure of the Internet. Though not obvious to most Internet users, every packet of data sent and received by every user on the Internet passes through routers and transmission infrastructure owned by a collection of private and public entities, including telecommunications companies, universities, and governments, suggesting that the Internet is not as independent as Barlow and others would like to believe. This is turning into one

of the most critical aspects of cyber law and has immediate jurisdictional implications, as laws in force in one jurisdiction have the potential to have dramatic effects in other jurisdictions when host servers or telecommunications companies are affected.

FREE SPEECH IN CYBERSPACE

In comparison to traditional print-based media, the accessibility and relative anonymity of cyber space has torn down traditional barriers between an individual and his or her ability to publish. Any person with an internet connection has the potential to reach an audience of millions with little-to-no distribution costs. Yet this new form of highly-accessible authorship in cyber space raises questions and perhaps magnifies legal complexities relating to the freedom and regulation of speech in cyberspace. Recently, these complexities have taken many forms, three notable examples being the Jake Baker incident, in which the limits of obscene Internet postings were at issue, the controversial distribution of the DeCSS code, and *Gutnick v Dow Jones*, in which libel laws were considered in the context of online publishing. The last example was particularly significant because it epitomized the complexities inherent to applying one country's laws (nation-specific by definition) to the internet (international by nature). In 2003, Jonathan Zittrain considered this issue in his paper, "Be Careful What You Ask For: Reconciling a Global Internet and Local Law".

In many countries, speech through cyberspace has proven to be another means of communication which has been regulated by the government. The Open Net Initiative, whose mission statement is "to investigate and challenge state filtration and surveillance practices" in order to "...generate a credible picture of these practices," has released numerous reports documenting the filtration of internet-speech in various countries. While China has thus far proven to be the most rigorous in its attempts to filter unwanted parts of the internet from its citizens, many other countries - including Singapore, Iran, Saudi Arabia, and Tunisia - have engaged in similar practices. In one of the most vivid examples of information-control, the Chinese government for a short time transparently forwarded requests to the Google search engine to its own, state-controlled search engines. These examples of filtration bring to light many underlying questions concerning the freedom of speech, namely, does the government have a legitimate role in limiting access to information? And if so, what forms of regulation are acceptable? The recent blocking of "blogspot" and other websites in India failed to reconcile the conflicting interests of speech and expression on the one hand and legitimate government concerns on the other hand. In the UK the case of *Keith-Smith v Williams* confirmed that existing libel laws applied to internet discussions.

LEGAL ASPECTS

An indispensable part of a successful business environment in any country is its legal aspects. They reflect the policy framework and the mind set of the Governmental structure of that country. They ensure that every company is functioning as per the

statutory framework of the country. Every enterprise must take into account this legal set up while framing the basic aims and objectives of its company. This is because, it is necessary for efficient and healthy functioning of the organisation and helps it to know about the rights, responsibilities as well as the challenges that it may have to face. In India, Companies Act, 1956 is the most important law regulating all aspects of a company.

It contains provisions relating to formation of a company, powers and responsibilities of the directors and managers, raising of capital, holding company meetings, maintenance and audit of company accounts, powers of inspection and investigation of company affairs, reconstruction and amalgamation of a company and even winding up of a company. All the transactions of a company are regulated by the Indian Contract Act, 1872. It lays down the general principles relating to the formation and enforceability of contracts; rules governing the provisions of an agreement and offer; the various types of contracts including those of indemnity and guarantee, bailment and pledge and agency. It also contains provisions pertaining to breach of a contract.

The other major legislations are: The Industries (Development and Regulation) Act 1951; Trade Unions Act; the Competition Act, 2002; the Arbitration and Conciliation Act, 1996; the Foreign Exchange Management Act (FEMA), 1999; laws relating to intellectual property rights; as well as laws relating to labour welfare.

NEW DIMENSIONS AND LEGAL REFORMS

The MRTP Act was amended in 1991 and the Government realized that pre-entry restrictions under the MRTP Act on the investment decision of the corporate sector outlived its utility, and became a hindrance to the speedy implementation of industrial projects. The Act was restructured with focus on curbing monopolistic restrictive and unfair trade practices. Ten years after this amendment, the Government realized that the whole setup had become an anachronism, and a committee was set up to suggest ways and means to promote competition.

On the basis of the S. V. S Raghavan Committee report, the Government came out with a new tax law styled the Trade Related Competition Bill. Amongst the first changes to be made were amendments to the MRTP Act to delete certain restrictive provisions that created pre-investment barriers. Since then, there has been a more competition-friendly regime, with greater freedom for business entities that has released the energies of the private sector and resulted in a more dynamic and faster growing economy. The forces of competition have brought benefits both for the business sector as well as for the consumers.

COMPETITION LAW

The Competition Law focuses on the firm's structure, not size and this is where the Competition law is different from the MRTP Act. The MRTP Act lists 14 offences. The Competition Law will recognize only four such offences. The MRTP Act's role was

only advisory. The proposed competition commission can initiate suo motu proceedings and levy penalties. The Competition Law does not consider the firm's dominance *per se* detrimental to competition. Frowning upon dominance, the MRTP Act laid down an arithmetical test.

It is only abuse of dominance that the Competition Law frowns upon. It seeks to regulate agreements that control production, supply, markets, technical developments or investment in provision of services. All such agreements are considered anti-competitive and penalty can be levied. The recent experience with cement cartels is an example of price rigging under agreements to share markets. Sources of production by way of geographical allocation of market will be considered anti-competitive.

Agreement arrived at between enterprises that directly result in bid rigging or collusive tendering is prohibited by the Competition Law. The Act defines bid rigging as agreements between enterprises or persons engaged in identical or similar manufacturing, trading or provision of service that has the effect of eliminating or reducing competition for bids, or adversely affecting/manipulating bidding. Apart from these guidelines, other restrictive trade practices will fall under the rule of reason test before the Competition Commission of India (CCI). Registration of agreements was mandatory under the MRTP Act, but the Competition Law has no such requirement. A combination is sought to be regulated beyond a threshold limit. The Government appointed the MRTP Commission's Chairman. A Collegium consisting of the Chief Justice, the Finance and other Cabinet Ministers, the Cabinet Secretary and the RBI Governor, will select the CCI chairman.

An important difference lies in the fact that the Competition Law leaves unfair trade practices to consumer fora and not to the CCI. The MRTP Act had definition of unfair trade practices u/S 36A that lists out various modes and possibilities of Unfair Trade Practice. The Competition Law will take care of cartelization that imposes unjustified cost on the consumers. Cases such as Sumitomo Corporation and others that quoted identical prices pursuant to a global tender floated by the Steel Authority of India will be easier to deal with under the new law.

The requirement of pre-merger notifications is dispensed with by the Competition Law. There will be a separate Bench of the CCI styled as the Merger Commission to deal with mergers. This part of the Law requires disposal of the case within 90 days. If there is no order, it will be presumed the merger has been approved. There is also possibility of power sharing conflict with SEBI, another body dealing with takeover bids of substantial shareholding of any listed company in India. The Competition Law also seeks to regulate predatory pricing though it is not clear how the Commission will determine pricing below the cost. There is bound to be some overlap between the functions of the High Court and the CCI when it comes to cases involving mergers, acquisitions and amalgamations.

The CCI is also expected by the Competition Act to promote competition through advocacy. This is in line with the thinking of the OECD. The OECD had pointed out

that in every member-country where significant reform efforts have been undertaken, the competition agencies have been active participants in the reform process. Advocacy can include persuasion behind the scenes and publicity outside. This part of the Act appears to suggest that the CCI should interact through the media to promote competition.

Objectives of Competition Law

Protecting the interests of consumers is one of the objectives of the Competition Law. They purchase the products of other firms as their inputs, and the price and quality of these inputs partly determine their own competitiveness and profits that they are able to make. Another objective is stimulating efficiency and competitiveness across the economy. Competition helps to achieve efficient allocation of resources and stimulates dynamism through the free entry and exit of firms into markets, raising productivity. At the same time Competition Law aims in preventing unfair competition. Businesses have a strong interest in ensuring that their competitors do not have an unfair advantage, and that their suppliers and customers are also operating in competitive markets. A large proportion of complaints of anticompetitive practices in firms are raised by other businesses. Competition policy relates to poverty reduction through these routes.

Many sectors, like water and electricity that are prone to anti-competitive behaviour, affect the poorest most. These take up a much higher proportion of the budget of poorer households. Furthermore, the poor benefit by the indirect route of economic growth and job creation. Of course, many other policies need to be in place if the poor are to benefit from economic growth, but under the right conditions it will help them.

Competition Law and Business

Competition policy is sometimes used as a weapon for businesses to use against each other. Rulings of the competition authority are seen as a redistribution of profits between businesses at different stages in the production chain or between horizontal competitors. However, it is the consumer who can and should be the ultimate beneficiary of competition law and policy. For the consumer to truly benefit, several requirements have to be met, both in the provisions of the law itself as well as an active well-organized and informed consumer movement.

Elements of Competition Culture

A competition culture promotes a well-regulated market; efficient allocation of resources and efficient production processes in the economy. It stimulates firms to compete with one another in price, quality and through the innovation of new products. It ensures that the benefits of this efficiency do not just accrue to firms, but lead to greater value and choice for consumers, environmental protection and sustainable development. It is made up of a number of elements, of which laws form only one part. All stakeholders have a role to play in creating this culture of competition. Among the most important elements are:

Active Consumers

Consumers can be the key beneficiaries of competition policy. But in order for them to reap these benefits, they need to take an active part in implementation of the law. This can be done by identifying product markets of concern, providing information to the competition authorities and perhaps bringing cases against offending businesses. Despite their numbers, consumers are often not well organized. When raising the profile of consumers, it is extremely important to strengthen consumer organisations and representative groups.

Responsible Business

For a competition culture to thrive, businesses need to respect certain standards of behaviour. This does not just mean refraining from cartelisation and the restrictive business practices that competition law is concerned with, but also maintaining standards of transparency, corporate governance, accounting practices and combating corruption as well as respecting social and environmental regulations. At the heart of fair competition is this respect for rules.

Sensitized Media

That public awareness and support necessary for competition laws to be implemented effectively has been noted above. The media has an extremely important role to play both in drawing attention to particular concerns and helping competition institutions and officials to demonstrate their role and public support when dealing with other public and private sectors institutions.

Institutional Cooperation

In any country, there are a large number of institutions that influence the competition culture. All these institutions and the people that work in them need to be sensitive to competition concerns and to cooperate with each other. Otherwise, the competition culture is in danger of being over-run by a bureaucratic culture and rent-seeking behaviour.

PRINCIPLES OF JURISDICTION

One of the advantages of the Internet over other methods of communication and commerce is that it enables access to a much wider, even a worldwide, audience. Spatial distance and national borders are irrelevant to the creation of an Internet business, many of which are conceived for the express purpose of expanding sales horizons across borders. In a sense, a person can be everywhere in the world, all at once. This ease of communication raises a vital legal question, however: when a person puts up a web site on his home server and allows access to it from all points on the globe, does he subject himself to the governance of every law- and rule-maker in the world? Under the current system, in order to decide what state's or nation's laws govern disputes that arise over

Internet issues, a court first must decide “where” Internet conduct takes place, and what it means for Internet activity to have an “effect” within a state or nation. Even apart from the Internet, this border-centric view of the law creates certain difficulties in an economy moving toward globalization

Entire bodies of law have been developed by every nation to deal with the resolution of international conflicts of law, conflicts that arise when geography and citizenship would allow a dispute to be decided by the laws of more than one country, and the laws of those countries are not consistent with each other. Conflicts of law are particularly likely to arise in cyberspace, where the location of an occurrence is never certain, where ideological differences are likely to create conflicting laws, and where rules are made not only by nations and their representatives, but also by sub-national and transnational institutions.

II. The Test Currently in Force

A. In the United States

A court does not have power over every person in the world. Before a court may decide a case, the court must determine whether it has “personal jurisdiction” over the parties. A plaintiff may not sue a defendant in a jurisdiction foreign to the defendant, unless that defendant has established some relationship with that forum that would lead him to reasonably anticipate being sued there.

In the U.S., the Due Process clause of the Constitution’s Fourteenth Amendment sets the outermost limits of personal jurisdiction. If a party has substantial systematic and continuous contacts with the forum, a court may exercise jurisdiction over a party for any dispute, even one arising out of conduct unrelated to the forum. This is known as general jurisdiction. For example, a corporation or person can always be sued in its state of residence or citizenship or its principal place of business, regardless of whether or not the claim arose there. If a party is not present in the state or does not have systematic and continuous contacts with the state, courts may exercise jurisdiction over a party for causes of action arising out of his contacts with the state, or arising out of activities taking place outside the state expressly intended to cause an effect within the state. This “effects” test is described from the American Law Institute’s Restatement (Second) of Conflict of Laws 37 (1971), which provides:

“A state has power to exercise judicial jurisdiction over an individual who causes effects in the state by an act done elsewhere with respect to any cause of action arising from these effects unless the nature of the effects and of the individual’s relationship to the state make the exercise of such jurisdiction unreasonable.” To do this, the court must look to the state’s “long-arm” statute, which sets the parameters for the state’s exercise of its constitutional power to govern conduct by non-citizens (including both Americans and foreigners). Long-arm statutes vary widely from state to state. For example, Arizona grants the broadest possible freedom to its courts: “Arizona will

exert personal jurisdiction over a nonresident litigant to the maximum extent allowed by the federal constitution.” New York, on the other hand, gives a more restricted and specific charge to its courts with its statute, which allows personal jurisdiction over those who transact business or commit a tortious act within the state of New York, and over those who commit an act outside the state that could reasonably be expected to have a tortious effect within New York.

The Federal courts have the equivalent of a long-arm statute of their own, in Federal Rule of Civil Procedure 4(k) (Rule 4(k)), which provides three basic grants of jurisdiction. First, it authorizes federal courts to “borrow” the long-arm statute of the state in which the federal court is located. Second, Rule 4(k) authorizes federal courts to exercise grants of personal jurisdiction contained in federal statutes, such as the federal securities and antitrust law, which have their own jurisdiction provisions. And third, Rule 4(k)(2) grants long-arm jurisdiction in an international context, within the boundaries of the Constitution, over parties to cases arising under federal law who are not subject to the jurisdiction of any particular state. The concept of being able to have minimum contacts with the United States as a whole has profound implications for the Internet and international jurisdiction. Users all over the world, without establishing contacts in a particular state, could establish contacts with the entire country with nearly every foray into cyberspace.

In order to be subject to personal jurisdiction in a state that is not his domicile, not only must a person fit under the ambit of the state’s “long-arm” statute, but also the state’s jurisdiction must be valid under the Due Process Clause of the Fourteenth Amendment. The Supreme Court set the standard for constitutional exercise of jurisdiction in *International Shoe Co. v. Washington*. Pursuant to the Due Process Clause, a nonresident defendant may not be sued in a forum unless it has first established sufficient “minimum contacts with [the forum] such that the maintenance of the suit does not offend traditional notions of fair play and substantial justice.” In addition, the nonresident’s “conduct and connection with the forum [must be] such that he should reasonably anticipate being haled into court there.” This test relies on courts to decide, according to “traditional notions of fair play and substantial justice,” what contacts are sufficient.

Courts will generally hold that contacts are sufficient to satisfy due process only if the nonresident “purposefully availed” itself of the benefits of being present in, or doing business in, the forum. According to a the plurality of the Supreme Court in *Asahi Metal Industry v. Superior Court*, a connection sufficient for minimum contacts may arise through an action of the defendant purposefully directed toward the forum State.

The placement of a product into the stream of commerce, without more, is not an act of the defendant purposefully directed toward the forum State, but advertising or marketing in the forum state may fulfill the deliberate availment requirement. There must be clear evidence that the defendant sought to serve the particular market. If the minimum contacts test is met, a court may only exercise jurisdiction if it is “reasonable” to do so. In determining reasonableness, a court must weigh and consider the burden on

the defendant to litigate in the forum, the forum state's interests in the matter, the interest of the plaintiff in obtaining relief, efficiency in resolving the conflict in the forum, and the interests of several states in furthering certain fundamental social policies.

In sum, under U.S. law, if it is reasonable to do so, a court in one state will exercise jurisdiction over a party in another state or country whose conduct has substantial effects in the state and whose conduct constitutes sufficient contacts with the state to satisfy due process. Because this jurisdictional test is ambiguous, courts in every state of the U.S. may be able to exercise jurisdiction over parties anywhere in the world, based solely on Internet contacts with the state.

B. Internationally

There is little dispute that nation-states can prosecute Internet users (or anyone else, for that matter), whatever their location, for revealing national secrets, falsifying official documents, or inciting war. These activities threaten national security, wherever they are committed, and therefore fall under international standards for jurisdiction. Similarly, it is a universal crime to publicly incite torture or genocide. These universal offences may be prosecuted extraterritorially by any nation, regardless of the citizenship or location of the user.

These are easy cases, however. Nations may also be interested in enforcing non-universal laws extraterritorially; for example, In Germany, it is illegal to import distribute material espousing a Nazi or Neo-Nazi viewpoint. Such material is not difficult to find in USENet or on the World Wide Web. German authorities may be interested not only in interpreting German laws to classify Internet viewing as "importation" of material, but also (in part because of the difficulty of locating those who break an importation statute without leaving their own homes) in prosecuting those who make such material available to Germans via the Internet.

If German authorities attempted to prosecute a U.S. citizen or resident for such an offence, however, they would be met with great opposition by the U.S., which certainly would not enforce any judgment against the U.S. citizen in such a case, because the German statute violates U.S. Constitutional principles. Under U.S. law, because it would be prohibitively difficult to prevent German users from viewing such a site and therefore the result of such a prosecution would be to chill otherwise legal (if unpleasant) speech in the U.S. Under the current system, it is possible to envision that German courts may have jurisdiction over Americans who publish such material, even though the material may not be "purposefully directed" (one interpretation of the American standard) toward Germany in the way a mailing of flyers would be.

As discussed above, U.S. courts apply the same "effects" test to foreign parties as to American parties. If minimum contacts exist, parties from other countries may be haled into court in the United States just as parties from one state may be haled into another. Similarly, Americans may be tried by courts in other countries depending on the rules of that country. Although each country's laws are different, most rely on some sort of

“effects” test resembling the U.S. test, whereby a party is subject to jurisdiction in a place where his conduct has an effect. This jurisdiction traditionally is subject to a “reasonableness” test. According to section 421 of the Restatement (Third) of the Foreign Relations Law of the U.S., exercise of jurisdiction is generally reasonable if the party is a citizen, resident, or domiciliary of the state, or if:

- The person, whether natural or personal, has consented to the exercise of jurisdiction;
- The person, whether natural or juridical, regularly carries on business in the state;
- The person, whether natural or juridical, had carried on activity in the state, but only in respect of such activity;
- The person, whether natural or juridical, had carried on outside the state an activity having a substantial, direct, and foreseeable effect within the state, but only in respect of such activity; or
- The thing that is the subject of adjudication is owned, possessed, or used in the state, but only in respect of a claim reasonably connected with that thing.

This standard differs somewhat from the U.S. standard for interstate exercise of jurisdiction; for example, transitory presence (known as “tag” jurisdiction), accepted in the U.S., is not generally accepted as a method of international jurisdiction.

Every nation has an obligation to exercise moderation and restraint in invoking jurisdiction over cases that have a foreign element, and they should avoid undue encroachment on the jurisdiction of other States. Although countries are given great discretion in deciding whether to exercise jurisdiction over conduct in other countries, international law dictates that a country exercising its jurisdiction in an overly self-centered way not only contravenes international law, but can also “disturb the international order and produce political, legal, and economic reprisals.”

Based on this traditional moderation, and the relatively high threshold of the “reasonableness” standard discussed above, it is unlikely that foreign nations will have the sort of long-arm power over citizens of other nations as states have over citizens of other states within the U.S. today. Scholars have suggested that individual persons and small commercial entities whose only contacts with a nation are on-line are, in all likelihood, more insulated from international jurisdiction than they are from interstate jurisdiction. This is largely speculative, however, because international Internet jurisdiction cases have thus far been rare, and nations have not hesitated to pass laws conferring global jurisdiction for Internet activities.

III. Application of the “effects” test to the Internet

A. in the United States

The Supreme Court has not discussed the impact that technology might have on the analysis of personal jurisdiction. Lower courts, on the other hand, have explored the

question of cyberspace jurisdiction. While most have held that merely creating and hosting a web site available to all does not subject a person to general jurisdiction everywhere in the U.S., they diverge widely as to whether the presence of such a site will lead to specific jurisdiction over the party for the purposes of disputes arising from the web site.

Some decisions suggest that a court may obtain personal jurisdiction over a non-resident defendant whose sole contact with the forum state arose through the Internet. Examples of these include: *CompuServe, Inc. v. Patterson*, *Zippo Manufacturing v. Zippo Dot Com, Inc.*, *Panavision International, L.P. v. Toeppen*, and *Maritz, Inc. v. Cybergold*. In each of these cases, Internet contacts with the forum state exceeded those of a passive web site: In *CompuServe*, the defendant knowingly reached out to and did business with CompuServe, knowing that CompuServe was an Ohio corporation. In addition, the dispute arose out of contacts with the forum state. In *Zippo*, the defendant's site required participants to submit address information in order to receive a news service; therefore, the site operators knowingly transacted business with residents of the forum state, where the plaintiff was headquartered.

In *Panavision*, the defendant had set up his web site as part of a "scam" to make the plaintiff purchase the domain name from him, and as such had intentionally directed his actions toward the plaintiff's home state. In *Maritz*, the defendant's site invited users to send and receive information about services it offered, and the defendant company had send information to over 100 users in the forum state. The court found that "[a]lthough [defendant] characterizes its activity as merely maintaining a 'passive web site,' its intent is to reach all Internet users, regardless of geographic location."

Two other recent decisions, in declining to exercise jurisdiction, support the notion that passive Internet sites are not sufficient to support jurisdiction. In *McDonough v. Fallon McElligott, Inc.*, a Minnesota defendant had displayed plaintiff's photographs on the Web without plaintiff's consent, in possible violation of California copyright and unfair competition laws. The Southern District of California held that: "Because the Web enables easy world-wide access, allowing computer interaction via the Web to supply sufficient contacts to establish jurisdiction would eviscerate the personal jurisdiction requirement as it currently exists Thus, [having] a Web site used by Californians cannot establish jurisdiction by itself." Similarly, in *Benusan Restaurant Corp. v. King*, the Southern District of New York held that the operator of a small Missouri jazz club called "The Blue Note" did not subject it to New York's trademark laws by erecting an advertising site on the Web.

The New York district court's holding in *Benusan* is at direct loggerheads with the District of Connecticut's holding in *Inset Systems, Inc. v. Instruction Set, Inc.* In *Inset*, a party utilizing the trademark of another company for its domain name and "800" number was subject to jurisdiction in the home of the party whose mark was infringed. Also in seeming conflict with *Benusan* and most other U.S. interstate Internet jurisdiction cases, the Federal Circuit found in *Graphic Controls Corp. v. Utah Medical Prods., Inc.*, that a

Utah corporation's activities, which included having an open-access web site for ordering goods, having an "800" number, having meetings in New York unrelated to the cause of action, and sending "cease and desist" letters to party in New York, did not constitute minimum contacts with New York. In similar conflict with the above cases, the Southern District of New York held that creating a commercial and interactive (though not yet operational at the time of litigation) web site that was available to, and used by, New York residents was not in itself enough contact to subject a publisher to New York jurisdiction in *Hearst Corp. v. Goldberger*. The District court found that exercising jurisdiction would violate traditional notions of fair play, and noted that the site operator did not purposefully direct his activities toward New York.

The disagreements between the cases above illustrate some of the variety among courts as to the proper approach to take when dealing with Internet jurisdiction. Approaches differed greatly, even among some of the above cases having similar final outcomes. States have not regularized an approach to the Internet, preferring to analogize it to real space. Erecting a web site has been compared to publishing in a widely distributed general-interest magazine or putting an item (with the capacity to travel) in the stream of commerce by selling it locally. As the above illustrate, courts seem to be taking an approach resembling that recently laid down by the Ninth Circuit Court of Appeals in *Cybersell, Inc. v. Cybersell, Inc.*, which held that the mere presence of a passive web site on the Internet does not constitute the minimum contacts needed to subject a person to the jurisdiction of every court and that "something more," either interactivity or purposeful direction, is needed to justify jurisdiction.

What degree of interactivity is required to constitute minimum contacts, however, remains largely unclear from case law. Under the rule set forth in *Cybersell*, a court would decide whether a web site creates minimum contacts by examining the degree to which the site is commercial and interactive, and the degree to which the site is directed at citizens of the forum state. The more interactive a site is (i.e. the more exchange of information is possible between the site and the user), and the more commercial the site's nature, the more likely a court is to find that contact exists between the site owner and the distant user. Similarly, the more the site is directed at an audience in the forum site or designed to harm citizens of the forum state, the more likely a court will be to find that purposeful availment has occurred. Still, the Supreme Court has not addressed the issue of persona jurisdiction in cyberspace and many details still remain unresolved.

B. Internationally

U.S. courts have, basically, shoehorned Internet cases into the same jurisdictional rules that they use for non-Internet cases, with the result that U.S. courts lean toward limiting jurisdiction, regulating only sites that intentionally direct themselves into the U.S. in some way. Other countries have not limited their courts so. Several examples illustrate that jurisdictional issues are at least as severe and jumbled in the international context as they are within the domestic U.S.

In the United Kingdom, the Financial Services Act of 1996 makes it a criminal offence to place investment ads in the U.K. unless they are issued or approved by the Financial Services Authority (FSA). In early 1998, the FSA notified the national U.S. mutual fund association that mutual fund Web sites which can be brought up on a screen in the U.K. are considered to have been issued in the U.K. This could have had a profound impact on the way in which U.S. mutual fund sites operated, however, the FSA stated that it would not take enforcement action against U.S. companies that complied with certain FSA regulations, including placing disclaimers or warnings on their Web sites.

Germany has passed a sweeping law that subjects any Web site accessible in Germany to German law, holding Internet service providers (ISPs) liable for violations of German content laws if the providers were aware of the content and were reasonably able to remove the content. This followed the settlement of a well-publicized incident between Germany and CompuServe, in which German authorities threatened to prosecute CompuServe for allegedly pornographic news groups. In response to the German threat, CompuServe blocked access to those newsgroups to all users, approximately 4 million worldwide. Later, CompuServe restored access and distributed free software for blocking pornography. This caused CompuServe's indictment for aiding in the distribution of pornography and computer games. Prosecutors charged that CompuServe did not do enough to block Germans from accessing the material.

Malaysia's new cyberspace law also extends well beyond the borders of Malaysia. The bill applies to offences committed by a person in any place, inside or outside of Malaysia, if at the relevant time the computer, programme, or data was either (i) in Malaysia or (ii) capable of being connected to or sent to or used by or with a computer in Malaysia. The offender is liable regardless of his nationality or citizenship.

IV. Conflicts of Law

As mentioned above, the Constitution and states' long-arm statutes may permit court jurisdiction over out-of-state conduct, depending on the specific long-arm statute and the conduct involved. This means that many states may have concurrent jurisdiction over the same conduct. A similar situation exists in the international context. Because it is generally accepted as a matter of international law that nations may govern conduct of citizens of the nation taking place outside the nation, conduct by non-nationals that take place elsewhere but has significant and intended effects in the state or nation, conduct that threatens the sovereignty or security of the nation, and conduct that constitutes a universal crime such as torture and genocide, many situations may arise in which several nations' laws could govern the same conduct.

To use a real-space example, imagine that A (an American shipping company) ships a batch of B's widgets from New York to B in Belgium, by way of France. The widgets are damaged during the French stopover and that this damage gave rise to a cause of action in tort between A and B. Assuming that A had significant enough contacts with

both France and Belgium to warrant jurisdiction in both courts, B could sue A in the U.S., in France, or in Belgium, depending on which legal system would treat B more favourably. Additionally, B could sue in U.S. court but request that the court apply Belgian law to the dispute, or sue in Belgian court but request that the court apply French law, or any other combination of courts and laws.

The many applicable laws will not necessarily be substantively compatible. Different states and nations will have different interests and each will want its laws to govern each dispute. This situation becomes extremely poignant when laws are not only inconsistent, but also incompatible; for example, in some states of the U.S., it is illegal to provide or engage in Internet gambling, but in Liechtenstein, such gambling is government-sponsored. Although the situation of inconsistent laws occurs with moderate frequency now (especially in the antitrust and securities fields) it is likely to become even more common as cyber-commerce becomes more prevalent. This is because, in cyberspace, cross-border transactions are no more difficult than transactions with local parties.

When conflicts of law arise, courts must decide which law will govern. A court need not decide a dispute according to its own law; for example, a court deciding a dispute arising out of an automobile accident in another state would be likely to apply the driving standards of the state where the dispute arose, rather than of the forum state. Several methods exist to aid courts in the decision between laws. Historically, U.S. courts decided a dispute according to the law in the *lex loci delicti*, the “place of the wrong.” In transnational cyberspace, however, the place of the wrong might be any of the nations that are on-line. There is no *lex loci delicti*.

The Restatement (Second) of Conflicts of Law rejected this historical formulation, preferring the so-called “most significant relationship” test, which values:

1. The needs of the international system;
2. Relevant policies of the nation in which the suit was brought;
3. The relevant policies of all interested states;
4. Justified expectations of the parties;
5. Certainty, predictability, and uniformity;
6. And ease of administration.

Several other approaches to choice of law have also been posited and accepted by some courts. The “center of gravity” approach, first adopted by the Court of Appeals of New York, might be characterized as a simplified version of the “most significant relationship” test of the Second Restatement. This approach authorizes courts to look at all the existing contacts between the various parties to a suit and various jurisdictions. Ultimately, the court should choose the law of whatever jurisdiction is most closely tied to the case.

Legal scholar Brainerd Currie espoused the “interest” approach, which encouraged courts to look to the history of the applicable laws and, if the laws of one state could be applied without impairing the other state’s interests, those laws were to apply. In the

case of a true conflict, in which one state's interests would always be impaired, Currie suggested using the law of the forum. California has accepted this approach, but instead of automatically applying the law of the forum in true conflicts cases, applies a "comparative impairment" analysis and applies the law of the state that creates the least impairment.

Finally, professor Robert Lefflar has devised a test in which courts consider 1) predictability of result, (2) maintenance of interstate and international order, (3) simplification of the judicial task, (4) advancement of the forums governmental interests, and (5) application of the better rule of law.

Currently, U.S. states and the U.S. itself take a variety of approaches; none of the above approaches have been universally accepted. Interestingly, most approaches other than the "place of wrong" approach eliminate the need to decide "where" the conduct in question occurred before deciding what law governs (although determining the location of an action may help create the list of nations' laws from which to choose). As the few reported cases show, however, courts may ignore traditional choice-of-law principles entirely and simply apply forum law to Internet-related disputes. Indeed, at least one state, responding to the problem of Internet-based gambling, has announced an intention to apply its own law to lawsuits resulting from in-state Internet contacts.

The Minnesota Attorney Generals office, has interpreted existing Minnesota law to prohibit all forms of on-line gambling, and noting that "[g]ambling is just one example of illegal activity on the Internet" and "the same jurisdictional principles apply with equal force to any illegal activity." Courts have tended to apply the law of the forum state in Internet cases, without discussion.

It should be noted that many Internet activities are commercial and that many of these involve contractual transactions. These contracts may contain choice-of-law clauses defining what state's law will govern any dispute arising out of the transaction. Most ISPs, for example, include choice of law clauses in their service agreements; such clauses may greatly simplify choice of law questions on the Internet, as choice of law clauses are, for the most part, honoured as a matter of international law. Many Internet activities are not commercial or even transaction-oriented, however, and choice of law clauses may not cure problems arising from non-transaction-oriented activities. Case law does not indicate what route courts might take in resolving true choice of law disputes arising from such activities. One commentator has suggested the creation of a choice-of-law treaty for the Internet.

V. Practical Implications of these Laws: Looking to the Future

Many questions remain about how courts will fit the Internet into the current system of jurisdiction. For this reason, people do not know what laws to live by. Although most people know the laws of their domicile state, many do not know the laws of states with which they will be interacting; therefore, under the current system, it remains entirely

possible that a person could be haled into court in a foreign land for something that is perfectly legal in his domicile. Simply put, the “effects” test doesn’t work. It is true that states and nations are perfectly within their power in prosecuting foreign parties who provide illegal services (for example, a Nevada site that provides gambling to a Minnesota resident), and foreigners who commit crimes (for example, an American who posts names of businesspeople on a site available to Chinese dissidents).

However, the architecture of the Internet makes it easy for people to obfuscate their identity and location and it is therefore impractical, under the current architectural regime, to make sites provide and deny service based on a person’s identity; yet, that is exactly what the current legal system requires. Most importantly, under the current regime, people can unwittingly open themselves to liability, by posting information on the Web that they consider proper.

For example, a local company with the same name as a different company across the country and thereby expose itself to trademark liability; or an American could put up a site containing photos of women in short skirts, thereby exposing himself to criminal liability in countries under Islamic law.

If the current legal system is to maintain effective and fair control over the Internet, courts all over the world will have to make a clear move toward a new test for jurisdiction, and a consistent test for resolving choice of law disputes. If courts could agree to exercise jurisdiction based on an effects test with a much stronger element of purposeful availment than exists in the current system, the current style of legal governance might be able to serve the Internet in an effective and consistent way, without the excessive and unpredictable elements that it currently suffers from. For the purposes of fairness, mere awareness that a site could be accessed at a location would not be enough to trigger jurisdiction; rather, in order to be subjected to jurisdiction in a place other than his domicile or its primary place of business, a party would have to display intent to reach the audience in that location through advertising or special targeting subject matter, or a positive awareness of an audience’s locations by way of interactions involving the exchange of information about realspace location.

Under such a system, a party would still be subject to jurisdiction in its home state or nation, but not in a foreign jurisdiction unless the party sought out an audience in that foreign jurisdiction. Such a system would put the burden on state and local authorities to prevent the viewing of illegal material and to focus on laws regarding the use of illegal material, rather than laws the provision of such material. As they are today, transactions would be susceptible to the jurisdiction of the domiciles of all parties involved and any jurisdiction in which the transaction was definitively intended to have an effect.

EFFECTIVENESS OF JUDICIAL SYSTEM

- How effective is the judicial or court system for the purpose of enforcing secured property rights?

The procedure under the judicial or Court system for the enforcing of security and recovery of debts is reflected in the filing of the recovery suits. The procedure is as follows:

Suit Procedure

Suits of a civil nature are filed in Civil Courts. Suits are required to be filed in the court having territorial and pecuniary jurisdiction over the subject matter in accordance with established legal principles. Filing of suits entails the presentation of a plaint, in accordance with the forms generally prescribed under the code of Civil Procedure, 1908. The plaint is required to set out names, description and addresses of the parties, facts constituting the cause of action, facts showing that the court has Jurisdiction, reliefs sought and a statement of the value of the subject matter of the suit for the purpose of Jurisdiction and court fees.

In actual practice many more details and facts are set out. Before commencement of an action by way of a suit for recovery, it is usual to issue a demand notice recalling/demanding outstanding amounts. Upon institution of the suit by filing the plaint and compliance with procedural requirements in the court's registry, summons is issued to the defendant to answer the plaint. In the case of summary suit, the form of the summons is different. The defendant is required to submit his defence in the form of a written statement before the returnable date of summons, complete discovery of documents including the process of inspection of documents. Leave to defend is required to be granted in a summary proceeding. The law of discovery of documents including the process of inspection of documents. The laws of discovery are not as wide ranging as those in the Western Countries. At the time of trial, the issues are framed, evidence recorded (oral and documentary) and usually, extensive verbal arguments are advanced. The Court would then record its judgements accepting or rejecting the case of the plaintiffs.

Summary Suits

The summary procedure is prescribed under Order 37 of the Civil Procedure Code, (CPC) for certain types of commercial actions. The summary procedure relates to recovery of dues on negotiable instruments, deeds (other than penalties) liquidated demands, arising on a written contract or under an enactment, and enforcement of guarantees (guarantee sums) on such debts and liquidated damages. Order 37 applies to all High Courts and City Civil Courts (including District Courts) the summary procedure is not applicable to suits involving the enforcement of a security and are applicable to a money suit or claim only.

Mortgage suits continue to be governed by Order 34 of the CPC. Accordingly, Order 37 has very limited utility to banks and financial institutions which have security and it is generally invoked only in respect of suits on guarantees or promissory notes. In a summary suit, (similar to provisions of Order 14 of U. K. Supreme Court practice) the

right of the defendant to defend a suit is not automatic. The plaintiff is required to issue a ‘summons for judgement’ within a prescribed time frame. In reply thereto, the defendant may file an affidavit setting out reasons why the suit should be permitted to be defended.

It is only if the court comes to the conclusion that, “triable issues” arise or that the defendant has a real and substantial defence that leave to defend is granted. Usually, such leave is refused or conditional leave to defend is granted with the condition being that the defendant is required to deposit a sizeable amount of the claim in court as a condition of being permitted to defend the action. In some cases, even unconditional leave to defend is granted if the defendant has cogent case. Summary suits generally operate on a faster time table than regular suit proceedings.

Suit for Enforcement of Security

Suits involving enforcement of security (whether on personal property or real property) form part of composite actions for recovery of the debts and for the enforcement of the security. The plaint would also contain suitable submissions and reliefs pertaining to the security. For mortgage suits, there are statutorily prescribed form in which a preliminary mortgage decree is to be passed, fixing a period of redemption during which the defendant may redeem the mortgage.

If not, a final mortgage decree would be passed extinguishing the equity of redemption. The reliefs sought in the suit would include those for enforcement of the mortgage, interlocutory reliefs (which would include appointment of a receiver, grant of an injunction against alienation of assets, attachment of receiver, grant of an injunction against alienation of assets, attachment before judgement, *etc.* In a mortgage suit, even if other *pari passu* lenders are not co- plaintiffs, they may be joined as formal defendants as the court has the power to ascertain and declare the correct amounts due in favour of each party who have an interest in the limitation for that party as a defendant, even though he can obtain a declaration.

Suitable reliefs are therefore sought. It is also the practice, to seek interlocutory relief such as the appointment of court receiver, attachment before judgement of properties and even injunctive reliefs. These suits get blocked by the ingenious and often fraudulent defences propagated with substantial degree of success by the borrowers and their lawyers. The typical defences are as follows:

Typical Defences

Typical defences in suits filed by banks/financial institutions include;

- Objections to jurisdiction on territorial or pecuniary grounds or inadequate payment of court fee.
- A bar of limitation,
- Disputing the authority, of competence of official declaring the plaint.
- Improper computation of the statement of claim including application of

incorrect interest rates (as periodic revisions of interest merely notified by the bank not having been approved bilaterally),

- Documents being incomplete by reason of defective execution or inadequate particulars and therefore being unenforceable.
- Security being invalid or unenforceable by reasons of inadequate stamping and for want of registration,
- Improper form of security for *e.g.* security by way of hypothecation over what is in reality immovable property.
- Official liquidator impeaching the security on the ground that the charge is not registered in accordance with company law,
- Want of statutory approvals required for creation of security *e.g.* ULCRA permission, approval under Income Tax, 1961 approval under Foreign Exchange Regulation Act, 1973 (where the lender is a non-resident),
- Bank having lost or parted with the security (bank's liability for negligence) and therefore a counter claim by the defendant against the Plaintiffs.
- Lender having been responsible for the sickness (or poor financial condition) of the borrower by reason of their the bank having committed various breaches of promise to lend or by virtue of excessive interference in the affairs of the bank or choking of working capital facilities leading to disruption of business and therefore causing losses.
- Surety claiming discharge, by material alteration in the guarantee or in direct negotiations by the bank with the principal debtor without the consent of surety.
- Surety claiming to be discharged by reasons of nationalization or Takeover of the principal debtor,
- Invocation of provisions of the SICA or Relief undertaking statutes in diverse states, which result in suspension of some aspects of the recovery action and execution, for a duration of time.
- Interest being questioned as being usurious.
- Exchange fluctuations being questioned as incorrect, and based on unfavourable.
- Liquidator questioning the rate of interest on the ground that it is penal,
- The scope and extent of the charge being questioned by other co- shares or banks and financial institutions questioning their respective securities.

The effectiveness of the suits is severely dented by the time frame involved.

Time Frame

Generally, the time frame with respect to suits is similar in most Courts, though it is possible that typical milestones in litigation may differ from court to court. A trial in the first instance usually takes 8 to 12 years to come up for hearing for the first time from the date of institution of the suit in most Courts. Presently, in the Bombay High Court the delay period for the first trial is about 12 –15 years.

Thereafter, an appeal is filed before a Division Bench of two judges of the High Court, a further period of 2 to 3 years at least is involved. If an appeal is filed before the Supreme Court thereafter (which is usually done if stakes are high), an additional delay of 10 to 12 years is clearly possible so that a delay of about 25 years from the date of original institution of a suit till final determination is usual. There are exceptions and sometimes, suits are expedited for earlier hearing where a question of public importance is involved. On the other hand a further slippage of the above time table is possible and a suit may well take around 30 years to be ultimately disposed off by the Supreme Court.

In certain commercial cities such as Bombay, the position regarding suit filed by banks and financial institutions is slightly better, and not as depressing, under Chapter XV of the Bombay High Court Rules”, commercial causes” are treated as a separate class of suits. Commercial causes, include causes arising out of ordinary transactions of merchants, bankers and traders whether of a simple or complicated nature” and include amongst others, “banking transactions”. There is a separate trial list of commercial causes and, one judge is specially assigned to hear such cases. The court, may be in respect of such cases, and for the speedy determination of such suits also the time- table is somewhat faster. However, this kind of special treatment to commercial causes is not available in all courts.

E-commerce: Contracts, Legal Issues and Agreements

INTRODUCTION

Before looking at the legal detail and common pitfalls of e-commerce, we will initially consider the nature of e-commerce — what exactly is it? We will also look at some terminology that is frequently used by e-commerce lawyers and business people, always bearing in mind that what we are dealing with here is not the technology, but rather the way the law applies to the transactions that are undertaken using the technology. E-commerce (or electronic commerce) is the term used to denote a commercial (usually contractual) transaction that takes place between two or more people using the communications infrastructure known as the Internet.

The first thing to note is that there is a range of e-commerce transactions, the following of which are examples:

- The purchase by an individual of a book from a commercial web site such as that run by Amazon.com
- The ordering by Company A of office stationery from Company B's web site
- The exchange of e-mails between two persons whereby it is agreed that a certain service will be carried out by one party in exchange for a fee from the other.

We do not usually think of the latter type of transaction as falling within the category of e-commerce, but it is just as much an electronically formed contract as the other two

examples: an exchange of e-mails can result in a legally binding contract. For example, if anyone send you an e-mail offering to sell you one dozen doughnuts, and you respond accepting my offer, this is just as much an e-commerce transaction as one taking place using the colourful pages and flash technology of the World Wide Web.

Initially it will be important for an e-commerce business to identify whether its e-commerce transactions will be classified as B2B (business-to-business) or B2C (business-to-consumer). This is because the law applies in a different way to each. Of course, it may be that the e-commerce business anticipates engaging in both types of transaction. The differences between these two types of transaction will be referred to throughout this Report.

Other terms commonly referred to throughout this Report are as follows:

- World Wide Web — that part of the Internet which uses the Hyper Text Transfer Protocol (HTTP) to display so-called web pages and to allow links between such pages anywhere on the Internet.
- ISP (Internet Service Provider) a provider of Internet connectivity.
- ASP (Applications Service Provider) — an online provider of computer applications, such as software.

LEGAL ISSUES IN E-COMMERCE

A comprehensive international legislation system for global e-Commerce does not exist at present and it is not expected in the foreseeable future. Even within the United States, which is so advanced in its use of the Internet, certain e-Commerce legal issues have caused significant disagreements, such as the validity of digital signatures.

While waiting for federal legislation, many states have set up their own laws, which have been widely different in substance from state to state. Organizations such as the UN conference on international trade law have been actively calling for global co-ordination of appropriate legal structures. The most important issues are copyright protection, contractual agreements and privacy laws.

E-COMMERCE GLOBAL STRATEGIES

The organizations practising in the area of e-Commerce are facing a number of decisions regarding their global activities. Having analysed the characteristics of the target foreign market, they now face the next stage of developing an international marketing programme.

This would include defining and selecting target segments, positioning the product or service and making the decision to modify the elements of the marketing mix to suit the conditions of the foreign market. For physical goods, the marketing mix is composed of four elements: product, price, place (distribution) and promotion. For services, the marketing mix extends to seven elements with the addition of people, processes and physical evidence.

Given the diversity of the global environment, with differing cultures, customs and competition, the decision facing the e-Commerce marketer is the degree of change necessary within the elements of the marketing mix.

There are a number of pressures affecting this decision, which can be summarized as:

1. Staying local versus going global,
2. Standardization or adaptation.

Local Versus Global

Ten years ago 'pan-European' advertising was all the rage among direct marketing companies in the United Kingdom. A precursor of global communication, it was based on a simple idea, namely that consumers in various countries fall into the same socio-demographic categories, with similar if not identical buying habits and tastes brought about by low-cost cross-border travel. It uses the same imagery, messages, product and brand positioning across all campaigns and all markets. Economies of scale could be achieved and global brands built up, as campaigns could be extended from one country to another.

More recently, globalization has received a few hard knocks, due to changes in customer tastes and timing. Some products achieved global status because they required very little effort in terms of product development in order to be successful in new markets. Others needed extensive redevelopment in order to localize their marketing mix. A good example is basic office software such as Microsoft Word, which works equally well in most countries, once it has been localized for language characteristics. Other examples are soft drinks like Coca-Cola or imaging products like Kodak, which need only minimal changes of their global campaigns to adapt to local taste. Alternatively, cars are more of a 'lifestyle' purchase.

Consumer tastes in cars vary more widely between the United States, Germany, Japan, Italy, the United Kingdom and India. Achieving substantial economies of scale either in product development or in advertising is more difficult. Ford has been one of the manufacturers most successful in using a modular approach to car manufacturing and marketing. Recognizing that there are strong differences in consumer preferences, Ford has reduced the number of components - and thereby has achieved economies of scale - without affecting the number of options available to customers in each of the seventy-five different national markets in which Ford sells cars. Ford manages to use a modular approach also in its advertising messages, emphasizing safety in Scandinavia, performance in Italy, design in France and handling in Germany. The Ford brand may be global, but the product and its supporting marketing effort are very much local.

For e-Commerce marketers with global ambitions, the requirements are to find out if the product or service they represent and the way of doing business can be exported in their 'domestic' forms or they will encounter consumer resistance and will need substantial adaptation to suit consumer tastes.

THE RISKS OF GLOBAL WEB MARKETING

For companies that have an established global distribution system the decision to exploit the Internet as a marketing tool needs to be considered with care. Once a Web site is established which has an online purchase facility, there is the risk that overseas customers will cut out the local distributor, place orders electronically and expect price discounts. This was the case of Millipore Corporation of Bedford, Massachusetts, a company with annual sales of \$600 million specializing in filtration products for water purification for laboratories and the detection of contaminants in semiconductor manufacturing. With a 'blue chip' list of customers worldwide the Internet clearly offered an effective medium to both communicate information and provide the customer with a purchasing mechanism.

However, Millipore faced a massive complication in that, similar to many other multinationals, it charges higher prices overseas in order to cover the additional cost of support services. Having faced the potential dilemma associated with inter-country pricing differentials, Millipore decided to postpone offering an online ordering facility and instead restrict its Web site to providing information with supporting pre- and post-purchase service activities.

Standardization Versus Adaptation

The decision whether to standardize or adapt the marketing mix elements is a major one for any organization operating outside its home environment. Standardization allows the organization to maintain a consistent image and identity throughout the world. It provides economies of scale and is particularly valuable for maximizing impact with the internationally mobile customer. It works best within an identified international segment and with a product of well defined complexity. Global branding and the transfer of new product ideas across boundaries are both possible with standardization.

Standardization has been used successfully in running global promotional campaigns. Transferring campaigns from one country to another is almost impossible, owing to language and cultural differences. Nevertheless, the clever use of pop music as a universal language has allowed Levi's advertising to be successful where others have failed.

Coca-Cola has also succeeded with its 'one sound, one sight, one appeal' philosophy which brings it closer to a full standardization position. Benetton is still running their 'United Colours of Benetton' global campaign based on multi-cultural, universal images that appeal globally. Adaptation is nevertheless necessary if the organization is under pressure to satisfy the multiplicity of needs of the global customer to the extent to which it will have to change the elements of the marketing mix.

CHANGING PATTERNS IN GLOBAL DISTRIBUTION

Distribution is the process by which all consumer and industrial goods are transferred from the producers to the end users. The process includes not only the physical handling and distribution of goods and the transfer of ownership from producer to consumer, but

also the whole range of buying and selling negotiations between producers and intermediaries, and between intermediaries and consumers. Distribution channels are the ways and means by which goods are distributed from the producer to the end user. The traditional distribution model has been linear.

The producer is the originator of the products, which it builds or manufactures. Wholesalers and distributors bring together the products from a number of manufacturers, then divide them and transport them in small lots to resellers or retailers who deal directly with the consumers. The value added of the distribution chain has been in the dispatch, warehousing and delivering of products. Sometimes distributors undertake the repackaging and marketing of the products. The chain of intermediaries, who are the wholesalers, distributors and retailers, adds substantial costs to the value chain, which can make the prices to the end user significantly higher than those of the producers. The Internet has been unique in its potential to revolutionize the value chain, as it enables producers to reach the end users directly.

As e-Commerce firms are developing, they will be choosing their entry and distribution modes on the basis of reducing their transaction costs even to the point of internalizing certain activities which have been traditionally performed by intermediaries. The Internet has raised expectations that, in time, producers will sell directly to end users and, in turn, end users will prefer buying direct from the producers. What is actually happening is that value chains are being deconstructed, reconstructed and transformed into value webs, giving rise to a new class of intermediaries, which are the collecting and distributing of information. For example, companies like Yahoo! or Netscape act as information collectors and disseminators, offering e-Commerce new possibilities of doing business.

e-Commerce Distribution System

For Web-based businesses focused on competing in global markets, there is every indication that representatives in the field are necessary in order to facilitate dispatch, the handling of returns and servicing. Companies that deal with consumer goods and have relatively low international volumes that require little service assistance may be able to dispense with any on-the-spot distribution requirements and rely on a global integrated carrier.

Global Exporting

The European Union in its present form has a larger population than the United States. Nevertheless, most of its producers and consumers live and work in a radius of 800 km, equivalent to 12 per cent of the land area of the United States, and for historical and economic reasons, transport in the European countries has been burdened with rules and regulations. For example, a trucker transporting goods from Glasgow to Athens used to spend 30 per cent of his time at border crossings, waiting and filling in up to 200 forms.

These inefficiencies are mercifully now a thing of the past. In order to move goods between EU member states only one simplified transit document is required and many of the custom formalities have been eliminated. However, consignment to countries outside the European Union is still burdened with an array of obstacles, cultural and linguistic, which lead to many companies failing to exploit the opportunities offered by overseas markets. The SME sector is affected in particular, as such companies do not have the resources to spend on overseas sales teams and trade-related computer systems.

MEASURING E-COMMERCE SUCCESS

CIOs often suggest that better measures of the payoffs of e-commerce operational and capital investments are necessary to demonstrate the value creation of e-commerce initiatives and to obtain additional resources for critical e-commerce projects. The measures are essential to monitor the key performance drivers (inputs and processes) and assess whether the e-commerce initiative is achieving its stated objectives (outputs) and thus contributing to the long-term success of the corporation (outcomes).

Companies often waste resources on e-commerce initiatives or do not invest when they should because they cannot effectively evaluate the potential payoffs of e-commerce investments. Measuring returns on e-commerce projects can be a daunting challenge. Predicting customer behaviour is difficult, because using the Web to do business is still relatively new to many businesses and thus forecasting sales and profits is typically imprecise. There is not much historical data and experience for managers to draw upon when developing or applying metrics, and many economic benefits of e-commerce projects are seen as difficult to measure.

Further, the pace of change in e-business and Web-based technologies has been so rapid that precise measurements are often difficult. Many senior managers have come to believe that further investment in new technology and e-commerce is an imperative that is required to maintain or develop a competitive position. They often make expenditures without completing a rigorous analysis.

However, today's more stringent economic environment and the widely publicized negative impacts of many e-commerce initiatives has caused many senior managers to question the payoffs of e-commerce investments. As companies assess the choice of appropriate measures to evaluate e-commerce initiatives, numerous potential issues arise. Since the choices are different for each company, because the strategies, structures, and systems are different, substantial customization is necessary.

Senior managers should consider six initial questions that can lead to the development of appropriate measures for e-commerce operations:

- What measurement systems are currently in place and being utilized within the organization?
- What are the important criteria to the company and its constituencies and stakeholders?

- What does the company desire to accomplish with the e-commerce initiative?
- What is the anticipated time frame associated with the e-commerce programme?
- Who are the parties involved in implementing the e-commerce project, and who will be affected by the results?
- What critical processes are associated with the successful execution of the e-commerce project?

To address these questions, it is imperative that companies not only specifically tailor their e-commerce measurement approach but also utilize multiple measures to fully analyse their situations. Different measurement criteria are important for companies that have different strategies or may be in a different stage of their life cycle or their e-commerce development. The multiple measures will typically include both financial and nonfinancial measures that are leading and lagging indicators of performance.

They may be used in a balanced-scorecard, shareholder-value-added, or other approach and can be developed specifically for IT or e-commerce or as a part of an overall corporate performance measurement system. Companies can also use a weighted scoring system to evaluate investments related to overall IT, e-commerce, or business strategy. There are many obstacles to implementing a successful measurement system, whether a lack of focus, a low priority, or just difficulty. It is the responsibility of senior managers to evaluate the e-commerce initiatives and decide on the right measures for their organization and ensure that the measures are captured and responded to properly. To obtain adequate resources for e-commerce and to effectively manage e-commerce initiatives, the payoffs of e-commerce investments must be calculated and integrated into management decision-making systems.

DEVELOPING APPROPRIATE METRICS

To closely monitor the cause and effect relationships evidenced in the e-commerce causal linkage model, appropriate metrics must be developed. These metrics must be consistent with and support the objectives and drivers and key success factors already defined. The selected metrics will likely include a combination of input, processes, output, and outcome metrics to effectively measure performance.

Senior managers involved in the e-commerce decision-making process should develop metrics appropriate to the strategy and objectives of the e-commerce initiative, the company, and its stakeholders. During the measure selection process, it is useful for the involved individuals to choose just a few measures, to focus those senior managers involved in the e-commerce initiative on the critical performance indicators.

The list of metrics presented here is not meant to be a comprehensive set of e-commerce performance measures. Rather, it is a selection and example of some metrics that may be appropriate. Managers must select those that most closely fit their strategy and adapt or develop others. There is no rule for the right number of metrics to include in a measurement system; however, including too many tends to distract managers from pursuing a focused strategy. Generally, a complete measurement system includes

perhaps three to six measures for each element being evaluated and no more than twenty measures in total. For each key success factor, a specific target should be identified and results should be measured against these targets. These results should be widely communicated among not only those senior managers directly involved in the e-commerce initiative, but also other individuals within the organization upon whom the initiative will have an impact.

One of the important contributions that e-commerce can make to an organization is an expanded communication and informational capability. Any e-commerce measurement system will have little impact if the results are not fully discussed. Results should be monitored regularly and used to identify areas of weakness, address the plans and systems in place, and establish new initiatives to improve deficiencies.

The measures chosen should be quantifiable, in either absolute or percentage terms, as well as complete and controllable. They should be complete in that the measure sums up in one number the contribution of all elements of performance that matter; for example, profitability is a summary measure of revenue generation and cost control. They should be controllable in that employees in the organization can actually influence improvement in the factor measured.

Some of the metrics shown here are evaluations of overall firm performance. Others are indicators of e-commerce performance that are derived through an aggregation of measures of individual business units and functions. It is important to evaluate the performance of both overall e-commerce performance and the specific aspects of e-commerce that lead to revenue enhancement or cost savings to determine the success of various operations and corrective action that can be taken to make improvements.

The measures should be of use to both senior and middle managers in the business units and functions. Thus, they must be disaggregated so each unit can examine its contribution to the achievement of the company's e-commerce strategy. These analyses ensure that each unit is making a contribution to the e-commerce initiative and improving corporate profitability. Additionally, these metrics can be used to provide a gap analysis that permits managers to determine what other inputs or processes are required to meet the company's e-commerce project objectives. Different tools and techniques are available to measure the different aspects of e-commerce performance.

For example, online surveys and polls are powerful tools to help e-commerce enabled companies to better understand the benefit of Internet usage for increasing revenue or decreasing costs related to their customers, thus providing valuable information regarding opportunities to improve overall profitability. Internally, surveys, focus groups, and other techniques are increasingly being used to measure and monitor employee, personnel, and stakeholder reactions and provide valuable feedback.

Once metrics have been developed, data on these indicators must be collected and statistical analysis, such as multiple regression, should be performed to analyse and test the validity of the customized e-commerce measurement system and causal relationships hypothesized by the company.

As companies evaluate the initial measurement system's performance, they will typically add some metrics and drop others because of a lack of evidence of a strong relationship. It is here that a final measurement system emerges, and the focus then shifts to applying the model to support improved decision making.

Outputs and outcomes: Overall-firm and e-commerce-specific performance. If the e-commerce initiatives are well designed and executed and the model of causal relationships properly specified, the identified inputs and processes should lead to improved performance. This should include increasing the success of the e-commerce initiative (outputs) and ultimately to improve corporate performance either through increased revenues or decreased costs (outcomes). To properly evaluate e-commerce performance, input, process, output, and outcome measures are all necessary and should be clearly linked in a causal relationship. These performance indicators empower senior managers with the information to evaluate whether the e-commerce programme is achieving its stated objectives and contributing to overall corporate profitability.

For example, metrics such as the percentage of customer attrition are indicators of the e-commerce customer service provided by the company and the related level of customer satisfaction. Since the goals relate to increasing corporate profits, not just improving customer satisfaction, both output and outcome measures are necessary.

A weak performance on the output metrics should signal a need to examine the inputs and processes and determine whether they have been misspecified or just poorly executed. It also can provide an opportunity to identify potential benefits to organizational effectiveness and profitability from e-commerce that may have been overlooked.

This is an opportunity to examine how well e-commerce programmes are contributing to corporate profits and should unveil specific opportunities, directions for improvements, and standards of performance. The e-commerce measurement system should highlight the specific contributions of the e-commerce activities, in addition to providing valuable feedback that can lead to future e-commerce programme and corporate improvements.

Results from the e-commerce evaluation and measurement process should be widely communicated throughout the organization. In a well-executed e-commerce venture, all units of the company will have some involvement in the e-commerce initiative. The evaluation and measurement of the e-commerce programme will have little impact if the results are not disseminated throughout the organization to the many disparate areas that both affect and are affected by it. Results should be monitored regularly and used to identify areas of weakness, challenge the plans and systems in place, and present new initiatives to improve deficiencies.

MEASUREMENT PRACTICES IN E-COMMERCE

Strong measurement practices form one of the cornerstones of good systems, particularly in e-commerce. Performance measures for e-commerce must overcome the uncertainty and unique dynamics associated with the Internet and must be more frequently adjusted in response to real-time information.

With these considerations, no company should simply extend its existing performance measures to an e-commerce venture. Still, long-term cost differentials must be balanced with other financial and nonfinancial measures and leading and lagging indicators that are particularly useful for successful e-commerce implementations. The information systems of the 1990s gave CEOs a new method of accessing, analyzing, and reporting on the accountability of their organization. The systems developed during this time of advancement in information technology helped to create a more streamlined capability for centralized accountability. The variety of data made available to a company at that time ranged from corporate-level results to the small-scale measurements of performance that enabled management to recognize advantages and potential problems in real-time. E-Commerce transforms these capabilities.

Like the root system of a massive tree, IT accountability systems helped management reach and observe every aspect of their business. With the addition of e-commerce to a pre-existing IT accountability system, companies can simplify access to previously collected information for those within the company and for external stakeholders, including partners, customers, and investors. New information could be created and both financial and nonfinancial measures could be integrated into the decision-making process. The information could also be easily and quickly aggregated and disaggregated to facilitate various decisions. Managers could now measure inputs, processes, outputs, and outcomes in ways never before possible. Cisco's new systems permit outsiders to view on its Web site not only its general business plan but also the company's performance statistics at any given moment.

Seamlessness between internal IT systems and e-commerce gives Cisco the ability to provide such information with relatively little effort. This transparency between internal and external systems also helped Cisco endure many of the pitfalls associated with the technology bust by expanding the role of e-commerce to replace human positions throughout the company.

But e-commerce can be used for more than simply replacing employees. Instead, the information made available through a well-developed e-commerce initiative can empower employees at every level of business. Improved measurement is a key component. Management receives information in a timely manner, which lets it act on up-to-date measures of performance, while lower-level employees can access information at any time and take the initiative based on that information, with or without direct managerial direction.

Among the most important aspects of e-commerce as it applies to system management and measurement is maintaining consistency across all company lines. A company implementing a new e-commerce solution should ensure consistency among accounting systems, information technology systems, and e-commerce systems and related measures. However, the implementation of an e-commerce solution also necessitates working to ensure compatibility between the systems of business partners using unified e-commerce solutions.

The uniquely advantageous relationship between Dell and its suppliers would not be possible without seamless internal and external systems that enable a free flow of information and measures between companies. E-commerce can facilitate consistency of information and measurements by cascading information throughout the organization and then externally to other stakeholders. In addition to measuring the performance of the business, e-commerce brings added importance to measuring the value and functionality of operations. Most e-commerce strategies will have a strong operational component, including cost savings from value chain management and cuts in labour costs for the online channel. Operational measures should be tracked by some dedicated resource and balanced between financial and nonfinancial assessments of operational performance. This analysis can lead to a better understanding of the payoffs of investment in e-commerce initiatives.

CUSTOMER RELATIONSHIPS

Under traditional business models, it may be sufficient to offer customer service during normal business hours, either at physical locations or over the phone. E-commerce dramatically changes those expectations both in terms of company strategy and the customer's perspective. Near-universal and constant customer service availability is expected for almost any viable e-commerce strategy. Companies must re-examine their customer service practices and make numerous key decisions in moving toward universal availability. Two of the main considerations are the distribution of response tasks between human and automated systems and the operating hours for human responses. Each of these include trade-offs among speed, cost, convenience, and efficiency.

Most e-commerce ventures have developed some FAQ (frequently asked questions) function on their Web sites. This function may consist of rudimentary inquiries only or a highly sophisticated taxonomy of nearly every conceivable question. The nature of the offering and the Web site dictates how sophisticated the FAQs should be, but no company can exhaust the list of possible questions. The challenge then becomes how to provide service to customers who are dissatisfied with the FAQs. A company may choose to provide immediate or delayed e-mail support to customers or may offer a phone line for further inquiries. The company must weigh the value of instant response and satisfaction to the customer against the added costs associated with phone banks or with rapid and 24/7 e-mail response.

Some pure-play companies like Half.com chose to keep the process entirely online and not offer phone service. For traditional companies, however, this strategy is more difficult, as it can inundate the traditional business phone lines with Internet inquiries. Delta Airlines, on the other hand, was able to reduce its phone volume and increase its status among elite customers by moving reservation information online while simultaneously implementing a wireless initiative.

Delta first identified such a programme as “nice to have” but not a necessity, but the company quickly recognized the appeal of wireless capabilities to its elite customers

by providing quick access to vast amounts of information. Implementing this programme was not only a cost-saving move, but also a successful effort to appeal to customers who desired this level of functionality.

For companies that do provide phone service in addition to their FAQs and Internet support, a decision must be made on availability. Limiting phone support to normal business hours reduces some of the value added associated with the Web site. On the other hand, labour costs may not justify 24/7 phone service for industries that offer commodity goods. More technical industries, B2B companies, and companies with customized and personalized Internet-related offerings should carefully consider offering a 24/7 service.

Another aspect of customer service practices is the development and implementation of policies on shipping, delivery, and returns. In particular, companies that use physical branches as part of a bricks-and-clicks strategy must not neglect proper training on the traditional commerce side necessary to facilitate e-commerce developments and service. Customer service representatives in physical branches must be thoroughly familiar with the Web site, its function, and its policies.

While e-commerce requires changes and enhancement to customer service policies, it also affords most companies excellent opportunities to refine marketing strategies, selling practices, pricing, and Web site design by observing customer behaviour. Physical branches have more limited opportunities to collect customer data, particularly data from sites where customers can impart complaints or grievances.

When customers shop online, however, their every mouse click is potentially revealing and useful for future improvements in e-commerce service. Although these methods require substantial analysis, companies can draw strong implications and direction about customer behaviour from this data. Unused search results may indicate a problem with the search engine.

Aborted purchases may indicate that the checkout procedure is unclear or too time-consuming. Consistent nonpurchase of a product may indicate that more precise information is needed to inspire consumer confidence. Corporations must be willing and able to redesign their Web sites and refine customer processes based on the gathered information as well as direct feedback from the customers. Since convenience is a prime source of competitive advantage, problems with customer service systems must be addressed immediately. Although shutting down the Web site is undesirable, important changes should be made by closing the site for a few hours in nonpeak time and by warning the customers in advance of this downtime. Thus, to effectively manage and enhance customer service attributes,

- Companies with an Internet presence should provide any information that a customer might need while online.
- Corporate Web sites should incorporate some form of direct contact mechanism, whether it is as simple as a service phone number or as advanced as online chat capabilities.

- Companies should ensure that the Web site easily connects to any other aspect of the company service that a customer might need information about, ranging from traditional brick-and-mortar locations to shipping policies.

INFORMATION PRACTICES IN E-COMMERCE

For e-commerce to be successful, it is necessary but not sufficient that members of the organization are competent and confident in their use of IT. Strong leadership and investment can bring an organization to that stage, but systems must be created to harness the power of IT for organizational integration. It has recently been emphasized that interactive control systems in traditional business must focus on constantly changing information. These systems provide the sensing and monitoring that is necessary whenever strategic uncertainties are high. In most businesses today, more interactive control systems are needed as strategic uncertainties have been increasing. In e-commerce, these systems become increasingly central to the successful implementation of strategy.

A new policy on information practices must become a central component of the company's systems. Information should be transparent and the sharing of information facilitated. Organizational boundaries must not impede the flow of information. An integration strategy cannot be properly implemented if information about e-commerce stalls within the e-commerce unit and is not disseminated widely. Ideally, internal processes are developed in-house with substantial consultation and communication; obstacles are easily recognized, understood, and dealt with, internally and quickly. Companies such as Amazon and Office Depot not only have garnered more efficient business practices because of their internal development, but also have been able to expand into other markets such as technology consulting and Web design because of their in-house expertise.

Companies can also use the implementation of an e-commerce strategy as a starting point for proper information practices. Prior to moving online, 3M had a fragmented information system that made it difficult to cross-sell to existing customers. Even having the same customer listed in different systems had additional costs. After moving online, the company created a \$20 million data warehouse to store all information, as part of a larger restructuring of information systems.

Senior management must set the example of open information practices and transparency, even when the information is bad news for the company or reflects negatively on an individual or business unit. Problems should be discussed openly, without fear of repercussions. As appropriate, information on the company's goals, measures, and progress should be made available to all employees and the public through the company's Web site. This information is increasingly available to interested parties, and by making it transparent, a company may build trust with its constituent groups. In addition, it is often beneficial to increase the information flow to the public to ensure that accurate information is communicated, rather than to risk inaccurate information

being made public by a third party. Further, many organizational processes should be restructured to fully take advantage of e-commerce.

In particular, e-commerce should encourage and facilitate companies to move away from strict hierarchical reporting in some areas and move toward the use of cross-functional remote teams. By collaborating with other business functions and units, the e-commerce initiative is likely to encounter less resistance in the traditional business. Processes should also be adaptable and flexible enough to respond to changes in real-time information.

Numerous virtual organization techniques serve these purposes, including:

- Cross-functional teams that form themselves
- Teams that may go into and out of existence regularly
- Use of both in-person and remote meetings
- Teams that span organizational boundaries
- Encouragement of innovation through mutual trust
- Self-governance
- Harnessing global resources to solve local problems.

An organization that utilizes these processes should then be able to assemble teams to innovate or to resolve challenges associated with e-commerce. Most important, all these techniques should be fully consistent with an integration strategy, eliminating any need for moving e-commerce outside the organization.

By upgrading these information practices, a company positions itself to make significant improvements throughout the value chain. For many companies, value chain management is an integral part of the e-commerce strategy. But even for those companies that focus primarily on selling online, subtle improvements can be made with minimal extra cost and effort. Thus, improving information practices can create significant benefits for sales, distribution, and procurement, along with various other processes throughout the organization.

Cisco has used its real-time capabilities to create supplier-side processes that vastly improve its supply-chain management. Cisco and its suppliers and manufacturers share extensive information on product quality. This allows Cisco to measure the number of defective products while they are still in the supplier's possession, rather than after they have been delivered. In turn, the supplier can make faster adjustments in its manufacturing process to minimize future quality problems.

Dell has mastered the use of e-commerce in limiting the amount of inventory kept on hand. Many component parts such as monitors and peripherals are never kept in inventory by Dell, and their movement and distribution is entirely handled by e-mail communication. But even for parts retained by Dell, real-time ordering forecasts are sent to suppliers to minimize the inventory and buffer stocks on a cycle as short as two hours. This relationship with suppliers, coupled with its information practices, also permits a *negative* cash conversion cycle. At the delivery end of the supply chain, Tesco spanned traditional organizational boundaries to create the most efficient delivery

process. While many in the grocery industry mimicked the Amazon model by attempting to deliver from large, high-tech warehouses, Tesco chose to deliver directly from stock at local physical locations. While this limited the potential delivery zone, it also saved Tesco the effort and resources of building expensive warehouses with uncertain profitability. Delivering from the store meant no separate inventories and no need for entirely new processes.

Nike, Inc. recently implemented a supply-chain system linking the company with its manufacturing partners. Before these system changes, 30 percent of Nike's total volume of shoe orders were based on estimates; now, only 3 percent of the orders are guesswork, because of better forecasting and planning.

Whirlpool Corporation has linked every Whirlpool factory and sales site worldwide through e-business software, allowing factories and sales sites to coordinate with suppliers and key retail partners. This has reduced inventories and increased vital communication. Krispy Kreme Doughnut is another company that has used e-commerce to its advantage by devising an intranet network linking its stores. The Web system tracks doughnut mix, doles out the right colours of sprinkles, monitors managers' decisions, and permits users to fix errors such as damaged goods, by allowing replacements to be sent. The system allows employees to focus more on customer service, reduce problem orders, and increase productivity.

For proper information practices to improve the success of e-commerce implementations:

- Organizational boundaries should not represent an impediment to the free flow of information in a company.
- Company leadership must set the example of open information practices.
- Companies must be willing to move away from strict hierarchical reporting systems.

ARCHITECTURE OF E-COMMERCE SYSTEMS

The buying and selling of products and services over the Internet is termed 'online trading'. Online trading or e-Commerce is important to businesses because it provides a flexible source of trading with customers, and with business partners. To enable online trading, companies need to combine existing computerized transaction processing systems and information systems with Internet and Web technology. Existing computerized systems, databases, and Internet and Web technology are the basic components of online trading and form the architecture of e-Commerce systems. Computer system architecture consists of hardware and software components that are configured in terms of the needs of organizations.

Companies connect computers to a LAN, WAN and the Internet, using TCP/IP, based on client-server architecture, to share computer software, printers or scanners. Basic Internet tools such as e-mail, browsers, search engines and protocols such as file transfer protocol (FTP) and Telnet are combined to form the system architecture for e-Commerce.

The systems architecture for a company will depend on its trading activities. A retail company will have a different architecture from a manufacturing company or a service company. As a company changes and evolves its mission or objectives, its e-Commerce system architecture will need to change too. Alternatively, the stimulus to change may come from competitors who use the most modern e-Commerce technology.

INTERNET, E-COMMERCE SYSTEMS AND INFORMATION SYSTEMS

Some of the terms used to describe a company's involvement with the Internet are that it has 'an online presence' or it has 'a Web site'. Such terms hide the complex IT and IS that form the infrastructure that underpin e-Commerce systems. An information system is the application of a computer to capture and process data to provide information for managers and executives for the purposes of decision making and management. A company will normally have various information systems, such as an airline reservation system or an inventory control system. These systems will be linked to a corporate database that stores relevant data, like product or customer details. A company's Web site is connected to such information systems and databases to provide the essential product or service information for customer and employee use.

Companies' existing client-server technology and its IS/IT infrastructure is connected to the Internet to form part of an e-Commerce architecture. The e-Commerce architecture is based on a company's business model, consisting of the basic logic of what it wants to produce or sell, how it will market it, how it defines its customers and other fundamental business issues. IT is combined with Internet technology to enable a company to personalize its service to a customer.

DATABASES

A computer database consists of records on specific items of interest, for example customers' contact details or the products they have bought. It is a collection of data that can be queried for specific purposes such as targeting a particular customer for certain products. Databases are used to market products and services. This is termed *database marketing*. Databases are used to store data on the purchasing habits of customers and enable a company to develop a tailored relationship with its customers. The data are then processed to provide tailored information on individual customers' preferences that are then sent information about products or services relevant to their current buying habits. Data-mining techniques are used to reprocess existing data in databases to extract previously unrealized information of potential commercial use.

E-COMMERCE SYSTEMS AND THE CUSTOMER

e-Commerce systems that interface with the customer need to be pragmatic. The selection and evaluation of Internet technology need to be appropriate to both the company's needs and the customer's ability to use the system. A good business model

will assume that the customer has little interest in technology and avoid the technology trap. *Avoiding the technology trap.* It is not user-friendly to use over-complex Internet and Web technology in e-Commerce systems. The technology needs to be appropriate to prevent prolonging the time it takes for customers' commands to be executed by the system.

WEB SITE EVALUATION SOFTWARE

It is possible to measure the popularity of a company's Web site using evaluation software that generates log files. Such log files record the number of visitors to the site or page impressions. They record which parts of the Web site are most popular, the times of the day people visit, record which search engines are being used by visitors and how long they spend on the site, known as the 'stickiness' of the site. These log files can subsequently be used to evaluate the performance of the site from a business perspective and the information can be used to make it easier for customers to make transactions.

INTERNET SERVICE PROVIDERS

A company wanting to trade over the Internet has to agree a contract with an Internet service provider (ISP). ISPs provide communication and hosting services for individuals and companies wanting to access the Internet. ISPs provide access to the Internet and enable Internet commerce by connecting individuals and companies to the Internet. There are thousands of ISPs that a company can choose from, all of them providing domain name services and electronic mail. The two most popular ISPs are UUNET Technologies and AT&T World Net. Other organizations, like a bank with mainframe computers, may also act as an ISP. ISPs also provide hosting services for Web sites by installing and operating the server computers and software for a company. The computers and software are located at the ISP's site. Transaction services such as payment systems or order capture and fulfilment are also provided by ISPs.

EXAMPLE OF E-COMMERCE ARCHITECTURE

The important considerations for designing e-Commerce architecture are the customer, the company, the company's existing information systems and how payment will be made. For example, a company with a Web site that includes an order form would need a Web server that can provide a catalogue, whose details would be retrieved from a catalogue database, and order form.

It would need to interface with an existing database to collect and record the order details entered by the customer. The payment might be transacted with a credit card whose details would be captured on the order form utilizing standard security features available on the Web. This example shows that an e-Commerce system is composed of existing information systems, databases and Internet technology. Other sophisticated e-Commerce architectures are possible for companies that need distributed transaction

processing. For example, the Open Buying Consortium (OBI) has proposed a system architecture consisting of six fundamental system components split into buy-side, the customer and sell-side activities. These components are: Browse, Request, Approve, Fill, Receive and Pay.

E-COMMERCE SECURITY TECHNOLOGY

The success of e-Commerce depends on the security of data like personal details and credit card numbers transmitted over the Internet. The domain name system Internet protocol that makes IP addresses readable by humans is insecure. Security measures need to be taken in e-Commerce systems to prevent compromising the systems. Some of these measures include building firewalls, incorporating cryptography and authentication, and using secure connections.

Firewall

A firewall is hardware and software that are used to secure a private computer network system from uninvited intruders. A firewall is used to control whether a client is permitted to connect to the private network it protects.

Cryptography

Cryptography is a science that provides secure communication over vulnerable channels. Cryptography is fundamental to the success of the Internet and e-Commerce. Governments regulate cryptographic technology because of its importance to national security. In cryptography a message, like a credit card number, is encrypted using a key and the encrypted message is transmitted. The receiver uses the key to decrypt the message and convert it back to its original form. The basic elements of a cryptographic system are algorithms, protocols and key management. An example of key management is the secret-key encryption algorithms. These algorithms are 'secret' because only the receiver and sender know the secret key. Cryptography is used to provide secure transmission of data over the Internet. Private data like credit card details or digital signatures are encrypted and then transmitted over the Internet. Cryptography can keep a message secret and act as a gateway for identifying senders and receivers. It provides the secure electronic transaction technology for credit card transactions on the Internet.

Authentication

Authentication procedures are used to establish the identity of an individual or another computer system. Authentication procedures can be hardware-or software-based. Authentication procedures make use of personal items of knowledge or possession such as secret names or birth dates. Good authentication systems make use of two-factor authentication, such as a place name and memorable date known to the user. Some banking systems make use of three-factor authentication before allowing customers to make online account transfers.

Secure Socket Layer

The secure socket layer (SSL) is a layer of security between the application and the transport protocol. The purpose of SSL is to enable secure and reliable data transmission and communication over the Internet. The SSL provides private connection, making use of encryption and secret-key cryptography. Authentication in SSL is achieved using public-key cryptography, which consists of a private key that is never made public chosen by one participant in the data exchange, and a public key chosen by the other participant in the exchange. Either key may be used for encryption. Reliability of data transmission is achieved by using secure hash functions like SHA or MD5. Secure hash functions check the integrity of a message. SSL is commonly used in e-Commerce systems.

E-COMMERCE CONTRACTS

Virtually all commercial transactions are undertaken in the setting of a legally binding contract. Indeed, without the presence of such a contract the parties would generally be unwilling to perform their obligations (such as the delivery of goods, the performance of a service or the payment of money) under the transaction. Contracts provide certainty as to the obligations of each party and, more importantly, a guarantee of the right to sue the non-performing party for breach. The law of contract dates back centuries, and has equal application to an ecommerce transaction as an offline contract.

But the nature of an e-commerce transaction gives rise to some special problems that do not arise offline:

- Formation of the contract — for each party to have confidence that the performance of its obligations will have legal effect, it is of vital importance that the contract has actually been formed at the time of that performance. In the offline world the existence of a contract can be evidenced in many ways — such as a signature on a printed order form or a note of a conversation — that are not available in the online world.
- Incorporation of contractual terms — commercial transactions take place under a set of contractual rules. Usually these rules are set out in written form and will apply to the contract by virtue of being ‘incorporated’ within it. In the offline world terms are included in a contract by agreement. There is a considerable body of caselaw that describes how and why such terms will be incorporated. That caselaw may not, and indeed in some cases cannot, apply to online transactions.
- Non face-to-face transactions — e-commerce transactions, by their definition, take place electronically. The advantage of this is that the parties do not need to be in each other’s presence at the time of formation of the contract. One disadvantage of this is that the law treats differently those contracts that are entered into between a business and a consumer ‘at a distance’ — the e-business must comply with a set of rules that do not generally apply to its offline competitors.

Formation of a Contract

It is generally well-known that a legally binding contract will arise where there is an offer to do something that is met by an unconditional acceptance of that offer. By way of example, where X offers to pay £250 for the delivery of a case of champagne to her home and Y agrees to perform that task for that sum of money, a contract is formed. To complicate matters, the law distinguishes an *offer* from an ‘invitation to treat’. An *invitation to treat* is something that might appear to be an offer, but in fact is not.

An example is the display of goods in a shop window or on the shelf of a supermarket. It has long been decided that such displays are not offers and that therefore they cannot be ‘accepted’. This is the rule that prevents a customer from being able to force a shopkeeper to sell goods to him at the displayed price, even where that price is clearly erroneous.

Whilst in the offline world it is easy to determine when a particular communication is an offer or an invitation to treat, such a distinction is unclear in a virtual transaction. Argos recently felt the effect of this when it mistakenly advertised televisions on its web site with a price tag of £2.99 instead of £299.

When several people placed orders for the televisions Argos realised its mistake and refused to supply the televisions at the price advertised. One ‘purchaser’ sued the company for failure to deliver on the contract that she said had been created. The case was settled before it reached trial so we do not know what the judge would have decided. Although we can speculate that, in English law at least, a web site is probably an invitation to treat as opposed to an offer, it would be wise for e-businesses to make clear the status of pricing and other information on their sites.

Incorporation of Terms

Of central importance to the e-commerce transaction are the terms and conditions on which the contract is based and by which it is governed. An e-business will always wish to trade on those terms that are favourable to it. For example, the business may wish to state that it is not to be held liable for late delivery of goods, or that its total liability for defective goods is to be limited to a certain monetary figure.

In most cases this will be done by the insertion of terms and conditions on the web site and by making some reference to them during the contracting process. By completing the transaction the customers will effectively be binding themselves to those terms and conditions. For the reasons mentioned above it is important for the terms and conditions of trade to form part of the contract. Without them we must rely on the ‘default position’ provided by the law and this is rarely ideal for a seller. The difficulty with e-commerce transactions is incorporating the terms into the contract is rarely a straightforward matter.

There are various techniques, such as those listed below:

- Click-through with acceptance — here the customer is required to click on a button which says, ‘I accept’ or similar. This is the best position for the e-commerce business because the acceptance by the customer is clear and

traceable. However, there is a perception that customers may be ‘put off’ by the need for this formality.

- Click-through without acceptance — the customer is required to scroll through the conditions but there is no acceptance button. Here there will be, at the very least, implied acceptance by the customer if they continue with the transaction after having scrolled through the terms and conditions.
- Reference with link—here the customer is referred to terms and conditions of trade but not forced to scroll through them. However, there is a link that will take the customer to the terms should they wish to see them. This is not ideal but may amount to ‘incorporation by reference’ under English law — this has not yet been tested by the courts in the context of online transactions.
- Reference without link—there is a reference to terms and conditions of trade but they do not appear on the site. Again this is an attempt at incorporation by reference. It is the least satisfactory method and is unlikely to have the desired effect in an e-commerce transaction as there is no reason why the terms and conditions could not appear somewhere on the web site.

The legal formalities for online trading are expected to grow. The European Union, for example, has recently proposed several new laws which will govern e-commerce in the future. With this in mind, businesses would be wise to adopt a policy of requiring those persons they contract with online to click an ‘acceptance button’ to demonstrate their assent to the terms and conditions displayed on the web site.

UNFAIR TERMS

E-businesses must be aware that since 1999 certain terms in their contracts with consumers will be void, i.e. unenforceable and of no legal effect. The Unfair Terms in Consumer Contracts Regulations 1999 apply, by definition, only to B2C contracts. For the purpose of the Regulations a ‘consumer’ is a person acting privately and not in the course of business. In contracts with such persons any unfair term is void. An ‘unfair’ term is one which, ‘contrary to the requirements of good faith, causes a significant imbalance in the consumer’s rights under the contract, to the detriment of the consumer’.

Examples of terms that would be regarded as unfair under the Regulations include:

- Allowing the business to change the characteristics of goods or services offered without recourse to the consumer; and
- Allowing the business to terminate the contract without reasonable notice of such termination being given to the consumer.

THE DISTANCE SELLING REGULATIONS

A consumer who purchases goods or services from an e-commerce business is protected to a greater degree than a business purchaser is. It is important therefore, at an early stage, to determine whether the e-business will engage in B2B or B2C transactions.

In many cases of course the e-business will wish to trade with both businesses and individual customers. Where the e-business anticipates that part of its customer base will be individual consumers, the so-called 'Distance Selling Regulations' become relevant. The Consumer Protection (Distance Selling) Regulations 2000 (reproduced in Appendix three) came into force on 31 October 2000.

They apply to contracts between a business and a consumer that are not 'face-to-face' *e.g.*, contracts concluded by way of coupons in newspapers, 'teleshopping', and of course the Internet. The Regulations impose two main obligations on suppliers of goods or services. The first is to provide certain information to the consumer. The second is to furnish the consumer with a 'cooling off' period of seven working days during which the consumer is able to return the goods to the e-business for a full refund. It is interesting to note that a recent survey of all 235 UK websites listed on Yahoo!, offering goods in the publishing, music and computer hardware sectors, found that 92% of the sites gave incomplete or incorrect advice on the right to withdraw. About half contained key contractual information that was not clear or prominent.

Most of those companies would be shocked to be informed that after each online sale has taken place the customer could contact the business at any time within three months and request the business to collect the goods from them. The e-business would then be required to refund the purchase price to the customer, whether or not it decided to collect the goods.

The Information Requirements

In most cases the 'information requirements' in the Distance Selling Regulations can be satisfied by posting the appropriate information online. Consumers must be provided with the information shown in the box below in a clear and comprehensible manner. The consumer must also be informed of any intention by the supplier to provide substitute goods if the goods ordered are not available, and of the fact that the supplier will meet costs of return of such substitute goods by the consumer to the supplier in the event of cancellation.

THE COOLING-OFF PERIOD

The requirement of a 'cooling-off' period for distance contracts puts e-commerce businesses at a disadvantage when compared with their offline competitors. For a consumer to be legally entitled to return goods purchased in the offline world (for example in a shop) for a full refund, they must prove that there is some defect in the goods or that the goods are not of satisfactory quality.

The Regulations however allow an online purchaser to return goods within seven working days for a full refund, without obliging the purchaser to furnish any reason for doing so. As far as the cooling off period is concerned, consumers must be given a period of seven working days, from the day after *receipt* of the goods, in which to

change their mind. During that period the consumer has the right to return the goods to the e-commerce business for a full refund. Crucially, if consumers are not informed of their right to cancel in this way, then the time period during which they have the right to cancel increases automatically to three months plus seven working days. Businesses should therefore be aware that their failure to inform consumers of the cooling-off period will effectively extend their potential liability to refund the purchase price by three months.

Excluded Contracts

The requirement to provide consumers with the above information, and to give a cooling-off period, does not apply to the following contracts:

- For the sale or other disposition of an interest in land except for a rental agreement
- For the construction of a building
- For financial services
- Contracts concluded by means of an automated vending machine
- Contracts concluded with a telecommunications operator by use of a public payphone
- Contracts concluded at an auction.

It should be remembered that the Distance Selling Regulations apply to contracts with consumers, but not to contracts with corporate and business customers. The information above must be stated clearly and concisely on the web site — this may mean some changes to existing sites. Additionally, the cooling-off period does not apply to bespoke or tailor-made products. Any attempt to exclude the operation of the Regulations will have no legal effect.

WEB SITE LINKING AGREEMENTS

To date operators of websites have provided links from their own sites to those of third parties with little thought for the legal consequences. Several legal actions in recent months have shown that this carefree attitude to linking is not good business practice. In one case, involving the web site of a Scottish newspaper, an e-business was sued for providing a link to a page within the site of the newspaper. This practice, known as deep linking, was challenged on the basis that it allowed users access to the site without being required to travel via the homepage — it was the homepage on which revenue-generating advertising appeared. In a case by Stepstone, the recruitment site, an injunction was obtained in early 2001 that forced OFiR, a Danish media company, to remove links to Stepstone's site. Recently a view has emerged that a link from one web site to another could infringe the database right in the linked-to site.

In any event, it seems that there is an emerging right for e-commerce businesses to control how users experience their sites, and a growing willingness by the courts to recognise such a right. E-commerce businesses should therefore consider, in appropriate

circumstances, putting in place a written contract that sets out the obligations of the parties to a linking agreement.

The contract should deal with the following issues:

- The link — consideration should be given as to how the link should be constructed, any technical requirements and where on the relevant web page the link should appear.
If deep linking is to be allowed then the requirements and specifications for this should be set out. The contract should set out those circumstances in which the parties are able to sever the link.
- Intellectual property — even where a copy of part of the site is not made when the link is used (this would raise copyright issues), there may be use of the linked site's trade mark on the link itself. A licence to use all relevant Intellectual Property Rights should appear in the contract.
- Commission — sometimes the motivation for providing a link will be in receiving revenue as a result of users travelling to a third party site via the link.
The commission arrangements should be clear in their terms — will commission be paid for example merely upon the site visit by a user or must the user first purchase goods from the site?
There should be appropriate provisions for monitoring traffic such that commission can be verified and charged.
- Data protection — if there is to be a sharing of customer information between the linked sites then it must be clear that each site is to obtain the data protection consent of its customers for that transfer to be able to take place
- Database right — the contract should make it clear that the links envisaged by the agreement will not constitute infringement of the database right.
investment.

This book provides the objectives, drivers, and measures of e-commerce success.

From a perspective that we didn't have a few years ago, we can now analyse why certain approaches to e-commerce were destined to fail. The twelve industry cases provide an opportunity to examine both successes and failures and identify specific lessons learned. The guidelines set out here characterize the most successful actions of established e-commerce leaders, providing pertinent advice to senior managers in all segments, industries, and stages of e-commerce.

The dynamic model centered on these guidelines focuses on the key factors for corporate success in e-commerce integration. It includes the critical inputs and processes that lead to success in e-commerce (outputs), as measured by its contribution to overall corporate profitability (outcome). We see how a company that lacks significant e-commerce operations can become a company that uses e-commerce operations to significantly enhance its profitability.

A company's existing strategy, structure, and systems represent important inputs to the model, leading to processes through which a strong leader can transform all three to

achieve e-commerce success. This book is grounded both in the latest academic and managerial research and in extensive research on company activities and experiences. This book is written for the many large corporations that are reexamining the formulation and the implementation of e-commerce strategies and their level of investment in e-commerce. Its target is the large manufacturing, distribution, and service firms that would like to improve their return on e-commerce investments.

The processes begin with leadership in the formulation and implementation of a successful e-commerce initiative. Leadership starts with a commitment by the CEO and other senior managers to consider a significant role for e-commerce in the organization. The CEO should initiate a dialogue within the company on the changes necessary for a full-scale e-commerce effort. The CEO, along with other key executives, must examine the emerging role of e-commerce within the company's industry, ascertain its current position with respect to its competitors, and determine an appropriate level of investment.

Next, proven guidelines for e-commerce strategies permit the company to build on its position and quickly increase its online business activities. Though these strategies do represent some different options, successful e-commerce operations will likely incorporate all of them into its activities. Senior managers must decide how to organize, finance, manage, and provide IT solutions for e-commerce. We encourage a perspective that looks at the various alternatives and generally propose a structure that leads to full integration of e-commerce throughout the fabric of the organization. The company should avoid the temptation of small benefits in one company segment in favour of the larger benefits of total or near-total integration.

Appropriate systems must also be implemented to ensure the successful integration of e-commerce. Corporate culture, performance and operational measures, and compensation systems must all be considered unique challenges. Customer data management and demand and supply chain management are vital to maximizing the value of e-commerce. The introduction of changes to these systems may not have any deleterious effect on traditional commerce, but systems designed for traditional commerce may fail to provide proper management control for e-commerce.

STRUCTURE OF E-COMMERCE

After formulating an e-commerce strategy, senior managers must develop a plan for implementation. The company's organizational structure may at times conflict with the goals of e-commerce, and senior managers must anticipate these conflicts and act accordingly, whether by changing the structure of the traditional organization or by creating new structures specific to e-commerce. These considerations span the areas of financing, management teams, and operations. A final consideration is the use of strategic alliances.

- *Internal Investment: Maintain Full-Equity Interest in E-Commerce.* The company faces a decision about equity for the e-commerce venture, whether

to retain complete ownership in the e-commerce venture or to spin it off as an independent company.

- The lessons are clear. Many high-profile failures occurred in traditional companies that tried to imitate the pure-play model by spinning off. Investment in the venture should be made from within the firm, and reliance on outside capital typically seems like an indicator of trepidation about e-commerce on the part of senior management.

Spinning off e-commerce was popular in the early period of e-commerce, and the results have generally been disastrous. Building up large market capitalizations through IPOs was a popular trend, but it has become less desirable in a weaker and more skeptical stock market. The vast majority of these companies, including Wal-Mart, Kmart, Barnes and Noble, and Staples, have bought back the stock from equity partners and folded the e-commerce spin-offs back into the company.

- Internal investment
- Integrated management teams
- IT know-how from within
- Strategic partnerships.

The other main justification for spinning off was that the company lacked the experience needed to pursue e-commerce or that additional investment capital was necessary. While it may be true that stand-alone firms sometimes find it easier to quickly develop e-commerce expertise, integrated e-commerce ventures in large organizations can also do this. Since e-commerce capabilities are becoming an increasingly important core competency, companies should develop the expertise and control the e-commerce operation inside the company through a full integration into business units and functions.

- *Integrated Management Teams: Innovate without Spinning Off.* From the distribution side, e-commerce should often be seen as simply another sales channel and thus changes in the management structure are kept to a minimum. In industries that already have multiple channels such as banks (i.e., physical branches, ATM) and catalogue businesses (e.g., Nordstrom), an integrated structure is particularly relevant. With full integration, business functions such as marketing are easy to coordinate and organizational territorial conflicts are reduced. A less ideal but still viable alternative is to create a separate business unit for e-commerce. Many companies that began with spin-offs and then brought e-commerce back within the company chose to adopt separate strategic business units (SBUs).

Many companies have started their e-commerce operations with entirely separate management teams, often proposed to increase the focus on innovation and e-commerce, under the premise that e-commerce will not be given enough attention or independence by senior management at the parent organization. These management teams have often been established far from company headquarters, have had loose organizational cultures, and have been led by independent leadership often drawn from pure-play or other

entrepreneurial companies. Certainly innovation, flexibility, and creativity are needed to drive a successful e-commerce operation. But we have seen that this can occur successfully within traditional organizational boundaries. A formal set of management control structures and systems is necessary to balance the desired empowerment. When separate business units are established, they should function primarily to coordinate the full integration of e-commerce throughout the business units.

As with other organizational functions, it is sometimes desirable initially to establish a central organization both to drive and to coordinate these new activities. But, as the function matures, it should generally be more fully integrated into the business units. The separate functional units and the integrated functions within the business units may need unique structures and systems including unique performance measures and compensation and reward systems, but this customization of the organizational structure can be accomplished within existing organizational boundaries. There is no need to spin off core capabilities and detach valuable corporate assets.

- *IT Know-How from Within: Building Future Capabilities.* Not every company, especially those late to e-commerce, will have invested well enough in IT to develop e-commerce independently. Many e-commerce companies, even those with strong IT departments, have chosen to outsource e-commerce because of the necessity of speed.

Speed is indeed a powerful motivation for outsourcing, but outsourcing should be used to catch up, not to give up. Although outsourcing may be used initially to create e-commerce solutions, the company can reassume control over e-commerce and IT and eventually can use the e-commerce systems for traditional commerce as well. Staples followed this approach in attempting to catch its IT-leader rival, Office Depot. Staples invested heavily in IT to build technology close to its core capabilities, but also created a sole-sourcing partnership to avoid delays in more peripheral areas of the business.

Another alternative is to acquire IT capabilities by purchasing a small IT firm and integrating the firm while the e-commerce solution is being implemented. UPS has followed this approach several times in its attempt to hold its IT leadership position over rival Federal Express, and UPS has been able to continually roll out new services while integrating its acquisitions.

Other companies, however, have made the mistake of selling equity to venture capitalists and then relied on these partners for all of their IT needs. Barnes and Noble made this mistake and then was hamstrung further when Amazon sued them for a patent violation related to their Web platform. Borders and Toys 'R' Us have taken the dramatic approach of having a competitor, Amazon, run their Web site for them. Companies must learn from these mistakes and not place themselves in such a precarious position with respect to IT.

- *Strategic Alliances: Moving beyond Core Competencies.* Although companies are advised to maintain equity interest, management control, and IT integration,

traditional businesses can benefit from alliances with online companies in other ways. These relationships are most beneficial when the arrangement gives the traditional company access to supply-chain management, peripheral offerings, and customer bases.

Supply-chain management is a particularly useful area for partnership when the company cannot simply supply online customers from the same distribution channels as physical stores. Wal-Mart, Kmart, and Target have all made such arrangements to deal with their general merchandising businesses. CVS also simplified its procurement process for its Web site by entering into a partnership with Merck.com.

To provide peripheral offerings that will be available only online, alliances have also proven to be a successful approach. Neiman Marcus partnered with several luxury retailers to create a limited-scope “luxury portal” of high-end goods within its Web site. CVS partnered with WebMD to provide medical advice to complement pharmacy services. Finally, traditional companies can gain access to large online customer bases by making strategic alliances with portal and ISP companies. Wal-Mart and Kmart entered agreements with AOL and Yahoo respectively to co-brand Internet service packages that would bring their customers online and to their site in particular.

Traditional companies may also profit from providing a physical presence to online companies that are realizing the inevitable trend toward integrated channels. Target has agreed to house E*Trade kiosks and customer representatives in its stores, in what has been a beneficial relationship for both parties. For E*Trade, the Target deal was just one aspect of a bricks-and-clicks strategy that culminated with the building of a New York super-centre.

STRATEGY OF E-COMMERCE

In designing an e-commerce strategy, it is critical to recognize what the Internet changes. Not every value proposition in traditional commerce can attract customers in e-commerce, and not every traditional strategy is a viable source of increased profitability. The best strategies do not ignore the fundamental properties of the Internet and the behaviour of its users.

The organization must develop strategies for e-commerce operations that are consistent with overall corporate strategy. (Since this is a dynamic model, the order of appearance of these elements may be different. Strategy may precede leadership and resource commitments and structure and systems already in place may constrain the strategy decision.) The goals should include an increase in customer acquisition, enhanced channel optimization, improved customer loyalty and retention, and capturing value for the organization.

- Well-positioned online brand
- Online-friendly offerings
- Reliable customer service
- Cross-channel coordination.

Four strategic moves are:

- *Well-Positioned Online Brand: Acquiring Customers and Building Trust.* The simplest way for companies to initiate e-commerce operations is to transfer the brand name to the Internet, using the company's name as the domain name so that existing customers and potential new customers can easily find the company's Web site.

The company can then advertise the Web site in various physical locations and in all print publicity materials. All letterheads, business cards, other types of business publications, and traditional advertising media should include the Web site address.

In-store signage, kiosks, advertising on bags, and other materials have also been successful in drawing customers to the online channel. Large online marketing campaigns are likely to produce click-throughs and hits, but not customers, and have typically proved to have poor payoffs.

A brand name will draw customers to the company's site and can help assure customers about the quality of the offerings, the accuracy of the information, and the security of the Web transactions on that site. The brand name can also promote customer loyalty. The company must continue to provide the level of service and security that the brand name implies, or it will fail to attract repeat online customers and may even damage the traditional brand.

If the company's traditional brand is centered on price competition, it must reevaluate this strategy for e-commerce. Price competition is rarely a viable source of profitability in e-commerce. Search costs for comparing prices are negligible online, and price wars have typically led to prices decreasing to just at or above cost. The Internet also offers easy access to stores that offer price differentials, because customers do not need to travel any physical distance to get to any given store. Only in limited contexts should companies expect to achieve long-term competitive advantage based on price.

- *Online-Friendly Offerings: Product Selection and Differentiation.* The company must decide what products to offer on the Web site. Companies such as Office Depot have found that they are able to provide more total products to customers through Web sites. Physical store space limits the number of inventory items available in stores, while the Web enables the company to provide a larger product offering. Even when integrating their offline and online brand management, a company should not feel compelled to offer precisely the same products online as in its physical stores. The ability to offer a large quantity of products, especially in retail industries, is a major inducement to acquiring an online channel.

Further, it is fully appropriate for a company to eliminate online offerings that are simply unprofitable or otherwise inappropriate. General retailers such as Wal-Mart and grocers have used price minimums, margin requirements, and category restrictions to limit their online offerings. Some companies also require a minimum total purchase

on their Web site. Companies should also consider peripheral offerings in addition to those provided by the company's physical facilities. Such additions are especially appropriate for companies in service industries and companies with large business-to-business (B2B) components. For example, Office Depot's business services offerings are a strong complement to its office supplies business.

Another differentiator is offering customers the ability to obtain unique or uniquely tailored products or information. Customization can be a powerful differentiator because it allows customers to control the product or service choices, and loyalty can be built partly due to high switching costs. Personalization gives customers their very own version of the site that can include information and recommendations based on the customers' prior purchases and demographics.

- *Reliable Customer Service: Inspiring Loyalty among Customers.* Building trust online requires more than a strong brand; it requires strong customer service practices that emphasize the convenience of the online channel. The customer must trust the company and its Web site. Trust can be built through assurances of privacy and transaction security and by providing accurate information. Trust in the transaction's completion is built through prompt and accurate fulfillment of orders. Many people purchase at Amazon rather than other Web sites because of its consistently highly rated fulfillment and customer service operations.

Web site design also contributes to satisfying customers. Basic characteristics of the site, such as simplicity, legibility, clarity, and a professional look, are necessities for all Web sites. Strong search engines are especially important for companies with a large number of product offerings. Information about physical stores, customer service, and delivery and return policies should be easily accessible from the home page.

Technology should be a facilitator and not an end in itself, however, and the company should not invest too heavily in design features without being confident of their appeal. Neiman Marcus found that its customers were not interested in viewing luxury products on a high-technology platform. Barnes and Noble discovered that they could not create interest in online parallels to their in-store coffee bars and author visits.

The site should also provide a number of services for customers who need information about the site, the offerings, or products they have purchased. These include comprehensive FAQ (frequently asked questions) sections, easily accessible information on delivery and returns, and rapid e-mail response and technical support, where appropriate. Numerous delivery options should be available to the customer to allow for both speed and low-cost preferences. Companies such as Nordstrom's, which offers products that need to be examined in person before the sale is complete, have provided particularly generous return policies to overcome the hesitation to purchase online inherent in their product line.

Physical stores can also be used for pick-up of items that are needed immediately in industries such as pharmacies. Customers should also be offered convenient opportunities

to return defective or unsatisfactory products, including returning such products to the physical stores.

- *Cross-Channel Coordination: Convenience and Revenue Stimulation.* The Web site and physical stores should not be seen as two separate entities that happen to share a brand name. Cross-marketing is an important component, but ultimately, the online and traditional stores must have complementary roles in the overall corporate strategy. Exploiting the advantages of integration means more than drawing customers to the Web site through promotion in the physical store.

Allowing online customers to pick up and return orders to physical stores provides customers with choices and convenience, bringing more traffic to the physical stores. Initially many traditional companies did not allow online customers to return purchased products to their stores. Today, virtually everyone allows customers to return undesired items to physical store locations. Numerous studies have shown the propensity of customers to make additional purchases when drawn to the store by online-generated functions. Therefore, the Web site should include an easily accessible store locator function that includes a map to the location, a phone number, and store hours.

The physical store and online operations can each be used to stimulate sales in the other channel. Some products that require high levels of interaction with the product can be introduced to the online customer but sold in the physical locations. Kiosks that provide a computerized access to the online site, on the other hand, have been a successful mechanism in physical stores to stimulate online traffic. Many retailers have encouraged the use of kiosks to order online products not available in the physical store or when delivery of the products is more convenient. Kiosks are also effective in overcoming concerns about online shopping for less computer literate customers.

The Web site can also provide more convenience for customers who wish to purchase specific items immediately rather than wait for delivery. For example, Office Depot can now direct customers to local stores that currently have the product in inventory. Customers need not call all the local stores and tolerate taped messages and long waits to find a product. Even Internet pure-play companies like Amazon, through its alliance with Borders, for which it provides Web site operations, can direct customers to the nearest Borders store that has a desired product. Customers need only provide a zip code. Integration of physical stores and online operations can also lead to greater efficiencies. Supply chain advantages that lead to more effective purchasing, inventory control, and logistics management may enhance the company's overall cost structure and result in greater profitability.

DEVELOPMENT OF e-COMMERCE

The literature on *e-service* has by and large followed the commercial thrust of Internet marketing development. Initially, the most common application by firms and use by users was through the web site itself, often a basic overview of the company and more

in keeping with having a public relations ‘presence’ in the e-commerce world. The next most common application was to post more detailed information about products, and conversely this was the second most common use by users. The third wave of application is the now dominant interest in e-selling, that is, the use of the Web to sell or buy goods. This takes us to the fourth and emerging wave, one that is still greatly underutilized, namely e-service.

Essentially, e-service offers the potential for some of the more advanced applications of Internet technology. For example, firms could use intelligent agents to provide extraordinary service, by tracking and datamining previous histories through Internet sites and developing known patterns of users’ requirements. Customer relationships can be developed, based on prior ‘modelled’ understanding of which offers a consumer is likely to respond to. This is a special case of market segmentation to a market of one. As another application of work by Brookes *et al.* referred to in Cravens *et al.* (2000, p. 4), a manufacturer of washing machines could include in its warranty agreement the electronic capability of monitoring the ongoing usage of the machine.

Suppose that, after the machine has been bought, the household has a new baby. If detergent dosage or load patterns place strain on the motor, the machine could have the built-in communication capability to automatically page the manufacturer’s service depot. The depot receives a message for a service representative to visit the household and adjust the washing machine before it becomes a problem. In this way, while both the information technologies and the servicing component are largely unobtrusive to the household, they are nonetheless essential to maintaining customer satisfaction and potential loyalty.

The literature has started to discuss e-service, but it is somewhat unstructured. For this reason, a review of the literature plays a particularly important front-end role and we provide a simple classification of the literature to assist. We have included a separate section debating whether self-service is a myth for e-service. Our own position on this matter is made clear. We follow the literature review with a proposed, more systematic, typology of e-service, which could be useful to readers.

THREE APPROACHES TO E-SERVICE

We have identified three types of e-service literature. The first is what we call the *macro* or *very broad view* of e-service, namely that e-service is effectively synonymous with e-commerce. An *intermediate view* of e-service is that we can study the provision of *specialist services* made available by *specialist service providers* that help service Internet users (both individuals and companies). The third perspective on e-service is what we term a *micro perspective*, namely the provision of particular and varied detailed customer services within a site as *part* of the web site - user interface.

At the broadest level is the view that e-commerce *per se* is an electronic service to customers - one that provides greater convenience. This view places e-commerce as an option available to customers, providing another channel of distribution or information.

For example, instead of spending, say, an hour to physically access and purchase from a bookstore, the customer has the option to purchase the same electronically. As a further example, a company may use the Web to provide information about the company's offering to assist consumers in their product search, without necessarily enabling the consumer to purchase through the Web.

A related macro view of e-service is the *services marketing* perspective on e-retailing. They argue that e-retail offerings are service offerings and exhibit many of the same characteristics as other non-Internet-based services. They further argue that Internet services can be evaluated by similar criteria, such as responsiveness, empathy and the establishment of trust through courtesy and competency. The principal service provided by e-retailers is a search and evaluation facility that potentially saves time and effort for the consumer. The task of the e-retailer is to provide a web site design that caters to different shopping styles, provides evidence to reduce risk and also educates the user in a shopping mode that may differ to what they are used to in conventional retail shopping.

There is now a considerable number of papers that have applied the macro perspective of e-service to a specific industry. For example, Muir and Douglas (2001) have studied how service delivery has changed in legal services with the rise of e-commerce. They argue that the quality of service is potentially improved with the Web. A Web presence allows legal practices to be more transparent and to offer greater access to information for customers by way of improving their services. It is suggested that this improved communication may lead to a reduction in complaints against solicitors. An *intermediate perspective* of e-service is the provision of *electronic services* from *specialist providers* to users of the Internet and intranets. Thus we have a market (external or internal) in which key electronic services are similar to products and sold or exchanged in a market to general users of the Web. here is a huge variety of specialist firms that offer their services (products) to Web users.

Web designers form a stereotypical group in this category, but also included are all types of suppliers of a wide range of Internet services, such as portal providers or providers of any specific link in the Internet network. For example, consider the commercial services offered by Compuserve, Prodigy, America Online and e-world. The pages of the national financial newspapers are filled with the advertisements of companies offering such Web-enabling services, some claiming to offer an integrated service. Electronic trust services are specialist e-services that provide reassurance and trust to the financial and privacy security of Internet information flow. There is a number of third-party commercial service providers who guarantee protection of either the financial security or the personal confidentiality of information flows.

A special case of this situation is the role of the electronic signature, an issue discussed by Travers (2001), who is particularly concerned with the status, planning and implementation of electronic signatures, in the context of the UK Electronic Communications Act 2000. He argues that electronic signatures can be considered within a knowledge management framework and proposes a six-part system that incorporates

people, clients, knowledge matters, business development and training. Travers notes that the Electronic Communications Act provides:

- An approvals scheme for businesses providing cryptography services such as electronic signature services and confidentiality services;
- For the legal recognition of electronic signatures and the process under which they are verified; and
- For the removal of obstacles in other legislation to the use of electronic communications and storage in place of paper.

A somewhat unusual example of a specialist e-service is the provision of electronic money (Buck, 1997), which could be redefined as a trust-service, but Buck did not do so in his paper. He notes that there is a range of online payment systems, including credit systems (*e.g.*, Payflow Pro), debit systems (*e.g.*, BankNet), token-based mechanisms (*e.g.*, Digicash) and electronic cash schemes (*e.g.*, Mondex). Such mechanisms vary considerably in terms of safety, privacy protection and trustworthiness.

An example of a specialist e-service within an internal market is that of an e-mail-mediated help service (Hahn 1998). Hahn analysed 265 help-service responses from service logs and found, among other things, that users and help-service staff held different internal models for ideal e-mail communication. Users desired a fairly simple exchange of communication, that is, a clear question followed by a quick, simple response. Staff, on the other hand, envisaged the need for a more complex interrelationship, over several messages. This takes us to the third perspective of e-service, namely the *micro* approach. Perhaps the dominant element in this field is the role of information. Some authors see information-based marketing as a potential competitive advantage. Other scholars see the Web as important for tracking and gathering customer feedback (Sampson, 1998; Sen *et al.*, 1998). Still other writers focus on the role of e-information as an aid to facilitating consumer search.

It is particularly in Merrilees (2002) that a broader perspective is given to interactivity. He embraces a more multi-dimensional approach to the concept of interactivity. Included factors are two-way communication between the e-retailer and the user; the ability of each party to communicate with the other including through e-mail; the ability to personalize the situation for the individual user; and the ability of the individual to control the communication and learn from it.

Finally, there are numerous other Internet studies that emphasize particular aspects of e-service besides information or interactivity. For example Mols (2000), in his study of Danish retail banking, examined the role of more individualized services for consumers and their need for a close relationship with the bank. As a final example of a difficult-to-classify study of providing services on the Internet see Mathur (1998) who takes a financial accounting approach to the topic.

In summary, we have used our three-part classification of macro, intermediate and micro as an initial way of structuring the literature, as it exists. This is not to say that we endorse all perspectives of the literature. In particular we have reservations about the

macro perspective. In a sense, the use of the World Wide Web by a retailer to market its organization as an online e-retailer is no more a services marketing exercise than the use of catalogues makes Lands End a services marketer. The *service component for an e-retailer* is the sum total of the ancillary support mechanisms provided by the retailer and the channel intermediaries to aid the Web prospective buyer to select, pay for and receive the merchandise. The e-retailer may provide services for the consumption of the prospective buyer through the Web channel, but this still does not make the channel a service unto itself. Thus we prefer the intermediate and micro perspectives of initial classification and our empirical research design is more in keeping with the micro perspective. A more refined taxonomy of e-services is developed below.

BRITISH LIBRARY

This is a free service provided by the British Library in London that includes a ‘turn the page’ function. This technology is new to websites and is only available for three articles at the time of going to press. The home page offers a range of services provided by the library and is easy to use with simple menu selection and navigation. Of interest is the ‘turn the pages’ of Leonardo da Vinci’s notebook. This technology displays a scanned image of the original notebook and allows the user to turn pages using the cursor in a way similar to a real book. To enter this site, select ‘turn the pages’ on the home page and follow the instruction. To use this facility a Shockwave driver is required and can be downloaded free in two minutes from the same site. This is an interesting site for those who are interested in history.

A SECOND TAXONOMY OF E-SERVICE

We initially used a three-way classification of the e-service literature, namely macro, intermediate and micro, as a way of sorting the literature in this fragmented domain. Our thinking on this topic has progressed to another proposed taxonomy of e-service. Not all of the aspects covered in this taxonomy are addressed in the empirical part. Taxonomy is put forward as an initial framework that can be debated by interested academics or practitioners.

THE SELF-SERVICE MYTH

Before leaving the literature, it is useful to discuss a crucial issue in the e-service area referred to by Moon and Frei (2000). They give the example of customers visiting a typical airline web site and being confronted by a self-service search engine. If they know exactly when and where they want to travel, the web site will generate a list of feasible flights. However, the search process becomes complex if they wish to find the cheapest airfare and are flexible on the dates and destinations. Moon and Frei are sceptical of the usefulness of the self-service approach to e-service and suggest that the customer is likely to have a much more satisfying experience if they call the airline’s call centre for handling more complex situations.

Co-production is proposed by Moon and Frei (2000:26) as a better model than self-service for e-commerce. In the co-production model the company undertakes many of the tasks in shopping and buying, relieving the burden on the customer. The new model recognizes that, although customers like having choices, they do not want too many and appreciate pre-screened alternatives geared to their needs. Co-production also understands that customers want to state their preferences only once. Moon and Frei conclude that e-commerce firms should focus on customer service *not* self-service, and give illustrative examples of companies like Dell Computer Corporation. Dell performs a host of back-end transactions that are invisible to customers, such as grouping products by customer segments and displaying only in-stock items.

The current authors tend to endorse the sentiment of Moon and Frei in that we agree that it would be foolish for owners of e-sites to provide a minimalist infrastructure and simply to let users do their own thing. Our only objection is what to call the 'co-production model'. We are content to keep calling it a *self-service model*, but one that is designed appropriately and optimally.

Indeed, we believe that the challenge is for e-commerce firms to strive for an optimal self-service design capability. For a very interesting and relevant study, see Dabholkar (2000). His review of research into what consumers want from technologically-based self-service options suggests that speed, control and privacy are generally required. However, two additional attributes that positively influence the attitude of consumers towards computer technology are ease of use and fun or enjoyment. Web designers searching for the most effective design are advised to consult Dabholkar (2000).

We should point out that the same issue has been debated with conventional retailing. Some *weak* retailers have taken the view that *self-service* equals *no service* and have provided very little help to consumers, using the opportunity to reduce their costs (in particular fewer sales staff). This approach will only leave consumers frustrated and unable to find what they want and so they are likely to leave the store very unhappy. In contrast, as Merrilees and Miller (1996) note, for better-performing stores 'self-service' does not mean 'no service'. The best self-service stores, like IKEA, Wal-Mart and many superstores, appreciate that self-service needs to be designed in such a way that it delivers good service, albeit of a kind other than personalized service.

Thus *good self-service design* in *conventional retailing* includes a well-organized store, with good layout, good signage, helpful visual displays and ready access to product information. Merrilees and Miller (2001) have argued that the approach taken by superstores represents a new self-service paradigm of retail service. Broadly speaking, the same good self-service design principles readily transfer from conventional retailing to e-retailing.

E-SERVICE PERFORMANCE

Knowing what is meant by e-service is an important first step in managing an e-retail business. However, it is necessary to take another step, namely the measurement

of how well an organization is performing in each e-service activity. At this point it is clear that we only have a limited understanding of which e-service functions are important to consumers and how well e-retailers deliver these services. Some much-needed new research, namely a study of critical incidents in e-service delivery. Our critical incident analysis is followed by an examination of the role of e-service metrics as a management tool. In turn we discuss an alternative measure of e-service performance, namely e-retail service quality.

A CRITICAL INCIDENT APPROACH TO E-SERVICE PERFORMANCE

This section outlines the results of new research initiated especially for this book. A content analysis of 138 e-retail sites was undertaken by four judges in order to assess the nature and effectiveness of e-service across those sites. It was suggested that a good way of getting started might be with book, music, gift and department store sites, because these represent a high share of the e-selling transaction activity. The final discretion as to sites chosen was left to the individual judges. The main criterion for selection as a judge was expertise in analysing e-retailer websites.

Notwithstanding each judge's expertise, further controls were built into the evaluation process through careful briefing and training of the judges. This briefing included careful instruction to make sure that all items were fully understood by each judge and that all judges had the same meaning for each. Another instruction was that each judge was to spend about 10 to 15 minutes moving through the site to understand its features and its content before answering the various set questions. Each site was analysed in the same way. A survey instrument (protocol) was designed for each judge to use with each site. A wide number of site attributes were assessed on a Likert scale of 1 to 7, depending on whether the judge agreed or not that the site performed well on a particular attribute. There was also an open-ended section of the survey instrument where the focus was on the overall level of e-service for the site, based on a 1 to 7 Likert scale. On the same page the judge was asked to describe up to three important critical incidents/areas that positively contributed to their assessment of the overall level of e-service. Additionally, up to three important critical incidents could be listed that detracted from the level of e-service.

POSITIVE CRITICAL INCIDENTS

Initially we present the incidents that positively contributed to e-service. We do this in two ways. First we can note those positive facilitators to e-service across the *total* sample of 138 sites. We also break the results into *high-service sites* and *low-service sites*. High-service sites are those sites that were rated highly on the basis of the perceived overall level of e-service, that is, those sites that scored a 5, 6 or 7 out of 7. Those sites that scored 1 to 4 were called low-service sites. Across the total 138 sites, the three most positive incidents that contributed to e-service were:

- Interactivity and communication (56 per cent of all sites);

- Special offers (33 per cent);
- information (20 per cent).
- The next most important batch included:
- Variety of items for sale (14 per cent);
- Frequently asked questions (FAQs) (a separate type of interactivity) (12 per cent);
- Ease of use (12 per cent).

Other less important facilitators of e-service included security/privacy references (8 per cent), goods delivery (7 per cent) and returns policy (4 per cent). If we compare high-service and low-service sites, we more or less get a similar picture of what is important. One key difference is that the number of positive incidents of good e-service is a lot lower in low-service sites. Thus interactivity and special offers remain the highest two facilitators of good e-service, but the rate of incidence is about twice as much in high-service sites.

For example, in low-service sites there is a 27 per cent incidence of interactivity (compared to 72 per cent in high-service sites) and a 20 per cent incidence of special offers (compared to 39 per cent in high-service sites). A number of key determinants of e-service in high-service sites are downgraded in low-service sites. These include information, FAQ, security/privacy and returns.

NEGATIVE CRITICAL INCIDENTS

Apart from coding those elements of e-service that contribute to the overall level of e-service of a site, we have also analysed those elements that have reduced the overall level of e-service. We have retained the same classification of elements, but now we refer to negative critical incidents. That is, interactivity now refers to a lack of interactivity or a low level of service from this element. The same applies to the other eight elements of e-service.

The pecking order of negative critical incidents that affect overall e-service. The two most important negative incidents are a narrow variety of goods for sale (15 per cent of sites had this problem) and poor delivery service (14 per cent of sites). Other problem areas of e-service included interactivity (11 per cent), information (10 per cent), ease of use (8 per cent) and returns policy (8 per cent). The other three elements, FAQ, security/privacy and special offers, were rarely mentioned as a negative incident.

We have extended this analysis to a comparison of high-service and low-service sites. As we would expect, the number of negative incidents is almost thrice as great in the low-service sites. Variety and delivery were the top biggest problem areas for both types of sites, but had an incidence rate of less than 10 per cent in the high-service sites (compared to more than a quarter of the low-service sites). For the rest of the e-service elements, the most notable difference in the rankings is that of ease of use. It was the equal third highest problem area among low-service sites, but the lowest (and almost non-existent with a 1 per cent incidence) among the high-service sites.

LOGISTIC REGRESSION OF HIGH-LOW SERVICE SITES

We can extend our analysis from an enumeration of the critical incident elements (as a form of Pareto analysis), to a binary logistic regression analysis in which we can predict which sites fall into the high-service or low-service categories. The dependent variable is binary (either one type of site or the other). The independent variables are the nine dummy variables denoting positive critical incidents and the nine dummy variables denoting negative critical incidents.

Overall the degree of explanation is high, with an adjusted coefficient of determination of 0.53. Nine of the dummy variables were significant at the 5 per cent level. The model predicts very well, with an 88 per cent hit rate.

The six major e-service elements that shunt a firm into either a high-service or low-service category were:

- Positive interactivity incidents
- Negative information incidents
- Negative delivery incidents
- Negative interactivity incidents
- Negative ease of use incidents
- Positive FAQ incidents.

In addition, less important influences were positive information incidents, positive special deals and negative variety of items for sale incidents.

Multiple Regression Analysis of Overall Service Rating

Given that we have recorded the actual overall service rating of the site, on a 1 to 7 scale, we can also analyse the data with conventional multiple regression analysis, with the overall service rating as the dependent variable and the same independent variables as before. This is another way of testing the robustness of our results, although the logistic regression and the multiple Ordinary Least Squares (OLS) are testing slightly different models and therefore we would not expect exactly the same results to be produced. The results. The adjusted coefficient of determination is good, at 0.56. Eight variables were significant.

The six most important e-service elements explaining the overall service rating were:

- Positive interactivity incidents
- Negative delivery incidents
- Negative ease of use incidents
- Negative information incidents
- Positive information incidents
- Negative interactivity incidents.

Additionally, positive variety of the offerings had a marginal influence on overall service rating, while positive deals and negative variety of items were not quite significant. If we compare we see that essentially the same elements are at work. However there are slight differences, with some elements appearing in one table only, and the

pecking order changing marginally. For example, a positive FAQ service might elevate an e-retailer into the high-service category, but did not have any discernible influence on the overall rating regression.

E-SERVICE METRICS: A MANAGEMENT TOOL

Our research in the previous section revealed that interactivity was overwhelmingly important for achieving a perception of high e-service. This finding is even stronger when we add frequently asked questions (FAQs) to its role because this is a form of interactivity. In a sense the emphasis on interactivity is akin to the importance of personal service in the conventional literature. Interestingly, personal service was found to be the most important type of customer service in conventional retailing. Interactivity in the e-context includes two-way communication, the ability of the e-retailer to communicate to the user and the ability of the user to communicate with the e-retailer, responsiveness in answering questions (including the special case of FAQ) and personalization of the process.

Even without measurement, interactivity is clearly a good candidate as a capstone element in a powerful e-service programme. Similarly, we recommend that an e-retailer should audit its interactivity every year or so, along the lines suggested in the previous section (using, say, a critical incident analysis). If interactivity is found to be too low or not as high as desired, then steps can be undertaken to increase it, through, for example, increased customization or other means. A second key finding of our research study is that interactivity needs to be supported by special offers, information, variety of items for sale and ease of use, as part of an integrated approach to e-service. An e-service audit of an e-retail site should incorporate all components of e-service.

A third key finding of our research is also very important for the practice of good e-service. We have shown that it is not sufficient to create and manage *positive e-service experiences*. Equally, the firm's e-commerce strategy needs to be able to handle *negative critical incidents*. The first point is that *even high-service sites* experience periodic problems in e-service. Show that high-service sites, while generally having a very high (72 per cent) positive incidence of interactivity, nonetheless had a 6 per cent (that is, non-zero) incidence of negative interactivity. The same pattern occurs in the areas of product variety and delivery. Firms need to take steps to continuously improve (that is, lower) the rate of negative incidents. Ideally, more interactivity may need to be built in if all other aspects of interactivity fail - this is the ultimate approach to service-failure recovery. Perhaps it is a toll-free phone service that is needed as a service in the last resort?

A fourth key finding of our research that needs to be carefully considered by websites is that the solution for sites attempting to increase their e-service capability is not simply to *add* more information, an FAQ service or similar facility. Such actions are a necessary, though insufficient condition to becoming a high-service site. Take information, for example. Information incidents, both positive and negative, were important in

determining membership of the e-retailer in a high-service or low-service category and the overall level of e-service of the site. Yet there was only a slight difference (and one that was *not* statistically significant) in the quantity of information across high-service and low-service sites. This suggests that the problem for some sites is not the *quantity* of information, but rather the *quality* and *relevance* of information. Thus the high rate (20 per cent) of negative incidents about information on the low-service sites may be due to overemphasis on the wrong information, that is, the wrong details, rather than the lack of information in general.

In summary, we suggest that e-retailers should regularly monitor or audit all of their e-service, at least as frequently as annually. The critical incident approach is a simple way of doing this and the method is robust because it captures both positive and negative incidents in e-service. Notwithstanding the merit of this approach, there are alternative ways of evaluating e-service and we turn to one of those in the next section.

E-RETAIL SERVICE QUALITY: AN ALTERNATIVE PERFORMANCE METRIC

Retail service quality entails the application of service quality as both a concept and measure to retailing. The landmark study in this respect is Dabholkar *et al.* (1996). One advantage of using a measure of (retail) service quality is that it represents a composite measure, pulling together a number of components of service, such as personal service, store design and problem-solving. So, instead of having to say that six or seven or whatever components of service are performing at the individual service level, we can combine our assessment into a composite service quality measure. More recently, researchers have extended their scope of retail service quality from conventional retailing to also include Internet or e-retailing. These studies include Zeithaml *et al.* (2000), Francis and White (2002a; 2002b), Janda *et al.* (2002) and Wolfinbarger and Gilly (2002).

Each study uses slightly different dimensions (or items under each heading) of e-retail service quality, but generally the dimensions include:

- Web site design
- Security
- Ordering system
- Delivery system
- Communication.

The five dimensions of e-retail service quality provide an umbrella approach for e-retailers wishing to use an alternative measure of e-service performance. We offer no view as to whether this approach or the critical incident approach is better. Indeed, e-retailers could quite easily use both sets of metrics to evaluate their e-service, as they complement each other. At the time of going to press a major new article on e-retail quality has appeared, namely Wolfinbarger and Gilly (2003). This article seems to be the most comprehensive and methodical of all of the articles in this field and so we should highlight its findings. Four components of e-retail quality were identified, namely:

- Web site design (navigation, order processing, personalization);
- Fulfilment/reliability (receipt of correct goods, delivery on time);
- Privacy/security (security of credit card payments and privacy of shared information);
- Customer service (responsive to customer inquiries).

In terms of the predictive power of these four components, two of them (web site design and fulfilment) were found to be the most important in contributing to overall quality, satisfaction and return purchases.

ADDITIONAL GUIDANCE ON PRACTICAL E-SERVICE PROVISION

A number of practical tools that could help the improved delivery of e-service for e-retailers. First, key concepts, such as interactivity and delivery, have been highlighted as having special importance. Second, the idea of e-service metrics is another practical tool ready for actual use by e-retailers. In addition to these ideas and tools, the reader might wish to consult a number of 'how to' books, including Sterne (1996), Cusack (1998) and Zemke and Connellan (2001).

We have analysed the total number of websites in the sample as well as sub-samples of *high-service* and *low-service* websites. We have also contrasted *positive* critical incidents from *negative* critical incidents. Perhaps an unexpected result, we found that the main e-service elements that drive positive e-service are *not* the same as the elements that cause negative critical incidents in e-service. Interactivity in particular, strongly supported by special offers and information about the product and firm, was the key component of positive e-service. In contrast, negative e-service experiences were most often associated with a lack of variety of items for sale and poor delivery arrangements.

We have suggested that interactivity could be the key capstone element for e-retailers trying to build a powerful e-service programme, with support from information, ease of use, variety and special offers. There is also a need to manage *service-failure recovery*, that is, the myriad of negative critical incidents. This may lead to the ultimate form of interactivity - namely an interactive service that handles the collective failure in all the other interactive mechanisms.

Two different taxonomies or classifications were used to explore the nature of e-services. We also argued that self-service is not a myth for e-services and should not be taken for granted. It needs to be properly designed to genuinely help the e-customer. Further, evaluating e-service performance is important for e-retailers, if they are to fully understand what attributes are needed by customers. Two different ways of measuring e-service performance were given, namely a critical incident approach and an e-retail service quality approach.

The Evolution of Cyber Crime

Cyber crime has been an artifact of computer systems for a number of decades. However, the phenomenon of cyber crime did not truly come into being until the advent of the computer network. Information moving from across physical distances was much easier to intercept than that on a standalone system. Moreover, attaching a system to a network provided would-be criminals an access point into other vulnerable systems attached to the same network. But even in the early days of networked computing, cyber crime was rare.

The relative rarity of computers, combined with the highly specialised knowledge needed to use them prevented widespread abuse. The cyber crime problem emerged and grew as computing became easier and less expensive. Cyber crime evolved from hacking of another system, the public switched telephone network. These phone “phreakers” developed methods of breaking into phone systems to make long distance calls for free. Perhaps, the most famous of these phreakers was John Draper, who discovered that toy whistles given away with Cap’n Crunch cereals generate a 2600-hertz sound, which can be used to access AT&T’s long-distance switching system. Draper proceeded to build a “blue box” which, when used together with the whistle, allowed phreakers to make free calls. Shortly after, wire fraud in the United States escalates. Draper was arrested on toll fraud charges in 1972 and sentenced to five years’ probation. In the 1970’s, the first affordable personal computers became available on the market, and it was shortly thereafter that the first bulletin board service, or BBS, was established.

Early hackers and phreakers seized on the BBS idea as a way to communicate with one another and share their tricks and techniques. Still, even as the Internet grew, getting

online was far from easy. Designers of operating systems at the time had no idea how important the Internet would be. They didn't design software with built-in functionality to connect to an Internet service provider. ISPs were few and far between, and very pricey.

For a user to connect to the Internet, they would have to obtain, install and configure a number of settings that could be tricky for the casual user. Online services such as CompuServe, AOL, and Prodigy helped to solve this problem. They provided their subscribers with software that would enable them to connect to their service with relative ease. In 1986, alarmed by the larger numbers of computer break-ins, the US government passes the Computer Fraud and Abuse Act.

This made it a crime to illegally break into computer networks. The law did not apply to juveniles. Robert Morris became the first person to be convicted under the new Computer Fraud and Abuse Act of 1986. Morris was punished for his Internet worm, which crashed 6,000 Net-linked government and university computers. Price was still an issue, though, but in the early 1990's, costs for the user dropped to around \$3 an hour, and eventually, to less than \$20 a month for unlimited usage, allowing not only the Internet to grow exponentially, but also for criminals to learn how to effectively exploit the system. Computers are now ubiquitous and many tasks performed in the daily lives of users depend on computers and computer networks.

The Internet has become a mission-critical infrastructure for governments, companies, and financial institutions. Computers and networks are used for controlling and managing manufacturing processes, water supplies, the electric power grid, air traffic control systems, and stock market systems, to mention a few. A benefit of online services that attracts criminals is the anonymity they offer, making it easier for criminals to change identities and cover their tracks. The rapid growth of the Internet in the mid 1990's gave rise to cyber crime as we know it today.

EVOLUTION OF MOTIVATION

Ten years ago, hackers were dabbling on other systems to only see how they were configured and operated. Most of the time they did not cause any damage. Unfortunately, the circumstances have changed and become incredibly malicious. Instead of being driven by curiosity, hackers today are driven mostly by financial motives. The value of Internet activities and the wealth stored on computers is the source of the attraction.

While e-commerce represents only a fraction of total commerce, it reached almost \$70 billion in the U.S. at the end of 2004, an increase of 24 per cent over 2003. A third of the U.S. workforce is online, roughly 50 million people, an important consideration since more than half of e-commerce transactions are made from work. Sixty million residents of North America, almost half of the Internet user population in Canada and the U.S., have online bank accounts. The combination of banking and commerce draws criminals more than anything else.

CATEGORIES OF CYBER CRIME

Cyber crime has manifested itself in many different forms over the years.

The following points are illustrative of some of the different categories that criminals have entered:

- **Spam** - Although for much of history, spam was not technically a crime, the 2003 CAN-SPAM Act changed legal definitions on what is acceptable. Spam now represents more than 50 per cent of all e-mail transmitted over the Internet. It's costs, which Internet service providers (ISPs) pass on to their customers, are enormous. With spam's ubiquity comes a whole culture and industry devoted to fighting it. Large groups of people, such as the Spamhaus Project, spend enormous effort to identify the sources of spam so as to block their activity. New technologies have been created to flag its sources, like blacklists, and spam identification through Bayesian filters, distributed checksum databases, and other advanced heuristics. Increasingly on the defensive, spammers are fighting back by becoming more sophisticated, generating unique messages, and using subverted computers to send messages.
- **Extortion and Damaging Reputations** - In the Internet variant of a blackmail, criminal gangs will threaten companies with disruption of their networks, through denial of service attacks, or the theft of valuable information, unless they pay ransom into offshore bank accounts. Defacement of a company's web site can cause not just embarrassment but loss of sales. In other cases, spite or a desire to inflict harm means that the attack will be executed without warning.
- **Fraud and Phishing** - The anonymity and opportunities for misrepresentation found on the Internet make fraud easy. Consumer Sentinel, a complaint database developed and maintained by the US Federal Trade Commission, has recorded more than 390,000 Internet-related fraud complaints regarding transactions involving over US\$540 million losses in 2004 alone. Fraud schemes are usually peddled by individuals who spam potential victims, such as the Nigerian, or 419, scam. But as the number of fraud cases has increased, so has the public's awareness of them; fraudsters are increasingly forced to resort to more intricate schemes. New practices like "phishing" are gaining popularity with fraudsters. Using this scheme, criminals create e-mail messages with return addresses, links, and branding that seem to come from trusted, well-known organisations with the hope to convince victims to disclose sensitive information. This practice originates in attempts to fool America Online users into parting with their screen names and passwords in the mid-1990s. The goal these days is to extract information from a victim that crackers can use for financial gain. A commonly targeted item is victim's credit card information. Criminals also want access to Internet payment systems such as e-Bullion, egold, or PayPal; online transaction services such as Authorise. Net, iBill, and Verotel; and Internet accessible banks which includes almost all major banks today.

- **Service Disruption** - A cyber criminal can use an Internet attack to disrupt a key service. Denial of service attacks are one method, worms and viruses containing malicious code are another. A major auto manufacturer was one of many companies that had to shutdown its e-mail network for a few days because of the Love Letter virus.
- **Information Theft** - The most damaging category of Internet crime, information theft can take several forms. Cyber criminals can extract personal identification information or credit information from a company's database and affect thousands of consumers. Cyber criminals can also extract a company's own financial information. Finally, cyber criminals can steal valuable intellectual property from a company. While the reported cost of information theft is declining, it remains one of the greatest Internet risks a company can face.
- **Money Laundering** - The growth of global financial services makes it easy to conduct banking operations across borders over the Internet. The Financial Action Task Force, a group of national law enforcement agencies, notes that "within the retail banking sector, services such as telephone and Internet banking allow customers to execute transactions on a non face-to-face basis from any location with telephone or Internet access." While use of the Internet provides law enforcement agencies a greater ability to trace transactions through electronic records, the volume of transactions, the anonymity, and the lack of consistent record-keeping make it attractive to criminals and terrorists.
- **Child Pornography** - The Internet has become an important tool for sex offenders in order to facilitate the making, collection, trading and distribution of abusive images engaging children. It constitutes a vehicle to simplify the contact between child pornographers mutually, on the one hand, and with their victims, on the other hand. Consequently, the Internet linked with other technological advances has an enormous impact on both the volume and the nature of child pornography.

CYBER CRIME TOOLS

Cyber criminals have developed a wide array of potential tools that have had varying degrees of success over the years.

The following are a short list of some of these techniques:

- **Bots** — A bot (short for robot) is a computer on which a worm or virus has installed programmes that run automatically and allow cyber criminals access and control. Cyber criminals use viruses or other bots to search for vulnerable computers where they can load their own programmes or store data. A botnet is a collection of these infected machines that can be centrally controlled and used to launch simultaneous attacks. Spammers, hackers, and other cyber criminals are acquiring or renting botnets, making it harder for authorities to track down the real culprits.

- **Keylogging** — Keyloggers are programmes that covertly recover the keys typed by a computer user and either stores the data for later access or secretly sends the information to the author. The advantage of a keylogger programme is that the cyber criminal does not need to trick a user into supplying sensitive information.
- **Bundling** — Covertly attaching a virus or spyware to a benign or legitimate download, such as a screensaver or a game. When the computer user downloads and installs the legitimate file, they are unwittingly also giving permission to install the criminal programme.
- **Denial of Service** — An attack specifically designed to prevent the normal functioning of a computer network or system and to prevent access by authorised users. A distributed denial of service attack uses thousands of computers captured by a worm or trojan to send a landslide of data in a very short time. Attackers can cause denial of service attacks by destroying or modifying data or by using zombie computers to bombard the system with data until its servers are overloaded and cannot serve normal requests.
- **Packet Sniffers** — Software programmes that monitors network traffic. Attackers use packet sniffers to capture and analyse data transmitted via a network. Specialised sniffers capture passwords as they cross a network.
- **Rootkit** — A set of tools used by an intruder after hacking a computer. The tools allow the cyber criminal to maintain access, prevent detection, build in hidden backdoors, and collect information from both the compromised computer.
- **Spyware** — Software that gathers information without the users' knowledge. Spyware is typically bundled covertly with another programme. The user does not know that installing one also installs the other. Once installed, the spyware monitors user activity on the Internet and transmits that information in the background to someone else.
- **Social Engineering** — Social engineering is not limited to cyber crime, but it is an important element for cyberfraud. Social engineering tricks deceives the recipient into taking an action or revealing information. The reasons given seem legitimate but the intent is criminal. Phishing is an obvious example, a certain percentage of users will respond unthinkingly to a request that appears to be from a legitimate institution.
- **Worms and Trojans** — A trojan is a malicious programme unwittingly downloaded and installed by computer users. Some trojans pretend to be a benign application. Many hide in a computer's memory as a file with a non-descript name. Trojans contain commands that a computer automatically executes without the user's knowledge. Sometimes it can act as a zombie and send spam or participate in a distributed denial of service attack. It may be a keylogger or other monitoring programme that collects data and sends it covertly to the attacker. Worms are

wholly contained viruses that travel through networks, automatically duplicate themselves and send themselves to other computers whose addresses are in the host computer. In the past, cyber criminals occasionally use worms and trojans to hijack a victim's Web browsers. They replace the victims' home and search pages with links to Web spam, as well as drop links to the spam in the victims' bookmarks and on their desktops. To make money, they infect computers with malicious code that generates fraudulent ad views.

- Virus—A programme or piece of code that spreads from computer to computer without the users' consent. They usually cause an unexpected and negative event when run by a computer. Viruses contaminate legitimate computer programmes and are often introduced through e-mail attachments, often with clever titles to attract the curious reader.
- Internet message boards – Internet message boards dedicated to stocks are fertile ground for impersonators. A habit of many posters to these boards is to cut-and-paste press releases and news stories from other electronic sources into their posts to alert other posters and visitors to that information. Frequently, posters will paste in a hyperlink to direct a reader to a source directly, as Hoke did in the PairGain hoax. In addition to the rising threat, as national level attacks become more plausible, the vulnerabilities have also increased.

TYPOLOGY OF CYBER CRIME

The term “cyber crime” includes a wide variety of crime. Recognised crimes cover a broad range of offences, making it difficult to develop a typology or classification system for cyber crime. An interesting system can be found in the Council of Europe Convention on Cyber crime.

The Convention on Cyber crime distinguishes between four different types of offences:

- Offences against the confidentiality, integrity and availability of computer data and systems;
- Computer-related offences;
- Content-related offences; and
- Copyright-related offences.

This typology is not wholly consistent, as it is not based on a sole criterion to differentiate between categories. Three categories focus on the object of legal protection: “offences against the confidentiality, integrity and availability of computer data and systems”; content-related offences; and copyright-related offences.

The fourth category of “computer-related offences” does not focus on the object of legal protection, but on the method. This inconsistency leads to some overlap between categories. In addition, some terms that are used to describe criminal acts (such as ‘Cyber terrorism’ or ‘phishing’) cover acts that fall within several categories. Nonetheless, the categories provided by the Convention on Cyber crime serve as a useful basis for discussing the phenomena of cyber crime.

ECONOMIC IMPACT OF CYBER CRIME

Without any doubt, the financial damage caused by computer and Internet crimes is significant. Various recent surveys have been published analysing the economic impact of cyber crime, highlighting its significant impact. The same general concerns about crime statistics also apply to estimates of financial damage – it is uncertain to what extent surveys provide accurate figures and statistics, as many victims may not report crimes.

RESULTS OF SELECTED SURVEYS

The Computer Security Institute (CSI) Computer Crime and Security Survey 2007 analysed the economic impact of cyber crime, based on the responses of 494 computer security practitioners in U.S corporations, government agencies and financial institutions. It is mainly relevant for the United States. Taking into account the economic cycle, the survey suggests that, after rising until 2002, the financial impact of cyber crime decreased over the following years.

The survey suggests that this finding is controversial, but it is unclear why the number of reported crimes and the average loss of the victims may have decreased. In 2006, the extent of losses climbed again. The survey does not explain the reduced losses in 2002 or the rise in 2006. From 21 categories identified by the survey, the highest dollar losses were associated with financial fraud, viruses, system penetration by outsiders and theft of confidential data.

The total losses for 2006 of all respondents amounted to some USD 66.9 million. After a number of years of decreasing average losses per respondent, a turnaround is taking place. In 2006, the average loss was USD 345,000. In 2001, the average loss was nearly ten times higher (USD 3.1 million). The average loss per respondent depends strongly on the composition of respondents - if mainly small and medium sized enterprises (SMEs) respond one year and are replaced by larger companies the next year, the change in participants strongly affects the statistical results.

The FBI Computer Crime Survey 2005 follows an approach similar to the CSI Survey, but with a greater and more extensive coverage. The FBI survey estimates that the cost of security incidents from computer and Internet crimes amounted to USD 21.7 million. The most popular offences that detected by respondents organisations were virus attacks, spyware, port scans and sabotage of data or networks. The FBI Computer Crimes Survey 2005 includes an estimate of the total loss for the United States economy.

Based on average losses and the assumption that some 20 per cent of US organisations are affected by computer crime, a total loss of USD 67 billion was calculated. However, there are concerns as to how representative these estimates are, and the consistency of participants year on year. The 2007 Computer Economics Malware Report focuses on the impact of malware on the worldwide economy by summing up total estimated costs caused by attacks. One of its key findings is the fact that offenders designing malicious software are shifting from vandalism to a focus on financial profits.

The report finds that the financial losses caused by malware attacks peaked in 2000 (USD 17.1 billion) and 2004 (USD 17.5 billion), but have reduced Since, 2004 to USD 13.3 billion in 2006. However, similar to the survey results, there is uncertainty as to whether the statistics on the impact of malware are realistic. There are large discrepancies between reported losses and proven damages – take the case of the Sasser Worm, for example. Millions of computer systems were reported to be infected. In the civil law suit against the software designer, very few companies and private individuals responded to the request to prove their losses and join the lawsuit. The case ended with a settlement that the designer of the virus should pay compensation of less than ten thousand US dollars.

DIFFICULTIES RELATED TO CYBER CRIME STATISTICS

It is unclear how representative the statistics on the economic impact of cyber crime are and whether they provide reliable information on the extent of losses. It is uncertain to what extent cyber crime is reported, not only in surveys, but also to law enforcement agencies. Authorities engaged in the fight against cyber crime encourage victims of cyber crime to report these crimes. Access to more precise information about the true incidence of cyber crimes would enable law enforcement agencies to better prosecute offenders, deter potential attacks and enact more appropriate and effective legislation. Several public and private sector organizations have tried to quantify the direct and indirect costs of malware.

While it is difficult to estimate the cost to businesses, it is even more difficult to assess the financial losses inflicted by malware and the like to individual consumers, although there is scattered evidence that damages can be very large. However, such costs have different components. They may result in direct damages to hardware and software as well as financial and other damages due to identity theft or other fraudulent schemes. The range of estimates differs, although the emerging overall picture is quite coherent. Businesses on the other hand may avoid reporting cyber crime offences for several reasons: Businesses may fear that negative publicity could damage their reputation.

If a company announces that hackers have accessed their server, customers may lose faith. The full costs and consequences could be greater than the losses caused by the hacking attack. However, if offenders are not reported and prosecuted, they may go on to reoffend. Targets may not believe that law enforcement agencies will be able to identify offenders. Comparing the large number of cyber crimes with the few successful investigations, targets may see little point in reporting offences. Automation also means that cybercriminals follow a strategy of reaping large profits from many attacks targeting small amounts (*e.g.*, as happens with advance fee fraud). For only small amounts, victims may prefer not to go through with time-consuming reporting procedures. Reported cases

are often based on extremely high fees. By targeting only small amounts, offenders design scams that will often not be followed up.

EVOLUTION AND PROFILE OF THE ATTACKER

There is a growing convergence of technically savvy computer crackers with financially motivated criminals. Historically, most computer crime on the Internet has not been financially motivated: it was the result of either curious or malicious technical attackers, called crackers. This changed as the Internet became more commercialised. Financially motivated actors, spammers and fraudsters, soon joined crackers to exploit this new potential goldmine. Criminals have fully adopted the techniques of crackers and malicious code authors. These are financially motivated people, who pursue their goals considerably more aggressively than an average cracker. They have the monetary means to buy the required expertise to develop very sophisticated tools to accomplish their goals of spamming and scamming the public.

The perpetrators of these attacks vary considerably. At the low end are script kiddies, who are usually unsophisticated users that download malicious software from hacker web sites and follow the posted instructions to execute an attack on some target. These attacks are often only annoyance attacks, but they can be more severe. At the next level are hackers who are trying to prove to their peers or to the world that they can compromise a specific system, such as a government web site.

Next are insiders, who are legitimate users of a system that either access information that they should not have access to or damage the system or data because they are disgruntled. Insiders are often less knowledgeable than hackers, but they are often more dangerous because they have legal access to resources that the hackers need to access illegally. Next are organisational level attacks. In this case, the organisation's resources are used to get information illegally or to cause damage or deny access to other organisations to further the attacking organisation's gain. These can be legitimate organisations, such as two companies bidding on the same contract where one wants to know the other's bid in order to make a better offer.

They could also be criminal organisations that are committing fraud or some other illegal activity. At the highest level is the nation state that is trying to spy on or cause damage to another state. This level used to be called "national lab" attackers, because the attackers have a substantial amount of resources at their disposal, comparable to those that are available to researchers at a national lab, such as Los Alamos Laboratory or Lawrence Livermore Laboratory. After the September 11, 2001 terrorist attacks on the World Trade Center, the idea of nation state level cyber attacks being carried out by terrorists became a big concern.

MALWARE AND THREAT EVOLUTION

Viruses started appearing on dedicated networks such as the ARPANET in the 1970s. The boom in personal computers, initiated by Apple in the early 1980s, led to a

corresponding boom in viruses. In 1981 the first virus in the wild came into being even before the experimental work that defines viruses of today. Founded on the Apple II operating system, it was spread on Apple II floppy disks containing the operating system. While the viruses of the 1980s targeted a variety of operating systems and networks, most viruses today are written to exploit vulnerabilities in the most commonly used software: Microsoft Windows. The increasing number of vulnerable users is now being actively exploited by virus writers.

The first malicious programmes may have shocked users, by causing computers to behave in unexpected ways. However, the viruses which started appearing in the 1990s present much more of a threat: they are often used to steal confidential information such as bank account details and passwords. Classic file viruses reigned supreme in the 90s; however they have almost totally disappeared today. There are currently about 10 file viruses that are still active.

They experience peaks of activity when they infect the executable files of worms: the file virus will then travel as far as the infected worm file. For instance, samples of MyDoom, Netsky and Bagle that are infected by file viruses such as Funlove, Xorala, Parite or Spaces. On the whole, there is very little danger that classic file viruses will cause any major epidemics. The trends in epidemiology that are observed today have their primary roots in the second half of 2003. Internet worms Lovesan, Sobig, Blaster, Slammer and Sobre all not only caused global epidemics, but also profoundly changed the malware landscape. Each of these malicious programmes set new standards for virus writers. In 2003, we witnessed the emergence of an attack type that combines exploitation of server and workstation vulnerabilities with the characteristics of virus and Trojan horses.

By using more efficient attack vectors and, therefore, minimizing the human effort required to deliver attacks and use the compromised systems, the risks related to newly discovered vulnerabilities moved up in the risk measurement scale. Optimising costs, achieving greater efficiency, and applying the minimum necessary effort to accomplish goal are central concepts to modern day life. Therefore, it is not difficult to identify the same approach in the vulnerability exploitation techniques and attack trends. The appearance of many efficient worms as a result of attackers' attempts to maximize their bang for the "bug" are examples. They compromise a very large number of systems with minimal effort. The steadily increasing amount of cross-site scripting and SQL injection vulnerabilities discovered and disclosed during 2003 point to another path of less resistance into vulnerable networks. These vulnerabilities have rather simple ways of exploitation and they provide casual attackers with a high yield, direct access to internal networks, compromise of database servers and their content, and indirect ways of attacking unsuspecting users of third-party systems.

The level of sophistication in worms seen in 2003 and the installation of backdoors and tools with elaborate communication protocols and auto update capabilities indicate that attackers are trying to optimise the management of large amounts of newly acquired

assets. Classic e-mail worms are on the decline, with network and instant messaging worms exploiting relatively lax security to take their place in early 2005. IM worms were at the peak of their development in spring and summer 2005, and showed the highest growth rate among all classes of network worms. In the first six months of this year, an average of 28 new IM worms were detected every month. It should be stressed here that when P2P worms were at the peak of their evolution in 2003, approximately 10 new variants were detected every week. However, the situation changed afterwards and the flood of IM worms suddenly dried up. AOL and MSN, both of which have proprietary IM clients, were the main targets for such worms. Both companies took measures to protect their users. Firstly, by blocking the transmission of files with names and extensions which were known to be used by IM worms. In spite of the fact that IM worms rarely use file transmission as a propagation method, the move did have a noticeable effect.

The next step was to block the worms' main method of propagation, hyperlinks leading to files containing the body of the worm. These actions closed the majority of security loopholes being exploited by virus writers. And most importantly, they closed the loopholes which IM worms based on source code circulating in the computer underground used. Most of the code used in IM worms is of fairly low quality. The majority of these worms are created by script kiddies who have no significant programming skills. When the off the shelf code was no longer effective, these self styled virus writers were unable to create new propagation methods on their own, and this led to a sharp drop in the number of new worms.

Improved antivirus technologies, and increased user awareness of security issues are clearly forcing virus writers and hackers to use new approaches to access users' information and systems, mostly in the form of phishing attacks. Malicious users are starting to use viruses which propagate by exploiting vulnerabilities within web applications, particularly Internet Explorer, rather than network and e-mail worms. One consequence of this is an increase in the number of compromised sites. Exploits for IE are placed on compromised sites, which means that users who visit these sites will have trojan programmes downloaded to their machines.

To date Linux-based platforms have mainly been the victims of rootkit attacks and simple file viruses. However, the growing number of publicised vulnerabilities means that the increased number of users switching to Linux will not remain untouched by new malware. Handheld devices, such as PDAs and cell phones are almost household appliances for many people. Virus writers have been quick to take advantage of their growing popularity. The first trojan for Palm OS appeared in September 2000. And finally, the increasing interest in on-line games, with the potential profits to be made in this area, make it more than likely that malicious code designed to steal such information will continue to evolve rapidly. The first Trojan for gaming consoles had also been discovered. Sony PlayStationPortable was the first victim - the Trojan targeting this device deleted system files causing the console to cease functioning correctly. This

behaviour is very similar to Trojans for mobile phones. It may be that these new Trojans for gaming consoles signal the start of a new interest among virus writers.

EVOLUTION OF EXPLOIT FRAMEWORKS

Cyber criminals increasingly rely on powerful exploitation frameworks to launch their attacks. Free tools like Metasploit and commercial tools like CORE IMPACT and Immunity CANVAS have revolutionised the attackers' methodology. Previously, upon finding a vulnerability, the attacker either had to create custom exploit code from scratch or scour the Internet to find such code to exploit the hole. Today, instead of scraping together a bunch of individual exploits, these integrated exploit frameworks include around one hundred or more exploits to compromise target systems. One property of the exploit tools is the separation of the exploit from the payload.

An exploit is the software that takes advantage of a flaw, letting the attacker load and execute a programme of the attacker's choosing. The code triggered by the exploit is known as the payload. Old-fashioned attacks tightly bundled exploits and payloads together. An attack might exploit a database buffer overflow with the purpose of adding a user for the attacker to the local administrators group. But, with this tight integration, the attackers were stuck with the given payload attached to the given exploit for the given vulnerability. Taking the payload from one attack and embedding it with another exploit required some serious machine-language fine tuning, and was often impossibly difficult. To remedy the situation, today's exploit frameworks include an arsenal of different exploits and an arsenal of different payloads, each offering a different effect the attacker wants to have on the victim.

So today, the attacker can use a tool like Metasploit to choose an exploit, such as a buffer overflow in lsass.exe, originally used by the Sasser worm last year. Then, the attacker can choose from more than a dozen different payloads. Metasploit packages the payload with the exploit, and then launches it at the target. The real effect of these frameworks in separating the exploits and the payloads is now reverberating through the industry. Developers who create fresh exploits for new flaws don't have to reinvent the payload wheel every time. Thus, they can focus their time on perfecting their exploits and producing them much more quickly. Moreover, those developers who don't focus on exploits can now zoom in on the production of high-quality payloads.

DEFENCE EVOLUTION

Computer security has been reactive for most part. That is, system administrators and security professionals are usually reacting to the latest attack. After they fix the vulnerability that allowed the attack, the attackers look for new vulnerabilities to exploit for new attacks. Trends in worm and virus delivery mechanisms and infection speed have also changed. Not long ago, a virus warning and the patch to vaccinate computers against it would appear days before the virus began spreading. Today, too often the first sign of a virus is that a part of the network goes down. Flash worms

such as SQL Slammer have paved the way for future worms to carry payloads that directly target their victims and wreak havoc on government, business, and societal structures. Existing technologies such as firewalls, intrusion detection systems, intrusion protection systems, virtual private networks (VPNs), and virus scanners provide integrated security solutions.

Not surprisingly, security has become a massive industry, and it is now a focal point for virtually every organisation. Proactively eliminating just the known threats places an impractical burden on existing server and network infrastructures. Eliminating unknown threats or zero day attacks, which as the name implies reveal themselves only when they first occur, requires real-time solutions that can identify unique attacks without overburdening the network with security and management overhead. The imagination of social engineers knows no bounds. Social engineers are highly aware of Internet user psychology and are well able to exploit current anxieties. In connection with this it should be stressed that the attempts of some companies to create a browser which is capable of determining the veracity of any site visited, or a browser which protects information stored on the potential victim machine is very hard to be one hundred per cent successful.

CYBER VICTIMS

Early exploits were mass attacks which affected the whole Internet community. Between 1996 and 2000, high-profile web sites such as eBay, the U.S. Department of Commerce, UNICEF, the New York Times and Microsoft all fell victim to hackers or defacers. The Melissa virus caused company e-mail servers to shut down. A fraudulent web page that was designed to appear to be a Bloomberg financial news story resulted in the shares of a small tech company increasing 31 per cent in response to the “news.” As the new millennium began, a huge, distributed DoS attack shut down major Web sites such as Yahoo! and Amazon. Apache, RSA Security, and Western Union were hacked.

The Code Red worm attacked thousands of web servers, and the Sircam virus hit e-mail accounts all over the world. As of today, spam accounts for fifty per cent of all e-mail sent, a staggering 12.4 billion messages a day, worldwide. Malicious users are now changing their focus from conducting mass attacks to targeting specific business structures, and these attacks are tailored to each individual case.

Identity thefts and credit card fraud are prevalent attacks affecting the public directly. Social engineering remains a threat, and the methods used are continuing to evolve. The biggest mass mailings were comparable in size to the activity shown in December of 2004 through and January, when cyber scammers exploited the tsunami in South East Asia. Cyber criminals target people who are new to the Internet gullible. With huge numbers of people connecting to the Internet for the first time every year, cyber criminals always have a fresh crop of Net newbies on which to prey. Elderly people, youngster and kids are also among the top targets.

CURRENT SITUATION

The Computer Security Institute (CSI) announced the results of its 10th annual Computer Crime and Security Survey. The survey showed that virus attacks continue as the source of the greatest financial losses, accounting for 32 per cent of the overall reported losses. Theft of proprietary information also showed a significant increase in average loss per respondent, more than double that of last year. Also unauthorised access showed a dramatic increase and replaced denial of service as the second most significant contributor to computer crime losses, accounting for 24 per cent of overall reported losses and showing a significant increase in average dollar loss. On a better note the total dollar amount of financial losses resulting from security breaches is decreasing, with an average loss of \$204,000 per respondent, down 61 per cent from last year's average loss of \$526,000. However the percentage of organisations reporting computer intrusions to law enforcement has continued its multiyear decline. Respondents cited the concern over negative publicity as the key reason for not reporting intrusions to law enforcement.

CONSEQUENCES FOR DEVELOPING COUNTRIES

Finding response strategies and solutions to the threat of cyber crime is a major challenge, especially for developing countries. A comprehensive Anti-Cyber crime Strategy generally contains technical protection measures, as well as legal instruments. The development and implementation of these instruments need time. Technical protection measures are especially cost-intensive. Developing countries need to integrate protection measures into the roll-out of the Internet from the beginning, as although this might initially raise the cost of Internet services, the long-term gains in avoiding the costs and damage inflicted by cyber crime are large and far outweigh any initial outlays on technical protection measures and network safeguards. The risks associated with weak protection measures could in fact affect developing countries more intensely, due to their less strict safeguards and protection. The ability to protect customers, as well as firms, is a fundamental requirement not only for regular businesses, but also for online or Internet-based businesses. In the absence of Internet security, developing countries could encounter significant difficulties promoting e-business and participating in online service industries.

The development of technical measures to promote cyber security and proper cyber crime legislation is vital for both developed countries and developing countries. Compared with the costs of grafting safeguards and protection measures onto computer networks at a later date, it is likely that initial measures taken right from the outset will be less expensive. Developing countries need to bring their anti-cyber crime strategies into line with international standards from the outset.

INTERNATIONAL CYBER CRIME

A significant problem that arises when working with cyber crime is that most crimes transit data through a multitude of international borders before reaching the final,

intended target. Such circuitousness has a deleterious effect on investigating cyber crimes as well as the application of laws. An illustrative example of the legal hurdles faced with international incidents comes from the “Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations” manual for the United State Department of Justice. The manual reports that when seeking assistance from ISPs overseas, officers must work “with the consent of that country,” which means certain formalities need to be resolved before proceeding.

First, prior permission of the foreign government must be obtained. Next, approval from the Justice Department’s Office of International Affairs, and finally a clear indication that the actions would not be objectionable in the foreign country. The process is long and unwieldy, especially since, by the time the necessary paper work is filed, ISPs may have already deleted the information. Or in a worse case, after the information is obtained, it will then be discovered that the attacker went through another country, forcing the process to be repeated.

Many developing countries are short on the resources and technical knowledge needed to expedite this process, causing the investigation to fail. By 1997, the problem was being recognised internationally and the G-8 Justice and Interior Ministers noted that to be “consistent with the principles of sovereignty and the protection of human rights, nations must be able to collect and exchange information internationally, especially within the short time frame so often required when investigating international high-tech crimes.” To aid this process they created a Point of Contact network which required participating countries to specify a specific group that could assist 24 hours a day, 7 days a week. By 2002, twenty countries were participating. These types of mutual legal assistance treaties (MLATs) have been effective where in the past law enforcement has been stymied.

For example, in 1992, the US government required assistance from Switzerland regarding an attack in the U.S., but since, Switzerland had no such laws regarding hacking on the books, they refused to help. In devising MLATs, a country can either create bilateral or multilateral relationships, each having its own benefits and drawbacks.

Traditionally, sovereign nations have entered bilateral agreements with countries that they trust and are willing to accept each other’s legal characteristics. They are quicker to negotiate, produce more detailed documents, are easier to change and allow nations to feel more comfortable sharing sensitive information. In fact, after the 2001 terrorist attacks, the US was eager to more quickly establish such ties and has concluded over 45 such agreements.

The drawbacks of course are that separate, and perhaps unequal, agreements must be reached, resulting in varying interpretations of crime and legal precedent. Multilateral pacts seem more suited to issues that are global in scale, much like cyber crime. Thus, it was with great fanfare that in November of 2001, thirty countries signed the Council of Europe’s Convention on Cyber crime. The convention had been five years in the making and represents the first truly multinational attempt at defining, regulating and

providing a framework for the legal issues in relation to cyber crime. Briefly, it established conduct that is prohibited, identified required national legal processes and addressed international cooperation. At the U.S., Senate hearings on ratifying the treaty, Swartz noted “in the past, if an electronic transmission’s trail led to another country, the chances were slim of successfully tracking the communication to its source or securing the evidence before deletion.

With the tools provided for under the Convention, however, the ability of U.S., law enforcement to obtain international cooperation in identifying major offenders and securing evidence of their crimes so that they can be brought to justice will be significantly enhanced.” Although the Senate Foreign Relations Committee approved the treaty, it has stalled in the Senate for nearly two years, as certain groups have opposed it for reasons related to civil liberties. The current state of multinational legislation thus remains a patchwork of bilateral treaties put together piece by piece. Establishing transnational treaties is a difficult task and remains as an open policy debate. What can be agreed upon is that all nations need multilateral assistance in a global sense, not just a limited group, as cyber criminals can route through any country. Treaties, then, need to harmonise laws, while building capabilities. Most importantly, such treaties should not be used to violate human rights, even though to do so may be legal in some countries.

For example, with the current Convention on Cyber crime, China could ask the U.S., to assist in finding political dissidents and supporters of democracy and the U.S., would be obliged, under the terms of the Convention, to provide assistance. More often than not, even if a successful conviction can be obtained, extraditing a criminal is still a tough legal battle. For example, in October of 2001, a Pakistani man was charged with defacing an American-Israeli organisation’s web site. The FBI, working with the U.S., Embassy in Pakistan, was able to identify the attacker and get a warrant issued for his arrest in Pakistan, yet three years later he is still at large. Clearly, there is a need for a more comprehensive international plan.

FUTURE TRENDS IN LEGISLATION

The direction of legislation has slowly been proceeding to more severe and serious punishments for cyber crime. November 3 saw the first prosecution for owning and operating a botnet system. It seems probable as legislatures, federal and state, become aware of threat posed by botnets, and as methods become more advanced in discerning botcontrollers, legislation aimed at the problem will follow.

Whether it will become an effective deterrent probably rests with the ability to investigate and prosecute. Another area of concern is identity theft, a process facilitated to a large degree through the Internet. California has been the first to create legislation aimed at companies with lax security regarding the protection of personal information they may store. The California Security Breach Information Act (SB-1386), which went into effect in July of 2003, forces organisations to notify individuals if there is such a security breach.

It has been a powerful method for not only making people aware of the issue, but also applying a force for change in policy within many organisations, lest they be branded as uncaring and incompetent. With more sensitive information being stored by a greater number of third parties, more states will come to the conclusion California has and indirectly apply pressure to organisations to reform. In another example, a recent piece of county legislation in Westchester, New York proposed to make it illegal for companies storing personal information to allow insecure access to their networks. In a sense, it would criminalise using a wireless network with no security measures. Although, many have pointed out specific weaknesses in the bill, the idea has been praised as a step in the right direction and an important conduit for educating the public.

Cyber crime presents a challenging position for lawmakers, as they struggle to keep up with changes in technology and in the methods used to exploit those technologies for maliciousness. Unfortunately, legal wrangling leaves the judicial system in a state that can be behind the times. It should be realised that in the end, laws can only do so much to regulate an activity. Proactive security, user education and vigilance, combined with effective forensics and enforcement remain the best remedies for combating cyber crime. Legislation still needs to enact appropriate punishments and establish frameworks, though and in that sense it has a crucial role to play in the mitigation of cyber crime.

THE EMERGING THREAT OF INTERNET BOTS

Network intrusions, data theft using Trojan horses, viruses and worms are among the threats security experts worry about on a regular basis. However, something more dangerous is emerging. Botnets, with their proliferation, sophistication and criminal use are emerging as the number one security threat. The recent arrest of 20-year old Californian man who made \$60,000 by selling access to botnets to spammers and hackers is evidence of the growing menace.

A bot is a malicious software programme that invades computer so that it can covertly be controlled by a remote attacker. A bot is seeded by attackers through worms, viruses or other means to exploit desktop and server vulnerabilities. They are then herded into botnets, which can then be controlled from a central command point that can force zombie machines to work together to perform any issued task. Botnets are evolving and getting nastier. Previously, they were controlled exclusively through Internet Relay Chat (IRC) channels, but are now increasingly being manipulated through other means, such as Web, instant messaging or peer-to-peer systems.

Moreover, bots are using rootkits to conceal itself from the user of the machine. “Kernel level rootkits are extremely dangerous as they conceal their malicious code and cannot be removed by most anti-virus or antispyware programmes,” says Martin Overton, security specialist at IBM Global Service. “The state of bot technology has reached the point that the state of Web technology has,” says Peter Tipett, CTO at Cybertrust, whose security experts found more than 12,000 people contributing to bots or renting out botnets. “Instead of fighting with each other, these guys are working

together and posting their code. We are getting a rich set of commands and capabilities used by the bad guys.” Apart from evolving as sophisticated security threats, their presence is growing exponentially. Network-security experts identify and shut down botnets with 10 to 100 compromised hosts several times a day. Crackdowns on large botnets with 10,000 or more hosts are rarer, but they still occur weekly, said Johannes Ullrich, chief technology officer for the Internet Storm Center, which detects, analyses, and disseminates information about Internet-related security problems. “Security investigators have even found one botnet of 100,000 computers,” Ulrich noted.

Research conducted by Symantec found that on average more than 60,000 botnets were activated each day in the first half of this year. They also noted that this is an increase of more than 140 per cent from the previous year’s semi-annual count. The following parts discuss how hackers profile and select their victims, attack techniques and their criminal usage and defences home users and system administrators can undertake to mitigate the risk of these attacks.

PROFILING AND TARGET SELECTION

Hackers are diligently profiling hosts and choosing targets that can provide them with longest survivability and carry out large scale attacks, and prevent their detection:

- *High Bandwidth:* One of the most sought after hosts are the machines connected to the Internet using high-bandwidth broadband. This can provide an attacker with an enormous cumulative bandwidth to carry out large scale DDoS attacks on target servers.
- *Availability:* Hosts with broadband connection are always connected to Internet and thus are the most sought after targets. This ensures hackers can carry out attacks round the clock without depending on whims of the users with dial-up connection which may connect to Internet at irregular intervals.

Low User Awareness and Monitoring Capability

Attackers prefer hosts where users have low security awareness and do not have access control mechanism like firewalls installed on their computers. The absence of such defences along with un-patched operating systems create ideal victims for hackers to break into and then install and maintain bots over a long period of time without being identified or traced.

Location

One of the prime goals of these cyber-criminals is to avoid detection after they commit crimes. They achieve this by selecting hosts that are geographically far away from their location. This makes very difficult for law enforcement officers to detect bots back to hackers. Also international prosecution being time consuming, expensive and non-standardised process that varies for each country, unfortunately ends up helping these cyber-criminals to go Scot-free. The typical profile that fits the criteria is that of residential broadband connection that has low or no access control mechanism or

university subnets connected to Internet with minimal monitoring, high bandwidth with high availability.

ATTACK TECHNIQUES

Bots generally employ one of several attack methods, but sometimes use multiple techniques to create a network of compromised computers. Some of these approaches are quite sophisticated, such as Phatbot, which can generate a new encryption for itself each time it infects a new system. This makes it difficult for the software to find a common code signature for and thus recognise Phatbot. Director of malicious code for Security Consultancy iDefence, Phatbot has successfully evaded detection by mutating itself from spyware to launch vitriolic DDoS attacks on compromised networks. The following are some of the ways that attackers use to create networks of bots for themselves.

Chat

IRC is the most common used technique, including those in the large Phatbot/Agobot and Sdbot/Robot families as a way to communicate and receive commands from hackers. IRC has a built in mechanism for multicast capabilities which let attackers quickly send commands to all parts of a botnet without writing new code for the bot.

Peer-to-Peer

Many bots take advantage of peer-to-peer communication to infect computers with vulnerabilities. They connect to open-source file sharing technology such as Gnutella and work with the WASTE file-sharing protocol. WASTE uses a distributed directory rather than a central server which lets bots easily find each other and communicate with one another. They can thus exchange hacker commands or other attack-related information among themselves. An attacker can initiate the process by serving as a peer in P2P network sending commands to one bot, which can then pass them onto the others. Thus, hackers don't have to communicate to bots via IRC multicasting. Decentralised-based bot systems are harder for security officials to trace or shutdown than systems using a single IRC source.

E-mail Attachments/Worms

Many hackers use methods such as e-mail attachments or worms to infect computers. Bots don't replicate or spread on their own, but they can use the worms' functionality to do so. In fact, hackers can spread bots more quickly with worms than with other methods. In addition, Botnets can spread worms faster than worms can spread on their own. The Symantec Security Response team said 2004's Witty worm, which infected and crashed tens of thousands of servers, was probably launched by a botnet. "We saw Witty break out more or less at the same time from a hundred or more machines. The machines were all over the world but they had something in common: they were on our bot list of compromised computers," he noted.

CRIMINAL USE OF BOTS AND BOTNETS

Bots can serve several purposes both legitimate and illegitimate. One legitimate purpose is to support the operation of IRC channels by conferring special administrative privileges or designated users. However, most of the common uses are criminally motivated for monetary gains or for destructive purposes.

Distributed Denial-of-Service Attacks

A DDoS attack is an attack on a computer system that causes a loss of service to users, typically the loss of network connectivity and services by consuming of the bandwidth of the victim network or overloading the computational resources of the victim's system. Most commonly implemented and often used are TCP SYN and UDP flood attacks. One of the most common uses of DDoS attacks is to wrest control of an IRC channel from its founder and founder's delegates. To take over an IRC channel, attackers conduct a DoS attack against one or more of the network's servers.

If they can succeed in downing a server they can split the network into two or more disconnected segments. If in a given segment there are no users joined to a particular channel of interest, the attacker can join that channel and seize the founder's privileges. Apart from the role in taking over IRC channels, attackers can launch successful DDoS attack against Internet sites. Let us assume if a given botnet has around 15,000 compromised hosts and has an associated bandwidth of 56kbps, a simultaneous attack by the entire botnet would direct almost 850 Mbps at its target – enough to cripple almost all e-commerce sites. These estimates are conservative because most of these compromised machines have cable modem and DSL hosts.

Moreover, because bots are widely distributed within the IP address space, filtering or blocking such DDoS attacks is not easy. At best, it requires cooperation between the target and multiple service providers. DDoS is not just limited to web servers; virtually any service available on the Internet can be a target of such an attack. Higher-level protocols can be used to increase the load even more effectively by using very specific attacks, such as running exhaustive search queries on the victim's web site. Recursive HTTP flooding means that the bots start from a given HTTP link and follow all links on the provided web site in a recursive way. This is also called spidering.

Further research also showed that botnets are used to run commercial DDoS attacks against competing corporations. Jay R. Echouafni and Joshua Schictel, alias EMP, ran botnets to send spam and carry out paid DDoS attacks to take a competitor's web site down. Echouafni was indicted on August 25, 2004 on multiple charges of conspiracy and causing damage to protected computers.

Spamming

Some bots enable SOCKS v4/v5 proxy – a generic proxy protocol for TCP/IPbased networking protocol on a compromised machine which allows them to launch spam attacks. Using bots and thousands of zombies (compromised machines) attackers can

send massive amounts of bulk e-mails. These bots can also add special functionality to harvest e-mail-addresses. Harvested e-mail addresses help them to send phishing mail which appears to victims to come from legitimate sources.

Sniffing Traffic

Bots can be used as a packet sniffer to watch for interesting clear-text data passing by compromised machine. The sniffers are mostly used to retrieve sensitive information like usernames and passwords. They can also provide information about other Internet bots if it has been compromised more than once. This allows one to “steal” another’s botnet.

Keylogging

If the compromised machine uses encrypted communication channels, then just sniffing the network packets on the victim’s computer is useless since, the appropriate key to decrypt the packets is missing. But most bots also offer features to help in this situation. With the help of a keylogger it is very easy for an attacker to retrieve sensitive information. An implemented filtering mechanism further helps in stealing secret data. If the keylogger runs on thousands of compromised machines in parallel, it is easy to imagine how quickly PayPal accounts are harvested.

Spreading New Malware

In most cases, botnets are used to spread new bots. This is very easy since, all bots implement mechanisms to download and execute a file via HTTP or FTP. But spreading an e-mail virus using a botnet is also attractive. A botnet with 10,000 hosts which acts as the starting base for a mail virus allows very fast spreading and thus causes more harm. The Witty worm, which attacked the ICO protocol parsing implementation in Internet Security System (ISS) products is suspected to have been initially launched by a botnet due to the fact that the attacking hosts were not running any ISS services.

Attacking IRC Networks

Botnets are also used for attacks against Internet Relay Chat (IRC) networks. Popular among attackers is especially the so called “clone attack.” In this kind of attack, the controller orders each bot to connect a large number of clones to the victim IRC network. The victim is flooded by a service request from thousands of bots or thousands of channel-joins by these cloned bots. In this way, the victim IRC network is brought down - similar to a DDoS attack.

Manipulating Online Polls/Games

Online polls/games are getting more and more attention and it is rather easy to manipulate them with botnets. Since, every bot has a distinct IP address, every vote will have the same credibility as a vote cast by a real person. Online games can be manipulated in a similar way.

Mass Identity Theft

Often the combination of different functionality can be used for large scale identity theft, one of the fastest growing crimes on the Internet. Phishing e-mails that pretend to be legitimate ask their intended victims to go online and submit their private information. These fake e-mails are generated and sent by bots via their spamming mechanism. These same bots can also host multiple fake web sites pretending to be Ebay, PayPal, or a bank, and harvest personal information. In addition, keylogging and sniffing of traffic can also be used for identity theft.

DEFENDING AGAINST BOTS AND BOTNETS

Defence against botnet infection and attack can be classified in three stages: prevention, detection and response. These stages need to be treated differently from home and system administrator perspective.

Prevention

The most common way for bots to compromise hosts is by exploiting the known vulnerabilities in the operating system or installed applications. Home users should follow guidelines regarding safe use by updating the installed OS and applications to defend their computers from being infected by attackers. If available, they should activate the auto-patch update facilities included in many popular operating systems and applications. Users should always install the latest version of antivirus software and practice safe handling of common web applications such as web browsers, e-mail, and instant messaging. In addition to this, every system administrator should be given training on online security and privacy issues. A high level of awareness on these issues is the best course in preventing malicious bots from infecting computers. They should implement access control measures and regularly monitor the generated logs on access control/peripheral devices.

Detection

Home users can use Microsoft Antispyware and Antivirus software, which are able to detect and respond to known types of bots, but are not effective for new bots on net. Online resources for scanning a system can also be employed, like the Symantec online security checker which will scan the system for commonly used Trojan ports. In addition to detection techniques used by home users, system administrators can employ network based tools to monitor perimeter defence devices to detect anomalies in Internet traffic. Slow network response, unexpectedly high volumes of traffic, traffic on unusual ports, and unusual system behaviour indicate the presence of malicious software including bots. Tools like network packet sniffers can be used not only to identify but also to isolate the subnet/machine which is generating malicious traffic. Analysis of the logs generated by network sniffer can also be used for finding IRC servers used, the names of the attacker's private channels and authentication keys.

Responses

As soon as the user realises that his/her computer has been compromised, the computer should be physically disconnected from the network. This denies access to the attackers and helps limit the potential damage both to user's own system and to other systems on the Internet. They should immediately update anti-virus software and check OS and application vendor sites for latest patches. If the user stores bank or credit card information on PC, the user should assume them to have been compromised and contact the appropriate organisation. Any passwords or secure data should be no more be used and changed at once.

Apart from response measures suggested for the home user, system administrators should isolate infected subnets to prevent the spread of bots. They can asses the damage with the help of a network packet sniffer by identifying the number of machines infected by bots within a subnet. They can assist the incident response team by preserving data on the affected system and relevant system logs like firewalls, mail servers, IDS, DHCP servers, and proxies.

END

Growth of network models like IRC and easily available tools to edit bots has provided attackers, many whom have very limited knowledge of the underlying technology, the ability to create large botnets that are scalable and automated. Sophisticated bots are incorporating encryption and shape-shifting polymorphism in their code, and finding wider uses for rootkits, code that allows a permanent and undetectable presence of computer, to conceal itself from the user of the machine and creating nightmarish scenarios for security experts. Moreover, hackers are also diligently picking victims with poorly implemented access control mechanisms, minimal monitoring to avoid detection, high bandwidth and software that is easy to infect and that allows for propagation. Bots are creating difficult challenges; nevertheless, users can fight back by proactively following best practices as recommended by the operating system and application vendors to prevent their machines from getting compromised in the first instance.

Some of the reactive methodologies outlined include using packet sniffers, monitoring firewalls and preserving critical logs to help incident response teams track down the attackers. However, none of these, in isolation are effective. High level security awareness among users and diligent monitoring of the systems are the most effective and real defences against the growing menace of bots.

Further, much research is being done at universities and institutions using honeynets to learn about attacker's tools, tactics, and motives, while developing ways to track these criminals. Government should encourage these research efforts as they may provide a future arsenal for law enforcement agencies against bots, the fastest emerging threat on net, which if left unchecked may jeopardise the safety of cyber world in coming years.

CONTENT-RELATED OFFENCES

This category covers content that is considered illegal, including child pornography, xenophobic material or insults related to religious symbols. The development of legal instruments to deal with this category is far more influenced by national approaches, which can take into account fundamental cultural and legal principles. For illegal content, value systems and legal systems differ extensively between societies.

The dissemination of xenophobic material is illegal in many European countries, but can be protected by the principle of freedom of speech in the United States. The use of derogatory remarks in respect of the Holy Prophet is criminal in many Arabic countries, but not in some European countries. These legal challenges are complex, as information made available by one computer user in one country can be accessed from nearly anywhere in the world. If “offenders” create content that is illegal in some countries, but not in the country they are operating from, prosecution of the “offenders” is difficult, or impossible. There is much lack of agreement regarding the content of material and to what degree specific acts should be criminalised. The different national views and difficulties in prosecuting violations committed outside the territory of an investigating country have contributed to the blocking of certain types of content on the Internet. Where agreement exists on preventing access to websites with illegal content hosted outside the country, states can maintain strict laws, block websites and filter content.

There are various approaches to filter systems. One solution requires access providers to install Programmes analysing the websites being visited and to block websites on a black list. Another solution is the installation of filter software on users’ computer (a useful approach for parents who wish to control the content their children can view, as well as for libraries and public Internet terminals). Attempts to control content on the Internet are not limited to certain types of content that are widely accepted to be illegal. Some countries use filter technology to restrict access to websites addressing political topics. OpenNet Initiative reports that censorship is currently practised by about two dozen countries.

EROTIC OR PORNOGRAPHIC MATERIAL (EXCLUDING CHILD PORNOGRAPHY)

Sexually-related content was among the first content to be commercially distributed over the Internet, which offers advantages to retailers of erotic and pornographic material including:

- Exchange of media (such as pictures, movies, live coverage) without the need for cost-intensive shipping;
- Worldwide access, reaching a significantly larger number of customers than retail shops;
- The Internet is often viewed as an anonymous medium (often erroneously) – an aspect that consumers of pornography appreciate, in view of prevailing social opinions.

Recent research has identified as many as 4.2 million pornographic websites that may be available on the Internet at any time.

Besides websites, pornographic material can be distributed through:

- Exchange using file-sharing systems;
- Exchange in closed chat-rooms.

Different countries criminalise erotic and pornographic material to different extents. Some countries permit the exchange of pornographic material among adults and limit criminalisation to cases where minors access this kind of material, seeking to protect minors. Studies indicate that child access to pornographic material could negatively influence their development. To comply with these laws, “adult verification systems” have been developed.

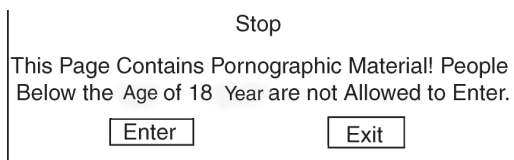


Fig. The Graphic Shows one Approach Used to Prevent Access of Minors to Websites with Pornographic Content. Since this Solution does not Provide Verification of the Answer Given by the User, it is Considered Inadequate in a Number of Countries.

Other countries criminalise any exchange of pornographic material even among adults, without focussing on specific groups (such as minors). For countries that criminalise interaction with pornographic material, preventing access to pornographic material is a challenge. Beyond the Internet, authorities can often detect and prosecute violations of the prohibition of pornographic material. On the Internet, however, as pornographic material is often readily available on servers outside the country, enforcement is difficult.

Even where authorities are able to identify websites containing pornographic material, they may have no powers to enforce removal of offensive content by providers. The principle of *National Sovereignty* does not generally permit a country to carry out investigations within the territory of another country, without permission from local authorities. Even when authorities seek the support of countries where offensive websites are hosted, successful investigation and criminal sanctions may be hindered by the principle of “dual criminality”. To prevent access to pornographic content, countries with exceptionally strict laws are often limited to prevention (such as filter-technology) to limit access to certain websites.

CHILD PORNOGRAPHY

In contrast to differing views on adult pornography, child pornography is broadly condemned and offences related to child pornography are widely recognised as criminal acts. International organisations are engaged in the fight against online child pornography, with several international legal initiatives including: the 1989 United Nations Convention on the Rights of the Child; the 2003 European Union Council Framework Decision on

combating the sexual exploitation of children and child pornography; and the 2007 Council of Europe Convention on the Protection of Children against Sexual Exploitation and Sexual Abuse, among others. Sadly, these initiatives seeking to control the network distribution of pornography have proved little deterrent to perpetrators, who use the Internet to communicate and exchange child pornography.

An increase in bandwidth has supported the exchange of movies and picture archives. Research into the behaviour of child pornography offenders shows that 15 per cent of arrested people with Internet-related child pornography in their possession had more than 1,000 pictures on their computer; 80 per cent had pictures of children between 6-12 years on their computer; 19 per cent had pictures of children younger than the age of 3; and 21 per cent had pictures depicting violence. The sale of child pornography is highly profitable, with collectors willing to pay great amounts for movies and pictures depicting children in a sexual context. Search engines find such material quickly. Most material is exchanged in password-protected closed forums, which regular users and law enforcement agencies can rarely access. Undercover operations are thus vital in the fight against child pornography. Two key factors in the use of ICTs for the exchange of child pornography pose difficulties for the investigation of these crimes.

The Use of Virtual Currencies and Anonymous Payment

Cash payment enables buyers of certain goods to hide their identity, so cash is dominant in many criminal businesses. The demand for anonymous payments has led to the development of virtual payment systems and virtual currencies enabling anonymous payment. Virtual currencies may not require identification and validation, preventing law enforcement agencies from tracing money-flows back to offenders. Recently, a number of child pornography investigations have succeeded in using traces left by payments to identify offenders. However, where offenders make anonymous payments, it is difficult for offenders to be tracked.

The Use of Encryption Technology

Perpetrators are increasingly encrypting their messages. Law enforcement agencies note that offenders are using encryption technology to protect information stored on their hard disks, seriously hindering criminal investigations. In addition to a broad criminalisation of acts related to child pornography other approaches such as the implementation of obligations of Internet Service to register users or to block or filter the access to websites related to child pornography are currently discussed.

RACISM, HATE SPEECH, GLORIFICATION OF VIOLENCE

Radical groups use mass communication systems such as the Internet to spread propaganda. Recently, the number of websites offering racist content and hate speech has risen - a study in 2005 suggested a rise of 25 per cent in the number of webpages promoting racial hatred, violence and xenophobia between 2004 and 2005.

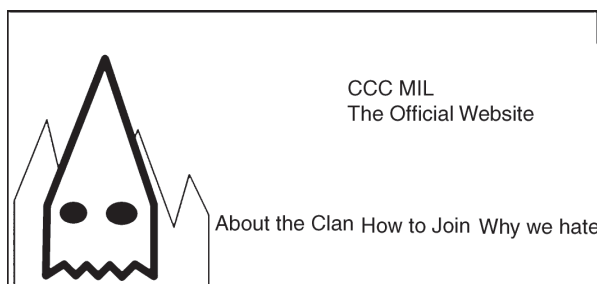


Fig. The Graphic Shows a Web site from a Radical Group. The Internet is Used Intensively by Such Groups to Inform People of their Aims and to Recruit New Members.

In 2006, over 6,000 such websites existed on the Internet. Internet distribution offers several advantages to offenders, including lower distribution costs, non-specialist equipment and a global audience. Examples of incitement to hatred websites include websites presenting instructions on how to build bombs. Besides propaganda, the Internet is used to sell certain goods *e.g.*, Nazi-related items such as flags with symbols, uniforms and books, readily available on auction platforms and specialised web-shops. The Internet is also used to send e-mails and newsletters and distribute video clips and television shows through popular archives such as YouTube. Not all countries criminalise these offences.

In some countries, such content may be protected by principles of freedom of speech. Opinions differ as to how far the principle of freedom of expression applies with regard to certain topics, often hindering international investigations. One example of conflict of laws is the case involving the service provider Yahoo! in 2001, when a French court ordered Yahoo! (based in the US) to block the access of French users to Nazi-related material. Based on the First Amendment of the United States Constitution, the sale of such material is legal under United States law.

Following the First Amendment, a US court decided that the French order was unenforceable against Yahoo! in the United States. The disparities between countries on these issues were evident during the drafting of the Council of Europe Convention on Cyber crime. The Convention seeks to harmonise cyber crime-related laws to ensure that international investigations are not hindered by conflicts of laws. Not all parties engaged in negotiations could agree on a common position on the criminalisation of the dissemination of xenophobic material, so this entire topic was excluded from the Convention and instead addressed in a separate First Protocol. Otherwise, some countries (including the United States) might have been unable to sign the Convention.

RELIGIOUS OFFENCES

A growing number of websites present material that is in some countries covered by provisions related to religious offences *e.g.*, anti-religious written statements. Although some material documents objective facts and trends (*e.g.*, decreasing church attendance in Europe), this information may be considered illegal in some jurisdictions. Other examples include the defamation of religions or the publication of cartoons.

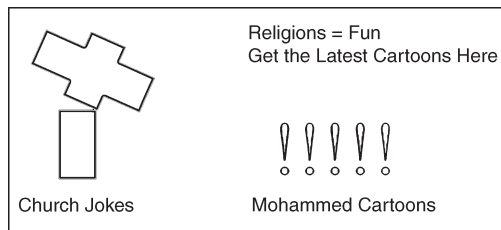


Fig. The Graphic Shows a Web site that makes Available Content with a Religious Background, which can be Accessed by Users Worldwide.

The Internet offers advantages for those who wish to debate or deal critically with a subject – people can leave comments, post material or write articles without having to disclose their identity. Many discussion groups are based on the principle of freedom of speech. Freedom of Speech is a key driver behind the Internet’s success, with portals that are used specifically for user-generated content.

Whilst it is vital to protect this principle, even in the most liberal countries, conditions and laws govern the application of principles of Freedom of Speech. The differing legal standards on illegal content reflect the challenges of regulating content. Even where the publication of content is covered by provisions relating to Freedom of Speech in the country where the content is available, this material can be accessed from countries with stricter regulations.

The “Cartoon Dispute” in 2005 demonstrated the potential for conflict. The publication of twelve editorial cartoons in the Danish newspaper Jyllands-Posten led to widespread protests across the Muslim world. As with illegal content, the availability of certain information or material is a criminal offence in some countries. The protection of different religions and religious symbols differs from country to country. Some countries criminalise the use of derogatory remarks in respect of the Holy Prophet or the defiling of copies of the Holy Quran, while other countries may adopt a more liberal approach and may not criminalise such acts.

ILLEGAL GAMBLING AND ONLINE GAMES

Internet games and gambling are one of the fastestgrowing areas in the Internet. Linden Labs, the developer of the online game Second Life, reports that some ten million accounts have been registered.

Reports show that some such games have been used to commit crimes including:

- Exchange and presentation of child pornography;
- Fraud;
- Gambling in online casinos; and
- Libel (*e.g.*, leaving slanderous or libelous messages).

Some estimates project growth in estimated online gambling revenues from USD 3.1 billion in 2001 to USD 24 billion in 2010 for Internet gambling (although compared with revenues from traditional gambling, these estimates are still relatively small). The regulation of gambling over and outside the Internet varies between countries - a loophole

that has been exploited by offenders, as well as legal businesses and casinos. The effect of different regulations is evident in Macau. After being returned by Portugal to China in 1999, Macau has become one of the world's biggest gambling destinations. With estimated annual revenues of USD 6.8 billion in 2006, it took the lead from Las Vegas (USD 6.6 billion). Macau's success derives from the fact that gambling is illegal in China and thousands of gamblers travel from Mainland China to Macau to play. The Internet allows people to circumvent gambling restrictions. Online casinos are widely available, most of which are hosted in countries with liberal laws or no regulations on Internet gambling. Users can open accounts online, transfer money and play games of chance. Online casinos can also be used in money-laundering and activities financing terrorism.

If offenders use online casinos within the laying-phase that do not keep records or are located in countries without money-laundering legislation, it is difficult for law enforcement agencies to determine the origin of funds. It is difficult for countries with gambling restrictions to control the use or activities of online casinos. The Internet is undermining some countries' legal restrictions on access by citizens to online gambling. There have been several legislative attempts to prevent participation in online gambling: notably, the US Internet Gambling Prohibition Enforcement Act of 2006 seeks to limit illegal online gambling by prosecuting financial services providers if they carry out settlement of transactions associated with illegal gambling.



Fig. The Graphic Shows the User Interface of an Online Casino. After a Registration Process and the Transfer of Money the User can Participate in Online Gambling. A Number of Online Casinos Enable the Use of Services Without a Formal Registration Progress.

LIBEL AND FALSE INFORMATION

The Internet can be used to spread misinformation, just as easily as information. Websites can present false or defamatory information, especially in forums and chat rooms, where users can post messages without verification by moderators.

Minors are increasingly using web forums and social networking sites where such information can be posted as well. Criminal behaviour can include (for example) the publication of intimate photographs or false information about sexual behaviours. In most cases, offenders take advantage of the fact that providers offering cheap or free publication do not usually require identification of authors or may not verify ID. This

makes the identification of offenders complicated. Furthermore, there may be no or little regulation of content by forum moderators.

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PETRO VOICE HOLDING
Hot Stock in Momentum play for week

OTC: PHVC
This company is nasdaq bound

Petro Voice in on a roll earning Contrac

This is not a Fly by Night
Real market Cap, Real Earnings

Here are some the latest news on PHVC.

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Fig. Spam e-mails are a Serious Problem. These e-mails Cover a Wider Range of Topics. In Addition to Promoting Different Products, Providing Information on Stocks and Shares is Very Popular.

These advantages have not prevented the development of valuable projects such as the online user-generated encyclopaedia, Wikipedia, where strict procedures exist for the regulation of content.

However, the same technology can also be used by offenders to:

- Publish false information (*e.g.*, about competitors);
- Libel (*e.g.*, leaving slanderous or libellous messages);
- Disclose secret information (*e.g.*, the publication of State secrets or sensitive business information).

It is vital to highlight the increased danger presented by false or misleading information. Defamation can seriously injure the reputation and dignity of victims to a considerable degree, as online statements are accessible to a worldwide audience. The moment information is published over the Internet, the author(s) often loses control of this information. Even if the information is corrected or deleted shortly after publication, it may already have been duplicated (“mirroring”) and made available by people that are unwilling to rescind or remove it.

In this case, information may still be available in the Internet, even if it has been removed or corrected by the original source. Examples include cases of ‘runaway e-mails’, where millions of people can receive salacious, misleading or false e-mails about people or organisations, where the damage to reputations may never be restored, regardless of the truth or otherwise of the original e-mail. Therefore the freedom of speech and protection of the potential victims of libel needs to be well balanced.

SPAM AND RELATED THREATS

“Spam” describes the emission of unsolicited bulk messages. Although various scams exist, the most common one is e-mail spam. Offenders send out millions of e-mails to users, often containing advertisements for products and services, but frequently also malicious software. Since the first spam e-mail was sent in 1978, the tide of spam e-mails has increased dramatically. Today, e-mail provider organisations report that as many as 85 to 90 per cent of all e-mails are spam. The main sources of spam e-mails in 2007 were: the United States (19.6 per cent of the recorded total); People’s Republic of

China (8.4 per cent); and the Republic of Korea (6.5 per cent). Most e-mail providers have reacted to rising levels of spam e-mails by installing anti-spam filter technology.

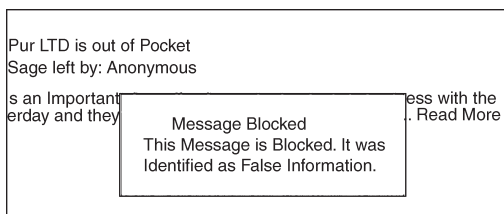


Fig. Internet Forums where Anybody can Leave Messages without Formally Registering are Popular Places to Leave Messages Containing False Information.

This technology identifies spam using keyword filters or black-lists of spammers' IP addresses. Although filter technology continues to develop, spammers find ways around these systems - for example, by avoiding keywords. Spammers have found many ways to describe "Viagra", one of the most popular products offered in spam, without using the brand-name.

Success in the detection of spam e-mails depends on changes in the way spam is distributed. Instead of sending messages from a single mail server (which is technically easier for e-mail providers to identify, due to the limited number of sources), many offenders use botnets to distribute unsolicited e-mails. By using botnets based on thousands of computer systems, each computer might send out only a few hundred e-mails. This makes it more difficult for e-mail providers to identify spam by analysing the information about senders and more difficult for law enforcement agencies to track offenders.

Spam e-mails are highly profitable as the cost of sending out billions of e-mails is low – and even lower, where botnets are involved. Some experts suggest the only real solution in the fight against spam is to raise transmission costs for senders. A report published in 2007 analysed the costs and profits of spam e-mails. Based on the results of the analysis, the cost of sending out 20 million e-mails is around USD 500. Since costs for offenders are low, sending spam is highly profitable, especially if offenders are able to send billions of e-mails. A Dutch spammer reported a profit of around USD 50,000 by sending out at least 9 billion spam emails. In 2005, the OECD published a report analysing the impact of spam on developing countries. Developing countries often express the view that Internet users in their countries suffer more from the impact of spam and Internet abuse. Spam is a serious issue in developing countries, where bandwidth and Internet access are scarcer and more expensive than in industrialised countries. Spam consumes valuable time and resources in countries where Internet resources are rarer and more costly.

OTHER FORMS OF ILLEGAL CONTENT

The Internet is not only used for direct attacks, but also as a forum for:

- Soliciting, offers and incitement to commit crimes;

- Unlawful sale of products; and
- Provision of information and instructions for illegal acts (e.g. how to build explosives).

Many countries have put in place regulations on the trade of certain products. Different countries apply different national regulations and trade restrictions to various products such as military equipment. A similar situation exists for medicines - medicines which are available without restriction in some countries may need prescription in others. Cross-border trade may make it difficult to ensure that access to certain products is restricted within a territory. Given the popularity of the Internet, this problem has grown.

Web-shops operating in countries with no restrictions can sell products to customers in other countries with restrictions, undermining these limitations. Prior to the Internet, it was difficult for most people to access instructions on how to build weapons. The necessary information was available (*e.g.*, in books dealing with chemical aspects of explosives), but timeconsuming to find. Today, information on how to build explosives is available over the Internet and ease of access to information increases the likelihood of attacks.

THE FUTURE OF CYBER CRIME ACTIVITY

During an eight day period in August of 2003, three separate worms cost the U.S., economy close to two billion dollars in lost production. Later that summer, the east coast of the United States experienced a massive blackout that might have been exacerbated by a worm called Blaster. Robert Cringley believed that same year that “the cost to society of identity theft is in the range of \$4-5 billion per year and may be even higher.” With the convergence of organised crime and technically savvy cyber criminals, the growing fear of cyberterrorism, the inability to understand security from major software vendors and the slow footedness of the government, the future seems to point to an Internet that is not only vulnerable to major attacks, but also infested with scams, thieves, and ever increasing opportunities for exploitation.

Although inherently difficult to predict trends within the field of technology, there are some eventualities that seem certain, like the growth of extortion and fraud through the Internet. These difficulties will of course result in new responses that better protect customers and force criminals into new realms. Although it is a hackneyed expression, the Internet is still in the Wild West stages. It is important to remember, though, that the Wild West eventually turned into suburbia after successful methods were found to control and police criminals, making their life extremely difficult. An examination of some of the potentialities will help to illustrate how the Internet and cyber crime can be envisioned, at least in the near term future.

FUTURE TRENDS IN CYBER CRIME ACTIVITY

The pace at which cyber crime is growing is one of the most disturbing trends. Valerie McNiven, a U.S., Treasury Advisor, has proclaimed “last year was the first year

that proceeds from cyber crime were greater than proceeds from the sale of illegal drugs, and that was, I believe, over \$105 billion.” She further added that “cyber crime is moving at such a high speed that law enforcement cannot catch up with it.” It seems clear that the issue will only become worse in the next few years, now that professionals have realised the potential windfalls if exploited properly. Recently, there has been significant discussion over the amalgamation of organised criminals and cyber crime. Such a pairing indeed forebodes an ill omen for the near term future. With most of the criminal groups operating out of eastern Europe, Russia and Asia, where laws and enforcement are scanty, there seems little hope in containing and neutralising the threat through traditional means. Phil Williams, a visiting scientist at CERT, summarised the issue succinctly. “The Internet provides both channels and targets for crime and enables them to be exploited for considerable gain with a very low level of risk.

For organised crime it is difficult to ask for more.” The result that can then be expected will be an increase in sophisticated phishing attacks and other means for identity theft that may be two pronged. For example, using call centers to notify “customers” ahead of time of some issue, and then following up with e-mails that request personal information. These types of social engineering attacks can be very damaging and have had a long history of success within cyber crime and fraud related transgressions.

The Internet makes it even easier since, a degree of authority can be imparted with official looking e-mails and fraudulent web sites that mimic their legitimate counterparts exactly. In the past, such social engineering relied on interpersonal skills like gaining trust. Now, the same idea has been transferred to technology, which can be done with greater ease by a wider group of people. Another aspect of fraud will result from the aggregation of personal information in many third party data centers, making them valuable targets to infiltrate. It is not hard to imagine criminals using data mining techniques to find the most gullible consumers, or tailoring phishing e-mails for specific people based on their medical, financial or personal history. Identity theft will also move in more automated directions.

For example, botnets will become vehicles not just for denial of service attacks and spam, but also as giant search platforms for finding personal information, like credit cards and social security numbers. Controllers of the botnets will then receive payment to run queries on their “database.” With professional criminals managing the money laundering and organisation of such schemes, it begs to ask where will all the technical know-how come from in order to perform cyber crime? Unfortunately, there are growing numbers of intelligent black-hats with university degrees spread around the globe, many of them operating in countries where legal employment does not pay as well and the chances of being caught are slim. But more troublesome is that it has become easier than ever before to be a hacker capable of inflicting great harm on networks and committing cyber crime.

The Internet has created a repository of knowledge where anyone is able to learn the fundamentals of subverting computer systems, with numerous tutorials available that

spell out in nearly layman's terms how to perform a buffer overflow or a man in the middle attack. Interestingly, the greatest problem is not those who will take the time to learn and find new exploits. In fact this group will probably remain a small, highly intelligent network of researchers and security groups focused solely on finding holes in software. In this, it is preordained, that even if someone is motivated to learn how exploits work, finding a new exploit takes a degree of investigation, skill and diligence that most are not willing to invest.

The real threat comes from the profound ease at which anyone can run a programme like "MetaSploit," a framework for running exploits against targets that allows new modules to be imported and run automatically. The attacker literally needs to know nothing about how computers work, besides how to operate one. In fact, for almost all attacks, the hard work is done by a small group of people, and then released into the public domain, allowing almost anyone to just run the attack. Botnets are no longer handcrafted software made by one group who truly understood the fundamentals, but instead are opensource collaborative efforts that aim to make it as easy as possible to control remote computers, such as BotNET, eggheads and CSharpBot, all available from SourceForge.

Thus, the barrier to entry to the field is so low that it allows almost anyone to experiment and join the swelling ranks of cyber criminals. With the learning curve so low, it should prompt discussion on the need for a new paradigm of thought in how to pre-empt and deal with criminals, in a way that is no longer tied to traditional methods. For example, for someone to break into a house, not only do they need to plan the opportune moment, but they may also have to be aware of lock picking, security system evasion and possess a degree of gumption to overcome moral thresholds.

In opposition, the ease of cyber crime seems inversely proportional to the lucrativeness that it bestows and moreover, these trends show signs of accelerating. Beyond the "who" and the "why" of future cyberattacks, the "how" will also change as operating systems become more secure and harder to exploit. With the damage that comes with each new security hole released, Microsoft, Apple and open source vendors have finally begun to seriously focus on security. Of course that hasn't stemmed the flow of vulnerabilities discovered, but techniques such as address space randomisation to stop buffer overflows, advanced and automatic code reviews and more training will reduce the ability to compromise a machine through operating system protocols over time.

The real danger in the future lies with user applications, which are created by individuals or small groups without the knowledge or training required to implement security correctly. Especially dangerous are web applications that can be installed on web servers. The traditional problem with these types of security vulnerabilities was finding susceptible hosts. But with Google, a programme can automatically search for sites with a specific version of a programme installed and then launch an attack.

If remote code can be executed, the programme may not be able to take over the whole system, but it can run programmes as the user which may be enough to install

bots and automatically replicate. The first instance of such an attack occurred in December of 2004 with the Santy worm that attacked a popular bulletin board system by searching through Google to find hosts with a specific file that was vulnerable. Such attacks aim their weapons at the least secure and vetted of software created. Although the installed base of such systems may be small, with “Google Hacking,” they can still be quickly located and exploited. Another avenue of attack that will open up will be through embedded systems, such as cell phones, mobile devices and other electronics that may connect to the Internet for the most mundane of purposes. Software is usually recreated for each iteration of a device as it is specifically designed for the hardware.

This allows for security problems to creep back in over time that may have been eliminated before. As these devices start to allow consumers to make purchases, while storing valuable information, they will become more attractive for criminals. The incentives now to do so are low, so security researchers have only seen “proof-of-concept” viruses, like the one that infected cellphones running a version of the Symbian OS that could spread automatically. It appears organised crime has not moved into this area due to the lack of research and understanding of how such attacks can be made profitable. Much as with the Internet, it will take time to exploit successfully.

In the same vein, eventually automobiles, home electronics devices, refrigerators and almost all devices can and will use the Internet in order to perform maintenance, download upgrades or monitor performance. These will present opportunities for maliciousness and blackmail that doesn’t equate in the same way as the purely virtual environment of the Internet. If someone enters their car, controlled by a foreign agent that demands a wire transfer or else the car will be crashed at high speeds, a situation arises that people will no longer accept. Whether such a scenario will occur is debatable, but the possibility will certainly exist.

MITIGATING CYBER CRIME ACTIVITY

Although it is inevitable that cyber crime will increase and continue to explore new vectors for undermining privacy, authentication and law enforcement, there will also be valid and useful attempts for mitigating the abilities of criminals, as well as the effects of cyber crime. These solutions will take form in better software, anti-spyware and anti-virus software integrated into operating systems and more user education regarding phishing and identify theft. These solutions will come primarily from software vendors themselves. On the other side, legislators will work with banks to reduce and prevent fraud, putting some of the liability with those most able to prevent it.

Finally, advanced solutions coming out of research and academia will try to inhibit the inherently anonymous and insecure nature of the Internet. With Microsoft’s upcoming release of Vista, the latest version of their operating system, they’ll have a new chance to focus on not only improving the general security of the system through fundamental changes, but also in providing methods for eliminating common problems, such as botnets, spyware and phishing attacks. In October of 2005, Microsoft began working

together with the FTC to educate customers about botnets and the danger of allowing a computer to turn into a zombie. To deal with the problem of phishing, Microsoft released a programme in July of 2005 called the “Microsoft Phishing Filter,” which aims to invalidate the ability of phishers to reach Microsoft customers by dynamically notifying them when there is a high chance that what is being viewed is a phishing attack.

Finally, Microsoft released their “AntiSpyware” programme in January of 2005, to be included with Vista as well, that automatically scans your computer for programmes that match spyware signatures or that try to perform suspicious actions, like modifying system functionality or trying to run upon computer start up. If cyber crime continues to grow to epidemic proportions, as all indications seem to point to, legislation will invariably step in, but more importantly, those with the most to lose will become more involved. This includes credit card companies, banks, lending operations and other organisations dealing with monetary transactions. Paypal.com has quickly come to dominate the online payment industry, while also serving as a bank in many capacities. With only an e-mail address and a password required to send money, this low hanging fruit has been one of the most heavily exploited realms for phishing attacks. In response, Paypal has offered at least a thousand dollars of purchase protection and a supposed one hundred per cent protection against unauthorised payments sent from an account.

A fraud investigation team responds to queries and according to their web site, they have software that automatically monitors every transaction for inconsistencies. This last measure used by Paypal has also become fertile ground for credit cards companies, as their systems have become powerful at identifying fraudulent purchases though the use of neural networks, a type of software emerging out of the field of artificial intelligence. In some cases, this software has been able to reduce fraud by thirty per cent or more. It’s important to remember that the systems are not perfect solutions, but do address a large portion of illegal activity. Combined with other efforts, the goal is to reduce the effect of fraud, while making it more difficult to achieve.

Legislation will attempt to do its part as well, even though it has moved notoriously slowly when dealing with cyberthreats. The past few years have seen laws specifically crafted for spam and dealing with attacks that threaten the integrity of the infrastructure of the Internet. If the botnet problem continues to grow, coupled with identify theft, surely more action will be taken. Although, it is still unclear how effective it will be without a significant contribution to cyberforensic development and funding for the various governmental enforcement agencies responsible for handling cyber crime matters.

Another issue discussed in the Legal Policies section is the need for more international cooperation in locating, extraditing and prosecuting foreign criminals when possible, as the current system leaves much to be desired. Finally, as with any dangerous and difficult problem, there will be new and inventive ways to handle security issues coming out of research. One contribution that has limited, but not eliminated many common security flaws that are exploited, is the use of randomisation in dealing with code, data

and other programmatic necessities. By introducing a factor of unpredictability, it can make the work of a hacker much more difficult and prone to error, limiting the ability of those who do not possess the skill to effect a novel attack. Other interesting proposals have included trace back systems that can remove the anonymous identity of data traveling through the Internet, devising a system for fast and accurate discovery of the source of even one packet of data.

Stopping distributed denial of service attacks and worm discovery has also been proposed as a method that can be automated and integrated into the backbone of the Internet, high speed routers. By analysing similar patterns coming from separate locations, such detectors can realise an attack while it is in its infancy and isolate infected hosts. There is also still room for ISPs to actively monitor and discourage botnets, spam and DDoS attacks from occurring. As the first link in the chain for many zombie hosts, as well as attackers, they are in a prime position for stopping spam, either by blocking outgoing mail, which most users have no need for, or by identifying when one host is sending out a large amount of data that does not match expected behaviour. Additionally, if they noticed that a number of hosts were acting in concert, with regards to the data being disseminated from those machines, they may assume with likelihood that they are being controlled remotely. Consequently, the ISPs can examine logs to find who is sending the commands and initiate a complaint with the F.B.I. The problem holding back this kind of proactive approach has not been technical in nature, but rather legalistic, as it can be considered an invasion of privacy. Furthermore, such methods are being used to track down minor copyright violations, instead of focusing on more substantial problems, such as cyber crime and identify theft.

LOOKING AHEAD

The future of the Internet is still up for grabs between criminals and normal users. Fears of a cyber-apocalypse still abound, while the potential extent of damage that can be caused by wide scale fraud is nearly unbounded. These anxieties should be rationally tempered with the knowledge that the problems are being addressed, although perhaps not fast enough. The usefulness of the Internet has proved itself in numerous and myriad ways that will hopefully be enough to ensure it does not become a wasteland of criminal activity and a bastion for the malicious. The government still has an important role to play, but most of the prevention needs to be done by commercial entities producing software and those with the ability to stop fraud. Relying on consumer education programmes will only affect a percentage of possible victims. The others need to be automatically protected through measures that do not stress and require considerable participation. Security needs to be easy and effective if it is to work. Whether cyber crime is still a pertinent issue ten years from now is unknowable in a sense, but if the Internet will continue to grow, it must be solved so that the realities of cyber crime will be proportional to real-world crimes, if not better.

Intellectual Property and Computer Crime

Although existing criminal laws will continue to provide effective weapons to combat cyber crimes committed upon the traditional service industries that offer their services and products through e-commerce, Congress and the global community must move swiftly to provide an effective array of substantive laws designed specifically for Internet transactions. That array should include laws and treaties to provide law enforcement and the private sector with the tools essential for the detection and investigation of cyber crime.

Some Congressional attention has focused on peripheral issues such as privacy and encryption, and there are signs that Congress is beginning to address Internet-specific issues. Five pending Congressional bills suggest that Congress is aware that law enforcement must be equipped with adequate resources and remedies to combat cyber crime. A High Tech Crime Bill was introduced on February 24, 2000 by senators Charles Schumer and John Kyl.

This bill proposes amendments to the computer fraud statute that would include fraudulent acts committed upon “protected computers” in the United States, by persons in foreign countries and would include damage done to computers and computer systems irrespective of property damage losses. That amendment would encompass damage caused by, for example, denial of service attacks. The proposed bill also would amend 18 U.S.C. § 3123 (c), enabling law enforcement to employ trap and trace devices to assist in the investigation of computer attacks. The Internet Security Act of 2000, introduced in April 2000 by Senator Leahy, would expand the jurisdictional scope of

the computer fraud statute to include international hacking. It would amend section 3123 to enable law enforcement to employ trap and trace devices, impose encryption standards, and authorise the prosecution of juvenile hackers. It would also appropriate \$25 million for each of the next four fiscal years for law enforcement training Programmes in computer fraud investigations. Both bills seek to react to recent attacks upon e-Commerce sites, including the denial of service attacks.

The Internet Integrity and Critical Infrastructure Protection Act of 2000, introduced on April 13, 2000 by Senator Hatch, co-sponsored by Senator Schumer, would impose criminal penalties for cyber hacking committed by persons under 18 years old, create a National Cyber Crime Technical Support Center to serve as a resource center for federal, state and local law enforcement and assist them in the investigation of computer-related crimes, and provide the implementation of computer crime mutual assistance agreements in order to enable reciprocal assistance for foreign authorities.

Also on April 13, 2000, Senator Hutchinson introduced S. 2451 that would increase the criminal penalties for computer fraud committed in violation of 18 U.S.C. § 1030 and establish a National Commission on Cyber security to study incidents of computer crimes and the need for enhanced methods of combating such crimes. Finally, on May 9, 2000, Congressman Boehlert introduced the Law Enforcement Science and Technology Act of 2000, co-sponsored by Congressman Stupak. This bill would establish an Office of Science and Technology in the Office of Justice Programmes of the Department of Justice.

The mission of that office would be:

- To serve as the national focal point for work on law enforcement technology; and
- To carry out Programmes to improve the safety and effectiveness of, and access to, technology to assist Federal, State and local law enforcement agencies.

The bill would direct the appropriation of \$40 million for regional National Law Enforcement and Corrections Technology Centers, \$60 million for research and development of forensic technologies and methods to improve crime laboratories, and \$20 million for the testing and evaluation of technologies. Additional governmental attention must be focused on extradition and mutual assistance treaties that will enable the United States to prosecute cyber crimes committed by international terrorists and hackers.

TYPES OF ECONOMIC CRIMES AND DEVELOPMENT OF STATUTORY LAW

There seems to be no limit to the types of economic crimes and the methods of committing them. However, certain crimes are unique to certain industries. For example, cloning applies to the wireless telecommunication industry and currency transaction reporting applies primarily to the banking and financial services industry. The discussion

that follows relates to specific economic crimes in nine different areas and the laws enacted by Congress that proscribe the illegal conduct.

MAIL AND WIRE FRAUD: GENERIC FRAUDS AND SWINDLES

The mail fraud statute was enacted in 1872. The law was designed specifically to enable special agents of the U.S. Mail to investigate frauds and swindles perpetrated through the use of the mail system and to seek federal prosecution for such offences. Because it applies to “any scheme or artifice to defraud, or for obtaining money or property by false pretenses, representations or promises,” the statute has been used to prosecute fraudulent insurance claims, fraudulent loan applications, securities frauds, and an unlimited variety of frauds and swindles.

It frequently is applied to new types of fraudulent acts in those situations where Congress has not had the opportunity to enact a specific statute to deal with the crime. Because mail fraud has generic appeal, it applies to conventional economic crimes in the eight areas on which this report focuses. The crime of mail fraud is committed by depositing or receiving matter with or from the Postal Service or a private interstate carrier or causing matter to be delivered by the Postal Service or such carrier for the purpose of furthering the fraudulent scheme.

There is no monetary threshold necessary for prosecution, although federal prosecutors informally may decline to handle minor cases. The wire fraud statute is patterned after the mail fraud statute. It proscribes the use of interstate communications by wire, radio or television to perpetrate a scheme or artifice to defraud. The statute is generic in application and applies to a wide range of criminal activity across industries.

Courts have not limited the reach of the statute to land line telephone communications; the statute has been applied to wireless communications using microwave technology, in part. Because transactions over the Internet usually involve interstate communications by telephone wire or cable, the wire fraud statute will continue to serve as an effective law enforcement and prosecutorial tool to combat cyber crimes against all major industries.

Banking Industry: Financial Institution Crimes

Like mail and wire fraud, the bank fraud statute applies generically to any scheme to defraud a financial institution or any fraudulent act designed to obtain money or property from a financial institution. The statute currently defines a financial institution as any depository institution insured by the FDIC, as well as credit unions and the Federal Reserve Bank. Unlike mail or wire fraud, which are limited to the medium of communication, the bank fraud statute applies to any fraudulent scheme designed to obtain money or property from the financial institution, including check forging, check kiting, stolen checks, credit card fraud, fraudulent loan applications, student loan fraud, and embezzlement. The financial statement statute applies to any fraudulent statement made to a financial institution for the purpose of obtaining money or property in the

custody or control of the institution. The quintessential financial statement fraud is a false statement made in an application for a loan or to obtain credit from a financial institution.

The Continuing Financial Crimes Enterprise (CFCE) statute was enacted in 1990 and is designed to impose criminal sanctions for large-scale frauds committed upon financial institutions, such as the massive frauds upon the savings and loan industry. A CFCE is a series of violations of numerous sections of title 18 of the U.S. Code, as well as the crimes of mail and wire fraud, if they affect a financial institution, committed by one who organises, manages or supervises the enterprise, and receives \$5 million in gross receipts from the enterprise over a 24-month period. It is limited to financial institutions as victims, but would include a series of credit card frauds, if those crimes were charged under 18 U.S.C. § 1341, 1343 or 1344.

The computer fraud statute currently prohibits accessing a “protected computer” without authority, or in excess of authority, to obtain information or to obtain something of value, hacking into a protected computer and transmitting a Programme, information, code, or command with the intent to damage the computer, or transmitting in interstate commerce any threat to cause damage to a protected computer in order to extort something of value. Because a “protected computer” includes a computer “exclusively for the use of a financial institution” or if not exclusively used, a computer “used by or for a financial institution and the conduct constituting the Offence affects that use by or for the financial institution”, the statute applies to specific criminal conduct directed at bank computers or computerised data storage facilities. The money laundering statutes were enacted in 1986 as part of the Money Laundering Control Act. These sections apply to the conduct of the customer of a financial institution who deposits the proceeds of criminal activity with the bank and uses the bank to layer or launder those proceeds and to facilitate the transportation of the proceeds into or out of the country. Banks providing online services and electronic wire transfers are particularly susceptible to such conduct.

In 1970, Congress enacted the Bank Secrecy Act, also known as the Currency and Foreign Transaction Reporting Act, to curb the laundering of cash through banking institutions. That Act requires the filing of currency transaction reports (CTR's) for any deposit or withdrawal of cash exceeding \$10,000. Subsequent amendments to the Act require the filing of reports for the transportation of currency or monetary instruments exceeding \$10,000 into or out of the U.S., cash transactions exceeding \$10,000 at casinos or as part of business transactions, and for suspicious activity transactions. The Treasury Department has promulgated several regulations providing the details of the reporting requirements, which apply to both electronic fund transfers and online banking transactions.

The Racketeering Influenced and Corrupt Organizations (RICO) statute has application to mail fraud, wire fraud, bank fraud, currency transaction reporting violations and money laundering because they are predicated acts constituting racketeering activity.

RICO provides an effective law enforcement weapon against those who engage in a pattern of racketeering activity and victimise a financial institution, or those financial institutions that engage in a pattern of racketeering activity.

Credit Card Crimes

In addition to mail and wire fraud, Congress has enacted two laws that specifically address the traditional means of committing crimes involving a credit card. The credit card fraud statute prohibits the sale, use or transportation of a counterfeit, altered, stolen, lost or fraudulently obtained credit card in a transaction affecting or using interstate or foreign commerce, or furnishing money obtained through the use of a counterfeit, altered, stolen, lost or fraudulently obtained card, or the receipt or concealment of goods or tickets for interstate or foreign transportation obtained through the use of such a card.

Federal courts disagree whether this crime can be committed without a plastic card. Because online transactions involve the use of the credit card number and not the card itself, it is unclear whether section 1644 applies to online transactions. The access device statute, however, defines an “access device” to include a card or an account number. This statute prohibits the production, use or trafficking in counterfeit credit cards or account numbers, the possession of 15 or more counterfeit cards or account numbers, producing, trafficking in or possessing equipment used to produce counterfeit cards, without authority from the card issuer, offering a card or selling information regarding applications to obtain cards, or attempts or conspiracies to commit any of the acts regarding credit cards or account numbers.

Health Care Fraud

Prior to 1996, federal prosecutors employed generic crimes such as the false statements statute, false claims statute, mail fraud statute or wire fraud statute to prosecute those charged with engaging in conduct encompassed by the term “health care fraud.” In some instances, prosecutors were able to use laws created to address specific methods of committing health care fraud relating to the Medicare and Medicaid Programmes, such as the false claims statute, the anti-kickback statute, and the self-referral statute. In 1996, Congress enacted the Health Insurance Portability and Accountability Act (HIPAA), which created five distinct health care fraud crimes. This enactment was significant in that it imposes criminal penalties for health care frauds perpetrated upon private health care benefit plans, as well as Medicare and Medicaid. “Health care fraud” under HIPAA is committed by executing or attempting to execute a scheme or artifice to defraud any health care benefit Programme or fraudulently obtaining the money or property of a health care benefit Programme.

HIPAA also proscribes the theft or embezzlement of funds, securities, premiums, property and other assets of a health care benefit Programme, concealment of a material fact or fraudulent statements in connection with the delivery or payment of health care

benefits or services, and the obstruction of or interference with the communication of records relating to a violation of HIPAA to a criminal investigator. Additionally, a conspiracy to violate any of those offences also is a crime. Federal financial statement fraud and false claims fraud have limited application to the health care industry because the victim in those statutes is the government. The enactment of HIPAA significantly enhances the prosecutorial arsenal available to combat health care fraud committed by traditional means because the victim of those crimes are health care benefit Programmes, which include private health insurance plans. The Act, however, does not address a myriad of other cyber crimes that are health care related, for example, pharmaceutical fraud, web sites that fraudulently purport to provide medical advice, and phony web sites. The only statutory tool currently available to prosecute such conduct effectively is the wire fraud statute.

Insurance Fraud

Most fraudulent conduct impacting upon the insurance industry constitutes application fraud or claims fraud. These frauds may be committed by the insured, an agent or employee of the company, a third party, or as is more often the case, a conspiracy between two or more of those groups. Historically, insurance fraud has been the subject of state regulation. Nevertheless, federal jurisdiction and laws may be invoked where the fraudulent activity constitutes a mail or wire fraud, which is frequently the case. Applications for insurance are commonly sent through the mail or processed online over the Internet. Proofs of loss, bills, invoices and receipts submitted in support of an insurance claim typically are transmitted to the insurer by mail or by fax transmission. Further, a continuing pattern of such activity may constitute a RICO crime. Federal law also regulates the conduct of persons or entities engaged in the business of insurance.

It imposes criminal sanctions for:

- The embezzlement or misappropriation of premiums, money or other property of insurance companies,
- The making of false statements or reports to an insurance regulatory agency or official involving the overvaluing of land, property or security for the purpose of influencing action by that agency,
- Making false entries in any book, report or statement of the insurance company to deceive the company or an insurance agency or official regarding the financial solvency of the business,
- Engaging in the insurance business by any person convicted of a felony involving dishonesty or a breach of trust, and
- Threatening, influencing or obstructing the administration of law in any pending proceeding before an insurance regulatory agency or official.

There is no published decision to date regarding prosecutions under this section. Section 1034 authorises the U.S. Attorney General to seek civil penalties or injunctions for violations of section 1033.

Securities Fraud

Congress created numerous securities fraud crimes by enactment of the Securities and Exchange Act of 1934. The major anti-fraud statute is section 10(b) of the Act, which is codified at 15 U.S.C. § 78j (b). That section and Rule 10b-5, which was promulgated by the Securities Exchange Commission, prohibit the use of any instrumentality of interstate commerce or stock exchange or mail for the commission of a fraudulent act, scheme or artifice to defraud, or the making of a false statement in connection with the purchase or sale of a security. Although that statute and rule have served as an effective prosecutorial weapon in combating securities fraud and insider trading, the statute was not drawn with the Internet and e-trading in mind.

Telecommunication Fraud

Fraud impacts all sectors of the communications industry—telephone, wireless, and cable. The types of fraud are numerous: toll fraud, including “clip-on” and “shoulder surfing” methods, call-sell operations established by organised crime groups who engage “phreaks” to hack into phone line, subscription fraud, which frequently also involves identity theft, PBX fraud, which also involves call-sell operations, calling card fraud, remote call forwarding, and computer terrorism/sabotage.

Because these frauds frequently encompass interstate wire communications, the wire fraud statute is an effective legal weapon to combat telecommunication fraud. The principal fraudulent activities impacting upon the wireless communication industry are subscription fraud and cloning. Subscription fraud occurs when an individual applies for and obtains a wireless telephone account.

Because a fraudulent application customarily involves the use of a counterfeit or stolen credit card or account number, or a stolen means of identification, the credit card fraud, access device fraud, and identity fraud statutes are available for the prosecution of such Behaviour. Most wireless service providers currently encourage customers to apply by accessing their web site.

The wire fraud statute serves as an additional prosecutorial weapon to combat online subscription fraud. Cloning entails the theft of an electronic serial number (ESN) and mobile identification number (MIN) assigned to a legitimate wireless phone, and the installation of those numbers on a stolen phone. Once accomplished, the cloned stolen phone behaves like the phone owned by the legitimate customer, and the monthly bill for charges are sent to the legitimate customer.

The access device statute specifically proscribes the theft of ESNs and MINs, the production or use of a cloned phone, and the possession, use, production or trafficking in scanning receivers or hardware or software utilised to clone phones for the purpose of obtaining unauthorised wireless services. Although cloning a phone and the use of a cloned phone historically has not involved the Internet, technology is now in place to enable wireless access to the Internet. The access device statute should continue to be an effective legal weapon to combat the use of cloned phones for Internet access.

Additionally, the computer fraud statute may also apply to such conduct. Congress initially sought to regulate abuses committed by the telemarketing industry by enactment of the Telephone Consumer Protection Act of 1991. That Act provides for statutory damages for each violation, but does not impose criminal sanctions. In 1994, Congress enacted the Senior Citizens Against Marketing Scams (SCAMS) Act. The SCAMS Act imposes criminal penalties for telemarketing frauds involving wire communications, including conspiracies to commit such crimes, and enhanced sentences where senior citizens are victimised. Because Internet scams encompass the use of wire communications in part, the wire fraud statute will remain an effective legal weapon for the prosecution of scammers.

VICTIMS OF ECONOMIC CRIME

CONSUMERS

Very few studies on fraud victimization have been conducted. Two that studied telemarketing are Harris and Associates and a study by the American Association of Retired Persons. The most comprehensive study to date is the National White Collar Crime Center's National Public Survey on White Collar Crime. The study's goal was, "to present a picture of what the average American thinks about white collar crime."

Its survey of 1,169 households throughout the United States found that:

- Over 1 out of 3 households had been victimised by white collar crime in the last year.
- Widely held opinions concerning the profile of typical white collar crime victims are divorced from the actual profile of victims found by recent research on victimization.
- There is a disparity between how Americans believe they will react if victimised and how they do react when they are actually victimised.
- Less than 1 in 10 victimizations were ever reported to law enforcement or consumer protection agencies.
- The public has a deep concern with increasing the apprehension and sanctioning of white collar criminals.

Consumer victimization usually results in three types of losses: privacy, good credit status, and funds or assets. Consumers are concerned that personal information disclosed to companies with which they do business may be compromised. Such compromises include unauthorised access and/or by the company's employees, lack of security to protect the information, providing the information to third parties, and the maintenance of accurate information. Any one of these breaches could result in the consumer's personal information falling into criminal hands, which could easily result in identity theft. Other consequences range from damage to an individual's credit rating to the loss of funds and/or assets.

INDUSTRY

The victimization of industry falls into four categories:

1. Profit losses
2. Damage to reputation
3. Loss of continuity of business
4. Loss of intellectual property.

According to The Credit Risk Management Report, “The average organization loses more than \$9 a day per employee to fraud and abuse. The average organization loses about 6 percent of its total annual revenue to fraud and abuse committed by its own employees. Fraud and abuse costs US organizations more than \$400 billion annually.” Early on, many corporations were able to take the position that fraud was a cost of doing business, and could make it up by passing the cost of fraud to the consumer through increased prices. In more competitive markets this is not possible. In those cases when the bottom line is hit hard by fraud, executives are less reluctant to commit funds to fraud management and computer security.

While big business can sustain a major loss to fraud, many small businesses have suffered severely and in some cases have gone out of business as a result of their fraud losses. This often occurs because these small organizations cannot afford sophisticated hardware and software to prevent and detect fraud. Because corporations are afraid that reporting fraud may damage their reputation, they are reluctant to do so.

They fear legal retaliation if they share or disclose too much, and are afraid that their consumers and stockholders will lose confidence in them. The actual amount of corporate victimization is not known, because of the unwillingness of corporations to report or admit that fraud has affected them. Many e-Businesses are concerned about the continuity of their business. That is, they do not want their services to customers to be disrupted. Although security remains a significant concern for business, consumers are paramount in e-Commerce; they want to shop quickly with no hassles. Recent distributed denial of service attacks on web sites such as e-bay and Amazon.com point to the vulnerabilities of e-Commerce. The lack of security and the intrusion of criminals (fraudulent element) both impede the growth of e-Commerce.

Intellectual property theft – in the form of trademark infringement, cyber squatters, typo squatters, trade-secret theft, and copyright infringement – has increased as Internet use and misuse has risen. It occurs across the seven industries detailed here, as well as most other businesses. “According to the American Society for Industrial Security, American businesses have been losing \$250 billion a year from intellectual property theft since the mid-1990’s.”

GOVERNMENT

Government suffers from several forms of victimization, much like corporations do, including theft of intellectual property, theft of assets, and loss of reputation. Several cases have been in the news where United States secrets have been compromised or

potentially compromised. These events have tarnished the reputation of several government agencies by pointing out the lack of, or loose, security procedures. Numerous federal governmental web sites have been defaced by hackers, including the CIA, FBI, and the United States Department of Justice. Several reports of intrusions have occurred with government computers. In many of these cases, systems have been penetrated, but no classified information was accessed. Fraud, waste, abuse, and mismanagement are generally reported together. While it is hard to get a handle on their size and scope, the Senate Governmental Affairs Committee, chaired by Senator Fred Thompson (R-Tennessee) reported on January 26, 2000, that "In 1998 alone, \$35 billion in taxpayer dollars was lost due to government waste, fraud, abuse, and mismanagement."

WAGING THE WAR ON ECONOMIC CRIME

LAW ENFORCEMENT

On the federal level, numerous regulatory and law enforcement agencies are authorised to combat specific economic crimes, including the Federal Bureau of Investigation (FBI), United States Secret Service (USSS), the United States Postal Inspection Service, Securities and Exchange Commission (SEC), and United States Customs.

Each of these agencies has jurisdiction over the following economic crimes/fraud.

- *FBI:* health care, financial institution, intellectual property, telemarketing, securities/commodities, bankruptcy, insurance, computer, and Internet
- *Secret Service:* credit card, cellular, and computer
- *Postal Inspector:* mail and consumer
- *Securities and Exchange Commission:* insider and online trading, stock manipulation, and fraudulent stock offerings
- *United States Customs:* money laundering, cyber crimes, including child pornography and the importing of dangerous substances

On the international level, Interpol recently announced its intention to become active in the investigation of international computer crimes. Interpol announced on June 30, 2000 that it is establishing an international intelligence network to inform the public and private sectors of impending cyber attacks and potential targets for malicious hacks.

The intelligence information will be relayed to Interpol by Atomic Tangerine, a venture consulting firm, using technology (Net Radar) developed by SRI International, the parent company of Atomic Tangerine. Local law enforcement capabilities for combating economic crime vary depending on the size and location of the department, and the allocation of resources. Some larger municipalities and state law enforcement agencies have formed economic and computer crime units. As resources, training, and awareness of the intensity of the problem increase, it is likely that more of these units will be formed.

NATIONAL FRAUD CENTER

The National Fraud Center (NFC) is an internationally recognised leader in global customised fraud and risk management solutions. Formed in 1982, its original mission was to combat insurance fraud, which, at that time, was becoming a societal concern. Since then the NFC has earned a reputation for combining expert knowledge and technology to produce solutions to fraud problems for vertical industries and the government. “Technological solutions developed with NFC’s expertise have saved clients tens of millions of dollars.” The NFC has an in-depth understanding of how economic crime affects businesses, consumers, and government agencies, as its researchers collect and analyse fraud data continuously.

ECONOMIC CRIME INVESTIGATION INSTITUTE

The Economic Crime Investigation Institute (ECII) was formed in 1988 by Dr. Gary R. Gordon, then Director of the Economic Crime Investigation Programme at Utica College of Syracuse University, and Mr. John Martin, Esq., then Chief of Internal Security for the United States Department of Justice. Part of the Institute’s mission is to support education and research in the areas of economic crime and computer security, by advising Utica College faculty on formal academic Programmes for pre-professional students and professionals in the field of economic crime investigation.

Its other goal is to develop the ECII into the premier educationally focused institute, providing a national forum that brings together all interested parties to develop solutions to economic crime problems faced by society. The ECII strives to provide a forum for individuals in government, corporate America, and higher education to discuss current economic crime issues and to promote the dissemination of information on economic crime and its investigations. The Economic Crime Investigation Institute’s annual conferences are its primary way of accomplishing this.

NATIONAL WHITE COLLAR CRIME CENTER

A non-profit organization that receives federal grant funding from the Bureau of Justice Assistance, Department of Justice, the National White Collar Crime Center (NW3C) “provides a nationwide support network for enforcement agencies involved in the prevention, investigation, and prosecution of economic and high-tech crime.” Founded in 1980, the Center’s focus is to support state and local agencies, using their needs as a guide for the projects and endeavors the NW3C undertakes.

In addition to its state-of-the-art financial and computer crime training course development and delivery, new projects include the development of the National Fraud Complaint Management Center (NFCMC) to leverage technology in the management of economic crime complaints and to bring added value to the prevention, investigation, and prosecution efforts surrounding complaints. A significant part of this project is the establishment of the Internet Fraud Complaint Center (IFCC) in partnership with the Federal Bureau of Investigation. The IFCC represents a unique approach to the growing

problem of fraud on the Internet. The NW3C has also been selected to serve as the Operations Center for the National Cyber crime Training Partnership (NCTP), an initiative of the United States Attorney General, headed by the Computer Crimes and Intellectual Property Section of the Justice Department.

THE NATIONAL COALITION FOR THE PREVENTION OF ECONOMIC CRIME

A non-profit organization established in 1996, the National Coalition for the Prevention of Economic Crime (NCPEC) provides support services to businesses in their fight against economic crime. Its mission is to reduce incidents of economic crime through cooperative, information-sharing efforts. Current training Programmes include instruction on fraud management, operational and strategic fraud management techniques, financial investigations practical skills, basic data recovery and analysis and instruction on how to use the Internet as an investigative tool. Hosted in partnership with the NW3C, the NCPEC has established an annual national conference entitled the Economic Crime Summit which brings together academics, government, private corporations, victims' interest groups, prevention specialists, and others to examine methodologies and share ideas to address economic crime on all levels.

INTERNET FRAUD COUNCIL

The Internet Fraud Council, a division of the National Coalition for the Prevention of Economic Crime, is composed of organizations from around the world that are interested in the prevention, investigation, and prosecution of Internet fraud. The Internet Fraud Council's mission is to provide research, education, best practices, and tools for the prevention of economic crime committed using the Internet.

INTERNET FRAUD COMPLAINT CENTER

The Internet Fraud Complaint Center (IFCC) is a partnership between the Federal Bureau of Investigation (FBI) and the National White Collar Crime Center (NW3C). IFCC's mission is to address fraud committed over the Internet. For victims of Internet fraud, IFCC provides a convenient and easy way to alert authorities of a suspected criminal or civil violation. For law enforcement and regulatory agencies, IFCC offers a central repository for complaints related to Internet fraud. The data from this source can then be analysed to identify and quantify fraud patterns, as well as statistics on the current fraud trends.

The IFCC's ultimate goal is to reduce Internet fraud victimization. As stated on its web site, "Long term benefits of this Programme will be substantial. Not only will its efforts reduce the amount of economic loss by Internet fraud throughout the United States, it will enable state and local law enforcement professionals to develop and successfully prosecute criminal Internet fraud cases. IFCC will also serve as the catalyst

that allows law enforcement and regulatory authorities to network and share fraud data.”

INDEPENDENT CORPORATIONS AND PRIVATE SECTOR INDUSTRY COALITIONS

As a result of limited law enforcement resources, corporations on their own or in cooperation with industry coalitions, such as BITS, the technology group for the Financial Services Roundtable, have had to initiate strategic economic crime management plans and investigative groups. There is a growing level of frustration among these corporations, because the monetary thresholds for law enforcement even to investigate a case, let alone prosecute, can be very high, depending on the jurisdiction. Coupled with this, is increased legislation requiring corporations to institute anti-fraud Programmes and compliance departments. While the protection of corporate assets and their consumers should be their responsibility, there are several consequences to this arrangement. Many economic crimes go unreported, fewer prosecutions of these offences occur, and perpetrators tend to be fired rather than prosecuted, leaving them free to move on to another organization and continue their victimization.

FUTURE NEEDS AND CHALLENGES

LAW ENFORCEMENT TRAINING

Specialised training in the areas of economic and computer crime, and how they affect specific industries, as well as computer forensics, needs to continue to increase for law enforcement personnel. Without an understanding of how specific industries function, it is difficult to investigate or prosecute economic crimes. New career paths within law enforcement organizations could be established before promotions and reassignments drain agencies of their limited skilled personnel in technically sophisticated areas. Often, individuals develop expertise and then are promoted or reassigned, making it necessary to train new people from ground zero. Unless the individuals who have expertise, experience, and contacts in industry are given an incentive to stay in their units, this cycle will continue and the investigation and prosecution of economic crime will not increase or improve.

LAWS, REGULATIONS AND REPORTING SYSTEMS

In the United States, government (federal, state and local), with limited exceptions, has allowed self-regulation of the Internet. Government regulation has, for the most part, focused on cyber crimes that are not economic crimes, such as child pornography and cyber stalking. Fortunately, that attitude appears to be changing. There are numerous bills pending in Congress that address criminal frauds committed on the Internet, identity theft, and issues involving Internet security and attacks upon web sites. This legislation should use language that will be easily adaptable to future technological changes to

help deter future economic crimes. However, there are other gaps in legislation. There are many regulations that require businesses to protect themselves by working to prevent fraud (*i.e.*, know your customer).

However, the government sends conflicting signals when it will not assist in prevention efforts by cleaning up regulations and enacting new supporting laws, as well as providing prosecutorial support. Current government regulations covering certain industries prohibit companies from sharing information with each other. This eliminates the possibility of an instrument, such as a central database of fraud, which companies could use in their procedures for preventing and detecting fraud. It is important that legislation addressing this be written and passed. Other legislation is also necessary, such as laws that keep pace with the changing nature of credit card payment and online payment systems.

PUBLIC-PRIVATE PARTNERSHIPS

No one group will be able to solve the complex problem posed by economic crime. Coalitions of private and public groups need to work together to combat economic and cyber crime. As more of these alliances develop, there will be more resources available to reverse the trend of economic crime. College and universities need to revamp their existing Programmes, *e.g.*, criminal justice, accounting, computer science, or create new ones to meet the changing needs of society in this area. At this time there are only two undergraduate Programmes and one graduate Programme addressing these issues – the Economic Crime Investigations Programmes at Utica College (Utica, NY) and Hilbert College (Hamburg, NY) and the Economic Crime Management Master's Programme at Utica College. These Programmes are supported by advisory boards consisting of individuals at the top of their fields from the credit card, banking, insurance, and telecommunications industries, as well as representatives from government agencies, such as the U.S. Secret Service and the FBI. Further Congress, through the Identity Theft Assumption and Deterrence Act requires the FTC and industry to work together. Presidential Directive 63 required industry and government to work together in combating Internet/e-commerce fraud.

BALANCING PRIVACY INTERESTS

The growth of e-commerce and the creation of new law enforcement techniques to combat cyber crime raise critical issues concerning consumer, business and governmental privacy. The protection of individual privacy, while considered almost sacred, in the world of economic and cyber crime can actually work to the criminal's advantage. The new FBI tool, Carnivore, is an attempt to gather intelligence information, without compromising privacy.

According to the FBI's web site,

- In recent years, the FBI has encountered an increasing number of criminal investigations in which the criminal subjects use the Internet to communicate

with each other or to communicate with their victims. Because many Internet service providers (ISP) lack the ability to discriminate communications to identify a particular subject's messages to the exclusion of all others, the FBI designed a diagnostic tool, called Carnivore.

- The Carnivore device provides the FBI with a “surgical” ability to intercept and collect the communications which are subject of the lawful order while ignoring those communications which they are not authorised to intercept.
- This is a matter of employing new technology to lawfully obtain important information, while providing enhanced privacy protection.

Carnivore provides law enforcement with the ability to keep pace with the technical advances in communication. However, this tool raises “Big Brother” concerns for the public. The controversy that Carnivore has evoked in its infancy points to the issue of trust that government, industry, and society as a whole need to resolve. BITS, Financial Services Roundtable adopted privacy principles in late 1997 that are guidelines for banking industry self-regulation concerning privacy. Industry, in general, sees self-regulation as preferable to government rule.

The BITS policy includes guidelines in each of these areas.

- Recognition of a customer's expectation of privacy
- Use, collection, and retention of customer Information
- Maintenance of accurate information
- Limiting employee access to information
- Protection of information via established security procedures
- Restrictions on the disclosure of account information
- Maintaining customer privacy in business relationships with third parties
- Disclosure of privacy principles to customers

Several other industry organizations are developing similar guidelines. Their aim is to have self-regulation rather than government intervention. By informing customers of privacy policies, industry is attempting to engender their trust. There is a delicate balance between protecting one's privacy, legitimate business use of personal data, and fraud prevention. However, the use of personal data for fraud prevention and interdiction is beneficial to society. Therefore, fraud and risk management exceptions should be built into any and all laws, regulations, and policies. In fact, the use of personal data for identity theft prevention directly reduces the number of identity theft victims. Further, many industries (*i.e.* insurance, banking and securities) require fraud prevention through regulation to protect consumers, customers, shareholders, and employees.

Effective authentication in an e-commerce transaction is not possible without the use of independent, personal verification data. Authentication is critical to the growth and confidence of e-commerce. Fraudsters have quickly learned to defeat our technical systems. If they are allowed to opt out of databases, they will rapidly exploit our vulnerabilities to the financial detriment of the general public. Global cooperation is

also needed in this area. The United States must take a leadership role in fostering cooperation throughout the global community in the development of uniform laws, meaningful and comparable privacy policies, effective assistance to prosecutions by foreign countries, and a sharing of information.

The U.S. already has surrendered a leadership role in the areas of privacy and information sharing. The European Union developed a comprehensive privacy directive applicable to all member nations in 1995, and the European Parliament recently refused to allow its member nations to share data and nonpublic information with the U.S. With respect to information sharing, Interpol has announced its intention to provide private industries throughout the world with intelligence information regarding the vulnerability of those industries or specific companies to cyber attacks. At this point, the global community perceives the U.S. as a reluctant partner, not a leader.

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Index

A

Abandonment 61
Abundance 49
Accessed 4, 5, 6, 174, 216, 270, 286, 290, 292, 302, 309, 313
Accountability 179, 230, 304
Accuracy 99, 186, 249
Acquiescence 47
Act 216, 217, 218, 219, 220
Adequate 7, 45, 113, 122, 128, 134, 135, 146, 159, 227, 300
Adherence 35
Administration 23, 36, 125, 127, 194, 214, 305
Affidavit 218
Allegiance 93
Alliances 36, 42, 46, 49, 167, 245, 254, 258, 259, 315
Ambiguity 59, 186
Amendments 58, 142, 144, 150, 203, 300, 303
Analogy 1, 3, 31, 88
Anonymous 13, 20, 60, 286, 288, 297, 299
Anthropomorphic 66
Antitechnological 49
Appellations 126

Applicability 10, 13, 22, 136
Approaches 4, 5, 8, 160, 162, 164, 193, 198, 212, 214, 215, 244, 273, 281, 286, 288
Architecture 21, 22, 164, 177, 187, 216, 235, 236, 237, 238
Assimilation 319
Asymmetry 42, 117
Attribution 297
Authority 6, 7, 14, 32, 35, 52, 83, 94, 96, 97, 99, 139, 143, 149, 204, 205, 213, 218, 295, 303, 304
Automation 22, 168, 182, 270
Autonomous 15, 16, 18, 23, 99, 164

B

Biopiracy Patent 36
Bipartisan 41, 42
Botnets 276
Bourgeois 31
Burdening 162
Bureaucratic 48, 49, 206

C

Capitalism 29, 64
Cartelisation 206
Civilization 19, 85, 201
Clusters 85
Combating 279
Communication 1, 5, 10, 11, 20, 93, 98, 146, 156, 162, 166, 177, 180, 181, 182, 183, 184, 185, 186, 187, 188, 190, 191, 194, 202, 206, 221, 223, 228, 233, 234, 235, 237, 238, 239, 240, 252, 253, 254, 257, 260, 261, 272, 278, 281, 283, 288, 302, 305, 306, 314
Community 2, 15, 16, 17, 18, 22, 23, 33, 83, 84, 85, 86, 87, 89, 90, 91, 92, 93, 95, 96, 97, 128, 130, 137, 138, 154, 156, 193, 196, 198, 201, 275, 300, 315, 317, 319
Comparable 13, 315
Conceptual 44, 50, 51, 152, 190
Conflict 2, 18, 33, 49, 74, 81, 82, 85, 90, 95, 96, 97, 113, 144, 164, 200, 201, 204, 207, 209, 211, 212, 215, 245, 289, 290
Congressional 300
Conservative 94, 282
Conspiracy 8, 282, 305
Consumer 204, 205, 206
Controversial 20, 21, 147, 202, 269
Controversy 147, 314

Conventional 21, 57, 253, 256, 259, 260, 261, 302

Cyber 1, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20, 22, 23, 76, 77, 103, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 166, 167, 193, 194, 195, 196, 198, 199, 201, 202, 214, 263, 264, 265, 266, 267, 268, 269, 270, 271, 274, 275, 276, 277, 278, 279, 280, 285, 289, 294, 295, 296, 297, 298, 299, 300, 301, 302, 305, 308, 309, 311, 312, 313, 315

Cyberspace 3, 19, 20, 21, 22, 23

Cyberutopianism 49

D

Demographic 195, 197, 223

Deterioration 12

Dignity 292

DMCA 35, 36

E

Electromagnetism 136

Embedded 155, 159, 160, 162, 297

Embraced 18

Emissions 10, 12

Empirical 92, 123, 141, 255

Encryption 288, 300, 301

Engaging 304, 305

Epiphenomenal 47, 51

Espionage 4, 7, 8, 9, 10

Europe 268, 288, 289

European 286, 287, 315

Explicit 23, 86, 89, 91, 92, 96, 143, 144, 149

Exploit 68, 91, 158, 159, 166, 197, 224, 226, 264, 271, 272, 274, 275, 279, 296, 297, 314

F

Facilitators 257, 258

Farsighted 160

Federal 26, 30, 41, 54, 60, 61, 103, 104, 157, 161, 162, 193, 194, 195, 208, 211, 222, 247, 265, 278, 301, 302, 304, 305, 309, 310, 311, 312

Forensics 279, 312

Fraudulent 268, 275

G

Gambling 290, 291

GDP 131
Geistigen 24
Glorification 288

H

Hacking 75, 103
Hegelia 137, 141
Hostile 157, 159
Humanitarian 36, 49

I

Illegal 4, 5, 6, 7, 9, 10, 11, 14, 18, 19, 35, 53, 73, 101, 172, 174, 175, 200, 201, 209, 214, 215, 216, 271, 279, 286, 289, 290, 291, 293, 294, 295, 298, 302
Implementation 9, 276, 288, 301
Implications 19, 47, 115, 121, 202, 208, 215, 232, 318
Implicitly 41
Intellectual 18, 21, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 36, 37, 39, 40, 42, 44, 46, 47, 50, 52, 53, 60, 63, 64, 65, 67, 73, 80, 104, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 144, 147, 151, 177, 203, 244, 266, 300, 308, 309, 311
Interception 7, 9, 10, 11, 12, 186
Interplay 161
Interpretation 2, 44, 45, 49, 58, 59, 68, 133, 137, 138, 209
Invasion 299
Investigative 311, 312

J

Jurisdiction 1, 18, 19, 21, 22, 29, 96, 98, 113, 126, 198, 200, 202, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 289, 300, 305, 309, 312

L

Legislation 5, 12, 30, 33, 46, 58, 64, 69, 118, 133, 134, 145, 146, 147, 148, 149, 150, 151, 217, 218, 270, 276, 278, 279, 291, 298, 312, 313
Legitimate 7, 13, 44, 46, 47, 130, 133, 144, 267, 268, 271, 306, 314
Liability 17, 82, 163, 167, 168, 195, 196, 216, 219, 240, 243, 297

M

Malicious 13, 158, 201, 266, 267, 268, 269, 271, 272, 273, 275, 279, 281, 284, 292, 299, 309
Manifest 36, 39, 141

Mitigation 201
Mobilization 34, 36, 40, 41, 43, 48, 49, 50, 51
Monopolization 109, 110, 136
Motivation 138, 140, 244, 247, 264

N

Nazirelated 289
Negligence 219

O

Obtaining 277
Organizations 270, 303, 308, 311, 312, 314

P

Paradoxically 43, 48
Pessimistic 51
Pharmaceutical 34, 35, 46, 47, 48, 109, 110, 113, 114, 116, 119, 124, 130, 148, 149, 150
Pornographic 286, 287
Pornography 286, 287, 288, 290, 309, 312
Potential 5, 13, 16, 19, 20, 21, 22, 23, 34, 39, 55, 56, 57, 69, 71, 73, 75, 79, 82, 86, 87, 91, 92, 93, 96, 97, 115, 125, 134, 137, 160, 171, 178, 198, 202, 224, 225, 226, 229, 230, 235, 236, 243, 249, 252, 254, 265, 266, 270, 271, 273, 275, 285, 290, 292, 295, 299, 309
Preventing 286, 287, 288, 313

R

Reduced 159
Religious 3, 93, 97, 286, 289, 290
Rigorous 158

S

Skepticism 136
Sniffing 283, 284
Software 1, 15, 16, 17, 18, 19, 21, 23, 75, 77, 78, 79, 80, 102
Sophisticated 4, 202, 271, 308, 312
Sovereignty 277
Spamming 282, 284
Stakeholders 205
Strategies 276

T

Tangible 12, 13

Tarnished 309

Taxonomy 231, 255

Technology 3, 9, 22, 35, 39, 40, 42, 52, 55, 65, 76, 78, 79, 80, 81, 82, 85, 112, 114, 115, 117, 119, 120, 121, 122, 123, 124, 125, 127, 132, 133, 147, 148, 149, 153, 154, 155, 156, 158, 161, 162, 168, 169, 178, 179, 185, 193, 197, 198, 199, 210, 221, 222, 226, 230, 233, 235, 236, 237, 238, 247, 250, 252, 255, 256, 279, 280, 281, 285, 286, 287, 288, 292, 293, 294, 295, 301, 302, 306, 309, 310, 312, 314, 316, 319, 320

TRIPs 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 125, 128, 129, 130, 133, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151

Typology 252, 268

U

UPOV 36

Utilitarian 26, 28, 68, 135, 137, 138, 139, 141

V

Validation 288

Victimization 307, 308, 311, 312

Violations 4, 286, 287, 303, 305

Virtual 13, 15, 22, 187, 234, 240, 275, 288, 297

Vulnerable 156, 158, 159, 294, 297

W

WIPO 35, 36, 39, 41, 121, 127, 128, 129